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MasterCard
International

Information about this Total Replacement

Total Replacement

The enclosed June 2003 *MDS Online Specifications* manual completely replaces your existing manual.

What is in the new version?

This new version reflects changes effective June 2003.

Please refer to:

- "Summary of Changes" for a comprehensive list of changes reflected in this update.
- "Using this Manual" for a complete list of the contents of this manual.

Billing

MasterCard will bill principal members for this document. Please refer to the *MasterCard Consolidated Billing System Manual* for billing-related information.

Questions?

If you have questions about this manual, please contact the Global Member Operations Support team or your regional help desk. Please refer to "Using this Manual" for more contact information.

MasterCard is Listening…

Please take a moment to provide us with your feedback about the material and usefulness of the *MDS Online Specifications* manual using the following e-mail address:

publications@mastercard.com

We continually strive to improve our publications. Your input will help us accomplish our goal of providing you with the information you need.

MasterCard International

Summary of Changes

***MDS Online Specifications*, June 2003**

Change Summary	Description of Change	Where to Look
New Publication Date	The next publication date of this manual is February 2004.	NA
Contact Information Update	Updated all US Debit Operations Support contact information (phone, email, fax)	All Chapters
GCMS Updates	Updated the settlement information to include GCMS information	All Chapters
New Message Description	Added a description of a Financial Transaction Request Response/0210 message received from an issuer that fails MDS edit checks and contains a format error	Chapter 2
Cash Back Additions	Included new U.S. Maestro Cash Back information	Chapter 3
DE 29, DE 31, and DE 97 Changes	The MDS online messages no longer include DE 29 and DE 31. Changed the contents of DE 97 within 05xx messages	Chapter 3 Chapter 4
DE 60 Clarification	Clarified Exception Advice (DE 60) for Acquirer Reversal Advice/0420 messages and Issuer Reversal Advice/0422 messages	Chapter 4
DE 38 Clarification	Clarified Authorization Identification Response DE 38	Chapter 4
Authorization Enhancements	Authorization enhancements include changes to DE 48 to support subelement 90 and subelement 95.	Chapter 4
Encryption Key Clarification	Clarified the Dynamic Working Key exchange and the Communication Key information	Chapter 6

MasterCard International

MDS Online Specifications

June 2003

Media

This document is available:

- On MasterCard OnLine®
- On the *MasterCard Electronic Library (CD-ROM)*
- On the *MDS Operations Suite (CD-ROM)*
- In printed format

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Using this Manual

Chapter 1 Implementation Planning

Chapter 2 ISO 8583–1987 Message Definition and Transaction Flow

Chapter 4 ISO 8583–1987 Data Element Definitions

Chapter 7 Database Forms

Appendix A North American State, Province and Territory Codes

Appendix B Country and Currency Codes

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Forms

Using this Manual

This chapter contains information that helps you understand and use this document.

Purpose

The MasterCard *MDS Online Specifications* is one of the four manuals comprising the MasterCard® Debit Switch Suite. The MasterCard® Debit Switch Suite consists of:

- *MDS Programs and Services* manual
- *MDS Settlement and Reports* manual
- *MDS Online Specifications* manual
- *NICS™ User's Guide*

The *MDS Online Specifications* manual serves as one of the primary technical references for all debit programs and services supported by MasterCard. It includes information relating to MasterCard's debit value-added services including ISIS, Stand-In, Member Level Settlement, and Gateway Services.

Audience

MasterCard provides this manual for members and their authorized agents. Specifically, the following personnel should find this manual useful:

- MasterCard members directly connected to the MasterCard® Debit Switch
- Third party processors directly connected to the MasterCard® Debit Switch

Overview

The following table provides an overview of this manual:

Chapter	Description
Table of Contents	A list of the manual's chapters and subsections. Each entry references a chapter and page number.
Using this Manual	A description of the manual's purpose and its contents.
1 Implementation Planning	An overview of all implementation tasks and testing requirements, including a task list to aid new participants in planning a successful and timely implementation of services.
2 ISO 8583–1987 Message Definition and Transaction Flow	Illustrates and describes the various message types used for transaction processing.
3 ISO 8583–1987 Message Formats	Identifies all of the required, conditional, optional, or switch-generated data elements within each individual ISO 8583–1987 message.
4 ISO 8583–1987 Data Element Definitions	Provides a detailed definition of all data elements used in online application messages.
5 Communication Protocols	Illustrates and defines options and requirements for establishing a link to the MasterCard® Debit Switch.
6 Encryption	Illustrates and defines procedures and requirements for key generation, key maintenance, and PIN encryption.
7 Database Forms	Defines the procedures for completing the forms required to create and maintain member records in the MasterCard® Debit Switch database.
A North American State, Province and Territory Codes	A list of state, province and territory codes of North America by state, province and territory name.
B Country and Currency Codes	A current list of country codes, currencies, and currency codes.
C Merchant Category Codes	A current list of merchant category codes (MCCs) assigned by MasterCard.
Glossary	A dictionary of terms and acronyms used by MasterCard in the MasterCard® Debit Switch.
Forms	The forms members use with the *MDS Online Specifications* manual.

Excerpted Text

At times, this document may include text excerpted from another document. A note before the repeated text always identifies the source document. In such cases, we included the repeated text solely for the reader's convenience. The original text in the source document always takes legal precedence.

Language Use

The spelling of English words in this manual follows the convention used for U.S. English as defined in *Merriam-Webster's Collegiate Dictionary*. MasterCard is incorporated in the United States and publishes in the United States. Therefore, this publication uses U.S. English spelling and grammar rules.

An exception to the above spelling rule concerns the spelling of proper nouns. In this case, we use the local English spelling.

Times Expressed

MasterCard is a global company with locations in many time zones. The MasterCard operations and business centers are in the United States. The operations center is in St. Louis, Missouri, and the business center is in Purchase, New York.

For operational purposes, MasterCard refers to time frames in this manual as either "St. Louis time" or "New York time." Coordinated Universal Time (UTC) is the basis for measuring time throughout the world. You can use the following table to convert any time used in this manual into the correct time in another zone:

	St. Louis, Missouri USA Central Time	**Purchase, New York USA** Eastern Time	**UTC**
Standard time (last Sunday in October to the first Sunday in April [a])	9:00	10:00	15:00
Daylight saving time (first Sunday in April to last Sunday in October)	9:00	10:00	14:00

[a] For Central European Time, last Sunday in October to last Sunday in March.

Revisions

MasterCard periodically will issue revisions to this document as we implement enhancements and changes, or as corrections are required.

With each revision, we include a Summary of Changes describing how the text changed. Revision markers (vertical lines in the right margin) indicate where the text changed.

Occasionally, we may initially publish revisions or additions to this document in the Global *Deposit Access Bulletin* or other bulletin. Revisions announced in another publication, such as a bulletin, are effective as of the date indicated in that publication, regardless of when the changes are published in this manual.

Related Information

The following documents and resources provide information related to the subjects discussed in this manual. Please refer to the *Quick Reference Booklet* for descriptions of these documents.

- *MDS Programs and Services* manual
- *MDS Settlement and Reports* manual
- *Banknet Systems Manual*
- *Chargeback Guide*
- *Cirrus® Worldwide Operating Rules*
- *Global Deposit Access Implementation Guide for Maestro and Cirrus*
- *Maestro® Global Rules*
- *MasterCard Test Simulator - Debit, User Guide*
- *MasterCard Member Testing Procedures - Debit*
- *Settlement Manual*
- *MasterCard Consolidated Billing System Manual*
- *NICS™ User's Guide*

Debit members that also process transactions using the Authorization (01xx) message format should reference the above manuals and the following manuals:

- *Authorization System Manual*
- *Account Management User Manual*

In addition to the documents listed previously in this chapter, the following international standards publications may also be useful if you are implementing a new ATM or POS program or if you are in the process of converting to the MasterCard® Debit Switch ISO-8583 CIS online message format interface.

- ISO 7810–1985: Identification Cards—Physical Characteristics
- ISO 7811–1985: Identification Cards—Recording Technique
- ISO 7812–1987: Identification Cards—Numbering System and Registration Procedure for Issuer Identifiers
- ISO 7813–1990: Identification Cards—Financial Transaction Cards
- ISO 3166–1988: Codes for the Representation of Names of Countries
- ISO 4217–1990: Codes for the Representation of Currencies and Funds
- ISO 9564–1991: Banking-Personal Identification Number Management and Security Part 1: PIN Protection Principals and Techniques

To order MasterCard manuals, please use the Ordering Publications service on MasterCard OnLine®, or contact the Global Member Operations Support team.

Support

Please address your questions to the Global Member Operations Support team as follows:

Phone:	1-800-999-0363 or 1-636-722-6176 1-636-722-6292 (Spanish Language support)	
Fax:	1-636-722-7192	
E-mail:	Canada, Caribbean, and U.S.	member_support@mastercard.com
	Asia/Pacific	apms@mastercard.com
	Europe	css@mastercard.com
	South Asia/Middle East/Africa	emeaap@mastercard.com
	Latin America (Spanish Language support)	lagroup@mastercard.com
Address:	MasterCard International Incorporated Global Member Operations Support 2200 MasterCard Boulevard O’Fallon MO 63366-7263 USA	
Telex:	434800 *answerback*: 434800 ITAC UI	

Member Relations Representative

Member Relations representatives assist U.S. members with marketing inquiries. They interpret member requests and requirements, analyze them, and if approved, monitor their progress through the various MasterCard departments. This does not cover support for day-to-day operational problems, which the Global Member Operations Support team addresses.

To find out who your U.S. Member Relations representative is, contact your local Member Relations office:

Phone: 1-404-459-2400 (Atlanta)

1-847-375-4000 (Chicago)

1-914-249-2000 (Purchase)

1-925-866-7700 (San Francisco)

Regional Representative

The regional representatives work out of the regional offices. Their role is to serve as intermediaries between the members and other departments in MasterCard. Members can inquire and receive responses in their own language and during their office's hours of operation.

To find out the location of the regional office serving your area, call the Global Member Operations Support team at:

Phone: 1-800-999-0363 or 1-636-722-6176

1-636-722-6292 (Spanish Language support)

1 Implementation Planning

This chapter provides an overview of the project planning and review process for new and existing participants.

Overview

This chapter provides an overview of the project planning and review process for new and existing participants.

An essential component to successful implementation is the review of the project among regional MasterCard/Maestro/Cirrus staff and all appropriate personnel within your organization. For best results, schedule this review early in the project-planning phase.

Note **Before initiating the planning process, MasterCard strongly encourages members to contact their MasterCard Regional Office to identify the requirements related to a particular project.**

The following steps should occur during the review meeting:

- MasterCard will answer all general questions regarding the MasterCard® Debit Switch (MDS) or this implementation guide.
- MasterCard and your organization will establish a detailed project plan with due dates and responsibilities to ensure that all parties agree on the process components.
- MasterCard and your organization will review and answer detailed technical/operational questions and establish specific timeframes for testing.
- MasterCard and your organization will schedule additional review sessions, as necessary. These additional review sessions will guarantee consistent, clear communication regarding the status of the implementation project, and generate a timely response to issues that may arise.

Implementation Task List

Use the Implementation Task List provided in the back of this chapter as a guide or tool when reviewing the project requirements with the MasterCard regional debit support staff. Incorporate these tasks directly into the Project Management process. The Implementation Task List provided is not all-inclusive, and not all tasks listed will apply to every implementation.

Objectives of Acceptance Testing

An integral part of the implementation plan is the system testing of all MDS processors. The MDS, as the provider of routing, message translation, and settlement services, tests all processors' system environments before implementation, to ensure the overall integrity of the MDS environment.

Each processor will receive the standard acceptance test scripts from the MasterCard Regional Office and their Product and Member Implementations Specialist. These scripts will assist each processor in complying with the standard processing requirements of the MDS. Using these scripts, makes it possible to test all conditions expected in the MDS production environment. MasterCard will review the completed acceptance tests, including receipt of test transactions and the documented results of each test transaction, to ensure that the processor meets all processor testing requirements.

Note **MasterCard will not allow any processor to connect to the MDS unless that processor has completed the minimum acceptance test requirements.**

MasterCard provides processor testing as a continuing service of the MDS. To test each processor's online interface, all processors will be required to accept acquirer and issuer processor functionality. All processors can use the test facilities to re-test their internal system changes and by scheduling convenient times with your MasterCard Regional Office, the Debit Payment Systems—Product and Member Implementations Specialist, or both.

MasterCard® Debit Switch Test Environment

The MDS maintains a test platform separate from the production system. This test environment enables processors to perform transaction processing with parameters and timeframes that mirror the production system. Settlement cut-over is included in cycle testing to give processors the opportunity to complete batch processing and balance back to the MDS before going live.

Testing Requirements

To obtain further information on test requirements for implementation, contact the MasterCard Regional Office assigned to your institution or the Product and Member Implementations Specialist assigned to your project.

Online Testing

Members use the MDS Test Facility to complete the testing process. After completing the initial testing with the simulator, members should have resolved any message format and procedural issues.

Background

The process ensures that online testing with a Product and Member Implementations specialist focuses on the message routing and processing aspects of testing, and testing the member interface to the MDS. Online testing also creates settlement reports and files allowing members to test file transfer and processing.

Requirements for New Members

All new members that will have a direct connection to the MDS network must complete online testing with the MDS Test Facility. Required online testing includes:

- Financial functionality for member-supported products and services
- Validation of processing settlement reports and files
- Reject Reason Code Mapping
- Administrative Messages

Warning **During acceptance testing, members must not make changes to their hardware or software environments. Any changes made by members will require starting the testing process over from the beginning.**

Requirements for Existing Members

Existing members have the option to conduct testing with the MDS Test Facility at any time. They can use the test facility to conduct authorization testing for MDS release changes or any changes to their authorization interfaces with MasterCard.

Members also can use the MDS Test Facility to test MDS release changes and other file transmissions between members and MasterCard.

Scheduling Test Time

To schedule test time, contact the MasterCard Regional Office assigned to your institution or the Product and Member Implementations Specialist assigned to your project.

Canceling Test Time

To cancel scheduled test time, contact the MasterCard Regional Office assigned to your institution or the Product and Member Implementations Specialist assigned to your project.

Note **Simulator testing is required before online testing can occur. For further information on member testing procedures, scheduling, canceling, or pricing, review the Global Deposit Access Bulletin No. 2, March 2000.**

Processor Debit Switch Test Environment

MasterCard highly recommends that processors establish a separate test platform for online testing with the MDS. This platform should consist of the following:

- One of each terminal type supported in the production environment and one host processor.
- A host processor capable of communicating directly with the MDS by using separate telecommunication lines and modems. The MDS can provide separate ports on a member MIP and maintain dial-up modems for testing. All processors must coordinate dial-up testing with MasterCard.
- Terminals and host that have the same functionality as the production system, and the capability to simulate production routing and authorization processes.
- Full batch processing functionality to test reconciliation, settlement, and reporting functions.

To ensure compliance with any applicable government regulations, MasterCard recommends verification of transaction activity (up to and including a customer's statement.)

Electronic Fund Transfer (EFT) Services

As a provider of Electronic Fund Transfer (EFT) Services, all processors must successfully meet and complete all test requirements before becoming "live" in the MDS production system.

Acceptance Testing Requirements

Each processor is required to successfully complete acceptance testing as an acquirer processor, an issuer processor, or both. A processor can only process transactions on the production system for which they have completed acceptance testing.

The acceptance process requires two consecutive error-free days of testing, therefore, testing cannot begin on a Friday. Failure of any portion of the testing requires the restarting of the entire testing process.

The processor must test only those transaction types supported by both the processor and the MDS, before the processor can participate in the MDS production environment.

Simulator Testing Requirements for New Members

New members can use the simulator to test their systems for compliance with MasterCard's processing requirements. All new members that will have a direct connection to the MDS must purchase the MasterCard Test Simulator—Debit software package.

MasterCard requires members to perform simulator testing before conducting online testing. New members must use the simulator to run predetermined scripts to test their issuing and acquiring host systems.

Members also must submit simulator trace files to Product and Member Implementations staff for review. If the trace files are error-free and meet testing requirements, MasterCard Product and Member Implementations staff will confirm the successful completion of simulator testing, and arrange for members to conduct online testing.

New Processor Test Scripts

Processors connecting to the MDS for the first time are required to complete all applicable acceptance test scripts. Members can obtain these scripts during the project planning phase. MasterCard makes every attempt to test the full variety of conditions that will occur in an operations environment in order to minimize production problems when the processor begins "live" processing. Each script requires the processor to accommodate specific conditions.

Personnel assigned to perform acceptance testing should carefully review the appropriate test plan to ensure that all identified conditions can be produced during any given test day. Conditions that cannot be accommodated in a timely manner during the testing process, must be brought to the attention of the MasterCard Regional Office, and the Product and Member Implementations Specialist.

The Product and Member Implementations Specialist will determine if the problem identified by the processor makes it unable to satisfy all acceptance testing requirements. The Specialist will document any variances. Significant variances from the approved test plan will only be allowed if the MasterCard Regional Office and the Product and Member Implementations Specialist give approval.

Note **Members must contact the MasterCard Regional Office and the Product and Member Implementations Specialist to initiate all scheduled test sessions. Therefore, members must ensure that their testing personnel have the necessary facilities and authority to dial-out from their Test Data Center.**

Simulator Testing Requirements for Existing Members

Existing members already should have the MasterCard Debit Financial Simulator software package. MasterCard strongly encourages members to use the simulator for all optional MDS release testing before conducting any online testing. MasterCard may **require** simulator testing for certain MDS releases.

MasterCard must verify simulator testing for all member-initiated system changes that may impact the member's interface to the MDS. Also, any existing members that implement new products or services, must complete appropriate simulator testing before performing any online testing.

Testing Multiple Terminals

If multiple terminals are available, the processor should perform test transactions simultaneously on multiple terminals to emulate a "live" environment.

Testing Unique Requirements

Testing of all applicable transaction processing scenarios for a specific processor may require the addition, modification, or deletion of test cases. All additions and deletions of test cases must be coordinated with your MasterCard Regional Office or your Product and Member Implementations Specialist. Your MasterCard contact will generate custom acceptance test scripts for all processors having unique requirements.

Test Card Requirements

Issuer Processor Testing

The processor must create a series of test card Track 2 data and PINs, to test and analyze multiple test case scenarios (both valid and invalid). Each processor must supply the MDS with a series of test card Track 2 layouts with PINs, for a single financial institution for which it performs issuer processing services.

Note **Processors must be sure to block test card information from use in the production system.**

Issuer processors should use the sample table included in the Issuer Functionality Form to provide the required Track 2 data to the MDS. Use the following rules to determine the data requirements for a particular card:

- Processors that do not support balance inquiry (BI) transactions should not provide data for any cards whose identifier begins with "BI."
- Processors should not provide test data for cards related to unsupported account types. For example, processors that do not support the savings account type are not required to provide data for any of the test cards with "SAV" in their identifier.

Note **All of the "VALID" cards relate to all account types, and processors must always provide Track-2 data and PINs for these cards.**

- Processors must provide test PINs in the PIN column for each test card required for acceptance testing.

Test Card Master Listing

The master listing describes any special conditions attached to a card. It also lists the type of balance (such as positive or negative, available or ledger account balance) associated with cards to be used for balance inquiry (BI) transactions. Each card, identified by a name, is associated with one or more cardholder account types (such as checking, savings, or credit card accounts), as indicated by the master list. To obtain the master listing of all test cards used to generate transactions during acceptance testing, contact the MasterCard Regional Office.

Acquirer Processor Testing

For acquirer processing systems MasterCard will provide Track 2 information. During the project planning phase, specific information will be available. Processors should contact their Regional Office or their Product and Member Implementations specialist.

Recommended ATM Screen Sets

The appropriate use of issuer-generated response codes is critical to accurately communicate to the cardholder the reason for the denial. Misrepresentation or a lack of information for the cardholder increases their frustration and impacts usage. This also affects processor performance because multiple declines occur when the cardholder does not receive meaningful feedback. Issuers should use the mapping of recommended screen messages as shown in Table 1.1.

It is the acquirer's responsibility to describe clearly to the cardholder the intent of the action taken by the issuer. Acquirers should use the following information to ensure that they are interpreting the response codes accurately.

ISO 8583 (1987) Financial Transaction Request/0200 Response Message Format

Response Codes are in DE 39 of the ISO 8583 (1987) Financial Transaction Request/0200 message format. The table below provides the valid response codes for the Financial Transaction Request/0200 message format as used by the MDS. It also lists definitions, expected acquirer actions, and examples of recommended English language screen messages. (Please contact your MasterCard Regional Office for examples in other languages.)

Table 1.1—ISO 8583 (1987) Financial Transaction Request/0200 Format—Financial Transaction Request Response/0210 Messages Response Code Mapping

Code	Response Code Definitions	Issuer Usage	Acquirer Action	Recommended ATM Screen Message
00	Approved or completed successfully	Transaction request approved	Approve	
04	Capture card	Transaction request declined The acquirer should retain the card. No reason is provided for this action.	Capture	• Your card has been retained. Please contact your card issuer. Or, • Your card issuer has declined your request and has instructed me to retain your card. Please contact your card issuer.

Code	Response Code Definitions	Issuer Usage	Acquirer Action	Recommended ATM Screen Message
12	Invalid transaction	Transaction request declined The transaction request is not supported or is not valid for the BIN. The MDS uses this response code exclusively. Card issuers or Intermediate Network Facilities (INF) should use response code 57.	Decline	• I am sorry you have selected an invalid transaction. Do you want to try another transaction? Or, • I am sorry you have selected an invalid transaction. Please try a different transaction type.
13	Invalid amount	Transaction request declined The requested amount is below the minimum limit set by the issuer for the type of transaction requested.	Decline	• You have selected an invalid amount. Please select amount in multiples ________ Or, • Your card issuer has declined your request because the amount requested is invalid. Please try a greater amount.
14	Invalid card number	Transaction request declined The presented card number is not valid on the issuer's file.	Decline	• I am sorry I am unable to process your request. Please contact your card issuer. Or, • Your request is declined because your card issuer did not recognize your card number.
15	Invalid issuer	Transaction request declined The transaction request contains a BIN that unsupported. This response code is valid for MDS usage only.	Decline	• I am sorry I am unable to process your request. Please contact your card issuer. Or, • Your transaction request is declined because your card is not supported at this location.
30	Format error	Transaction request declined Improper format	Decline	I am sorry I am unable to process your request. Please contact your card issuer.
41	Lost card	Transaction request declined The acquirer should retain the card. This is a reported lost card.	Capture	• Your card has been retained. Please contact your card issuer. Or, • Your card issuer has declined your request and has instructed me to retain your card because it has been reported lost. Please contact your card issuer.

Code	Response Code Definitions	Issuer Usage	Acquirer Action	Recommended ATM Screen Message
43	Stolen card	Transaction request declined The acquirer should retain the card. This is a reported stolen card.	Capture	• Your card has been retained. Please contact your card issuer. Or, • Your card issuer has declined your request and has instructed me to retain your card because it has been reported stolen. Please contact your card issuer.
51	Insufficient funds/over credit limit	Transaction request declined The request will result in an over credit limit or insufficient funds condition.	Decline	• I am unable to process for insufficient funds. Please contact your card issuer. Or, • Your request is declined due to insufficient funds. Please contact your card issuer.
54	Expired card	Transaction request declined The card number presented is expired.	Decline	Your request is declined because your card has expired. Please contact your card issuer.
55	Invalid PIN	Transaction request declined The cardholder-entered PIN is incorrect.	Decline	• You have entered your PIN incorrectly. Do you want to try again? Or, • Your request is declined because the PIN you entered is incorrect.
57	Transaction not permitted to issuer/ cardholder	Transaction request declined The card issuer or INF declines the transaction request because it is not supported or is not permitted for the card number presented. The MDS generates response code 12 when a transaction is declined for invalid transaction selection.	Decline	• I am sorry you have selected an invalid transaction. Do you want to try another transaction? Or, • Your request is declined because it is not supported. Please try a different transaction type.
61	Exceeds withdrawal amount limit	Transaction request declined The withdrawal amount is in excess of daily defined maximums.	Decline	• You have exceeded the withdrawal limit. Do you want to select another amount? Or, • You have exceeded the daily withdrawal limit. Please contact your card issuer.

Code	Response Code Definitions	Issuer Usage	Acquirer Action	Recommended ATM Screen Message
62	Restricted card	Transaction request declined The card number has been restricted for the type of use requested.	Decline	• I am sorry I am unable to process your request. Please contact your card issuer. Or, • Your request has been denied by your card issuer because your card has been restricted. Please contact your card issuer.
75	Allowable number of PIN tries exceeded	Transaction request declined The cardholder has incorrectly entered the PIN in excess of the allowable number of tries established by the issuer.	Decline	You have exceeded the amount of times you can enter your PIN. Please contact your card issuer.
78	Invalid or nonexistent account specified (general)	Transaction request declined The from (debit) or to (credit) account specified in the transaction is non-existent or is not associated with the card number presented. issuers should use this response code to decline a transfer request because of a nonexistent or invalid account.	Decline	Your request is declined because the account selected is invalid.
79	Invalid business date	Transaction request declined The business date is not valid. This response code is valid for MDS usage only.	Decline	I am sorry I am unable to process your request. Please contact your card issuer.
80	System not available	Transaction request declined The issuer system is not available. INF can generate this response code when the issuer systems are down, or when it cannot complete a transaction request because the issuer's applications or files are not available.	Decline	• I am sorry I am unable to process your request. Please contact your card issuer. Or, • Your request is declined because your card issuer's systems are not available. Please try again later.

Code	Response Code Definitions	Issuer Usage	Acquirer Action	Recommended ATM Screen Message
91	Destination processor (customer processor system [CPS] or INF) not available	Transaction request declined The destination processor is not available. The MDS generates this response code when it cannot deliver a transaction request because the destination processor does not have on online connection to the MDS.	Decline	• I am sorry I am unable to process your request. Please contact your card issuer. Or, • Your request is declined because your card issuer's systems are not available. Please try again later.
92	Unable to route transaction	Transaction request declined Insufficient information for routing	Decline	I am sorry I am unable to process your request. Please contact your card issuer.
96	System error	Transaction request declined A system failure has occurred or the files required for the authorization are not available.	Decline	I am sorry I am unable to process your request. Please contact your card issuer.

2 ISO 8583–1987 Message Definition and Transaction Flow

This chapter provides a definition of all ISO 8583–1987 message types employed by the MasterCard® Debit Switch (MDS).

Overview

This chapter of the *MDS Online Specifications* manual provides a definition of all ISO 8583–1987 message types employed by the MasterCard® Debit Switch (MDS). Also included are transaction flow diagrams that illustrate both standard and exception (such as error condition) message processing requirements at the online application programming level.

This chapter also discusses the basic transaction processing techniques employed by the MDS in its implementation of the ISO 8583–1987 message standard. It provides processors with a general overview of the underlying logic for all transaction flow scenarios.

Summary of Message Types Supported

The MasterCard® Debit Switch (MDS) supports the online message types listed in Table 2.1.

Table 2.1—Message Types

Code	Description
Authorization (01xx) Messages	
Supported for credit card issuer-only processing. The MDS passes Authorization Request/01xx messages to and from the Banknet network on behalf of Banknet processors; however, the MDS only communicates with MDS processors using the Financial Transaction/02xx message formats.	
Financial Transaction (02xx) Messages	
0200	Financial Transaction Request
0210	Financial Transaction Request Response
0280	Financial Transaction Acknowledgment [a]
0290	Financial Transaction Negative Acknowledgment
0220	Financial Transaction Advice
0230	Financial Transaction Advice Response
File Update (03xx) Messages	
0302	File Update Request
0312	File Update Request Response
Reversal Advice (04xx) Messages [b]	
0420	Acquirer Reversal Advice
0430	Acquirer Reversal Advice Response
0422	Issuer Reversal Advice
0432	Issuer Reversal Advice Response
Reconciliation Advice (05xx) Messages	
0520	Acquirer Reconciliation Advice
0530	Acquirer Reconciliation Advice Response
0522	Issuer Reconciliation Advice
0532	Issuer Reconciliation Advice Response

Code	Description
Administrative Advice (06xx) Messages	
0620	Administrative Advice
0630	Administrative Advice Response
0644	Administrative Advice [c]
Network Management (08xx) Messages	
0800	Network Management Request
0810	Network Management Request Response
0820	Network Management Advice

[a] The MDS does not support the Financial Transaction Acknowledgment/0280 message for new MDS member implementations. Existing member support is available until members convert their systems.

[b] The word reversal is often used generically to mean any change or exception made to an original transaction, and the name of the 042x message type, i.e., "Reversal Advice," suggests this generic meaning is the appropriate usage. However, a 042x Reversal Advice message can contain one of the following:

- An acquirer-generated reversal due to a terminal error or to cancel a financial request at the terminal.
- An acquirer-generated adjustment to the original request, made on a subsequent day using one of the MDS adjustment processes.
- An issuer-generated chargeback made using one of the MDS adjustment processes.
- An acquirer-generated representment made using one of the MDS adjustment processes.
- The common understanding is a reversal (as a specific kind of transaction exception request) occurs from the acquirer closely following the original transaction, and an adjustment is a non-automatic exception made on a subsequent day by a processor using one of the MDS adjustment processes.

[c] Debit Pass-through only for members connected to the Banknet® telecommunications network.

Authorization (01xx) Message Definitions

This message series is only available for MasterCard credit card issuer-only processors.

An issuer processing system (IPS) that is not planning on adding acquirer capabilities may choose to have all its transactions processed as credit card "cash advances" using Authorization Request/0100 and Authorization Response/0110 messages.

Authorization (01xx) messages do not contain sufficient information to post to a cardholder's account at the IPS. Processors post through receipt of a Global Clearing Management System (GCMS) batch file at the end of the business day.

IPS processors using the authorization (01xx) messages have the option of electing Stand-In PIN verification using the MDS and Stand-In authorizations through MasterCard St. Louis (available only if PIN verification has been selected). (Refer to chapter 2 of the *MDS Programs and Services* manual, for further discussion of this option.)

The network will handle any subsequent exception and reversal situations through the GCMS system for MasterCard credit card issuer-only processors.

Note **Authorization (01xx) messages may support balance inquiry transactions. The MDS Authorization Request/0100 message interface process (the MDS interface to the Banknet network and its credit processors) does not support Reconciliation Advice messages. Reversal Advice/04xx messages are not sent outbound by the MDS to the 0100 interface, however, reversals can be received inbound by the MDS, through the 0100 interface, from a debit MasterCard acquirer.**

Refer to the *Customer Interface Specification* manual, for further discussion on message formats and definitions for the ISO 8583–1987 Authorization (01xx) messages.

Financial Transaction (02xx) Message Definitions

The term “financial transaction” applies when there is sufficient data contained within the individual transaction message to provide actual posting of accounts at the issuer processing system (IPS). The MDS processes all financial transaction (02xx) messages under the assumption that when the transaction is successfully completed, no subsequent information (such as “paper” or “transaction tickets”) is required to perform actual cardholder account posting and cardholder billing. Financial transaction (02xx) messages are used for the following:

- Financial transaction requests
- Financial transaction responses
- Financial transaction response acknowledgments
- Financial transaction advices

The following information lists the definitions of all ISO 8583–1987 financial transaction (02xx) messages supported for the MasterCard® Debit Switch.

Financial Transaction Request/0200 Message

Type: Interactive

Routing: From an acquirer to the MDS

From the MDS to an issuer

Purpose: Requests approval of a transaction that, if approved, will permit the application of the transaction financial data to the cardholder’s account for issuing a bill or statement.

Response: A Financial Transaction Request Response/0210 message is required.

Financial Transaction Request Response/0210 Message

Type: Interactive

Routing: From an issuer to the MDS.

From the MDS to an acquirer.

Purpose: Must be sent in response to a Financial Transaction Request/0200 message. It carries the response information required to service (such as approve or deny) the request.

Response: The MDS will provide a Financial Transaction Negative Acknowledgment/0290 message to an issuer when the issuer processing system responds with a late Financial Transaction Response/0210 message to a Financial Transaction Request/0200 message.

If used, the acquirer **must** provide the MDS with a Financial Transaction Acknowledgment/0280 message to acknowledge positive receipt of the Financial Transaction Request Response/0210 message from the MDS.

The MDS will provide a Financial Transaction Negative Acknowledgment/0290 message to an acquirer when the acquirer processing system fails to provide, or provides a late Financial Transaction Acknowledgment/0280 message.

Note **The MDS does not support the Financial Transaction Acknowledgment/0280 message for new MDS member implementations. Existing member support is available until members convert their systems.**

Financial Transaction Advice/0220 Message

Type: Non-Interactive

Routing: From an acquirer to the MDS.

From the MDS to an issuer.

Purpose: The MDS forwards a Financial Transaction Advice/0220 message to an affected issuer when:

- An authorization of a Maestro or Cirrus transaction request occurred during Stand-In processing
- A debit MasterCard force post transaction message is received from an acquirer
- A Maestro preauthorization completion message is received from an acquirer

Response: A Financial Transaction Advice Response/0230 message is **required**.

Financial Transaction Advice Response/0230 Message

Type: Non-Interactive

Routing: From the MDS to an acquirer.

From an issuer to the MDS.

Purpose: Must be sent in response to a Financial Transaction Advice/0220 message. Indicates positive receipt of a Financial Transaction Advice/0220 message.

Response: None

Financial Transaction Acknowledgment/0280 Message

Type: Non-Interactive

Routing: From an acquirer to the MDS.

Purpose: For acquirers configured to use this message type, the Financial Transaction Acknowledgment/0280 message indicates positive receipt of a Financial Transaction Response/0210 message from the MDS.

Response: None

Note **The MDS does not support the Financial Transaction Acknowledgment/0280 message for new MDS member implementations. Existing member support is available until members convert their systems.**

Financial Transaction Negative Acknowledgment/0290 Message

Type: Non-Interactive

Routing: From the MDS to an acquirer.

From the MDS to an issuer.

Purpose: The MDS primarily uses the Financial Transaction Negative Acknowledgment/0290 message to inform an issuer that the MDS did not receive a Financial Transaction Response/0210 from the issuer in the required time interval.

The MDS also uses the Financial Transaction Negative Acknowledgment/0290 message to inform an acquirer that the MDS was unable to deliver a Financial Transaction Response/0210 message to the acquirer.

If an acquirer is configured to use the Financial Transaction Acknowledgment/0280 message, the MDS will send a Financial Transaction Negative Acknowledgment/0290 message to the acquirer if the MDS does not receive the Financial Transaction Acknowledgment/0280 message in the required time interval.

Response: None

File Update (03xx) Message Definitions

Issuers may use file update messages to update individual account files such as "hot card" files or system parameter files defined within the MasterCard Account Management System (AMS) or MDS Stand-In processing. MasterCard uses these account files to control the operation of standard and optional features that members may select when they participate in MasterCard programs.

The following information defines the File Update Request/0302 message and the File Update Request Response/0312 message.

File Update Request/0302

Type: Interactive

Routing: Directly from an issuer or through NICS™ to the MasterCard AMS. For MDS Stand-In processing, routing is to the MDS. Messages will be routed to both systems if the issuer participates in both services.

Purpose: Requests update of a file, typically a file used to minimize fraudulent usage of, or give preferential treatment to, the financial transaction cards provided by the issuer to its account holders or other customers.

Response: A File Update Request Response/0312 is **required**.

Note **In cases where the file update request is routed to both AMS and Stand-In, the MDS must receive an approved response from both systems in order to send an approved response to the issuer. If either system does not provide this response, the MDS will return a File Update Request Response/0312 message to the issuer containing a declined/failed response in Response Code (DE 39).**

File Update Request Response/0312

Type: Interactive

Routing: From the MasterCard AMS via the MDS to the issuer. For MDS Stand-In processing, routing is from the MDS to the issuer.

Purpose: Carries response information to the File Update Request/0312.

Response: None

Reversal Advice (04xx) Message Definitions

Reversal advice (04xx) messages reverse the impact of a previous authorization (01xx) or financial transaction (02xx) message.

The ISO 8583–1987 online specification employs only “non-interactive” reversal advice messages. These messages come under the general category of “Advice” messages; therefore, they are subject to the guaranteed advice delivery procedures that are standard for all Advice messages. If, for any reason, these messages cannot be immediately delivered to their intended destination, the MDS automatically assumes responsibility for storing and forwarding them to the proper destination when communication has been reestablished with the appropriate destination processor.

There are two basic categories of reversal advice (04xx) messages:

1. Acquirer Reversal Advice/0420 messages
2. Issuer Reversal Advice/0422 messages

Acquirer Reversal Advice/0420 messages may be generated in one of two ways:

1. By an acquirer (an “acquirer-generated reversal”)
2. By the MDS (a “network-generated reversal”)

Issuer Reversal Advice/0422 messages may be generated in one of two ways:

1. By an issuer (an “issuer-generated reversal”)
2. By the MDS (a “network-generated reversal”)

Acquirer Reversal Advice/0420 Message

Type: Non-Interactive

Routing: From an acquirer to the MDS

From the MDS to an issuer

Purpose: Reverses (partially or wholly) an earlier authorization (01xx) or financial transaction (02xx) message. The acquirer processing system usually generates this message upon detection of a malfunction at the point of interaction (POI).

The MDS sends this message to an issuer upon receipt of a reversal or upon receipt of an adjustment from the acquirer.

For settlement purposes, this message contains "force-post" information. Thus, the reversal will be processed regardless of message receipt acknowledgment.

Response: An Acquirer Reversal Advice Response/0430 message is **required**.

Acquirer Reversal Advice Response/0430 Message

Type: Non-Interactive

Routing: From an issuer to the MDS.

From the MDS to an acquirer.

Purpose: Must be sent in response to an Acquirer Reversal Advice/0420 message, to acknowledge positive receipt of that message.

Response: None

Issuer Reversal Advice/0422 Message

Type: Non-Interactive

Routing: From an issuer to the MDS (for a subsequent day NICS™ adjustment or issuer online adjustment)

From the MDS to an acquirer.

Purpose: Reverses (partially or wholly) an earlier transaction.

The MDS generates this message upon notice of an adjustment, chargeback, or representment for the acquirer.

For settlement purposes, this message contains "force-post" information. Thus, the reversal will be processed regardless of message receipt acknowledgment.

Response: An Issuer Reversal Advice Response/0432 message is **required**.

Issuer Reversal Advice Response/0432 Message

Type: Non-Interactive

Routing: From the MDS to an issuer.

From the acquirer to the MDS.

Purpose: Must be sent in response to an Issuer Reversal Advice/0422 message, to acknowledge positive receipt of that message.

Response: None

Reconciliation Advice (05xx) Message Definitions

The reconciliation advice (05xx) messages advise issuers and acquirers of their total financial transaction activity and their net settlement position, with respect to the MDS.

The ISO 8583–1987 online specification uses only "non-interactive" reconciliation advice (05xx) messages. These messages fall under the general category of "Advice" messages; therefore, they are subject to the guaranteed advice delivery procedures that are standard for all Advice messages. If for any reason these messages cannot be immediately delivered to their intended destination, the MDS automatically assumes responsibility for storing and forwarding them to the proper destination when communication has been reestablished with the destination processor.

There are two basic categories of reconciliation advice (05xx) messages:

1. Acquirer Reconciliation Advice/0520 messages. (Excludes Maestro ISIS fees.)
2. Issuer Reconciliation Advice/0522 messages.

The MDS will generate reconciliation advice (05xx) messages for all products for issuing or acquiring processors upon the end of day cutover. The MDS segregates all reconciliation data into acquirer activity (0520) and issuer activity (0522).

Note

At the processor's option, the MDS will generate the reconciliation advice (05xx) messages. The processor will receive both the Acquirer Reconciliation Advice/0520 message and the Issuer Reconciliation Advice/0522 message or no messages at all.

If the processor elects to receive the reconciliation advice (05xx) messages, the processor will receive both the Acquirer Reconciliation Advice/0520 message and the Issuer Reconciliation Advice/0522 message regardless of level of activity (such as issuer only, acquirer only, or both).

If the processor elects to receive the reconciliation advice (05xx) messages, the processor is required to respond.

Acquirer Reconciliation Advice/0520 Message

Type: Non-Interactive

Routing: From the MDS to the member

Purpose: Advises the member of settlement and reconciliation information ("counts," "amounts", and "net position") for all transactions processed during a specified time period. The MDS sends this message every business day to the member—even if settlement values are zero. The MDS sends these messages if the member is issuer-only, acquirer-only, or both.

Response: An Acquirer Reconciliation Advice Response/0530 message is **required**.

Acquirer Reconciliation Advice Response/0530 Message

Type: Non-Interactive

Routing: From the member to the MDS

Purpose: Must be sent in response to an Acquirer Reconciliation Advice/0520 to acknowledge receipt of that message.

Response: None

Issuer Reconciliation Advice/0522 Message

Type: Non-Interactive

Routing: From the MDS to the member.

Purpose: Advises the member of settlement and reconciliation information ("counts," "amounts," and "net position") for all transactions processed during a specified time period. The MDS sends this message every business day, even if settlement values are zero. The MDS sends these messages if the member is issuer-only, acquirer-only, or both.

Response: An Issuer Reconciliation Advice Response/0532 message is **required**.

Issuer Reconciliation Advice Response/0532 Message

Type: Non-Interactive

Routing: From the member to the MDS

Purpose: Must be sent in response to an Issuer Reconciliation Advice/0522 message, to acknowledge positive receipt of that message.

Response: None

Administrative Advice (06xx) Message Definitions

The administrative advice (06xx) messages may be used between any two parties (processors) connected to the MDS.

The originator routes the messages to a destination; no distinction is made as to whether or not the originator or destination is an issuer or an acquirer.

The ISO 8583–1987 online specification employs only "non-interactive" Administrative Advices. These messages fall under the general category of "Advice" messages; therefore, they are subject to the guaranteed advice delivery procedures that are standard for all Advice messages. If for any reason the intended destination does not immediately receive these messages, the MDS will automatically assume responsibility of storing and forwarding them to the proper destination when network delivery-point communication has been reestablished.

The MDS uses Administrative Advice (06xx) messages to return indecipherable messages to a message originator with an appropriate error condition code indicating the point at which the MDS terminated message parsing or message processing. The types of messages returned in an Administrative Advice/0620 message would either have, improperly coded Message Type Indicator fields or improperly coded bit maps.

Note **In all cases, the Advice Reason Code within the Administrative Advice/0620 message determines the specific reason for the Advice message.**

Administrative Advice/0620 Message

Type: Non-Interactive

Routing: Between the MDS and a card payment system, or between any two processors participating on the MDS.

Purpose: To transmit administrative or informational messages for various reasons, as indicated in the Advice Reason Code of the message.

Response: An Administrative Advice Response/0630 message is **required**.

Administrative Advice Response/0630 Message

Type: Non-Interactive

Routing: From the receiver to the originator of the related Administrative Advice.

Purpose: Must be sent in response to an Administrative Advice/0620 message to acknowledge receipt of that message.

Response: None

Administrative Advice/0644 Message

Type: Non-interactive

Routing: Between the debit Virtual Private Network (VPN) or the Banknet® telecommunications network and a processor connected to the Banknet telecommunications network.

Purpose: To return undelivered messages to the message originator with an appropriate error condition code.

Response: None

Network Management (08xx) Message Definitions

The MDS, acquirer processing systems, issuer processing systems, or intermediate network facilities use the network management messages to coordinate network events and tasks, and to communicate network status conditions.

Typical uses of network management (08xx) messages include:

- Sign-in/sign-out from the MDS
- Inquire on card payment systems or MDS status
- Perform encryption key management tasks
- Perform communication echo tests
- Advise of store-and-forward (SAF) end-of-file (EOF) condition
- SAF request

Within each Network Management Request/0800 message and Network Management Response/0810 message is a Network Management Information Code (DE 70) used to determine the specific purpose or function of each Network Management message. Refer to chapter 4 for detailed information on the Network Management codes used within Network Management (08xx) messages.

The MDS routes all Network Management (08xx) messages are from an originator to a destination; no distinction is made as to whether the originator or destination is an issuer or acquirer.

The following information defines all Network Management (08xx) messages supported by the MDS.

Network Management Request/0800 Message

Type: Interactive

Routing: Between the MDS and any other party (such as card payment system, acquirer processing system, issuer processing system, or intermediate network facility) communicating directly with the MDS. Either party may originate the message.

Purpose: To control the interchange network by communicating or coordinating system condition or system security. The Network Management Information Code, a mandatory data element within all Network Management (08xx) messages, determines the specific Network Management (08xx) messages functions.

Response: A Network Management Request Response/0810 message is **required**.

Network Management Request Response/0810 Message

Type: Interactive

Routing: From destination to originator of the related Network Management Request/0800 message.

Purpose: Must be sent in response to a Network Management Request/0800 message to acknowledge receipt of that message.

Response: None

Network Management Advice/0820 Message

Type: Non-Interactive

Routing: From the MDS to any other party (such as, card payment system, acquirer processing system, issuer processing system, or intermediate network facility) communicating directly with the MDS. This message originates from the MDS only.

Purpose: To provide advisory information to processors connected to the MDS.

Response: None

ISO 8583–1987 Message Processing Conventions

The ISO 8583–1987 online interface uses several basic processing conventions that are implemented uniformly for all financial products. Review these conventions thoroughly before the development of an ISO 8583–1987 interface, as they provide the foundation from which all transaction flow logic is derived.

Some of the most important ISO 8583–1987 message processing concepts are discussed on the following pages, including:

- Issuer Post-On-Authorization Concept
- Acquirer Response Acknowledgment Concept
- Guaranteed Advice Delivery Concept

Issuer Post-On-Authorization Concept

ISO 8583–1987 online application message processing uses the “post-on-authorization” technique for handling issuer processing system transaction processing for both Authorization (01xx) messages and financial transaction (02xx) messages. This technique ensures MDS integrity and minimizes resource use.

Note **As defined within the ISO 8583–1987 specification, the term “post-on-authorization” does not refer to the actual posting of cardholder accounts for billing purposes. “Post-on-authorization” refers only to the technique used to maintain accurate settlement reconciliation totals between the MDS and any attached issuer or acquirer processing system. The issuer’s cardholder account billing subsystem(s) handles the actual posting of cardholder accounts for billing purposes. The online processing procedures described in this manual do not include the issuer billing function.**

The issuer post-on-authorization procedure does not require the use of Completion Confirmation or Completion Response messages for processing of an Authorization Request/0100 messages or Financial Transaction Request/0200 messages at the issuer processing system. This makes it significantly more efficient than the alternative technique known as “post-on-completion.” “Post-on-completion” is often used in other EFT environments. It requires Completion Confirmation and Completion Response messages transmitted between the acquirer and issuer.

Upon receipt of a Financial Transaction Request Response/0210 message from an issuer, the MDS assumes the transaction approval will affect the cardholder's account, and handles any subsequent exception and reversal situations, accordingly. The MDS does not return a Financial Transaction Confirmation message back to the issuer. The issuer always assumes that the acquirer normally completed the transaction, **unless otherwise advised** with a Financial Transaction Reversal Advice/0420 message from the MDS or the acquirer processing system.

Late Responses

If the MDS receives a late issuer response to an interactive request message (such as a late Financial Transaction Request Response/0210 message), **the MDS will have timed-out the issuer processing system**.

For financial transactions, the MDS responds to the late issuer processing system Financial Transaction Request Response/0210 message with a Negative Acknowledgment/0290 message. This indicates to the issuer that the MDS identified the Financial Transaction Request Response/0210 message as a late response and any financial impact to the cardholder's account must be reversed.

When a Negative Acknowledgment/0290 message is received, the issuer processing system must assume that either the MDS timed-out the issuer message and invoked Stand-In processing (if applicable), or sent a denial to the acquirer. The issuer processing system must always **reverse** any previous update to the cardholder's account. The issuer may receive one or more advice messages (for example, Financial Transaction Advice/0220 message or Acquirer Reversal Advice/0420 message, as appropriate) to indicate the specific action taken and the completion status of the transaction.

If the issuer has not selected Stand-In processing options, the MDS automatically transmits a Financial Transaction Request Response/0210 message to the acquirer with a negative (transaction denied) response code.

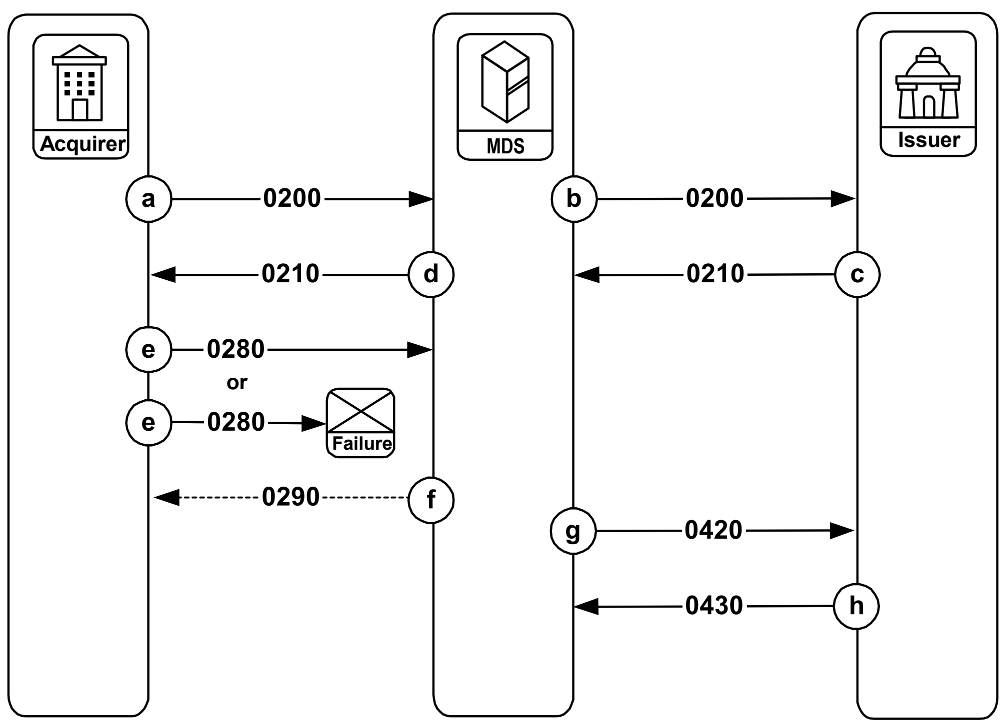
Acquirer Response Acknowledgment Concept

The ISO 8583–1987 online interface optionally supports a positive response acknowledgment technique to help ensure that acquirer processing systems acknowledge receipt of Financial Transaction Request Response/0210 messages at the online application level.

Simple acknowledgment of interactive Financial Transaction messages at the telecommunications protocol level may not provide necessary network integrity for transactions that may be "in progress" when an acquirer processing system fails. Consequently, transaction flows for Financial Transaction/02xx messages may include the option for the acquirer to acknowledge receipt of Financial Transaction Request Response/0210 messages from the MDS using Financial Transaction Acknowledgment/0280 messages.

Figure 2.1 illustrates the acquirer response acknowledgment concept optionally supported for Financial Transaction (02xx) messages.

Figure 2.1—Acquirer Response Acknowledgment



Note **The Financial Transaction Acknowledgment/0280 message is no longer allowed for new member implementations. Continued support is, however, available until all members are able to convert their systems.**

The Financial Transaction Acknowledgment/0280 message is supported only for Maestro and Cirrus ATM transactions. If the MDS receives a Financial Transaction Acknowledgment/0280 message from a Maestro POS transaction, the MDS truncates it and no further action is taken.

The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The issuer generates a Financial Transaction Request Response/0210 message and sends it to the MDS.

d. The MDS forwards the Financial Transaction Request Response/0210 message to the acquirer.

e. The acquirer forwards the Financial Transaction Request Response/0280 message to the MDS.

   Or,

   If the MDS is configured to expect a Financial Transaction Acknowledgment/0280 message, and the acquirer does not forward it to the MDS within the required 120 second time-frame, the following processing occurs:

f. The MDS sends a Financial Transaction Negative Acknowledgment/0290 message to the acquirer. In this message, Response Code (DE 39) contains a value of "96".

g. The MDS sends an Acquirer Reversal Advice/0420 message to the issuer.

h. The issuer responds with an Acquirer Reversal Advice Response/0430 message to the MDS.

Guaranteed Advice Delivery Concept

The ISO 8583–1987 online specification, as implemented on the MDS, employs a guaranteed advice message delivery concept for all advice messages transmitted through the MDS.

The guaranteed advice message delivery facility provides a message routing capability such that when an advice message is forwarded from any processor (acquirer or issuer) to the MDS, the MDS will do the following:

1. Secure the advice message for future delivery.
2. Respond to the originator of the advice message with an Advice Response to indicate that the advice message has been received and secured by the MDS.
3. Forward the advice message to the appropriate receiving entity.

If the MDS cannot deliver an advice message immediately (for example, due to a system or communication failure at the receiving destination) the MDS will store the message using an integrated store-and-forward (SAF) processing facility. The SAF process automatically delivers the message to its proper destination when communication with the endpoint destination has been restored. Thus, the delivery of all advice messages routed through the MDS is **guaranteed**.

Recipients of an advice message must acknowledge receipt with an appropriate Advice Response message. When the MDS has received the appropriate Advice Response message, the MDS considers advice delivery to be complete, and removes the advice message from any pending SAF processing queues.

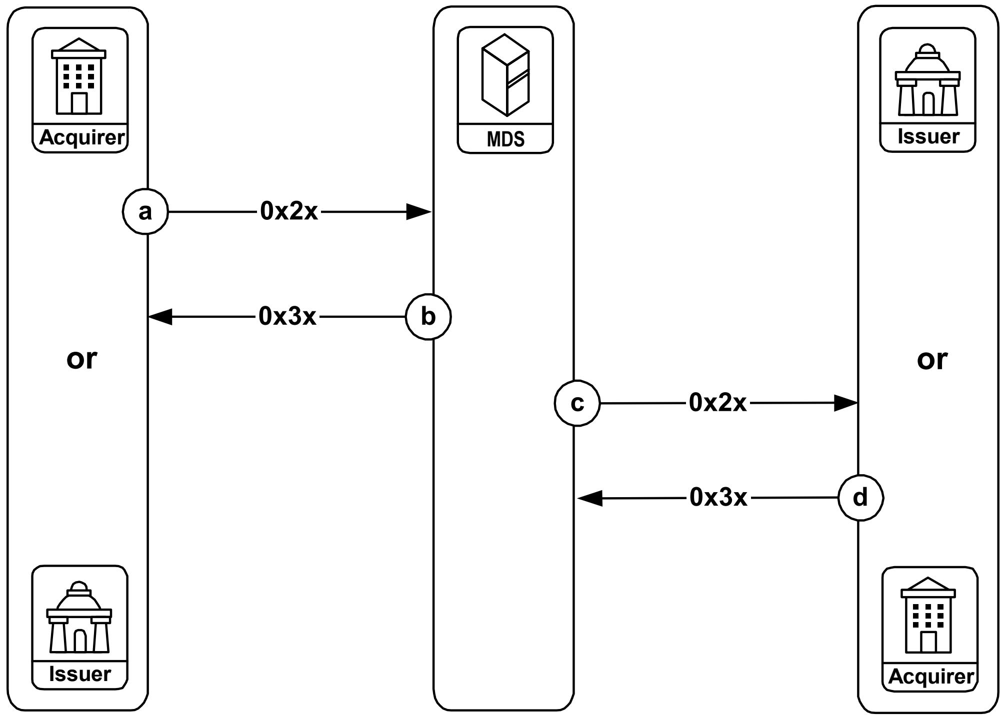
The MDS processes ISO 8583-1987 advice messages using the guaranteed advice delivery technique. These advice messages and their response messages are as follows:

- Financial Transaction Advice/0220
- Financial Transaction Advice Response/0230
- Acquirer Reversal Advice/0420
- Acquirer Reversal Advice Response/0430
- Issuer Reversal Advice/0422
- Issuer Reversal Advice Response/0432

Note **Other ISO 8583 transaction messages have a message type designation of "advice" but are not delivered by the MDS from one processor to another. The description of these other advice message types—0520, 0522, 0620, 0820—and their flows are found in the detailed message flow descriptions later in this chapter.**

Figure 2.2 illustrates the standard transaction flow requirements applicable to advice messages originated by customer processing systems connected to the MDS.

Figure 2.2—Guaranteed Advice Delivery



The sequence of events:

a. The acquirer or issuer processor forwards an Advice (0x2x) message to the MDS.

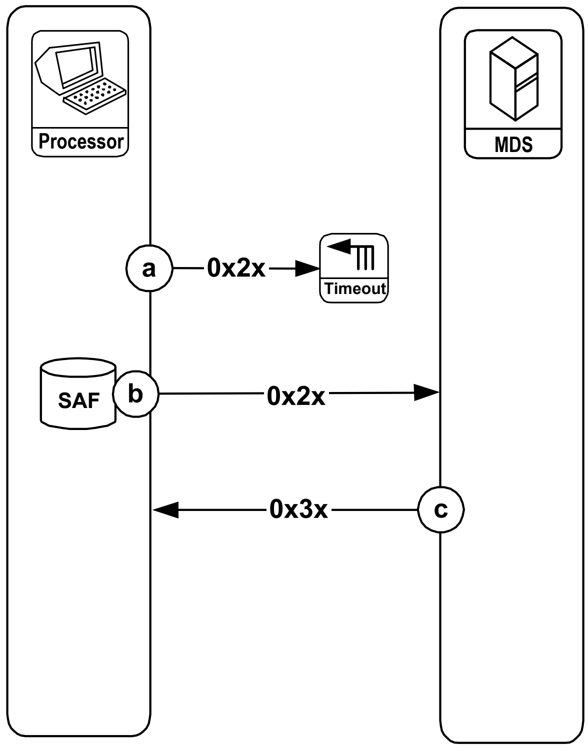
b. The MDS returns an Advice Response (0x3x) message after it has received the original advice message.

c. The MDS generates a corresponding Advice (0x2x) message to the receiving processor.

d. The receiving processor returns an Advice Response (0x3x) message as positive acknowledgment of receipt after it has received the advice message. Guaranteed advice delivery is now complete.

Figure 2.3 illustrates the exception condition transaction flow scenarios for advice messages with acknowledgments.

Figure 2.3—Exception Condition, Advice Delivery from Processor Following Timeout



The sequence of events:

a. An issuer or acquirer generates an Advice (0x2x) message. If it cannot be transmitted within the processor's configured timeout values, it should be stored by an appropriate store-and-forward (SAF) facility at the customer processing system. The Advice (0x2x) message will be transmitted later when communication has been reestablished with the MDS.

b. When communication is reestablished, the customer processing system forwards the Advice (0x2x) message from the SAF facility to the MDS.

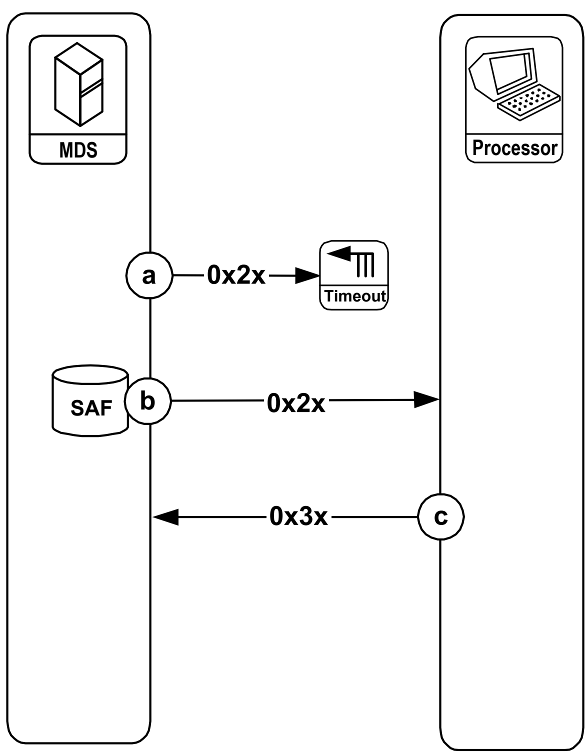
c. The MDS returns an Advice Response (0x3x) message after it has received the advice message.

Note The diagram does not illustrate the only scenario for advice messages: The MDS can initiate and send some types of advice messages to an individual processor. An individual processor can initiate and send advice messages to the MDS, and the MDS will forward these messages to another processor. However, the MDS will not forward some types of advice messages initiated by an individual processor to another processor.

The above figure and message flow description apply to processing during a given settlement period. If an advice message arrives at the MDS following the settlement day of the original transaction, reconciliation should be accomplished using one of the following methods:

- NICS™ (refer to the *NICS™ User's Guide* for more information)
- Online adjustment processing (see chapter 3 and chapter 4 for more information)
- Manual adjustments (refer chapter 7 of the *NICS™ User's Guide* for more information)

Figure 2.4—Exception Condition, Advice Delivery from the MDS Following Timeout



The sequence of events:

a. The MDS attempts to deliver the advice message to the intended destination. If it cannot be delivered, it is stored at the MDS SAF facility for later delivery.

b. The MDS forwards the Advice (0x2x) message from the SAF facility to the receiving destination when communication has been reestablished with the receiving processor.

c. The receiving processor returns an Advice Response (0x3x) message after it has received the advice message.

Note **The diagram does not illustrate the only scenario for advice messages: The MDS can initiate and send some types of advice messages to an individual processor. An individual processor can initiate and send some types of advice messages to the MDS and these messages will be forwarded to another processor. However, some types of advice messages initiated by an individual processor will not be forwarded to another processor by the MDS.**

The above figure and message flow description apply to processing during a given settlement period. If an advice message arrives at the MDS following the settlement day of the original transaction, reconciliation should be accomplished using one of the following methods:

- **NICS™ (refer to the *NICS™ User's Guide* for more information)**
- **Online adjustment processing (see chapter 3 and chapter 4 for more information)**
- **Manual adjustments (refer chapter 7 of the *NICS™ User's Guide* for more information)**

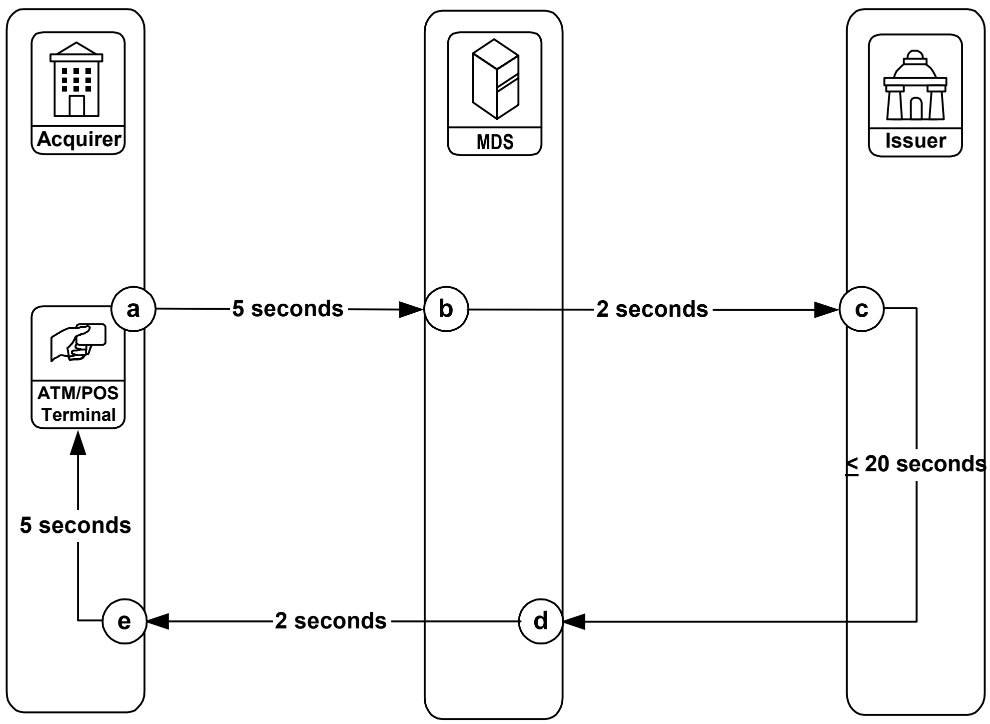
Detailed Transaction Flows

The transaction flows provided throughout the remainder of this chapter define all of the ISO 8583–1987 transaction flow procedures implemented on the MDS. These flows sometimes depict a "timeout" or late response situation.

Maximum Response Times

Figure 2.5 illustrates the maximum time interval available to each processor.

Figure 2.5—Maximum Time Intervals



The MDS expects the following time intervals in processing a transaction:

a. The acquirer processor delivers a Financial Transaction Request/0200 ATM or point-of-service (POS) message to the MDS within five seconds.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer processor within two seconds.

c. The issuer processor generates a Financial Transaction Request Response/0210 message to the MDS. Response is required within the following time-frames:

   – **Maestro/Cirrus** (20 seconds)

   – **Visa/Plus** (20 seconds)

   – **debit MasterCard** (10 seconds)

d. The MDS forwards the Financial Transaction Request Response/0210 message to the acquirer processor within two seconds.

e. The acquirer processor returns a Financial Transaction Request Response/0210 message to the ATM or POS device within five seconds.

The following additional maximum response times apply to other message types processed by the MDS:

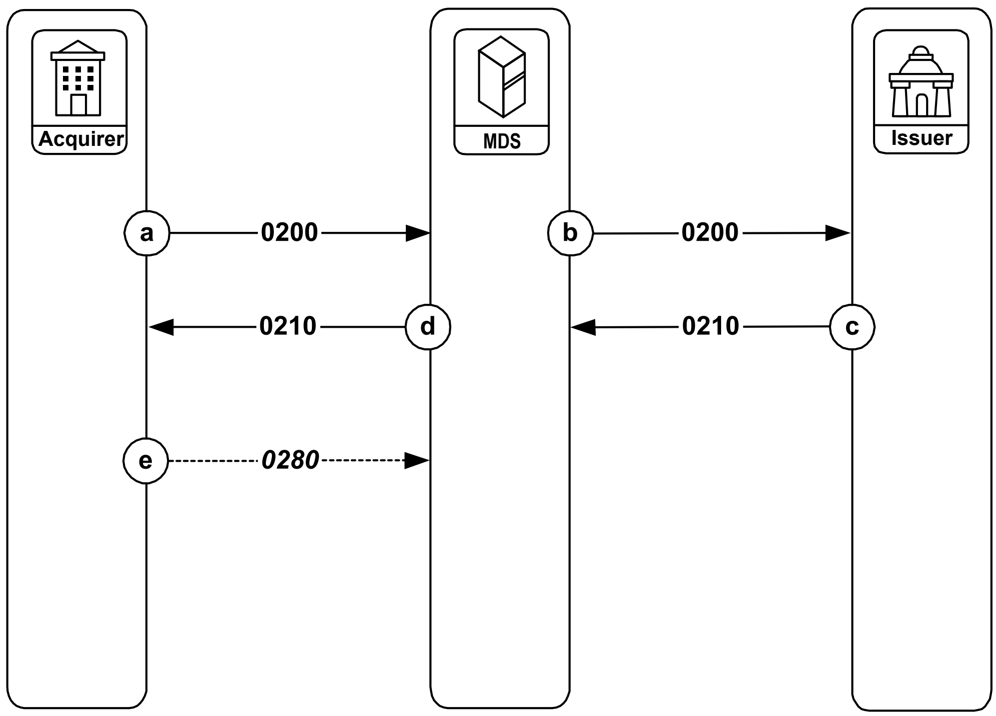
- 120 seconds for:
  – 0280 responses (when configured) to 0210 financial responses
  – 0430 responses to 0420 adjustment advices
  – 0432 responses to 0422 adjustment advices
  – 0530 responses to 0520 settlement advices
  – 0532 responses to 0522 settlement advices
- 60 seconds for:
  – 0810 responses to 0800 network management requests

The requirement for MasterCard acquirers is the terminal shall wait for (without timing out) a Financial Transaction Response/0210 message a minimum of 45 seconds after submitting a Financial Transaction Request/0200 message. **MasterCard recommended practice for minimum terminal timeout is 60 seconds.**

Financial Transaction Request/0200 and Financial Transaction Request Response/0210 Sequence—Standard Transaction Flow

Figure 2.6 illustrates the standard message flow for interactive financial transaction processing.

Figure 2.6—Standard Message Flow, 0200/0210 Messages



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The issuer generates a Financial Transaction Request Response/0210 message and sends it to the MDS.

d. The MDS forwards the Financial Transaction Request Response/0210 message to the acquirer.

e. If the acquirer is configured for 0280 acknowledgment messages, the acquirer sends a Financial Transaction Acknowledgment/0280 message to the MDS.

Note **The Financial Transaction Acknowledgment/0280 message is no longer allowed for new member implementations. However, support is available until all members are able to convert their systems.**

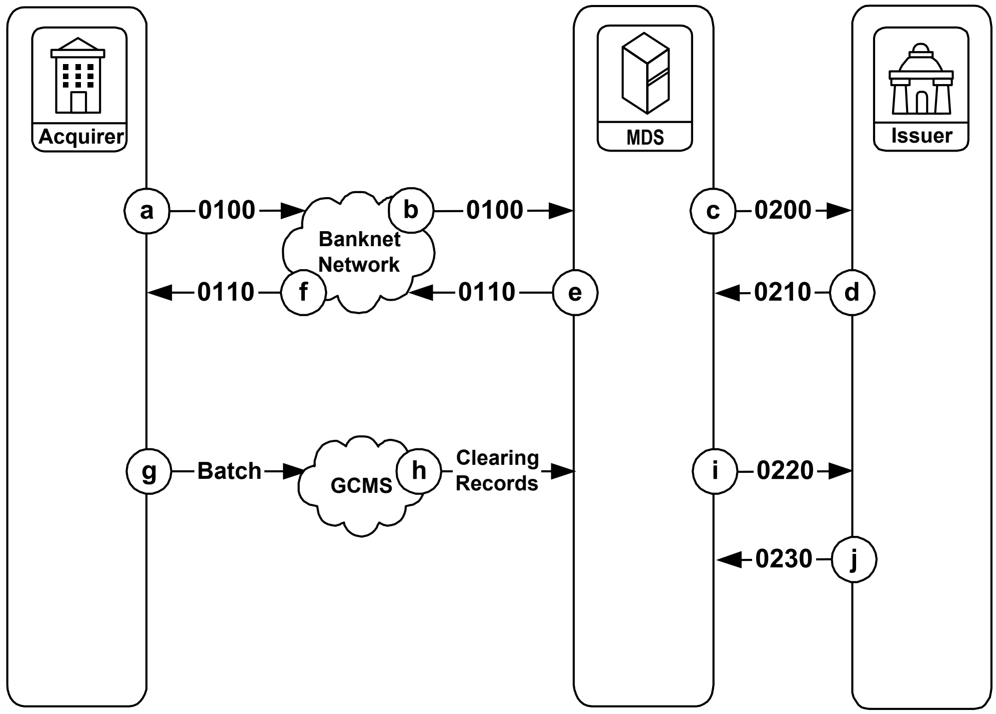
The Financial Transaction Acknowledgment/0280 message is supported only for Maestro and Cirrus ATM transactions. If the MDS receives a Financial Transaction Acknowledgment/0280 message for a Maestro POS transaction, the MDS truncates it and no further action is taken.

Debit MasterCard Preauthorization and Clearing Processing—Standard Transaction Flow

Figure 2.7 illustrates the standard message flow for a debit MasterCard preauthorization and clearing transaction when the debit MasterCard issuer is connected to the MDS.

Note **A transaction using debit MasterCard is designed to work like a credit card transaction where the completion occurs following a subsequent-day batch clearing process initiated from the acquirer.**

Figure 2.7—Standard Message Flow, debit MasterCard Preauthorization



The sequence of events:

a. The acquirer submits the Authorization Request/0100 message to the Banknet network.

b. The Banknet network forwards the Authorization Request/0100 message to the MDS.

c. The MDS converts the Authorization Request/0100 message to a Financial Transaction Request/0200 message and forwards it to the issuer. The Point of Service Data (DE 61) position 7, POS Transaction Status Indicator, will contain a value of “4” indicating that this is a preauthorization request.

d. The issuer responds with a Financial Transaction Request Response/0210 message to the MDS.

e. The MDS converts the Financial Transaction Request Response/0210 message to an Authorization Request Response/0110 message and sends it to the Banknet network.

f. The Banknet network forwards the Authorization Request Response/0110 message to the acquirer.

g. The acquirer processing system batches the clearing records and sends them to the Global Clearing Management System (GCMS) at MasterCard. This typically occurs within 2-5 days of the preauthorization.

h. GCMS groups all clearing records bound for the MDS and transmits them to the MDS.

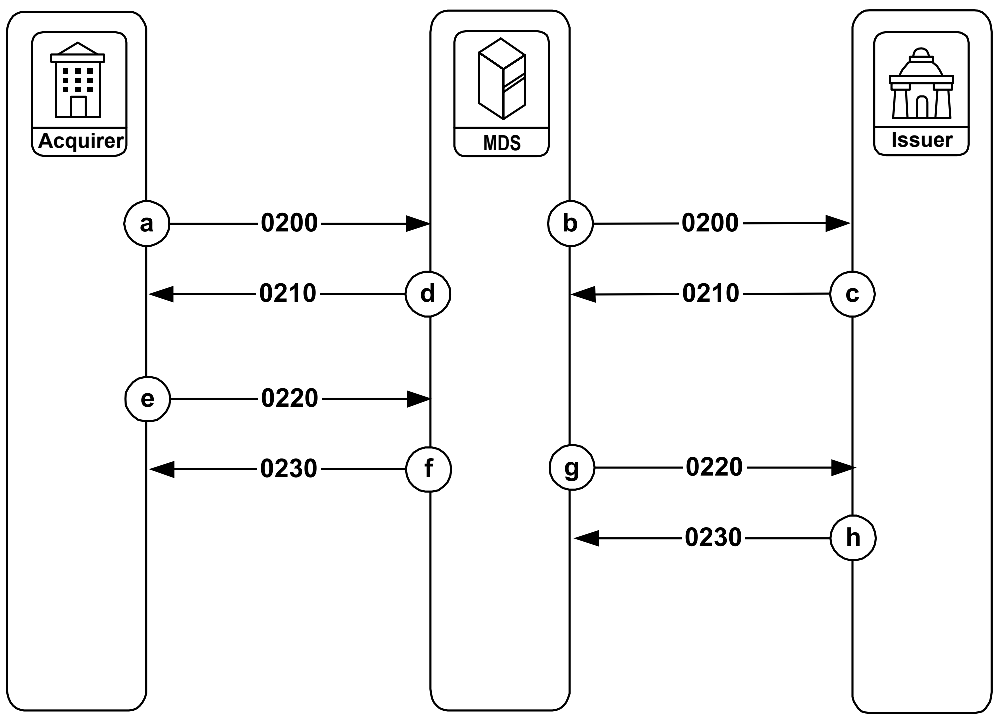
i. The MDS converts each detail record into a Financial Transaction Advice/0220 clearing message and forwards it to the issuer.

j. The issuer responds with a Financial Transaction Advice Response/0230 message to the MDS.

Maestro Preauthorization and Completion—Standard Transaction Flow

Figure 2.8 illustrates a POS preauthorization and completion transaction.

Figure 2.8—Standard Message Flow, Maestro Preauthorization and Completion



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS. The Point of Service Data (DE 61) position 7, POS Transaction Status Indicator, will contain a value of “4” indicating that this is a preauthorization request. The Amount, Transaction (DE 4) will contain either the acquirer’s standard predetermined requested amount or the cardholder’s requested amount.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. If approved, the issuer puts a conditional hold on the cardholder's account and returns a Financial Transaction Request Response/0210 message to the MDS.

d. The MDS returns the Financial Transaction Request Response/0210 message to the acquirer.

e. Within 20 minutes of the original Financial Transaction Request/0200 message, the acquirer must send a Financial Transaction Advice/0220 completion message to the MDS. This completion message must be provided with the actual completed amount of the transaction to be posted to the cardholder's account.

f. This completed amount **must** be provided in Replacement Amounts (DE 95). The Point of Service Data (DE 61) position 7, POS Transaction Status Indicator and the Amount, Transaction (DE 4) will contain the same values as in the original Financial Transaction Request/0200 message.

g. The MDS responds with a Financial Transaction Advice Response/0230 message.

h. The MDS forwards the Financial Transaction Advice/0220 message to the issuer.

i. The issuer responds with a Financial Transaction Advice Response/0230 message.

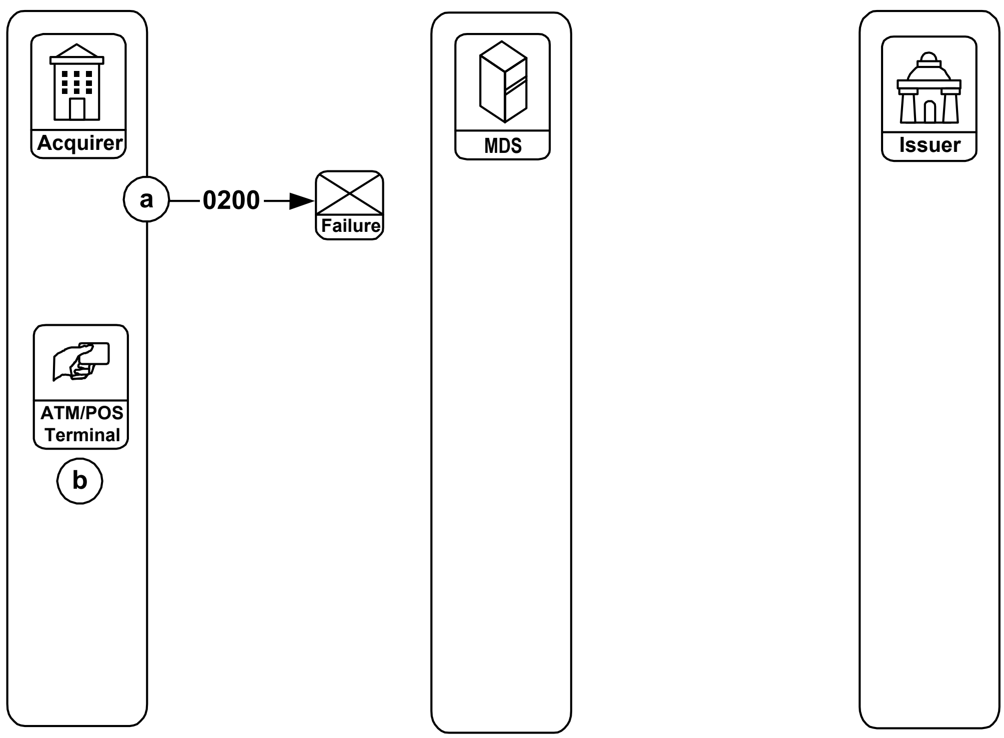
Note **Please note that if the MDS does not receive a Financial Transaction Advice/0220 message from the acquirer, the MDS assumes the transaction was not completed and no further message processing takes place.**

Note **The MDS treats the Maestro preauthorization and completion cycle as a single transaction. This is unlike the debit MasterCard preauthorization and completion cycle, which is treated as two separate transactions. This difference is because the Maestro processing cycle is completed within 20 minutes on the same settlement day, while the debit MasterCard completion message is processed within 2-5 calendar days of the preauthorization message.**

Financial Transaction/02xx Exception Condition Transaction Flow—System Failure During Acquirer Financial Transaction Request/0200 Message

Figure 2.9 illustrates exception condition processing for a system or communication failure condition during the transmission of an acquirer Financial Transaction Request/0200 message.

Figure 2.9—Exception Condition, System Failure During Acquirer Request



The sequence of events:

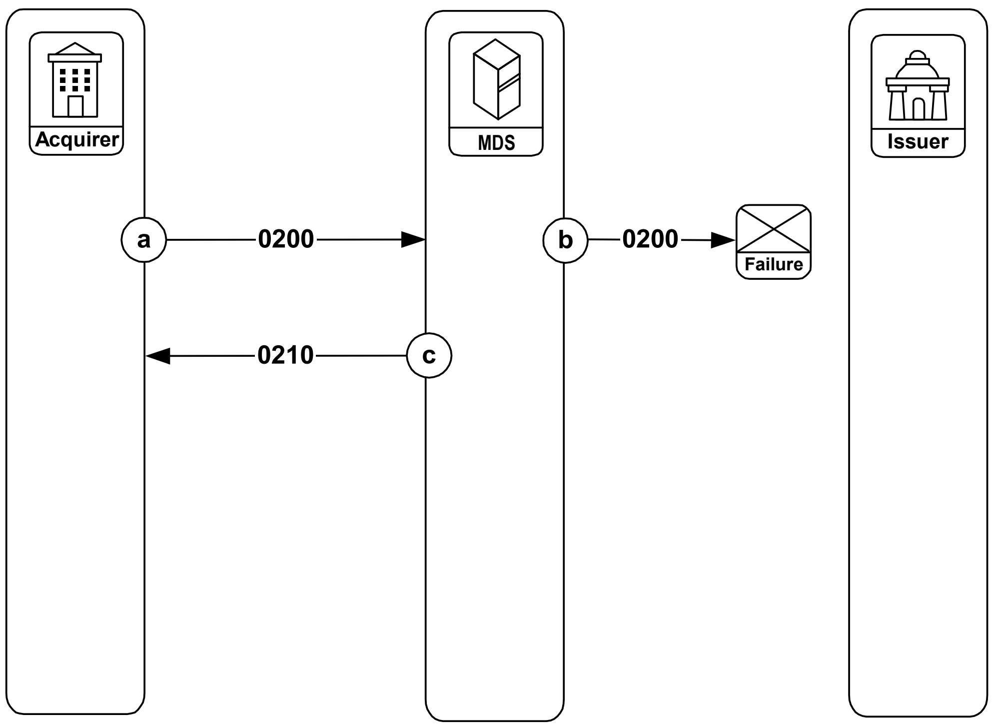
a. The acquirer initiates a Financial Transaction Request/0200 message, but it cannot be delivered to the MDS because of some type of system failure.

b. The acquirer processing system is unable to transmit the Financial Transaction Request/0200 message to the MDS, but must complete the transaction at the point of service. The MDS requires that the acquirer deny the transaction request at the point of service. Processing terminates at this point.

Financial Transaction/02xx Exception Condition Transaction Flow—System Failure During Issuer Financial Transaction Request/0200 Message

Figure 2.10 illustrates exception procedures for a system or communication failure condition during the transmission of a Financial Transaction Request/0200 message.

Figure 2.10—Exception Condition, System Failure During MDS Request to Issuer



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

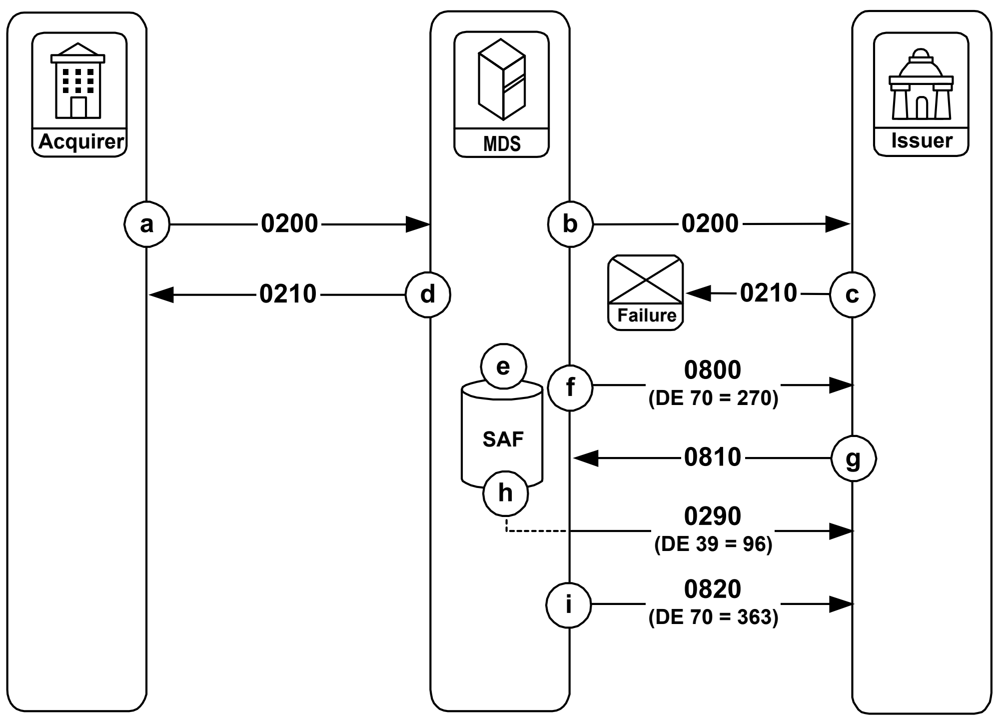
b. The MDS attempts to forward the Financial Transaction Request/0200 message to the issuer, but is unable to complete the message transmission due to a communication link failure or other problem at the issuer processing system.

c. The MDS will then generate a Financial Transaction Request Response/0210 message to the acquirer, indicating a request denial.

Financial Transaction/02xx Exception Condition Transaction Flow—System Failure During Issuer Financial Transaction Request Response/0210 Message

Figure 2.11 illustrates exception condition processing for a system or communication failure condition encountered during the transmission of an issuer Financial Transaction Request Response/0210 message.

Figure 2.11—Exception Condition, No Issuer Response



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The issuer cannot return the Financial Transaction Request Response/0210 message because of a communications failure between the issuer processing system and the MDS. The issuer must assume that the MDS or the acquirer will take appropriate action at this point, and should immediately reverse the impact to the cardholder's account file if the request was approved.

d. The MDS detects a timeout condition because of the issuer processing system failure on the Financial Transaction Request Response/0210 message. The MDS then generates a Financial Transaction Request Response/0210 message to the acquirer, indicating a request denial.

e. The MDS also creates a Transaction Negative Acknowledgment/0290 message with Response Code (DE 39) containing a value of "96", indicating that no Financial Transaction Request Response/0210 message was received. This message is placed in the Store-and-Forward (SAF) file for later delivery to the issuer.

f. The MDS initiates, at regular intervals determined programmatically, a Network Management Request/0800 "echo test" message to verify or establish communication with the issuer.

g. The issuer responds with a Network Management Request Response/0810 message.

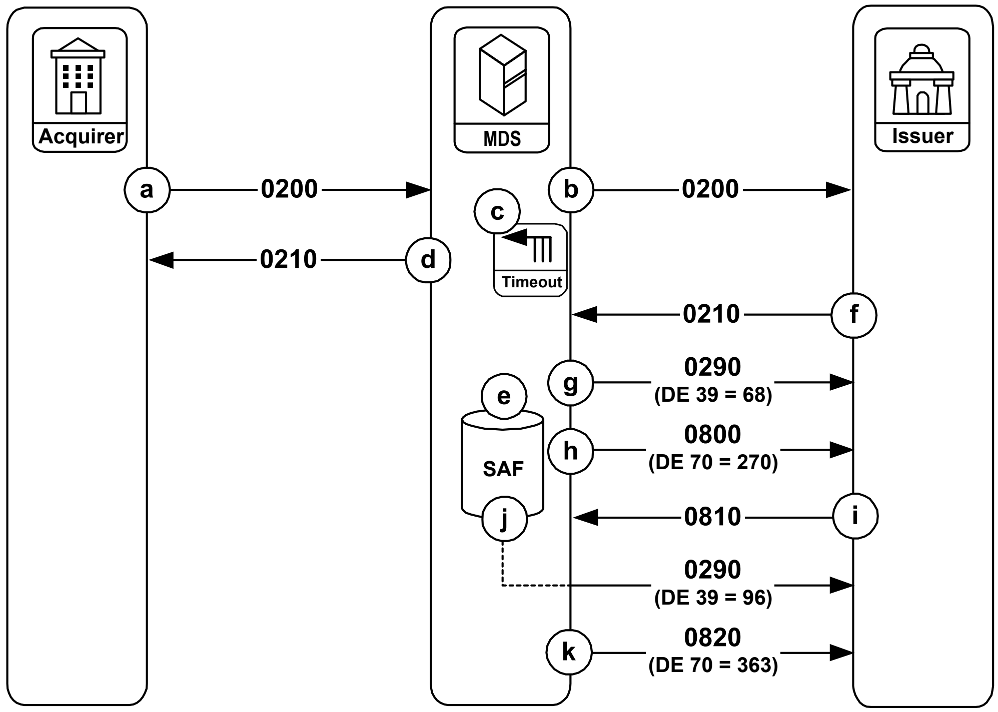
h. The MDS sends the Transaction Negative Acknowledgment/0290 message with Response Code (DE 39) containing a value of "96" from the SAF file to the issuer. Any remaining messages stored in the SAF file for the issuer will also be sent by the MDS to the issuer.

i. The MDS sends a Network Management Advice/0820 message to indicate the SAF facility has reached an end-of-file (EOF) condition.

Financial Transaction/02xx Exception Condition Transaction Flow—Late Issuer Financial Transaction Request Response/0210 Message

Figure 2.12 illustrates exception condition processing for a late Issuer Financial Transaction Request Response/0210 message.

Figure 2.12—Exception Condition, Late Issuer Response



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The MDS detects a timeout condition on the Financial Transaction Request Response/0210 message that is expected from the issuer.

d. The MDS then generates a Financial Transaction Request Response/0210 message to the acquirer, indicating a request denial.

e. The MDS also creates a Transaction Negative Acknowledgment/0290 message with Response Code (DE 39) containing a value of “96”, indicating that no Financial Transaction Request Response/0210 message was received. This message is placed in the Store-and-Forward (SAF) file for later delivery to the issuer.

f. The MDS receives an unsolicited (late) Financial Transaction Response/0210 message from the issuer.

g. If the late Financial Transaction Request Response/0210 message indicates an approval from the issuer, the MDS responds with a Financial Transaction Negative Acknowledgment/0290 message with Response Code (DE 39) containing a value of "68". This indicates to the issuer that this Financial Transaction Request Response/0210 message is late and rejected. The issuer must assume the MDS or the acquirer will take appropriate action at this point, and should immediately reverse any impact to the cardholder's account file.

Note **If the late Financial Transaction Request Response/0210 message has a response code indicating a request denial, then the MDS will not take action (the Financial Transaction Negative Acknowledgment/0290 message is not sent).**

h. The MDS initiates, at regular intervals determined programmatically, a Network Management Request/0800 "echo test" message to verify or establish communication with the issuer.

i. The issuer responds with a Network Management Request Response/0810 message.

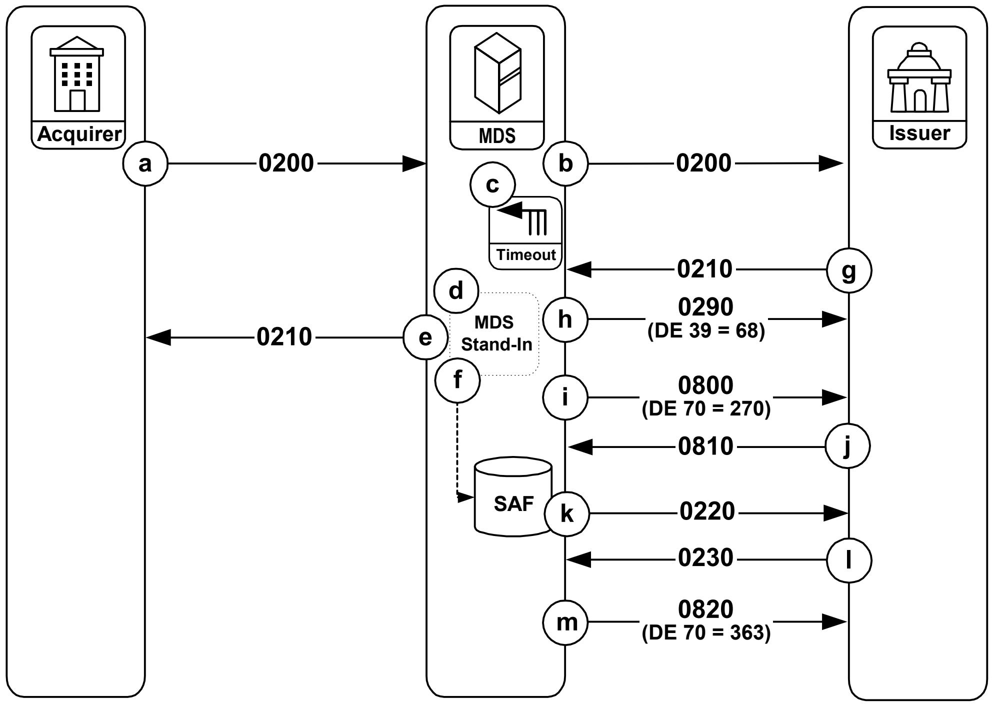
j. The MDS sends the Transaction Negative Acknowledgment/0290 message with Response Code (DE 39) containing a value of "96" from the SAF file to the issuer. Any remaining messages stored in the SAF file for the issuer will also be sent by the MDS to the issuer.

k. The MDS sends a Network Management Advice/0820 message to indicate the SAF facility has reached an end-of-file (EOF) condition.

Financial Transaction/02xx Exception Condition Transaction Flow—MDS Stand-In Processing—Late Response From Issuer

Figure 2.13 illustrates exception condition processing for a late issuer Financial Transaction Request Response/0210 message. This example assumes that the issuer subscribes to the MDS Stand-In processing service.

Figure 2.13—Exception Condition, Late Issuer Response, MDS Stand-In



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The MDS detects a timeout condition on the Financial Transaction Request Response/0210 message that is expected from the issuer.

d. If the issuer processor is configured for Stand-In processing at the MDS, the MDS creates an internal financial request message and sends it to Stand-In processing for authorization. The Stand-In processing service validates the request and formulates an internal response message.

e. The MDS uses the internal response to create a Financial Transaction Request Response/0210 message, and sends it to the acquirer.

f. A record of this Financial Transaction Request Response/0210 message is placed in the Store-and-Forward file on the MDS for later delivery to the issuer.

g. The MDS receives an unsolicited (late) Financial Transaction Request Response/0210 message from the issuer.

h. If the late Financial Transaction Request Response/0210 message indicates an approval from the issuer, the MDS responds with a Financial Transaction Negative Acknowledgment/0290 message with Response Code (DE 39) containing a value of “68”. This indicates to the issuer that the Financial Transaction Request Response/0210 message is late and rejected. The issuer must assume that the MDS or the acquirer will take appropriate action at this point, and should immediately reverse any impact to the cardholder’s account file.

Note **If the late Financial Transaction Request Response/0210 message has a response code indicating a request denial, then the MDS will not take action (the Financial Transaction Negative Acknowledgment/0290 message is not sent).**

i. The MDS initiates, at regular intervals determined programmatically, a Network Management Request/0800 “echo test” message to verify or establish communication with the issuer.

j. The issuer responds with a Network Management Request Response/0810 message.

k. When communication is established, the MDS sends a Financial Transaction Advice/0220 authorization completion message with Authorization Identification Response (DE 38) containing a value of “**STND**” (followed by 2 blank characters) to the issuer from the SAF facility.

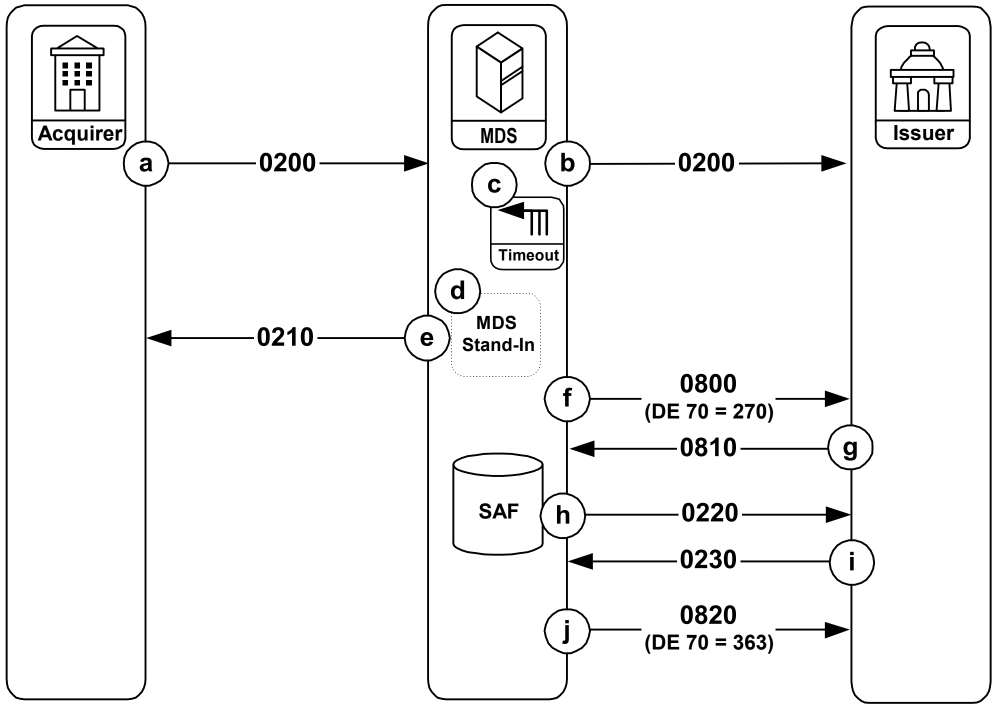
l. The issuer responds with a Financial Transaction Advice Response/0230 message.

m. Any remaining messages stored in the SAF file for the issuer will also be sent by the MDS. When completed, the MDS will send a Network Management Advice/0820 message to indicate the SAF facility has reached an end-of-file (EOF) condition.

Financial Transaction/02xx Exception Condition Transaction Flow—MDS Stand-In Processing—No Response From Issuer

Figure 2.14 illustrates the MDS Stand-In processing procedures when the issuer cannot complete the transaction. This example assumes that the issuer has subscribed to the MDS Stand-In processing service.

Figure 2.14—Exception Condition, No Response from Issuer, MDS Stand-In



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The MDS detects a timeout condition on the Financial Transaction Request Response/0210 message that is expected from the issuer.

d. If the issuer processor is configured for Stand-In processing at the MDS, the MDS creates an internal financial request message and sends it to Stand-In processing for authorization. The Stand-In processing service validates the request and formulates an internal response message.

e. The MDS uses the internal response to create a Financial Transaction Request Response/0210 message, and sends it to the acquirer.

The MDS can receive another Financial Transaction Request/0200 message for the issuer.

If the issuer processing system is still not online, the MDS repeats steps **d** and **e** above, as additional Financial Transaction Request/0200 messages arrive at the MDS destined for this issuer.

f. The MDS initiates, at regular intervals determined programmatically, a Network Management Request/0800 “echo test” message to verify or establish communication with the issuer.

g. The issuer responds with a Network Management Request Response/0810 message.

h. The MDS sends a Financial Transaction Advice/0220 authorization completion message with Authorization Identification Response (DE 38) containing a value of “**STND”** (followed by 2 blank characters) to the issuer from the SAF facility.

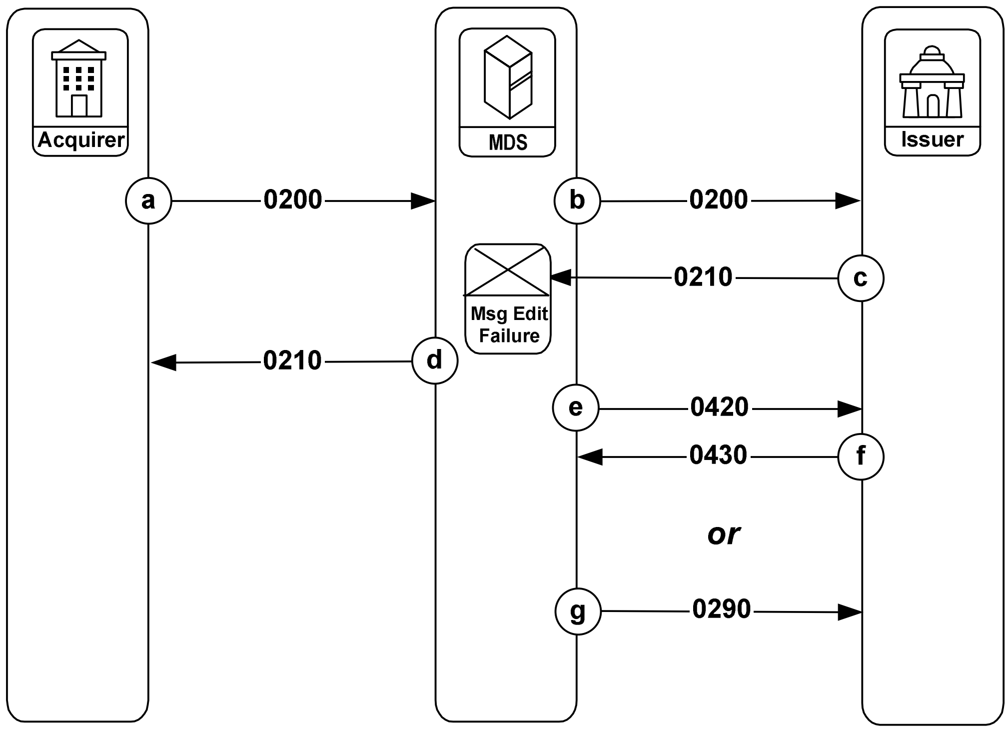
i. The issuer responds with a Financial Transaction Advice Response/0230 message.

j. Any remaining messages stored in the SAF file for the issuer will also be sent by the MDS to the issuer. When completed, the MDS will send a Network Management Advice/0820 message to indicate the SAF facility has reached an end-of-file (EOF) condition.

Financial Transaction/02xx Exception Condition Transaction Flow—Financial Transaction Request Response/0210 Failure of MDS Edit Checks

Figure 2.15 illustrates exception procedures when the MDS processes the Financial Transaction Request Response/0210 message from the issuer, and the message does not pass the edit checks.

Figure 2.15—Exception Condition, Message Edit-Check Failure



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The issuer responds with a Financial Transaction Request Response/0210 message and sends it to the MDS. The message does not pass one of several required edit checks.

d. The MDS then generates a Financial Transaction Request Response/0210 message to the Acquirer, indicating a request denial.

If the issuer's Financial Transaction Request Response/0210 message fails the edit check:

e. The MDS generates a Financial Transaction Advice/0420 to the issuer with an advice Reason Code that indicates the response could not be processed and sent to the acquirer.

f. The issuer responds by returning a Financial Transaction Advice Response/0430 to the MDS.

Or,

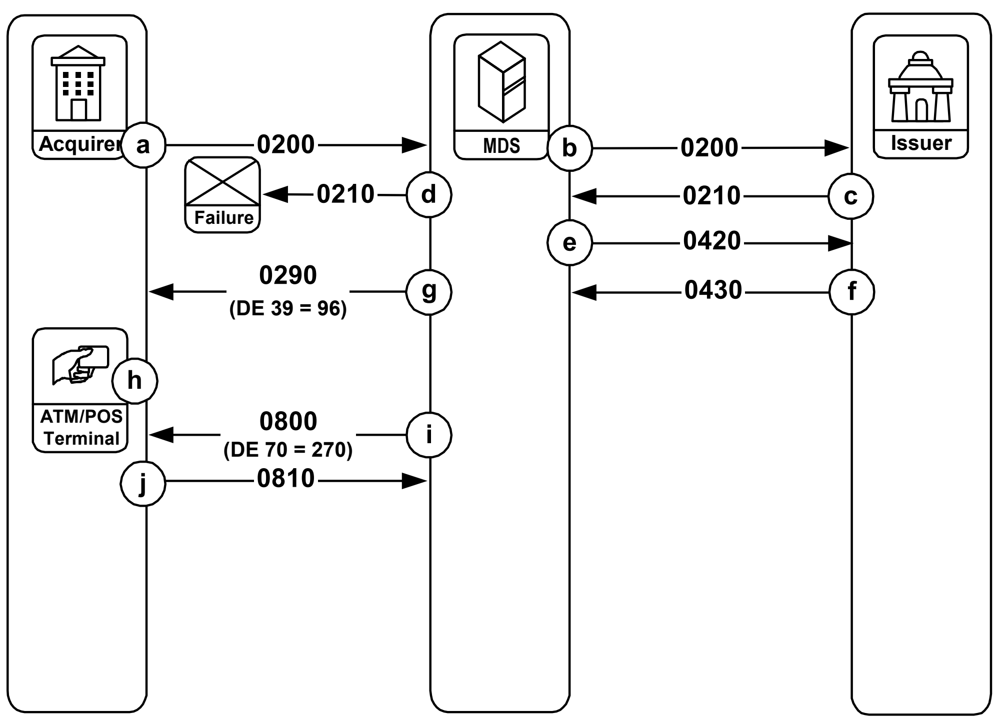
g. The MDS will generate a Financial Transaction Negative Acknowledgment/0290 to the issuer. The 0290 message requires no response from the issuer.

Note **IF the MDS cannot parse the Financial Transaction Request Response/0210 message, the MDS generates an Administrative Advice/0620 to the Issuer (see Figure 2.26).**

Financial Transaction/02xx Exception Condition Transaction Flow—System Failure During Acquirer Financial Transaction Request Response/0210 Message

Figure 2.16 illustrates exception procedures for a system or communication failure condition encountered during the transmission of a Financial Transaction Request Response/0210 message.

Figure 2.16—Exception Condition, MDS Response to Acquirer Failure



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The issuer responds with a Financial Transaction Request Response/0210 message and sends it to the MDS.

d. The MDS attempts to forward the Financial Transaction Request Response/0210 message to the acquirer, but cannot successfully complete the transmission due to a communications failure between the MDS and the acquirer processing system.

e. The MDS determines that the issuer's Financial Transaction Request Response/0210 message is undeliverable and, only if the response indicates a request approval, immediately generates an Acquirer Reversal Advice/0420 message to the issuer.

f. If the issuer receives an Acquirer Reversal Advice/0420 message from the MDS, the issuer responds with a Reversal Advice Response/0430 message to the MDS.

g. The MDS sends a Financial Transaction Negative Acknowledgment/0290 message with Response Code (DE 39) containing a value of "96" to the acquirer. This informs the acquirer that the MDS generated a reversal to the issuer for the failed transaction.

h. If the acquirer processing system is operational, it will detect a timeout condition on the Financial Transaction Request Response/0210 message that it is expecting from the MDS. When the timeout occurs, the MDS requires that the acquirer deny the transaction request at the point of service. Processing terminates at this point.

i. The MDS initiates, at regular intervals determined programmatically, a Network Management Request/0800 "echo test" message to verify or establish communication with the acquirer.

j. The acquirer responds with a Network Management Request Response/0810 message.

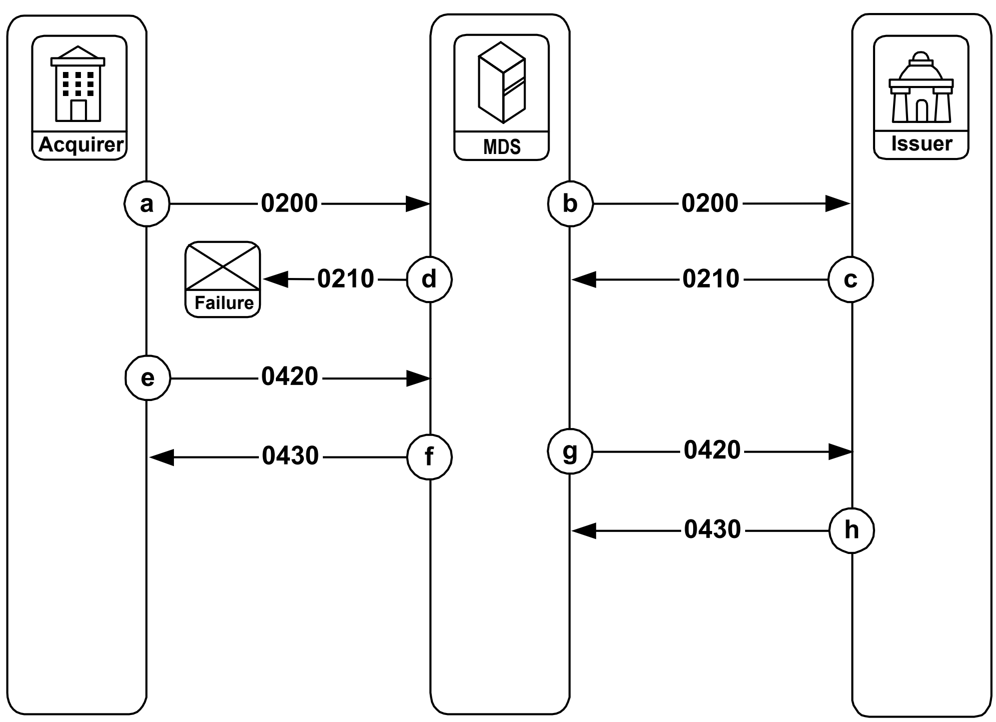
Financial Transaction/02xx Exception Condition Transaction Flow—Timeout of Financial Transaction Request Response/0210 Message to Acquirer

Figure 2.17 illustrates exception procedures for a message delivery failure condition encountered on the acquiring side when a Financial Transaction Request Response/0210 message sent to the acquirer by the MDS is not received by the acquirer application.

In this situation, the acquirer's timeout limit (typically in the range of 45-60 seconds) has been exceeded for receiving the Financial Transaction Request Response/0210 message. The acquirer has the option of responding to the timeout by sending a Timeout-Induced Reversal/0420 message to the MDS (for the description and message format, refer to chapter 3).

This Timeout-Induced Reversal/0420 message will not include the Date, Settlement (DE 15) or Network Data (DE 63). Further, Response Code (DE 39), Advice Reason Code (DE 60), and Replacement Amounts (DE 95) will have specific requirements as chapter 3 shows in the message format table and usage notes within the data element descriptions of chapter 4.

Figure 2.17—Exception Condition, Timeout Reversal



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The issuer generates a Financial Transaction Request Response/0210 message and sends it to the MDS.

d. The MDS forwards the Financial Transaction Request Response/0210 message to the acquirer, but the message fails at the acquirer application interface. It fails such that the MDS is not aware of a delivery problem.

e. The acquirer times out on receipt of the Financial Transaction Request Response/0210 message.[1] The acquirer sends a Timeout-Induced Reversal/0420 message, also known as an unsolicited message reversal, to the MDS. This reversal is distinct from a normal acquirer-generated reversal in the following manner:

   – Date, Settlement (DE 15) is not present in the message from the acquirer

   – Network Data (DE 63) is not present in the message from the acquirer

   – Response Code (DE 39) is "00"

   – Advice Reason Code (DE 60) is "4500018"

   – Replacement Amounts (DE 95) contains all zeroes

f. The MDS responds with an Acquirer Reversal Advice Response/0430 message.

---

1 The acquirer should permit no more than five (5) failures of this type before marking the MDS down and attempting to recover the connection. The acquirer may periodically send Network Management Request/0800 "echo test" messages to the MDS to test its connection and/or it may attempt delivery of the corresponding Timeout-Induced Reversal/0420 message at a later date.

Note **Reversal advice processing is complete when the acquirer receives the Acquirer Reversal Advice Response/0430 message, regardless of the Response Code (DE39) value contained in the response message. In cases where the Response Code (DE39) in the response message contains a value other than "00", the acquirer should not re-transmit the reversal advice to the MDS. The acquirer must clear their timer for the Acquirer Reversal Advice/0420 message, and consider processing to be complete.**

g. If the Financial Transaction Request Response/0210 message received at step **c** indicates an approval, the MDS generates a standard Acquirer Reversal Advice/0420 message and sends it to the issuer.

h. The issuer responds with an Acquirer Reversal Advice Response/0430 message.

Note **The acquirer should set a 120-second timer on the Timeout-Induced Reversal/0420 message. If an Acquirer Reversal Advice Response/0430 message is not received within 120 seconds, then the acquirer should place the Timeout-Induced Reversal/0420 message into its store-and-forward facility for later delivery to the MDS.**

When sending SAF messages, the acquirer should wait for the MDS response to each message before sending the next SAF record. This method of transmission is known as single-threaded mode.

Note **For acquirers that support the Financial Transaction Acknowledgment/Negative Acknowledgment (0280/0290) message flow, the above Timeout-Induced Reversal/0420 messaging is not necessary. However, the acquirer has the option of using the Timeout-Induced Reversal/0420 message instead of the 0280/0290 message flow.** [2]

Acquiring processors that connect to the MDS through the virtual private network (Banknet) and use the Timeout/Reversal/0420 process do not need to support 0644 message processing. The acquirer has the option of turning off 0644 messages at its DIU or MIP and using the Timeout/Reversal/0420 message instead.

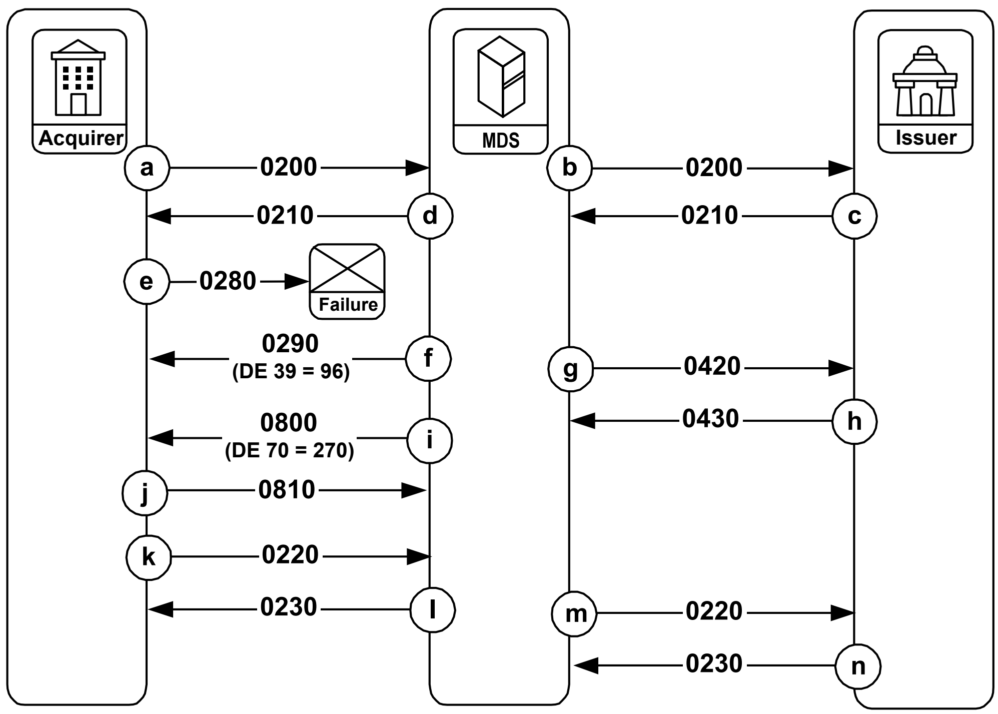
The above exception procedures apply to issuer-approved financial transactions only.

2 For members that support 0280/0290 messages, if the MDS sends an 0290 message to the acquirer with response code 96, it is not necessary to send the 0420 timeout reversal message. In those circumstances, the 0290 message indicates to the acquirer that the MDS has already reversed the transaction.

Financial Transaction/02xx Exception Condition Transaction Flow—System Failure During Financial Transaction Acknowledgment/0280 Message

Figure 2.18 illustrates exception procedures for an Acquirer Financial Transaction Acknowledgment/0280 message that is not received by the MDS due to a system failure or other condition that prevents the 0280 message from being generated or delivered.

Figure 2.18—Exception Condition, 0280 Message Not Received



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The issuer generates a Financial Transaction Request Response/0210 message and sends it to the MDS.

d. The MDS forwards the Financial Transaction Request Response/0210 message to the acquirer.

e. The MDS detects a timeout condition on the Financial Transaction Acknowledgment/0280 message it has been configured to expect from the acquirer.

f. The MDS sends a Financial Transaction Negative Acknowledgment/0290 message with Response Code (DE 39) containing a value of "96" to the acquirer. This informs the acquirer of the timeout of the Financial Transaction Acknowledgment/0280 message.

g. Because of the 0280 message timeout, the MDS generates an Acquirer Reversal Advice/0420 message to the issuer if the Financial Transaction Request Response/0210 message indicated an approval from the issuer.

h. The issuer responds with an Acquirer Reversal Advice Response/0430 message.

i. The MDS initiates, at regular intervals determined programmatically, a Network Management Request/0800 "echo test" message to verify or establish communication with the issuer.

j. The acquirer responds with a Network Management Request Response/0810 message.

k. If the acquirer has already processed the transaction approval, the acquirer must send a Financial Transaction Advice/0220 (force post) message to the MDS indicating the actual disposition of the transaction.

Note **If the acquirer does not send a Financial Transaction Advice/0220 message in this situation, it will not receive funds settlement for a successfully completed financial transaction message. If this occurs, the acquirer must reconcile with the MDS using one of the following methods:**

- **NICS™ (refer to the *NICS™ User's Guide* for more information)**
- **Online adjustment processing (see chapter 3 and chapter 4 for more information)**
- **Manual adjustments (refer chapter 7 of the *NICS™ User's Guide* for more information)**

l. The MDS responds with a Financial Transaction Advice Response/0230 message.

m. The MDS forwards the Financial Transaction Advice/0220 message to the issuer.

n. The issuer responds with a Financial Transaction Advice Response/0230 message. The issuer uses the information in the Financial Transaction Advice/0220 message to correctly update the cardholder's account file.

Note The exception procedures above apply to issuer-approved financial transactions only.

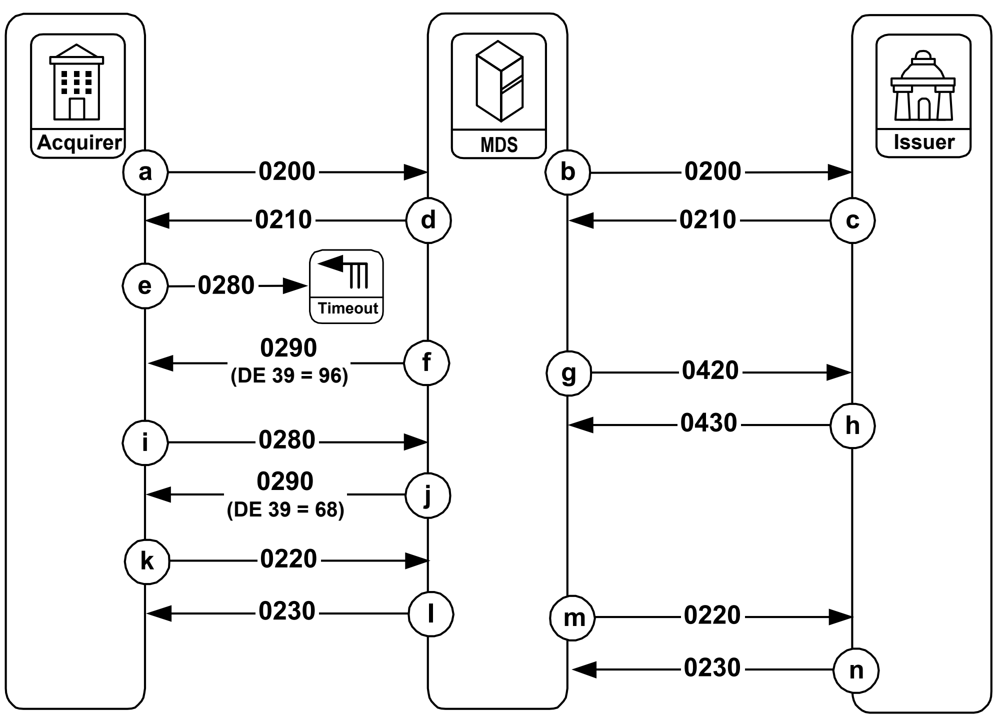
Note The Financial Transaction Acknowledgment/0280 message is no longer allowed for new member implementations. Continued support is, however, available until all members are able to convert their systems.

The Financial Transaction Acknowledgment/0280 message is supported only for Maestro and Cirrus ATM transactions. If the MDS receives a Financial Transaction Acknowledgment/0280 message for a Maestro POS transaction, the MDS truncates it and no further action is taken.

Financial Transaction/02xx Exception Condition Transaction Flow—Late Acquirer Financial Transaction Acknowledgment/0280 Message

Figure 2.19 illustrates exception procedures for a late Acquirer Financial Transaction Acknowledgment/0280 message.

Figure 2.19—Exception Condition, 0280 Message Late



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The issuer generates a Financial Transaction Request Response/0210 message and sends it to the MDS.

d. The MDS forwards the Financial Transaction Request Response/0210 message to the acquirer.

e. The MDS detects a timeout condition on the Financial Transaction Acknowledgment/0280 message it has been configured to expect from the acquirer.

f. The MDS sends a Financial Transaction Negative Acknowledgment/0290 message with Response Code (DE39) containing the value of "96" to the acquirer. This informs the acquirer of the timeout of the Financial Transaction Acknowledgment/0280 message.

g. When the 0280 message times out, the MDS generates an Acquirer Reversal Advice/0420 message to the issuer if the Financial Transaction Request Response/0210 message indicated an approval from the issuer.

h. The issuer responds with an Acquirer Reversal Advice Response/0430 message.

i. The acquirer sends a late Financial Transaction Acknowledgment/0280 message to the MDS.

j. The MDS responds with a Financial Transaction Negative Acknowledgment/0290 message with Response Code (DE 39) containing a value of "68".

k. If the acquirer has processed the transaction approval at the POS, the acquirer must send a Financial Transaction Advice /0220 (force post) message to the MDS indicating the actual disposition of the transaction.

Note **If the acquirer does not send a Financial Transaction Advice/0220 in this situation, it will not receive funds settlement for a successfully completed financial transaction message. If this occurs, the acquirer must reconcile with the MDS using one of the following methods:**

- **NICS™ (refer to the *NICS™ User's Guide* for more information)**
- **Online adjustment processing (see chapter 3 and chapter 4 for more information)**
- **Manual adjustments (refer chapter 7 of the *NICS™ User's Guide* for more information)**

l. The MDS responds with a Financial Transaction Advice Response/0230 message.

m. The MDS forwards the Financial Transaction Advice/0220 message to the issuer.

n. The issuer responds with a Financial Transaction Advice Response/0230 message. The issuer uses the information in the Financial Transaction Advice/0220 message to correctly update the cardholder's account file.

Note **The exception procedures above apply to issuer-approved financial transactions only.**

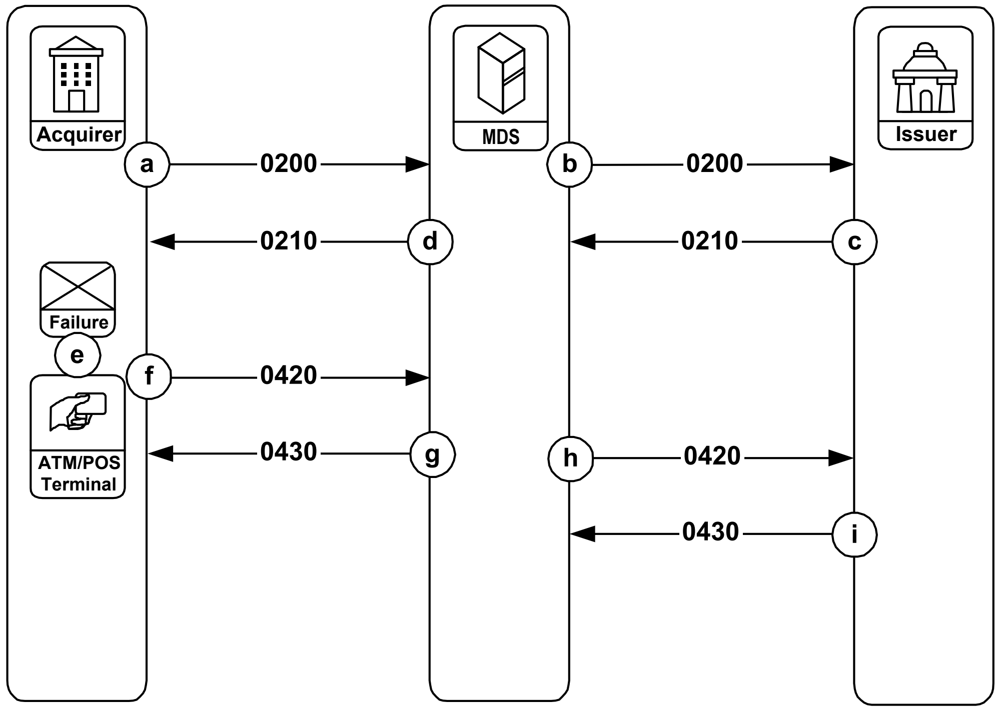
Note **The Financial Transaction Acknowledgment/0280 message is no longer allowed for new member implementations. However, support is available until all members are able to convert their systems.**

The Financial Transaction Acknowledgment/0280 message is supported only for Maestro and Cirrus ATM transactions. If the MDS receives a Financial Transaction Acknowledgment/0280 message for a Maestro POS transaction, the MDS truncates it and no further action is taken.

Financial Transaction/02xx Exception Condition Transaction Flow—Acquirer Unable to Complete Transaction

Figure 2.20 illustrates exception procedures for a situation where the acquirer is not able to complete the cardholder transaction. This situation may develop due to an ATM terminal failure resulting in partial or non-dispense of cash, communication failure within the acquirer's own network, or because a cardholder canceled the transaction prior to receiving a response. In all of these situations, the acquirer's processing system will reverse the transaction to the issuer.

Figure 2.20—Exception Condition Acquirer Unable to Complete Transaction



The sequence of events:

a. The acquirer initiates a Financial Transaction Request/0200 message to the MDS.

b. The MDS forwards the Financial Transaction Request/0200 message to the issuer.

c. The issuer generates a Financial Transaction Request Response/0210 message and sends it to the MDS.

d. The MDS forwards the Financial Transaction Request Response/0210 message to the acquirer.

e. The acquirer determines the transaction cannot be successfully completed.

f. The acquirer generates a Financial Transaction Reversal Advice/0420 message to the MDS. The reversal amount may be for the entire amount of the original transaction or for some partial amount (in the event of an ATM partial dispense).

g. The MDS responds with an Acquirer Reversal Advice Response/0430 message.

h. The MDS forwards the Acquirer Reversal Advice/0420 message to the issuer.

i. The issuer responds with an Acquirer Reversal Advice Response/0430 message. The issuer uses the information in the Acquirer Reversal Advice/0420 message to correctly update the cardholder's account file.

Note **Error condition processing for reversal advice (04xx) messages is not illustrated. If a reversal advice message does not transmit successfully, it should be re-transmitted. Issuer and acquiring processors must assume the responsibility for identifying any message as a possible "duplicate" transaction.**

Acquirer Reversal Advice/0420 and Acquirer Reversal Advice Response/0430 messages are used in conjunction with Financial Transaction/02xx messages when transaction flow exception (error) situations are encountered during financial transaction processing. For specific exception conditions, the MDS may directly generate Acquirer Reversal Advice/0420 messages. Refer to the Financial Transaction/02xx message flow schematics to determine proper use of Acquirer Reversal Advice/0420 and Acquirer Reversal Advice Response/0430 messages in these exception processing situations.

File Update Request/0302 and File Update Request Response/0312

Figure 2.21 and Figure 2.22 illustrate the message flow for file update messages. Issuers use file update messages to maintain fraudulent card-use ("hot card") or VIP databases that are available for users at the MDS and at the MasterCard Banknet Account Management System (AMS).

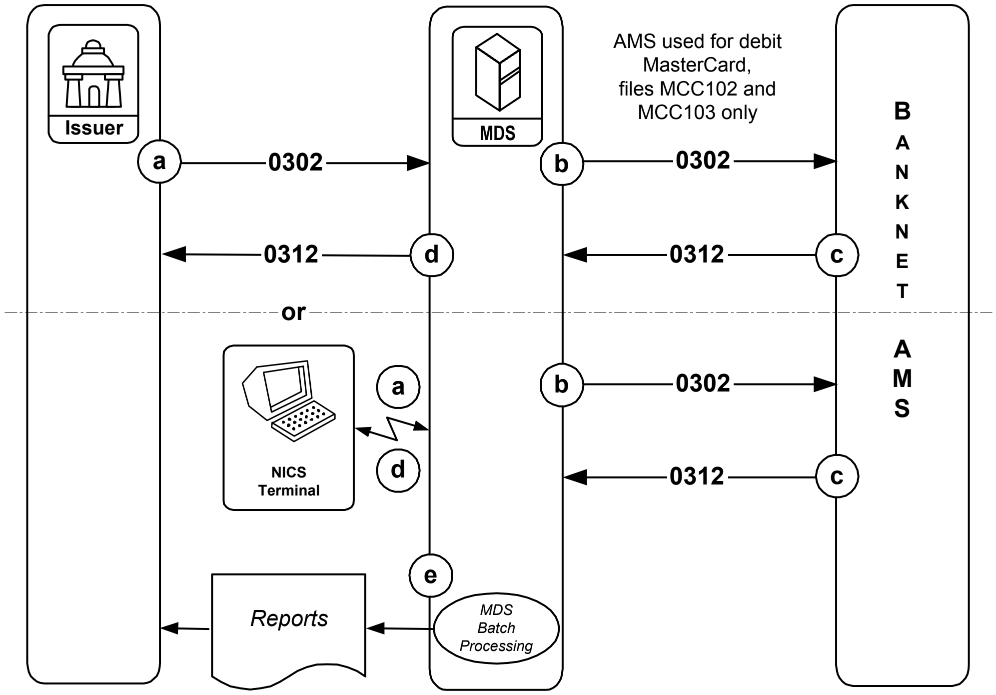
Two file update services are available:

- For debit MasterCard accounts which require access to the MasterCard Banknet AMS through use of MCC102 and MCC103 updates.
- For Maestro and Cirrus accounts that use the MDS negative file (MCCNEG) updates.

In the first case, the files being updated are the MCC102 and MCC103 files, which are maintained through the Banknet Account Management System (AMS). These updates apply to debit MasterCard cards only. For debit MasterCard issuers participating in the MDS Stand-In service, accepted AMS updates will also be updated in the MCC NEG file.

The following figure shows two separate flow paths: one originating from the issuer's online transaction processing (OLTP) system and the other from the NICS™ terminal by the issuer or issuer's authorized personnel.

Figure 2.21—0302/0312 File Update Message Flow, Case 1—Debit MasterCard



The sequence of events:

a. Issuers send File Update Request/0302 messages from their OLTP systems or from their NICS™ terminal. When issuers send file updates from their OLTP system, the File Update Request/0302 message is sent to the MDS online interface. When issuers send file updates from a NICS™ terminal, the file update request information is in an internal message format (IMF), but contains the same essential data as the 0302 message.

b. The MDS receives the File Update Request/0302 message through its file update processing facility, which passes the 0302 message to the Banknet AMS.

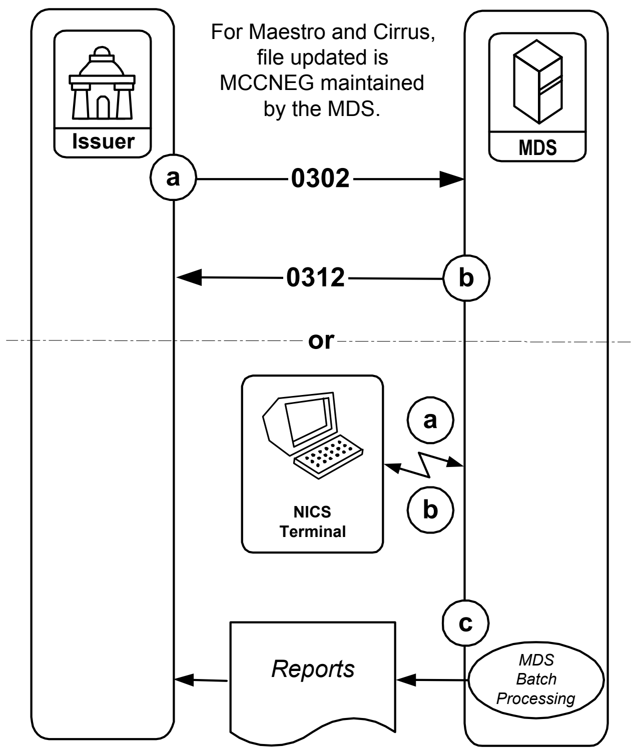
c. The Banknet AMS responds to the File Update Request/0302 message with a File Update Request Response/0312 message.

d. The MDS returns the response to the issuer. For the OLTP connection, the issuer will receive a File Update Request Response/0312 message. For the NICS™ connection, the issuer will obtain a screen image update reflecting the response. (The MDS sends an IMF message, which contains the File Update Request Response/0312 data, to the terminal).

e. Before sending the response to the issuer, the update processing facility logs the completed transaction data. From this update log file, the MDS batch processing facility generates update reports and makes them available to the issuer. These reports indicate to the issuer all the file updates that have been processed both through OLTP and NICS™.

Figure 2.22 shows how the file update process works for the Cirrus and Maestro negative card files, which are maintained on behalf of the issuer by the MDS.

Figure 2.22—0302/0312 File Update Message Flow, Case 2—Maestro and Cirrus



In this case, the issuer is updating the negative file that contains card numbers that should not be accepted. The MDS maintains and reads for Stand-In authorization of financial requests. The name of this file is MCCNEG. These updates apply to Maestro and Cirrus cards only.

The figure shows two separate flow paths: one originating from the issuer's online transaction processing (OLTP) system and the other from the NICS™ terminal by the issuer or issuer's authorized personnel.

The sequence of events:

a. From the OLTP system, the File Update Request/0302 message is sent to the MDS. From the NICS™ terminal, the file update request information is in an internal message format (IMF), but contains the same essential data as the 0302 message.

b. The MDS receives the File Update Request/0302 message through its file update processing facility, which updates the MCCNEG file. Then the MDS returns a response to the issuer: For the OLTP connection, the issuer will receive a File Update Request Response/0312 message. For the NICS™ connection, the issuer will obtain a screen image update reflecting the response. (The MDS sends an IMF message, which contains the File Update Request Response/0312 data, to the terminal.)

c. Before sending the response to the issuer, the update processing facility logs the completed transaction data. From this update log file, the MDS batch processing facility generates update reports and makes them available to the issuer. These reports indicate to the issuer all the file updates that have been processed both through OLTP and NICS™.

Reversal Advice/(042x) Transaction Exception Processing, Standard Transaction Flow

Figure 2.23 and Figure 2.24 illustrate Reversal Advice/(042x) MDS adjustment processing. MasterCard provides members access to certain functions and data within the MDS application in two ways:

- NICS™—An authorized member representative (or authorized representatives of MasterCard) can use NICS to adjust a previous transaction by initiating an exception message. These exceptions are normally performed following the settlement day of the original transaction (in the case of the issuer-initiated NICS exception, they must be performed following the settlement day of the original transaction).
- Online interface—Processors can access to the MDS transaction exception facility by sending appropriate online Reversal Advice/(042x) Exception messages to the MDS. The functionality is known as online exception processing. Online exceptions are available to the processors on a day following the settlement day.

Each method initiates one of the following types of exception message:

- **Chargeback**: An issuer-generated reversal advice message that informs an acquirer that a previously completed charge to the cardholder's account is not valid, and that the acquirer will be "charged back" that amount. A chargeback results in a credit to the issuer and a debit to the acquirer.

- **Adjustment**: An acquirer-generated reversal advice message that corrects the amount settled in a previously completed transaction. An adjustment may result in either a debit or a credit to the issuer.
- **Representment**: An acquirer-generated reversal advice message that informs an issuer a previous chargeback from the issuer is not valid, and the transaction is being "represented" for settlement. A representment results in a debit to the issuer and a credit to the acquirer.

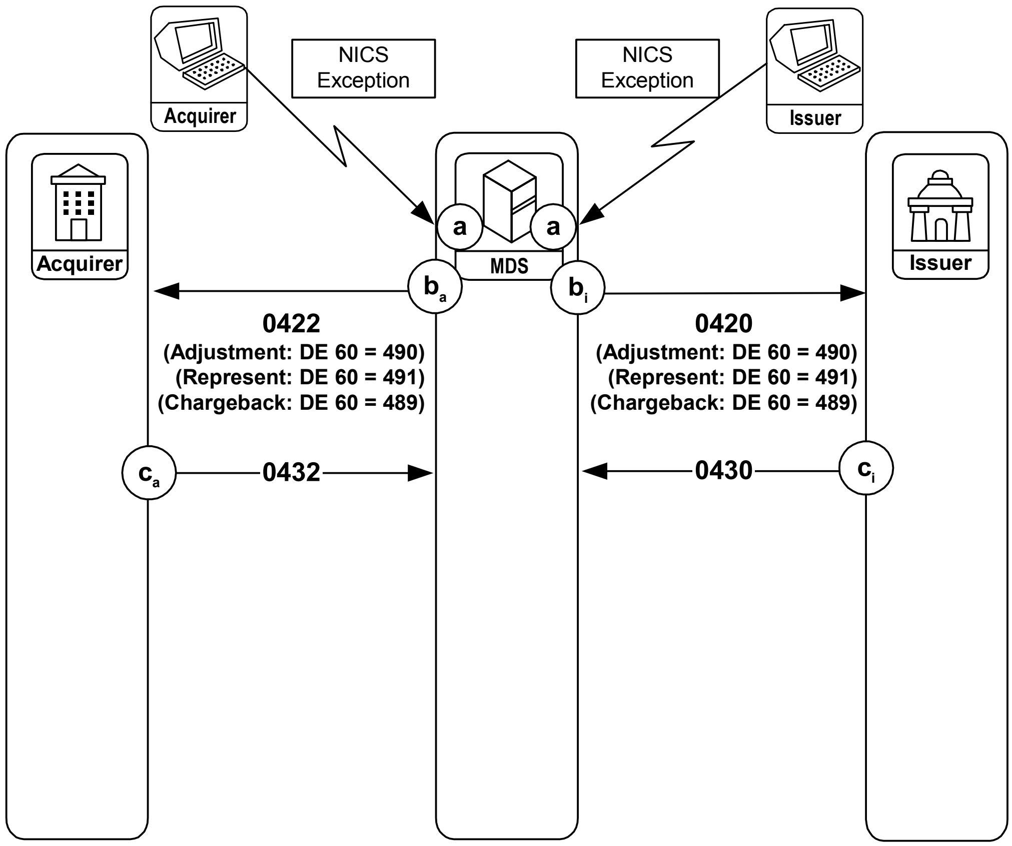
Each of the above forms of a transaction exception-item processing message is represented in an Acquirer Reversal Advice/0420-Exception message or an Issuer Reversal Advice/0422 message.

Note **The names of the message types at this point still use "Reversal" to connote the change being made.**

NICS Exception Advices

Past editions referred to these messages as MDS-generated because the external ISO 8583 message is produced by the Adjustment Manager process within the MDS application. The messages are the result of initial request made through the NICS terminal. Figure 2.23 illustrates the capability.

Figure 2.23—NICS Exception Processing



The sequence of events:

a. Authorized representatives of an acquirer, an issuer, or the MDS use NICS to create one of the following exception items:

   – Chargeback (originates from issuer)

   – Adjustment (from acquirer)

   – Representment (from acquirer)

b. After the MDS generates a chargeback, adjustment, or representment, it forwards these messages to the issuer and the acquirer processors. The acquirer always receives an Issuer Reversal Advice/0422; the issuer always receives an Acquirer Reversal Advice/0420. Both the Acquirer Reversal Advice/0420 and Issuer Reversal Advice/0422 messages (DE 90) have the same data elements as the original transaction in order to identify the financial transaction impacted by the reversal advice.

   – Chargeback: DE 60 = 489nnnn[3]

   – Reversal: DE 60 = 490nnnn[4]

   – Representment: DE 60 = 491nnnn[5]

c. In all cases, the acquirer's processing system must acknowledge the Issuer Reversal Advice/0422 with an Issuer Reversal Advice Response 0432. Similarly, the issuer's processing system must acknowledge receipt of the Acquirer Reversal Advice Response/0430 message. All reversal advices contain settlement amount and transaction fee data that affect MDS reconciliation and settlement.

Online Exception Messages

The exception in these circumstances may be an abbreviated set of the normal 0420/0422 reversal advice message. The following data elements are required:

- DE 2: Primary Account Number (PAN)
- DE 7: Transmission Date and Time
- DE 11: System Trace Audit Number
- DE 15: Date, Settlement
- DE 60: Advice Reason Code
- DE 95: Replacement Amounts

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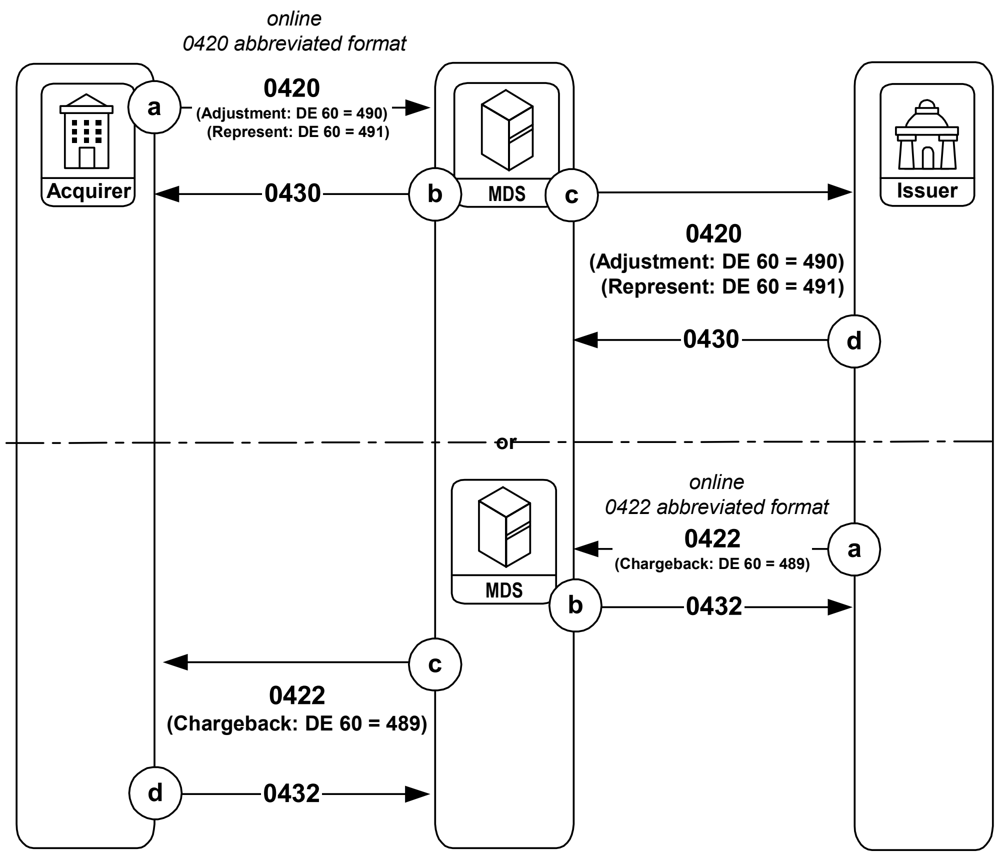
[3] nnnn = 4 digit advice detail code (for further information, see the description in chapter 4 for DE 60)

[4] nnnn = 4 digit advice detail code (for further information, see the description in chapter 4 for DE 60)

[5] nnnn = 4 digit advice detail code (for further information, see the description in chapter 4 for DE 60)

Online exceptions are available to the processors on a day following the settlement day. Figure 2.24 displays the basic flow for online exception messages.

Figure 2.24—Online Exception Processing



All Reversal Advices contain settlement amount and transaction fee data that are included with MDS reconciliation and settlement.

The sequence of events:

a. The processor initiates an online exception message:

   – Chargeback/0422 (issuer)

   – Adjustment/0420 (acquirer)

   – Representment/0420 (acquirer)

b. The MDS responds with the appropriate (0430 to an acquirer, 0432 to an issuer) Reversal Advice Response/043X message.

c. If the MDS receives a valid adjustment or a representment from the acquirer the MDS creates an Acquirer Reversal Advice/0420–Exception message to the issuer. If the MDS receives a valid chargeback from the issuer, the MDS creates an Issuer Reversal Advice/0422 message to the acquirer. The following advice reason codes (DE 60) apply:

– Chargeback: DE 60 = 489nnnn[6]

– Adjustment: DE 60 = 490nnnn[7]

– Representment: DE 60 = 491nnnn[8]

d. Issuing processors reply to the Acquirer Reversal Advice/0420 message with an Acquirer Reversal Advice Response/0430 message. Acquiring processors reply to the Issuer Reversal Advice/0422 message with an Issuer Reversal Advice Response/0432 message.

Note **Refer to the *NICS™ User's Guide* for specific information on the procedures for processing chargebacks, adjustments, and representments.**

To use the MDS online exception facility, the processor's online interface application must format and send the appropriate 042x advice message to the MDS online interface. Refer to chapter 3 for the definition of these reversal—exception message records.

Reconciliation Advice/05xx Messages—Standard Transaction Flow

Figure 2.25 illustrates the standard message flow for reconciliation advice (05xx) messages to the MDS.

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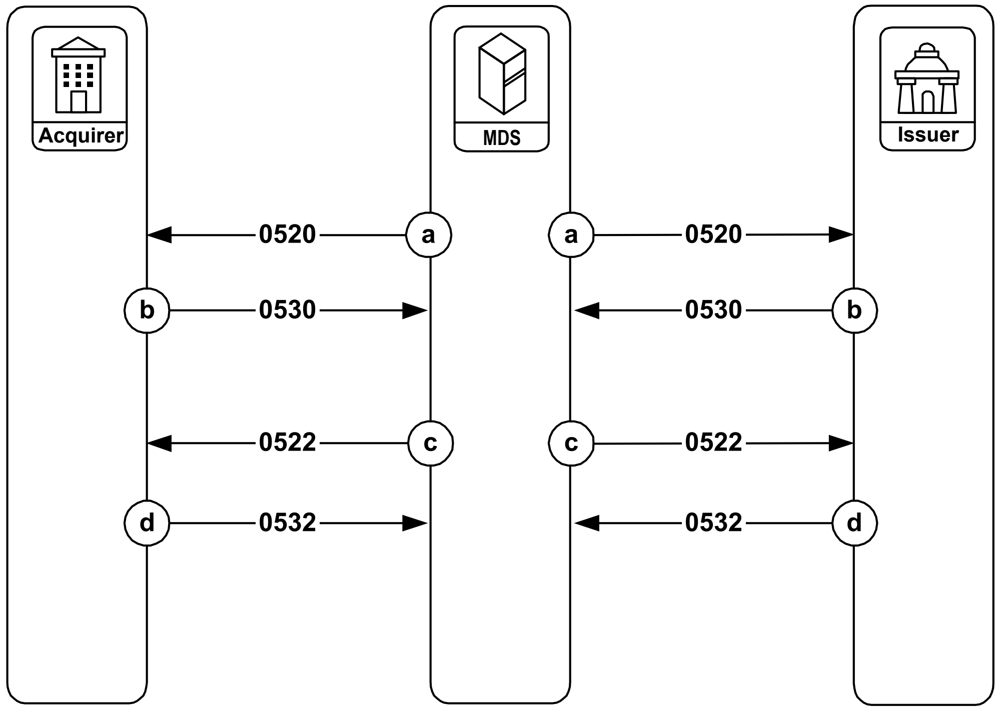
6 nnnn = 4 digit advice detail code (for further information, see the description in chapter 4 for DE 60)

7 nnnn = 4 digit advice detail code (for further information, see the description in chapter 4 for DE 60)

8 nnnn = 4 digit advice detail code (for further information, see the description in chapter 4 for DE 60)

Figure 2.25—Standard Message Flow, Reconciliation Advices 05xx



The sequence of events:

a. The MDS generates an Acquirer Reconciliation Advice/0520 message and forwards it to the processor.

b. The processor responds with an Acquirer Reconciliation Advice Response/0530 message.

c. The MDS generates an Issuer Reconciliation Advice/0522 message and forwards it to the processor.

d. The processor responds with an Issuer Reconciliation Advice Response/0532 message.

Note **If the MDS cannot immediately deliver a reconciliation advice (05xx) message to the required destination due to a communication failure or other problems, it will automatically save the message in its internal store-and-forward (SAF) facility. When communication is restored, the network will transmit the SAF file.**

At the processor's option, the MDS will generate the reconciliation advice (05xx) messages. The option is for a processor to receive both the Acquirer Reconciliation Advice/0520 message and the Issuer Reconciliation Advice/0522 message or no messages at all.

If the processor elects to receive the reconciliation advice (05xx) messages, the processor will receive both the Acquirer Reconciliation Advice/0520 message and the Issuer Reconciliation Advice/0522 message regardless of the level of activity (such as issuer only, acquirer only, or both.)

If the processor elects to receive the reconciliation advice (05xx) messages, the processor is required to respond to these messages.

Administrative Advice/06xx Messages—Standard Transaction Flow

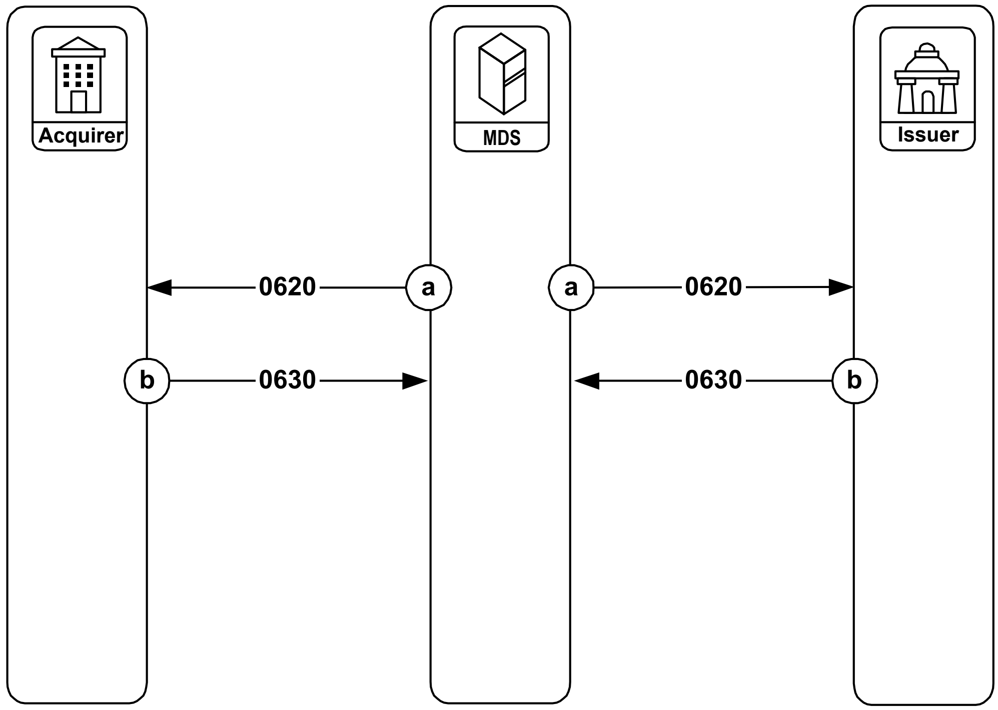
Figure 2.26 illustrates the standard message flow for administrative advice (06xx) messages to the MDS. The MDS uses the Administrative Advice/0620 messages to reject invalid messages received from the processor.

The process begins by receipt of a message from a processor that the MDS interface cannot process. The interface process generates the 0620 message, and the 0620 message envelops the data of the message the interface process just received. Table 2.2 lists all message types that can be rejected.

Table 2.2—Rejected Message Types

0200	0302	0530
0210	0420	0532
0220	0422	0800
0230	0430	0810
0280	0432	

Figure 2.26—Standard Message Flow, Administrative Advices 06xx



The sequence of events:

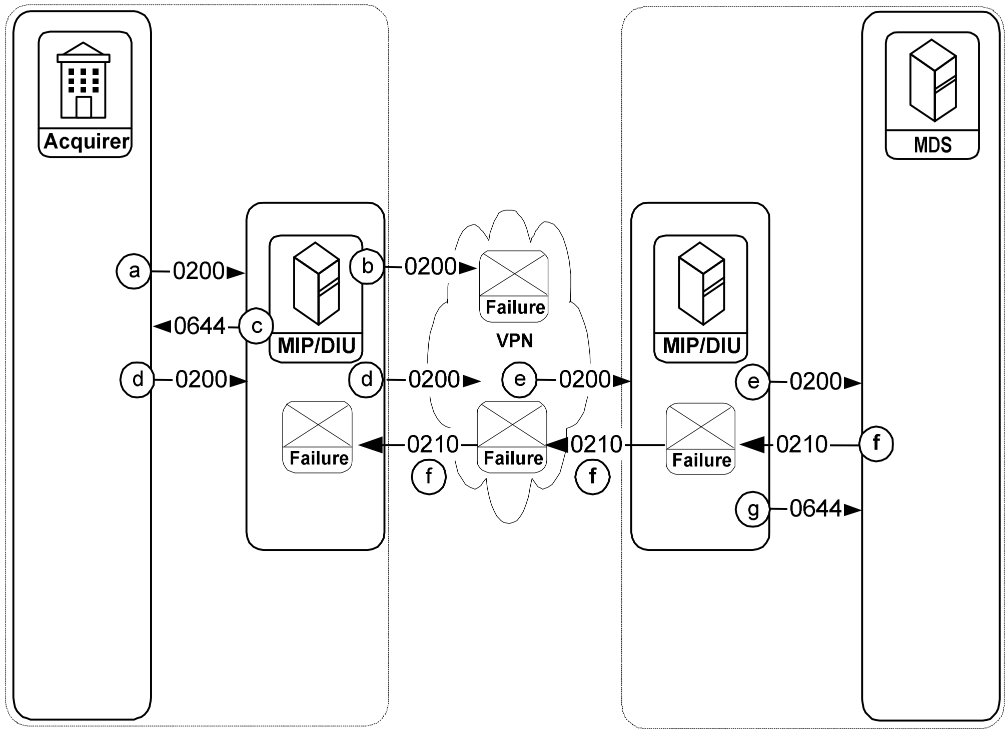
a. The MDS sends an Administrative Advice/0620 message to a processor.

b. The processor responds with an Administrative Advice Response/0630 message.

Administrative Advice/0644 for Virtual Private Network Connected Members—Standard Transaction Flow

Processing between the MDS and an acquirer connected through the virtual private network (VPN) is the same as for an issuer connected through the VPN. Acquirers and issuers may be connected through either the debit virtual private network or the Banknet telecommunications network.

Figure 2.27—Administrative Advice/0644—VPN Acquirer



Consider the following two scenarios (illustrated above) for VPN acquirers:

a. The acquirer sends a Financial Transaction Request/0200 message to the debit port on the MIP or to the Debit Interface Unit (DIU).

b. The acquirer's MIP or DIU is unable to deliver the Financial Transaction Request/0200 message to the VPN.

c. The acquirer's MIP or DIU generates an Administrative Advice/0644 message to the acquirer. The value contained in Advice Reason Code (DE 60) indicates the reason for the failure. (See the data element description in chapter 4 for a detailed list of values.)

   Or,

d. The acquirer sends a Financial Transaction Request/0200 message to the debit port on the MIP or the DIU, which forwards the message to the VPN.

e. The VPN forwards the Financial Transaction Request/0200 message to the MDS MIP. These MIPs are found at the site(s) of the MDS application and are known as "central-site MIPs", which forward the message to the MDS.

f. If the Financial Transaction Request Response/0210 message indicates an approval and the central-site MIP cannot deliver it [9], then:

g. The central-site MIP generates an Administrative Advice/0644 message to the MDS. The value contained in Advice Reason Code (DE 60) indicates the reason for the failure. (See the data element description in chapter 4 for a detailed list of values.)

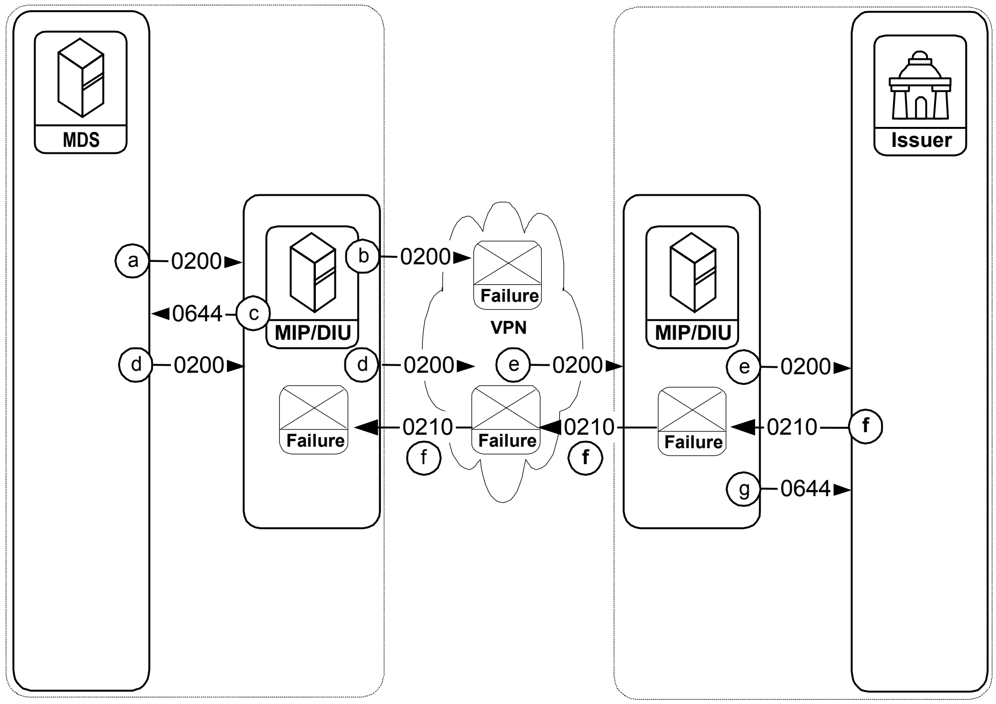
Note **Figure 2.27 shows the internal details of the Administrative Advice/0644 message processing. Following the perceived failure of delivery of the Financial Transaction Response/0210 message by the MDS, the MDS continues to resolve the failure in accordance with normal exception processing (see Figure 2.16).**

Administrative Advice/0644 for Issuing Virtual Private Network Connected Members—Standard Transaction Flow

Processing between the MDS and an issuer connected through the virtual private network (VPN) is the same as for an acquirer connected through the VPN.

9 This can occur because of a failure at the central-site VPN interface or at the VPN acquirer interface (typically, the failure would be the MIP does not have confirmation that the remote MIP delivered the 0210 to the acquirer host.)

Figure 2.28—Administrative Advice/0644—VPN Issuer



Consider the following two scenarios (illustrated above) for VPN issuers:

a. The MDS sends a Financial Transaction Request/0200 message to the MDS MIP—these MIPs located at the site(s) of the MDS application are often referred to as central-site MIPs.

b. The central-site MIP is unable to deliver the Financial Transaction Request/0200 message to the VPN.

c. The central-site MIP generates an Administrative Advice/0644 message to the MDS. The value contained in Advice Reason Code (DE 60) indicates the reason for the failure. (See the data element description in chapter 4 for a detailed list of values.)

Or,

d. The MDS sends a Financial Transaction Request/0200 message to the central-site MIP that forwards the message to the VPN.

e. The VPN forwards the Financial Transaction Request/0200 message to the issuer. The issuer receives the transaction through the debit port on the MIP or the DIU.

f. If the Financial Transaction Response/0210 message indicates an approval and the issuer MIP or DIU cannot deliver the 0210 to the VPN (e.g., the issuer MIP or DIU times-out), or because of a failure at the MDS interface to the VPN, then:

g. The issuer MIP or DIU generates an Administrative Advice/0644 message to the issuer. The value contained in Advice Reason Code (DE 60) indicates the reason for the failure. (See the data element description in chapter 4 for a detailed list of values.)

Note **Figure 2.28 shows the internal details of the Administrative Advice/0644 message processing. Following the failure of reception of the Financial Transaction Response/0210 message from the issuer by the MDS, the MDS continues to resolve the failure in accordance with normal exception processing (see Figure 2.11).**

Network Management Request/08xx Messages—Standard Transaction Flow

The MDS system uses Network Management/08xx messages provide control mechanisms between the MDS and a processor for performing the following actions:

- Signing on and signing off by the processor to/from the MDS.
- Establishing that communications exist between a processor and the MDS.
- Initiating and concluding sessions for delivery of SAF messages.
- Changing a PIN encryption key.

Three types of network management messages exist:

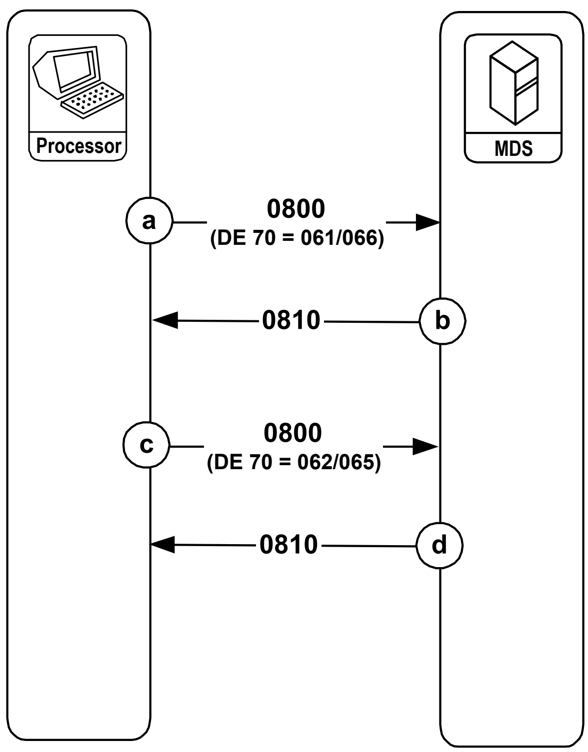
- Network Management Request/0800—the initiating message in the sequence, which identifies the purpose of the session.
- Network Management Request Response/0810—the response to the Network Management Request/0800, which indicates whether the request was received and approved.
- Network Management Advice/0820—a concluding message in some 08xx message sequences, which either indicates the end of a store-and-forward file delivery cycle from the MDS, or confirms a PIN encryption key update from the MDS to the processor.

The value of the Network Management Information Code (DE 70) of the initial Network Management Request/0800 message distinguishes the function of the 08xx message sequence. The following descriptions show how the type of network management session is related to the value of DE 70.

Network Management Message Session, Case Number 1: Sign-on and Sign-off

The processor can sign-on to the MDS by initiating a session, and can also sign-off from the MDS, which is illustrated in Figure 2.29.

Figure 2.29—08xx Signon and Signoff Sessions



The sequence of events:

a. The acquirer processing system or the issuer processing system originates the Network Management Request/0800 message for signing-on to the MDS. For a sign-on message, the Network Management Information Code (DE 70) contains one of the following values:

   – **061** General sign-on by processor to the MDS.

   – **066** If the processor is an issuer, then the processor can use this code to sign on to the MDS and direct the MDS to cease Stand-In processing for the processor.

b. The MDS responds to the sign-on request with a Network Management Request Response/0810 message, which contains the same value in DE 70 as was received in the Network Management Request/0800 message.

c. The acquirer processing system or the issuer processing system originates the Network Management Request/0800 message for signing off the MDS. For a sign-off message, DE 70 contains one of the following values:

   – **062** General sign-off by processor off the MDS.

   – **065** If the processor is an issuer, then the processor can use this code to sign off the MDS and direct the MDS to begin Stand-In processing for the processor.

d. The MDS responds to the sign-off request with a Network Management Request Response/0810 message, which contains the same value in DE 70 as was received in the Network Management Request/0800 message.

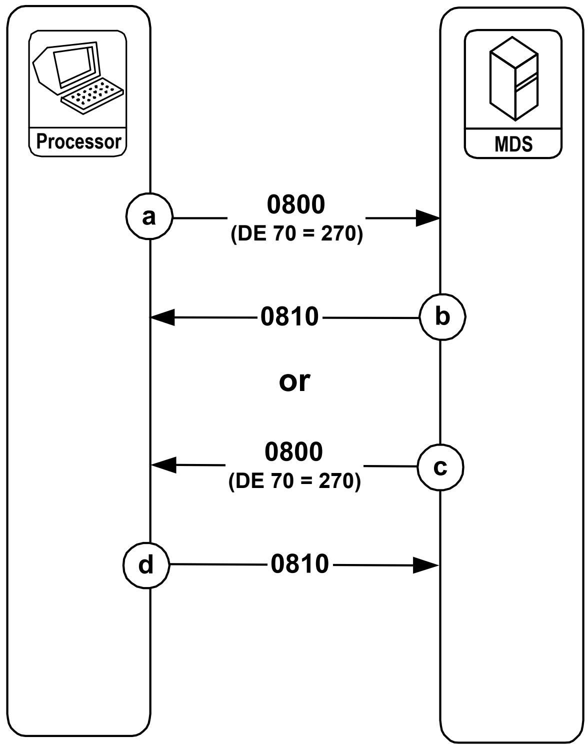
The Network Management Advice/0820 message is not used in any sign-on or sign-off sequence. The sign-on/sign-off sequence of messages was previously known as “class 0”.

Warning **Error condition message flow for Network Management (08xx) messages is not illustrated. Unsuccessful Network Management Request/0800 message transmissions should be retransmitted.**

Network Management Message Session, Case Number 2: Echo Test

The processor can initiate an echo test to the MDS and the MDS can initiate an echo test to the processor. The echo test is a means of establishing whether a processor or the MDS is connected and available for processing messages. Figure 2.30 shows the two situations:

Figure 2.30—08xx Echo Test Sessions



The sequence of events:

Processor to MDS

a. The acquirer processing system or the issuer processing system originates the Network Management Request/0800 message for an echo test to the MDS. For an echo test, the Network Management Information Code (DE 70) value in the request is "**270**".

b. The MDS responds to the echo test request with a Network Management Request Response/0810 message, which contains the same value in DE 70 as that received from the processor in the Network Management Request/0800 message. Receipt of the Network Management Request Response/0810 message by the processor indicates the MDS is operating and can process message traffic.

Note **After sending the Network Management Request Response/0810 message to the processor, the MDS sends any messages in the Store and Forward file to the processor. The MDS sends a Network Management Advice/0820 message to indicate the final message has been sent. The value for the Network Management Information Code (DE 70) in this 0820 message will be "363", SAF Delivery Complete (refer to Case Number 3 for additional details).**

Or,

MDS to Processor

c. The MDS originates the Network Management Request/0800 message for an echo test to the processor. For an echo test, the Network Management Information Code (DE 70) value in the request is "**270**".

d. The processor responds to the echo test request with a Network Management Request Response/0810 message, which contains the same value in DE 70 as that received from the MDS in the Network Management Request/0800 message. Receipt of the Network Management Request Response/0810 message by the processor indicates the processor is operating and can process message traffic.

   After sending the Network Management Request Response/0810 message to the MDS, the processor must send whatever messages exist for the MDS in the processor's Store and Forward file.

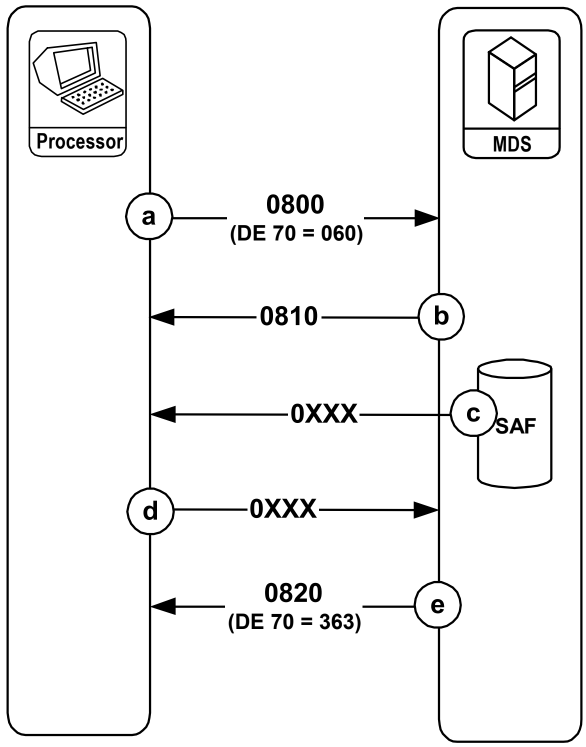
   The member-initiated echo test sequence of messages was previously known as "class 2," and the MDS-initiated echo test sequence of messages was previously known as "class-3."

Note **The echo test is also the mechanism by which the MDS requests the processor to send any SAF messages the processor has for the MDS. The MDS will send the appropriate response to each request. When the processor sends these messages, the processor does not have to end the session by delivering a Network Management Advice/0820 message.**

Network Management Message Session, Case Number 3: SAF Request by Processor to the MDS

The processor can explicitly request a delivery of the messages the MDS has in its SAF file for the processor. Figure 2.31 shows the sequence.

Figure 2.31—08xx SAF Request



The sequence of events:

a. The acquirer processing system or the issuer processing system originates the Network Management Request/0800 message explicitly requesting delivery of messages from the MDS SAF file for the processor. For a SAF request, the Network Management Information Code (DE 70) value is "**060**".

b. The MDS responds to the SAF request with a Network Management Request Response/0810 message, which contains the same value in DE 70 as that received in the Network Management Request/0800 message.

c. If the MDS has any messages for the processor in the SAF file, the MDS delivers these messages.

d. The processor responds to each advice, individually, with the appropriate message.

   – The MDS continues to send SAF messages to the processor until no messages for the processor remain.

   – Store-and-forward messages to an acquirer processing system may include any of the following message types:

     - Issuer Reversal Advice/0422
     - Acquirer Reconciliation Advice/0520
     - Administrative Advice/0620
     - Financial Transaction Negative Acknowledgment/0290 (special case—even though this is not an advice message)

   – Store-and-forward messages to an issuer processing system may include any of the following message types:

     - Financial Transaction Advice/0220
     - Acquirer Reversal Advice Response/0420
     - Issuer Reconciliation Advice/0522
     - Administrative Advice/0620
     - Financial Transaction Negative Acknowledgment/0290 (special case—even though this is not an advice message)

e. The MDS delivers a Network Management Advice/0820 message to the processor indicating that either it has no SAF messages for the processor or that it has completed delivery of the SAF messages that were requested. The value of the Network Management Information Code (DE 70) in this message is "**363**" (SAF delivery complete).

The member-initiated SAF sequence of messages was previously covered under a designation of "class 2."

Note **The processor can also receive its SAF messages from the MDS by sending an echo test (DE 70 = 270) to the MDS as illustrated above for Case Number 2.**

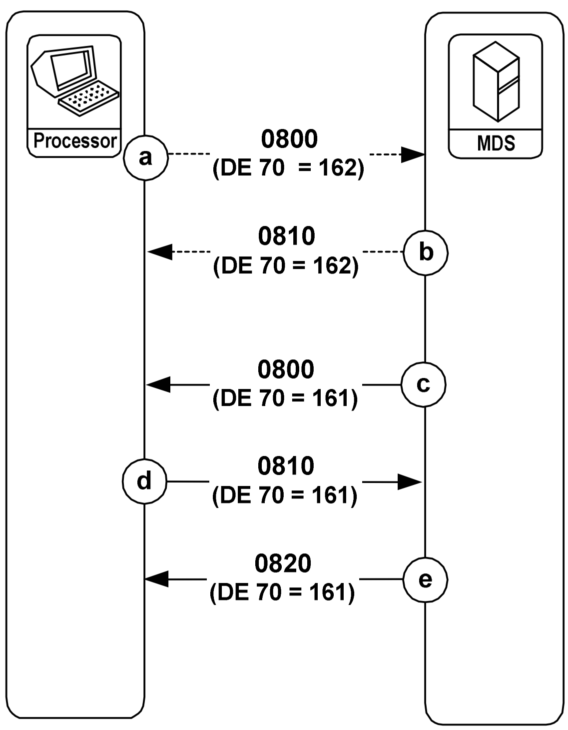
Network Management Message Session, Case Number 4: PIN Encryption Key Change

The MDS changes the PIN encryption key used between the MDS and the processor every 24 hours or after 2,500 transactions (whichever comes first). This key is also referred to as the “working key.”

Previously, only the MDS could initiate a PIN Encryption Key Request/0800 message and processors would have to contact the MDS application monitoring group to request the generation of a new PIN encryption key outside of the normal schedule.

Processors can also initiate the key change process by sending an 0800 message to the MDS with DE 70 = 162 (Initiate Key Change). This is an optional feature shown with dotted lines as steps a and b below. When the MDS receives the 0800 request to initiate key change, the MDS responds with an 0810, then immediately starts the normal key change sequence (steps c thru e below) as shown in Figure 2.32.

Figure 2.32—08xx Dynamic PIN Encryption Key Change



The sequence of events:

a. If the processor wants to initiate the key change sequence, the processor sends a Network Management Request/0800 with a Network Management Information Code (DE 70) value of 162:

b. If the processor has initiated a key change sequence by sending an 0800 request with a DE 70 value of 162, the MDS responds with a Network Management Response/0810 with the same value in DE 70.

c. The MDS originates a Network Management Request/0800 message to change the PIN encryption key (this is sometimes referred to as the "working key"). The 0800 request contains the following data:

   – DE 70 = 161 (PIN Encryption Key Change Request)

   – DE 48 subelement 11 contains the MDS Key Exchange Data Block, including the length prefix (indicating whether this is a single, double, or triple-length DES key), key cycle number, the actual PIN encryption key, and a key check value.

d. The processor responds with a Network Management Request Response/0810 message, which contains the following data:

   – DE 39 = 00 if the response is an approval (if there is a problem then DE 39 contains "96" indicating a denial).

   – DE 70 = 161 (PIN Encryption Key Change Response)

   – DE 48 subelement 11 may be returned at processor's discretion, or some portion of the subelement, but it is not required.

e. The MDS completes the sequence by sending a Network Management Advice/0820 message to the processor, indicating confirmation of the working key change. This message contains the following data:

   – DE 70 = 161 (PIN Encryption Key Change Confirmation)

   – DE 48 subelement 11 contains the first characters of the original Subelement 11 up to the beginning of the actual key. (Please refer to chapter 4, for a description of DE 48.)

The key change sequence of messages was previously referred to as "class 1."

Note **If the Network Management Request Response/0810 message indicated a denial, then the Network Management Advice/0820 message will not contain DE 48 and DE 70.**

Note **With release 2002.3, the MDS can receive key change requests from the processor. Figure 2.32 illustrates these processor initiated 0800 messages.**

The above description for 08xx message flows concludes the chapter for detailed transaction flows of ISO financial transaction series messages. chapter 3 contains message data element listings, and brief usage notes, for each of the message types described in the message flows of this chapter.

3

ISO 8583–1987 Message Formats

This chapter describes all ISO 8583–1987 message formats employed by the MasterCard® Debit Switch (MDS).

Overview

This chapter describes all ISO 8583–1987 message formats employed by the MasterCard® Debit Switch (MDS).

The MDS supports all of the following ISO 8583–1987 messages. The message format specification charts on the following pages identify all of the required, conditional, optional, or network-generated data elements employed within each individual message.

Data Element Flow

Several entities may insert or modify the data elements in an MDS message as it flows from the message origin to the MDS system, and from the MDS system to the message destination. These entities typically include the issuer or acquirer at the origin, the MDS system, and the issuer or acquirer at the destination.

In the message format layouts, the following three columns provide information to the originator, MDS system, and destination related to the data element requirements:

Table 3.1—Message Layout Format

Entity	Description
Org	**Originator Requirements.** The message originator must satisfy this data element's requirements before sending the message. A Financial Transaction Request/0200 from an acquirer is an example of an originator message.
Sys	**MDS System Requirements.** The MDS system may insert, correct, modify, or echo this data element while, for example, routing a message from the origin to the destination. The MDS system may overwrite the data element and thereby destroy any previous content.
Dst	**Destination Requirements.** The message destination must expect this data element (read it) and accept this data element (process it) if the originator requirements are satisfied. A Financial Transaction Request/0210 from an issuer is an example of a destination message.

Message Format Data Element Requirements

The following notations describe the requirements for each data element. These notations appear in the Org, Sys, and Dst columns.

Table 3.2—Data Element Requirements

Usage Code	Description
M	**Mandatory**. The data element is required in the message.
C	**Conditional**. The data element is required in the message if the conditions described in the accompanying text are applicable.
O	**Optional**. The data element is not required but may be included in the message at the message initiator's option.
X	**MDS System**. The MDS system may (or will) insert or overwrite the data element, depending on specific system support services provided for individual programs and services.
PR	**Private Data**. This is an "optional-use" data element. Proprietary data may be included within this data element at the discretion of the message initiator. It will not pass through the MDS to the message destination; however, the MDS will return this data element to the message initiator (with contents intact), in any subsequent response message.
●	**Not Required** or **Not Applicable**. The data element is not required or not applicable. The transaction originator should not include it if present in the "Org" column.

Data Element Usage in Messages

The layouts in this chapter describe the data elements in each message. Table 3.3 shows all the possible usage combinations:

Table 3.3—Data Element Usage

Usage			
Org	**Sys**	**Dst**	**Description**
M	•	M	Mandatory—Originator must provide the data element.
M	X	M	Mandatory—Originator must provide the data element; the MDS adjusts/appends data.
M	•	•	Mandatory—Originator must provide the data element; MDS does not forward data.
C	•	C	Conditionally required—Conditions are described in the Comments column.
C	X	C	Conditionally required—The MDS adjusts/appends data.
C	•	•	Conditionally required—The MDS does not forward data.
•	X	C	The MDS provides the data element conditionally.
•	X	M	The MDS always provides the data element.
•	X	•	MasterCard internal use.
•	PR	PR	Private proprietary data.
O	•	C	Optional—Originator may provide the data element.
O	X	C	Optional—Originator may provide the data element; MDS adjusts/appends data.
O	•	•	Optional—The MDS does not forward data.
PR	•	PR	Private proprietary data.
PR	PR	•	Private proprietary data.

Summary of Message Types Supported

The MasterCard® Debit Switch (MDS) supports all of the following ISO 8583–1987 message types.

Table 3.4—Message Types

Code	Description
Financial Transaction (02xx) Messages	
0200	Financial Transaction Request
0210	Financial Transaction Request Response
0220	Financial Transaction Advice
0230	Financial Transaction Advice Response
0280	Financial Transaction Acknowledgment
0290	Financial Transaction Negative Acknowledgment
MasterCard no longer allows the Financial Transaction Acknowledgment/0280 message for new member MDS implementations. Support is available until all existing members are able to convert their systems.	
File Update (03xx) Messages	
0302	File Update Request
0312	File Update Request Response
Reversal Advice (04xx) Messages	
0420	Acquirer Reversal Advice
0430	Acquirer Reversal Advice Response
0422	Issuer Reversal Advice
0432	Issuer Reversal Advice Response

Code	Description
Reconciliation Advice (05xx) Messages	
0520	Acquirer Reconciliation Advice
0530	Acquirer Reconciliation Advice Response
0522	Issuer Reconciliation Advice
0532	Issuer Reconciliation Advice Response
Administrative Advice (06xx) Messages	
0620	Administrative Advice
0630	Administrative Advice Response
0644	Administrative Advice (Debit Pass-through only for members connected to the Banknet network)
Network Management (08xx) Messages	
0800	Network Management Request
0810	Network Management Request Response
0820	Network Management Advice

Financial Transaction Request/0200 Message Format

Table 3.5—Financial Transaction Request/0200 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Value must be 0200.
-	Bit Map, Primary	M	•	M	Mandatory for all messages.
1	Bit Map, Secondary	C	•	C	Required only if any of DE 65 through DE 128 are present in the message.
2	Primary Account Number	M	•	M	Contains a cardholder's Primary Account Number (PAN).
3	Processing Code	M	•	M	Indicates type of transaction and the affected cardholder account type.
4	Amount, Transaction	M	•	M	Transaction amount in the currency of the acquirer's card acceptor. Currency Code, Transaction (DE 49) must also be present in conjunction with this data element to identify the currency of the transaction.
5	Amount, Settlement	•	X	C	Transaction amount in the currency of settlement. The MDS provides this data element when the currency of the transaction is other than U.S. dollars (840). When this data element is present, DE 9, DE 16, and DE 50 must also be present in the message.
6	Amount, Cardholder Billing	•	X	C	Amount billed to the cardholder in the currency of the cardholder account exclusive of cardholder billing fees. When present, DE 10, DE 16, and DE 51 must also be present in the message. DE 6, DE 10, and DE 51 are present if the issuer participates in ICCR.
7	Transmission Date and Time	M	X	M	Date and time in Universal Time (UTC) that the originator initiates the message. Upon receipt, the MDS updates this data element with its time stamp.
9	Conversion Rate, Settlement	•	X	C	DE 5, DE 9, DE 16, and DE 50 are included when the currency of transaction differs from the currency of settlement.
10	Conversion Rate, Cardholder Billing	•	X	C	Factor used in the conversion from transaction to cardholder billing amount. Amount Transaction (DE 4) is multiplied by DE 10 to determine Amount, Cardholder Billing (DE 6).

No.	Data Element Name	Org	Sys	Dst	Comments
11	System Trace Audit Number	M	•	M	Transaction trace number. Contents of this data element must be unique for each transaction initiated by a message originator on any single UTC date.
12	Time, Local Transaction	M	•	M	Local time of the transaction at the point-of-service as printed on all cardholder receipts and statements.
13	Date, Local Transaction	M	•	M	Local date of the transaction at the point-of-service as printed on all cardholder receipts and statements.
14	Date, Expiration	C	•	C	May be present in a debit MasterCard manually keyed authorization request.
15	Date, Settlement	O	•	M	Contains the settlement date of the transaction. Remains the same for all subsequent messages.
16	Date, Conversion	•	X	C	This field contains the effective date of any currency conversion performed for this transaction. Must be present in the message whenever the Amount, Settlement (DE 5) is present.
18	Merchant Type (MCC)	M	•	O	Must be valid on all transactions and accurately reflect the business product or service provided.
22	Point-of-Service Entry Mode	M	•	M	Indicates the method used to enter the transaction into the interchange.
23	Card Sequence Number	C	•	C	Must be present for integrated circuit card transactions.
26	Point-of-Service PIN Capture Code	C	•	C	Required only if PIN data is present and the terminal PIN capture capability is **other than** 12 characters.
28	Amount, Transaction Fee	C	•	C	Must contain ATM access fee, if applied.
32	Acquiring Institution Identification Code	M	•	M	Must contain an acquirer's identification number. Acquirer's Federal Reserve Routing and Transit number or a MasterCard assigned pseudo number.
33	Forwarding Institution Identification Code	M	•	M	Must contain the processor ID number of the CPS forwarding this message to the MDS.
35	Track 2 Data	M	•	M	Information encoded on Track 2 of the magnetic stripe. If the MDS must create Track 2 by using Track 1 data, the processors must be prepared to accept ANY character that would have been present in the Track 1.

No.	Data Element Name	Org	Sys	Dst	Comments
37	Retrieval Reference Number	C	•	C	Acquirer may use this as a document retrieval access key. If present, the system must send it back to the acquirer in any subsequent chargeback.
41	Card Acceptor Terminal Identification	M	•	M	Must contain a terminal or merchant number. The member must return it in any subsequent response.
42	Card Acceptor Identification Code	C	•	C	Must contain a merchant identifier for Maestro® ATM/POS and debit MasterCard POS transactions.
43	Card Acceptor Name and Location	M	•	M	Mandatory for all transactions. Provides location data.
45	Track I Data	C	•	C	May be present in debit MasterCard Authorization Request.
48	Additional Data	C	•	C	Contains a variety of subelements depending on the card and purpose of the request. Refer to the detailed description in chapter 4.
49	Currency Code, Transaction	M	•	M	Identifies the currency of the transaction, such as the currency used at the point-of-service.
50	Currency Code, Settlement	•	X	C	Required if present in the message.
51	Currency Code, Cardholder Billing	•	X	C	Only present if the issuer participates in ICCR
52	Personal Identification Number (PIN) Data	C	•	C	Used to contain encrypted PIN information. Required for Maestro, Cirrus, and ATM Gateway transactions. Not present for debit MasterCard POS transactions.
54	Additional Amounts	C	•	C	If used may contain cash back amount on Maestro POS transactions.
55	Integrated Circuit Card (ICC) System-Related Data	C	•	C	May be present for Chip Card transactions.
61	POS Data	M	•	M	Describes the conditions present at the point-of-service at the time the originator initiates the transaction.
62	INF Data	O	•	O	If used, may contain INF network information for use in any future on-line retrieval request, chargeback transaction, or both. When present in a request message, the destination must return it in the subsequent response message.

No.	Data Element Name	Org	Sys	Dst	Comments
63	Network Data	•	X	M	Provided by the MDS. Contains the financial network code and the switch serial number for the transaction. For debit MasterCard transactions, the Banknet reference number is also included.
112	Additional Data (National Use)	O	•	O	Parcelas data.
120	Record Data	C	•	C	May contain billing address data for debit MasterCard Address Verification Service (AVS) request.
127	Private Data	PR	•	PR	Available for private use by the message originator. Data does not pass through the MDS.

Financial Transaction Request Response/0210 Message Format

Table 3.6—Financial Transaction Request Response/0210 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Value must be 0210.
-	Bit Map, Primary	M	•	M	Mandatory for all messages.
1	Bit, Map Secondary	C	•	C	Required only if any of DE 65 through DE 128 are present in the message.
2	Primary Account Number(PAN)	M	•	M	Must contain the same value from the original request message.
3	Processing Code	M	•	M	Generally, contains the same value from the original request message. Refer to detailed description in chapter 4.
4	Amount, Transaction	M	•	M	Must contain the same value from the original request message.
5	Amount, Settlement	C	•	C	Must contain the same value from the original request, if present **and** the destination approves the transaction.
7	Transmission Date and Time	M	X	M	With limited exceptions, will contain the same value from the original request message.
9	Conversion Rate, Settlement	C	•	C	Must contain the same value from the original request, if present **and** the destination approves the transaction.

No.	Data Element Name	Org	Sys	Dst	Comments
11	System Trace Audit Number	M	•	M	Must contain the same value from the original request message.
12	Time, Local Transaction	M	•	M	Must contain the same value from the original request message.
13	Date, Local Transaction	M	•	M	Must contain the same value from the original request message.
15	Date, Settlement	M	•	M	Must contain the same value from the original request message.
16	Date, Conversion	C	•	C	Must contain the same value from the original request, if present **and** the destination approves the transaction.
28	Amount, Transaction Fee	C	•	C	Must contain the same value from the original request message.
32	Acquiring Institution Identification Code	M	•	M	Must contain the same value from the original request message.
33	Forwarding Institution Identification Code	M	•	M	Must contain the same value from the original request message.
37	Retrieval Reference Number	C	•	C	Must contain the same value from the original request message, if present.
38	Authorization Identification Response	C	•	C	May contain an Authorization ID code generated by an IPS.
39	Response Code	M	•	M	Response Code for this message.
41	Card Acceptor Terminal	M	•	M	Must contain the same value from the original request message.
44	Additional Response Data	C	•	C	Indicates the data element where the field edit error occurred.
48	Additional Data	C	•	C	Some subelements returned in response with same value as in 0200 request. Refer to detailed description in chapter 4.
49	Currency Code, Transaction	M	•	M	Must contain the same value from the original request message.
50	Currency Code, Settlement	C	•	C	Must contain the same value from the original request, if present **and** the destination approves the transaction.
54	Additional Amounts	C	•	C	If used, may contain balance inquiry and account information for Maestro and Cirrus transactions.

No.	Data Element Name	Org	Sys	Dst	Comments
55	Integrated Circuit Card (ICC) System-related Data	C	•	C	May be present for Chip Card transactions and must **not** contain the same value from the original request message.
62	INF Data	C	•	C	Must contain the same value from the original request message, if present.
63	Network Data	M	•	M	Must contain the same value from the original request message.
102	Account Identification-1	C	•	C	May contain the actual "FROM" account number.
103	Account Identification-2	C	•	C	May contain the actual "TO" account number.
112	Additional Data (National Use)	O	•	O	Parcelas data.
120	Record Data	C	•	C	Must contain the same value from the original request message.
127	Private Data	PR	•	PR	Available for private use by the message originator. Does not pass through the MDS.

Financial Transaction Advice/0220 Message Format

Table 3.7—Financial Transaction Advice/0220 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Value must be 0220.
-	Bit Map, Primary	M	•	M	Mandatory for all messages.
1	Bit Map, Secondary	C	•	C	Required only if any of DE 65 through DE 128 are present in the message.
2	Primary Account Number (PAN)	M	•	M	Must contain the same value from the original (PAN) request message.
3	Processing Code	M	•	M	Must contain the same value from the original request message.
4	Amount, Transaction	M	•	M	Must contain the same value from the original request message.
5	Amount, Settlement	•	X	C	Must contain the same value from the original request message, if present.
6	Amount, Cardholder Billing	•	X	C	Only present if the issuer participates in ICCR. Contains the same value as the original request.
7	Transmission Date and Time	M	X	M	Date and time in Universal Time (UTC) that the originator initiates the message. Upon receipt, the MDS updates this data element with its time stamp.
9	Conversion Rate, Settlement	•	X	C	Must contain the same value from the original settlement request message, if present.
10	Conversion Rate, Cardholder Billing	•	X	C	Only present if the issuer participates in ICCR.
11	System Trace Audit Number	M	•	M	Transaction trace number.
12	Time, Local Transaction	M	•	M	Must contain the same value from the original request message.
13	Date, Local Transaction	M	•	M	Must contain the same value from the original request message.
14	Date, Expiration	C	•	C	Must contain the same value from the original request message, if present.
15	Date, Settlement	M	•	M	Must contain the same value from the original request message.
16	Date, Conversion	•	•	C	Must contain the same value from the original request message, if present.

No.	Data Element Name	Org	Sys	Dst	Comments
18	Merchant Type (MCC)	M	•	O	Must be valid on all transactions and accurately reflect the business product or service provided.
22	Point-of-Service Entry Mode	M	•	M	Must contain the same value from the original request message.
23	Card Sequence Number	C	•	C	Must be present for integrated circuit card advices.
26	Point-of-Service PIN Capture Code	C	•	C	Must contain the same value from the original request message, if present.
28	Amount, Transaction Fee	C	•	•	Must contain the same value from the original message, if present.
32	Acquiring Institution Identification Code	M	•	M	Must contain the same value from the original request message.
33	Forwarding Institution Identification Code	M	•	M	Must contain the same value from the original request message, if present.
35	Track 2 Data	C	•	C	If the MDS receives DE 35 in a debit MasterCard force post from GCMS, the MDS will pass the data element to the issuer in the corresponding 0220 message. DE 35 can be present in a “chip clearing” 0220 message from an acquirer, and, if so, the MDS will send DE 35 to the issuer in the corresponding outbound 0220 message.
37	Retrieval Reference Number	C	•	C	Must contain the same value from the original request message, if present.
38	Authorization Identification Response	C	•	C	Must contain the same value from the original response message, if present.
39	Response Code	M	•	M	Must contain the same value from the original response message.
41	Card Acceptor Terminal Identification	M	•	M	Must contain the same value from the original request message.
42	Card Acceptor Identification Code	C	•	C	Must contain a merchant name for ATM/POS and debit MasterCard POS transactions.
43	Card Acceptor Name and Location	M	•	M	Must contain the same value from the original request message.
44	Additional Response Data	C	•	C	Must contain the same value from the original response message, if present.
49	Currency Code, Transaction	M	•	M	Must contain the same value from the original request message.

No.	Data Element Name	Org	Sys	Dst	Comments
50	Currency Code, Settlement	•	•	C	Must contain the same value from the original request message, if present.
51	Currency Code, Cardholder Billing	•	X	C	Only present if the issuer participates in ICCR
54	Additional Amounts	C	•	C	If used may contain cash back amount on Maestro POS transactions.
55	Integrated Circuit Card (ICC) System-Related Data	C	•	C	May be present for Chip Card transactions.
60	Advice Reason Code	M	•	M	The Advice Reason Code (ARC) indicates the specific purpose of this advice message.
61	POS Data	C	•	C	Must contain the same value from the original request message, except for debit MasterCard force post messages.
62	INF Data	C	•	C	Must contain the same value from the original request message, if present.
63	Network Data	M	•	M	Must contain the same value from the original response message.
90	Original Data Elements				Mandatory for all MDS 0220 Advices. The system uses subfields within this data element to identify the original referenced transaction.
	Original MTI	M	•	M	
	Original STAN (DE 11)	M	•	M	
	Original D & T (DE 7)	M	•	M	
	Original AID (DE 32)	M	•	M	
	Original FID (DE 33)	M	•	M	
95	Replacement Amounts				Required for partial completions only.
	Actual Amount, Transaction	C	•	C	Actual completion or adjusted amount.
	Actual Amount, Settlement	•	X	C	Actual Settlement Amount in U.S. dollars, except for those transactions labeled as ISIS.
	Actual Amount, Cardholder Billing	•	X	C	Actual Cardholder Billing Amount in U.S. dollars for issuers using ICCR.
102	Account Identification-1	C	•	C	Must contain the same value from the original response message, if present.
103	Account Identification-2	C	•	C	Must contain the same value from the original response message, if present.

Financial Transaction Advice Response/0230 Message Format

Table 3.8—Financial Transaction Advice Response/0230 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Value must be 0230.
-	Bit Map, Primary	M	•	M	Mandatory for all messages.
1	Bit Map, Secondary	C	•	C	Required only if any data elements in the range DE 65 through DE 128 are present.
2	Primary Account Number (PAN)	M	•	M	Must contain the same value from the original advice message.
3	Processing Code	M	•	M	Must contain the same value from the original advice message.
4	Amount, Transaction	M	•	M	Must contain the same value from the original advice message.
5	Amount, Settlement	C	•	C	Must contain the same value from the original advice message, if present.
7	Transmission Date and Time	M	X	M	With limited exceptions, will contain the same value from the original advice message.
9	Conversion Rate, Settlement	C	•	C	Must contain the same value from the original advice message, if present.
11	System Trace Audit Number	M	•	M	Must contain the same value from the original advice message.
12	Time, Local Transaction	M	•	M	Must contain the same value from the original advice message.
13	Date, Local Transaction	M	•	M	Must contain the same value from the original advice message.
15	Date, Settlement	M	•	M	Must contain the same value from the original advice message.
16	Date, Conversion	C	•	C	Must contain the same value from the original advice message, if present.
28	Amount, Transaction Fee	•	•	C	Must contain the same value from the original request message, if present.
32	Acquiring Institution Identification Code	M	•	M	Must contain the same value from the original advice message.
33	Forwarding Institution Identification Code	M	•	M	Must contain the same value from the original advice message.

No.	Data Element Name	Org	Sys	Dst	Comments
37	Retrieval Reference Number	C	•	C	Must contain the same value from the original advice message, if present.
39	Response Code	M	•	M	Response code for this message.
41	Card Acceptor Terminal Identification	M	•	M	Must contain the same value from the original advice message.
44	Additional Response Data	C	•	C	Indicates the data element where the field edit error occurred.
49	Currency Code, Transaction	M	•	M	Must contain the same value from the original advice message.
50	Currency Code, Settlement	C	•	C	Must contain the same value from the original advice message, if present.
54	Additional Amounts	C	•	C	If used, may contain balance inquiry and account information for Maestro® and Cirrus® transactions.
55	Integrated Circuit Card (ICC) System-related Data	C	•	C	Must contain the same value from the original advice message, if present.
62	INF Data	C	•	C	Must contain the same value from the original advice message, if present.
63	Network Data	M	•	M	Must contain the same value from the original advice message.
95	Replacement Amounts				Required if present in original advice message:
	Actual Amount, Transaction	C	•	C	Actual completion or adjusted amount in local currency.
	Actual Amount, Settlement	•	X	C	Actual Settlement Amount in U.S. dollars, except for those transactions labeled as ISIS.
	Actual Amount, Cardholder Billing	•	X	C	Actual Cardholder Billing Amount in U.S. dollars for issuers using ICCR.

Financial Transaction Acknowledgment/0280 Message Format

Note **MasterCard no longer allows the Financial Transaction Acknowledgment/0280 for new member MDS implementations. Support is available until all existing members are able to convert their systems.**

Table 3.9—Financial Transaction Acknowledgment/0280 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	•	Value must be 0280.
-	Bit Map, Primary	M	•	•	Mandatory for all messages.
1	Bit Map, Secondary	C	•	•	Required if any data elements in the range DE 65 through DE 128 are present in the message.
7	Transmission Date and Time	M	•	•	Must contain the same value from the original response message.
11	System Trace Audit Number	M	•	•	Must contain the same value from the original response message.
32	Acquiring Institution Identification Code	M	•	•	Must contain the same value from the original response message.
33	Forwarding Institution Identification Code	M	•	•	Must contain the same value from the original response message.
39	Response Code	M	•	•	Response code for this message.
63	Network Data	M	•	•	Must contain the same value from the original response message.
127	Private Data	PR	•	•	Available for private use by the message originator. Does not pass through the MDS.

Financial Transaction Negative Acknowledgment/0290 Message Format

Note **The MDS can send the Financial Transaction Negative Acknowledgment/0290 Message to either the issuer or the acquirer.**

Table 3.10—Financial Transaction Negative Acknowledgment/0290 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Value must be 0290.
-	Bit Map, Primary	M	•	M	Mandatory for all messages.
1	Bit Map, Secondary	C	•	C	Required if any data elements in the range DE 65 through DE 128 are present in the message.
7	Transmission Date and Time	M	•	M	Must contain the same value from the original response message.
11	System Trace Audit Number	M	•	M	Must contain the same value from the original response message.
32	Acquiring Institution Identification Code	M	•	M	Must contain the same value from the original response message.
33	Forwarding Institution Identification Code	M	•	M	Must contain the same value from the original response message.
39	Response Code	M	•	M	Response code for this message.
44	Additional Response Data	C	•	C	Used to indicate the data element location where the edit error or format error occurred.
63	Network Data	M	•	M	Must contain the same value from the original response message.
127	Private Data	PR	•	PR	Available for private use by the message originator. Does not pass through the MDS.

File Update Request/0302

Table 3.11—File Update Request/0302

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI) [a]	M	•	M	Value must be 0302.
-	Bit Map, Primary	M	•	M	Mandatory.
1	Bit Map, Secondary	M	•	M	Mandatory.
2	Primary Account Number (PAN)	M	•	M	Contains the primary account number to be listed by the issuer.
7	Transmission Date and Time	M	•	M	The transmission date and time expressed in Universal Time (UT).
11	System Trace Audit Number	M	•	M	Transaction trace number that must have a unique value for each transaction initiator within each UT day.
33	Forwarding Institution Identification Code	M	•	M	Contains the MasterCard customer ID number that identifies the entity to which this file update action applies.
91	File Update Code	M	•	M	File function code that describes appropriate action: add, change, delete, or inquire.
96	Message Security Code	C	•	C	File update password or security code that can be required to enable the file update.
101	File Name	M	•	M	Name of the file to be updated. Refer to chapter 4 description of DE 101 for file names, descriptions, and permissible updates.
120	Record Data	M	•	M	Contains record location for file update action, and if file add/change, contains new/changed data. Refer to chapter 4, DE 120 for a detailed description.
127	Private Data	PR	•	PR	Private use data element. Available for the message initiator's optional use.

[a] File update messages may originate from the issuer's online transaction processing system or the issuer's authorized representative via NICS™.

File Update Request Response/0312

Table 3.12—File Update Request Response/0312

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI) [a]	M	•	M	Value must be 0312.
-	Bit Map, Primary	M	•	M	Mandatory.
1	Bit Map, Secondary	M	•	M	Mandatory.
2	Primary Account Number (PAN)	M	•	M	Contains the primary account number to be listed by the issuer.
7	Transmission Date and Time	M	•	M	Must contain the same value from the original request message.
11	System Trace Audit Number	M	•	M	Must contain the same value from the original request message.
33	Forwarding Institution Identification Code	M	•	M	Must contain the same value from the original request message.
39	Response Code	M	•	M	Indicates whether the file update was successful. Refer to chapter 4 for a description of valid values for DE 39.
44	Additional Response Data	C	•	C	Provides additional information in the event of a format error. Refer to chapter 4 for a description of valid values for DE 44.
63	Network Data	M	•	M	Banknet reference number.
91	File Update Code	M	•	M	Must contain the same value from the original request message.
96	Message Security Code	C	•	C	Must contain the same value from the original request message.
101	File Name	M	•	M	Must contain the same value from the original request message.
120	Record Data	M	•	M	Must contain the same value from the original request message.
122	Additional Record Data	C	•	C	A free form field used to return additional data as a result of a file inquiry. Refer to chapter 4, DE 122 for a detailed record description and valid values.
127	Private Data	PR	•	PR	Must contain the same value from the original request message.

[a] File update response messages originate at the MasterCard Account Management System or, in the case of MDS Stand-In, from the MDS. The responses are ultimately returned to the issuer.

Acquirer Reversal Advice/0420 Message Format

Table 3.13—Acquirer Reversal Advice/0420 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Value must be 0420. Type "A" format: Generated by APS only.
-	Bit Map, Primary	M	•	M	Mandatory.
1	Bit Map, Secondary	M	•	M	Mandatory.
2	Primary Account Number (PAN)	M	•	M	Must contain the same value from the original request message.
3	Processing Code	M	•	M	Must contain the same value from the original request message.
4	Amount, Transaction	M	•	M	Must contain the same value from the original request message.
5	Amount, Settlement	•	X	C	If present, must contain the same value from the original request message.
6	Amount, Cardholder Billing	•	X	C	Contains the same value from the original request message, if present in the original request message.
7	Transmission Date and Time	M	X	M	Date and time in Universal Time (UTC) that the originator initiates the message. Upon receipt, the MDS updates this data element with its time stamp.
9	Conversion Rate, Settlement	•	X	C	Required if DE 5 is present. If present, must contain the same value from the original request message.
11	System Trace Audit Number	M	•	M	Must contain the same value as the original request message.
12	Time, Local Transaction	M	•	M	Must contain the same value from the original request message.
13	Date, Local Transaction	M	•	M	Must contain the same value from the original request message.
15	Date, Settlement	M	X	M	Contains the settlement date of this transaction. From original response, if available.
16	Date, Conversion	•	X	C	Required if DE 5 is present. If present, must contain the same value from the original request message.
23	Card Sequence Number	C	•	C	Must be present for Integrated Circuit Card (ICC).

No.	Data Element Name	Org	Sys	Dst	Comments
28	Amount, Transaction Fee	C	•	•	Must contain the same value from the original request message, if present.
32	Acquiring Institution Identification Code	M	•	M	Must contain the same value from the original request message.
33	Forwarding Institution Identification Code	M	•	M	Must contain the same value from the original request message.
37	Retrieval Reference Number	C	•	C	Must contain the same value from the original request message, if present.
38	Authorization ID Response	C	•	C	From original response, if available.
39	Response Code	M	•	M	Must contain the same value from the original transaction response message.
41	Card Acceptor Terminal Identification	M	•	M	Must contain the same value from the original request message.
49	Currency Code, Transaction	M	•	M	Must contain the same value from the original request message.
50	Currency Code, Settlement	•	X	C	Required if present in the message. If present, must contain the same value from the original request message.
54	Additional Amounts	C	•	C	Must be present if contained in the original cash back transaction.
60	Advice Reason Code	M	•	M	Indicates the specific reason for this Reversal message. The system uses the three-digit Advice Reason Code and four-digit MDS Advice Detail Code.
62	INF Data	C	•	C	Must contain the same value from the original request message, if present.
63	Network Data	M	•	M	Must contain the same value from the original transaction response message.
90	Original Data Elements				Mandatory for all Acquirer Reversal Advice/0420 reversals. The system uses the subfields within this data element to identify the original reversed transaction.
	Original MTI	M	•	M	
	Original STAN (DE 11)	M	•	M	
	Original D&T (DE 7)	M	•	M	
	Original AID (DE 32)	M	•	M	
	Original FID (DE 33)	M	•	M	

No.	Data Element Name	Org	Sys	Dst	Comments
95	Replacement Amounts				Required:
	Actual Amount, Transaction	C	•	C	Actual completion or adjusted amount.
	Actual Amount, Settlement	•	X	C	Actual Settlement Amount in U.S. dollars, except for those transactions labeled as ISIS.
	Actual Amount, Cardholder Billing	•	X	C	Actual Cardholder Billing Amount in U.S. dollars for issuers using ICCR.
	Actual Amount, Transaction	M	•	M	Actual completion or adjusted amount, in local currency.
	Actual Amount, Settlement	•	X	C	Actual Settlement Amount in U.S. dollars, except for those transactions labeled as ISIS.
127	Private Data	PR	•	PR	Available for private use by the message originator. Does not pass through the MDS.

Acquirer Reversal Advice/0420-Timeout-Induced Message Format

Table 3.14—Acquirer Reversal Advice/0420-Timeout-Induced Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Value must be 0420. Type “A” format: Generated by APS only.
-	Bit Map, Primary	M	•	M	Mandatory.
-	Bit Map, Secondary	M	•	M	Mandatory.
2	Primary Account Number (PAN)	M	•	M	Must contain the same value from the original request message.
3	Processing Code	M	•	M	Must contain the same value from the original request message.
4	Amount, Transaction	M	•	M	Must contain the same value from the original request message.
5	Amount, Settlement	•	X	C	The MDS supplies currency conversion data, if required.
7	Transmission Date and Time	M	X	M	Date and time in Universal Time (UTC) that the originator initiates the message. Upon receipt, the MDS updates this data element with its time stamp.
9	Conversion Rate, Settlement	•	X	C	Required if DE 5 is present.
11	System Trace Audit Number	M	•	M	Must contain the same value as the original request message.
12	Time, Local Transaction	M	•	M	Must contain the same value from the original request message.
13	Date, Local Transaction	M	•	M	Must contain the same value from the original request message.
15	Date, Settlement	•	X	M	Provided by MDS in the message to the issuer.
16	Date, Conversion	•	X	C	Required if DE 5 is present.
23	Card Sequence Number	C	•	C	Must be present for Integrated Circuit Card (ICC).
28	Amount, Transaction Fee	C	•	•	Must contain the same value from the original request message, if present.
32	Acquiring Institution Identification Code	M	•	M	Must contain the same value from the original request message.

No.	Data Element Name	Org	Sys	Dst	Comments
33	Forwarding Institution Identification Code	M	•	M	Must contain the same value from the original request message.
37	Retrieval Reference Number	C	•	C	Must contain the same value from the original request message, if present.
38	Authorization ID Response	C	•	C	From original response, if available.
39	Response Code	M	•	M	Contains the code “00” for the Timeout-Induced Reversal message.
41	Card Acceptor Terminal Identification	M	•	M	Must contain the same value from the original request message.
49	Currency Code, Transaction	M	•	M	Must contain the same value from the original request message.
50	Currency Code, Settlement	•	X	C	Required if DE 5, DE 29, or DE 31 are present in the message.
54	Additional Amounts	C	•	C	Must be present if contained in the original cash back transaction
60	Advice Reason Code	M	•	M	Has the value “4500018” for the Timeout-Induced Reversal message.
62	INF Data	C	•	C	Must contain the same value from the original request message, if present.
63	Network Data	•	X	M	Provided by the MDS in the message to the issuer.
90	Original Data Elements				Mandatory for all Acquirer Reversal Advice/0420 reversals. The system uses the subfields within this data element to identify the original reversed transaction.
	Original MTI	M	•	M	
	Original STAN (DE 11)	M	•	M	
	Original D&T (DE 7)	M	•	M	
	Original AID (DE 32)	M	•	M	
	Original FID (DE 33)	M	•	M	
95	Replacement Amounts	M	•	M	Contains all zeroes for the Timeout-Induced Reversal message.
127	Private Data	PR	•	PR	Available for private use by the message originator. Does not pass through the MDS.

Acquirer Reversal Advice Response/0430 Message Format

Table 3.15—Acquirer Reversal Advice Response/0430 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	•	M	Value Must be 0430. (Response to acquirer-generated 0420 advice)
-	Bit Map, Primary	M	•	M	Mandatory.
1	Bit Map, Secondary	M	•	M	Mandatory.
2	Primary Account Number (PAN)	M	•	M	Must contain the same value from the original advice message.
3	Processing Code	M	•	M	Must contain the same value from the original advice message.
4	Amount, Transaction	M	•	M	Must contain the same value from the original advice message.
5	Amount, Settlement	C	•	C	Must contain the same value from the original advice message, if present.
7	Transmission Date and Time	M	X	M	With limited exceptions, will contain the same value from the original advice message.
9	Conversion Rate, Settlement	C	•	C	The MDS will provide currency conversion data, if required.
11	System Trace Audit Number	M	•	M	Must contain the same value from the original advice message.
12	Time, Local Transaction	M	•	M	Must contain the same value from the original advice message.
13	Date, Local Transaction	M	•	M	Must contain the same value from the original advice message.
15	Date, Settlement	M	•	M	Must contain the same value from the original advice message.
16	Date, Conversion	C	•	C	Must contain the same value from the original advice message, if present.
23	Card Sequence Number	C	•	C	Must be present for integrated circuit card transactions.
28	Amount, Transaction Fee	C	•	•	Must contain the same value from the original request message, if present.
32	Acquiring Institution Identification Code	M	•	M	Must contain the same value from the original advice message.

No.	Data Element Name	Org	Sys	Dst	Comments
33	Forwarding Institution Identification Code	M	•	M	Must contain the same value from the original advice message.
37	Retrieval Reference Number	C	•	C	Must contain the same value from the original advice message, if present.
39	Response Code	M	•	M	Response code for this message.
41	Card Acceptor Terminal ID	M	•	M	Must contain the same value from the original advice message.
44	Additional Response Data	C	•	C	Indicates the data element location where the field edit error occurred.
49	Currency Code, Transaction	M	•	M	Must contain the same value from the original advice message.
50	Currency Code, Settlement	C	•	C	Must contain the same value from the original advice message, if present.
54	Additional Amounts	C	•	C	If used, may contain balance inquiry and account information for Maestro® and Cirrus® transactions.
62	INF Data	C	•	C	Must contain the same value from the original advice message, if present.
63	Network Data	M	•	M	Must contain the same value from the original advice message.
95	Replacement Amounts	M	•	M	Contains the same subelement values from the original advice message.
127	Private Data	PR	•	PR	Available for private use by the message originator. Does not pass through the MDS.

Acquirer Reversal Advice/0420-Adjustment Message Format

Table 3.16—Acquirer Reversal Advice/0420-Adjustment Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	•	M	Value must be 0420. Type “S” format: generated by MDS only.
-	Bit Map, Primary	•	•	M	Mandatory.
1	Bit Map, Secondary	•	•	M	Mandatory.
2	Primary Account Number (PAN)	•	•	M	Must contain the same value from the original request message.
3	Processing Code	•	•	M	Must contain the same value from the original request message.
4	Amount, Transaction	•	•	M	Must contain the same value from the original request message.
5	Amount, Settlement	•	•	C	MDS supplies currency conversion data, if required.
6	Amount, Cardholder Billing	•	X	C	Must contain the same value from the original advice message, if present.
7	Transmission Date and Time	•	•	M	The system initiates the date and time, in UTC format of this message.
9	Conversion Rate, Settlement	•	•	C	Required if DE 5 is present.
10	Conversion Rate, Cardholder Billing	•	X	C	Must contain the same value from the original advice message, if present.
11	System Trace Audit Number	•	•	M	Transaction trace number. The contents of this data element must be unique for each transaction initiated by a message originator on any single UTC date.
12	Time, Local Transaction	•	•	M	Must contain the same value from the original request message.
13	Date, Local Transaction	•	•	M	Must contain the same value from the original request message.
15	Date, Settlement	•	•	M	Contains the settlement date of this transaction.
16	Date, Conversion	•	•	C	Required if DE 5 is present.
32	Acquiring Institution Identification Code	•	•	M	Must contain the same value from the original request message.

No.	Data Element Name	Org	Sys	Dst	Comments
33	Forwarding Institution Identification Code	•	•	M	Must contain the same value from the original request message.
37	Retrieval Reference Number	•	•	C	Must contain the same value from the original request message, if present.
39	Response Code	•	•	M	Must contain the same value from the original request.
41	Card Acceptor Terminal Identification	•	•	M	Must contain the terminal ID to which this transaction applies.
49	Currency Code, Transaction	•	•	M	Must contain the same value from the original request message.
50	Currency Code, Settlement	•	•	C	Required if present in the message.
51	Currency Code, Cardholder Billing	•	X	C	Must contain the same value from the original advice message, if present.
60	Advice Reason Code	•	•	M	Indicates the specific reason for this reversal message. The system uses the 3-digit Advice Reason Code and 4-digit MDS Advice Detail Code.
62	INF Data	•	•	C	Must contain the same value from the original request message, if present.
63	Network Data	•	•	M	Must contain the same value from the original transaction response message.
90	Original Data Elements				Mandatory for all Acquirer Reversal Advice/0420 reversals. The system uses subfields within this data element to identify the original reversed transaction.
	Original MTI	•	•	M	
	Original STAN (DE 11)	•	•	M	
	Original D&T (DE 7)	•	•	M	
	Original AID (DE 32)	•	•	M	
	Original FID (DE 33)	•	•	M	

No.	Data Element Name	Org	Sys	Dst	Comments
95	Replacement Amounts				Mandatory for all MDS 0420 Reversals.
	Actual Amount, Transaction	C	•	C	Actual completion or adjusted amount.
	Actual Amount, Settlement	•	X	C	Actual Settlement Amount in U.S. dollars, except for those transactions labeled as ISIS.
	Actual Amount, Cardholder Billing	•	X	C	Actual Cardholder Billing Amount in U.S. dollars for issuers using ICCR.
127	Private Data	•	•	PR	Available for private use by the message originator. Does not pass through the MDS.

Acquirer Reversal Advice Response/0430-Adjustment Message Format

Table 3.17—Acquirer Reversal Advice Response/0430-Adjustment Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	•	M	Value must be 0430.
-	Bit Map, Primary	•	•	M	Mandatory for all messages.
1	Bit Map, Secondary	•	•	M	Mandatory.
2	Primary Account Number (PAN)	•	•	M	Must contain the same value from the original advice message.
3	Processing Code	•	•	M	Must contain the same value from the original advice message.
4	Amount, Transaction	•	•	M	Must contain the same value from the original advice message.
5	Amount, Settlement	•	•	C	Must contain the same value from the original advice message, if present.
6	Amount, Cardholder Billing	•	•	C	Must contain the same value from the original advice message, if present.
7	Transmission Date and Time	•	•	M	Must contain the same value from the original advice message.
9	Conversion Rate Settlement,	•	•	O	The MDS provides currency conversion data, if required.
10	Conversion Rate, Cardholder Billing	•	•	C	Must contain the same value from the original advice message, if present.
11	System Trace Audit Number	•	•	M	Must contain the same value from the original advice message.
12	Time, Local Transaction	•	•	M	Must contain the same value from the original advice message.
13	Date, Local Transaction	•	•	M	Must contain the same value from the original advice message.
15	Date, Settlement	•	•	M	Must contain the same value from the original advice message.
16	Date, Conversion	•	•	C	Must contain the same value from the original advice message, if present.
32	Acquiring Institution Identification Code	•	•	M	Must contain the same value from the original advice message.

No.	Data Element Name	Org	Sys	Dst	Comments
33	Forwarding Institution Identification Code	•	•	M	Must contain the same value from the original advice message.
37	Retrieval Reference Number	•	•	C	Must contain the same value from the original advice message, if present.
39	Response Code	•	•	M	Response code for this message.
41	Card Acceptor Terminal ID	•	•	M	Must contain the same value from the original advice message.
44	Additional Response Data	•	•	C	Indicates the data element location where the field edit error occurred.
49	Currency Code, Transaction	•	•	M	Must contain the same value from the original advice message.
50	Currency Code, Settlement	•	•	C	Must contain the same value from the original advice message, if present.
51	Currency Code, Cardholder Billing	•	•	C	Must contain the same value from the original advice message, if present.
62	INF Data	•	•	C	Must contain the same value from the original advice message, if present.
63	Network Data	•	•	M	Must contain the same value from the original advice message.
95	Replacement Amounts	•	•	M	Contains the same subelement values from the original advice message.
127	Private Data	•	•	PR	Available for private use by the message originator. Does not pass through the MDS.

Issuer Reversal Advice/0422 Message Format

Table 3.18—Issuer Reversal Advice/0422 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	•	M	Value Must be 0422. Type “S” format: generated by MDS only.
-	Bit Map, Primary	•	•	M	Mandatory.
1	Bit Map, Secondary	•	•	M	Mandatory.
2	Primary Account Number (PAN)	•	•	M	Must contain the same value from the original request message.
3	Processing Code	•	•	M	Must contain the same value from the original request message.
4	Amount, Transaction	•	•	M	Must contain the same value from the original request message.
5	Amount, Settlement	•	•	C	The MDS supplies currency conversion data, if required.
7	Transmission Date and Time	•	•	M	The system initiates the date and time in UTC format of this message.
9	Conversion Rate, Settlement	•	•	C	Required if DE 5 is present.
11	System Trace Audit Number	•	•	M	Transaction trace number. Contents of this data element must be unique for each transaction initiated by a message originator on any single UTC date.
12	Time, Local Transaction	•	•	M	Must contain the same value from the original request message.
13	Date, Local Transaction	•	•	M	Must contain the same value from the original request message.
15	Date, Settlement	•	•	M	Contains the settlement date of this transaction.
16	Date, Conversion	•	•	C	Required if DE 5 is present.
28	Amount, Transaction Fee	•	•	C	Must contain the same value from the original request message, if present.
32	Acquiring Institution Identification Code	•	•	M	Must contain the same value from the original request message.
33	Forwarding Institution Identification Code	•	•	M	Must contain the same value from the original request message.

No.	Data Element Name	Org	Sys	Dst	Comments
37	Retrieval Reference Number	•	•	C	Must contain the same value from the original request message, if present.
41	Card Acceptor Terminal Identification	•	•	M	Must contain the same value from the original request message.
49	Currency Code, Transaction	•	•	M	Must contain the same value from the original request message.
50	Currency Code, Settlement	•	•	C	Required if present in the message.
60	Advice Reason Code	•	•	M	Indicates the specific reason for this reversal message. The system uses the 3-digit Advice Reason Code and 4-digit MDS Advice Detail Code.
62	INF Data	•	•	C	Must contain the same value from the original message, if present.
63	Network Data	•	•	M	Must contain the same value from the original transaction response message.
90	Original Data Elements				Mandatory for all Acquirer Reversal Advice/0420 reversals. The system uses subfields within this data element to identify the original reversed transaction.
	Original MTI	•	•	M	
	Original STAN (DE 11)	•	•	M	
	Original D&T (DE 7)	•	•	M	
	Original AID (DE 32)	•	•	M	
	Original FID (DE 33)	•	•	M	
95	Replacement Amounts				Mandatory for all MDS 0422 Reversals.
	Actual Amount, Transaction.	•	•	M	Actual completion or adjusted amount, in local currency.
	Actual Amount, Settlement.	•	X	C	Settlement Amount in U.S. dollars, except for those transactions labeled as ISIS.
	Actual Amount, Cardholder Billing	•	X	C	Actual Cardholder Billing Amount in U.S. dollars if ICCR service is applicable.
127	Private Data	•	•	PR	Available for private use by the message originator. Does not pass through the MDS.

Issuer Reversal Advice Response/0432 Message Format

Table 3.19—Issuer Reversal Advice Response/0432 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	•	M	Value must be 0432. (Response to MDS generated 0422 advice).
-	Bit Map, Primary	•	•	M	Mandatory.
1	Bit Map, Secondary	•	•	M	Mandatory.
2	Primary Account Number (PAN)	•	•	M	Must contain the same value from the original advice message.
3	Processing Code	•	•	M	Must contain the same value from the original advice message.
4	Amount, Transaction	•	•	M	Must contain the same value from the original advice message.
5	Amount, Settlement	•	•	C	Must contain the same value from the original advice message, if present.
7	Transmission Date and Time	•	•	M	Must contain the same value from the original advice message.
9	Conversion Rate, Settlement	•	•	O	MDS provides currency conversion data, if required.
11	System Trace Audit Number	•	•	M	Must contain the same value from the original advice message.
12	Time, Local Transaction	•	•	M	Must contain the same value from the original advice message.
13	Date, Local Transaction	•	•	M	Must contain the same value from the original advice message.
15	Date, Settlement	•	•	M	Must contain the same value from the original advice message.
16	Date, Conversion	•	•	C	Must contain the same value from the original advice message, if present.
28	Amount, Transaction Fee	•	•	C	Must contain the same value from the original request message, if present.
32	Acquiring Institution Identification Code	•	•	M	Must contain the same value from the original advice message.
33	Forwarding Institution Identification Code	•	•	M	Must contain the same value from the original advice message.

No.	Data Element Name	Org	Sys	Dst	Comments
37	Retrieval Reference Number	•	•	C	Must contain the same value from the original advice message, if present.
39	Response Code	•	•	M	Response code for this message.
41	Card Acceptor Terminal ID	•	•	M	Must contain the same value from the original advice message.
44	Additional Response Data	•	•	C	Indicates the data element location where the field edit error occurred.
49	Currency Code, Transaction	•	•	M	Must contain the same value from the original advice message.
50	Currency Code, Settlement	•	•	C	Must contain the same value from the original advice message, if present.
62	INF Data	•	•	C	Must contain the same value from the original advice message, if present.
63	Network Data	•	•	M	Must contain the same value from the original advice message.
95	Replacement Amounts	•	•	M	Contains the same subelement values from the original advice message.
127	Private Data	•	•	PR	Available for private use by the message originator. Does not pass through the MDS.

Acquirer Reconciliation Advice/0520 Message Format

Table 3.20—Acquirer Reconciliation Advice/0520 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Value must be 0520 for all Acquirer Reconciliation Advice/0520 messages.
-	Bit Map, Primary	•	M	M	Mandatory.
1	Bit Map, Secondary	•	M	M	Mandatory.
7	Transmission Date and Time	•	M	M	The system initiates the date and time (in UTC format) of the message.
11	System Trace Audit Number	•	M	M	Transaction trace number. Contents of this data element must be unique for each transaction initiated by a message originator on any single UTC date.
15	Date, Settlement	•	M	M	Contains the effective settlement date for all financial information provided in this message.
33	Forwarding Institution Identification Code	•	M	M	Contains the processor ID of the CPS receiving this advice message.
50	Currency Code, Settlement	•	M	M	Contains a numeric code to indicate the currency of settlement. The system provides all “amount” data elements in this message in this currency.
60	Advice Reason Code (ARC)	•	M	M	The ARC indicates the specific reason for this reconciliation advice.
63	Network Data	•	M	M	Contains financial network ID and the MDS reference number assigned to this message.
74	Credits, Number	•	C	C	Number of acquirer credit transactions.
75	Credits, Reversal Number	•	C	C	Number of acquirer reversal credit transactions.
76	Debits, Number	•	C	C	Number of acquirer debit transactions.
77	Debits, Reversal Number	•	C	C	Number of acquirer reversal debit transactions.
78	Transfers, Number	•	C	C	Contains all zeros.
79	Transfers, Reversal Number	•	C	C	Contains all zeros.
80	Inquiries, Number	•	C	C	Contains all zeros.
81	Authorizations, Number	•	C	C	Contains all zeros.
82	Credits, Processing Fee	•	C	C	Contains all zeros.

No.	Data Element Name	Org	Sys	Dst	Comments
83	Credits, Transaction Fee Amount	•	C	C	Contains all zeros.
84	Debits, Processing Fee Amount	•	C	C	Contains all zeros.
85	Debits, Transaction Fee Amount	•	C	C	Contains all zeros.
86	Credits, Amount	•	C	C	Total amount of all acquirer credit transactions due to acquirer.
87	Credits, Reversal Amount	•	C	C	Total amount of all acquirer reversal credit transactions due to acquirer.
88	Debits, Amount	•	C	C	Total amount of all acquirer debit transactions due to acquirer.
89	Debits, Reversal Amount	•	C	C	Total amount of all acquirer reversal debit transactions due to acquirer.
97	Amount, Net Settlement	•	M	M	Total Acquirer Net Settlement amount. The system calculates this as the sum of all amounts specified in this message, and includes all transaction and switch fees. If this computation results in a positive number, the “x” portion of this data element is set to “C” indicating that the acquirer has a net credit position with respect to the financial network entity (such as funds are due to the acquirer). Otherwise, the acquirer has a net debit position and the “x” portion of this data element is set to “D” (funds are due from the acquirer).
99	Settlement Institution Identification Code	•	M	M	MDS Settlement agent, contains the value 1072000326.
100	Receiving Institution Identification Code	•	M	M	Processor ID of the system to which the MDS transmits this settlement message.

Acquirer Reconciliation Advice Response/0530 Message Format

Table 3.21—Acquirer Reconciliation Advice Response/0530 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Value must be 0530 for all Acquirer Reconciliation Advice Response/0530 messages.
-	Bit Map, Primary	•	M	M	Mandatory.
1	Bit Map, Secondary	•	M	M	Mandatory.
7	Transmission Date and Time	•	M	M	Must contain the same value from the original advice message.
11	System Trace Audit Number	•	M	M	Must contain the same value from the original advice message.
15	Date, Settlement	•	M	M	Must contain the same value from the original advice message.
33	Forwarding Institution Identification Code	•	M	M	Must contain the same value from the original advice message.
39	Response Code	•	M	M	Response Code for this message.
44	Additional Response Data	•	C	C	Used to indicate the data element location where a message format error occurred.
50	Currency Code, Settlement	•	M	M	Must contain the same value from the original advice message.
63	Network Data	•	M	M	Must contain the same value from the original advice message.
66	Settlement Code	•	O	O	Optional DE to indicate receipt of reconciliation message without performance of balancing (value of “9”).
97	Amount, Net Settlement	•	M	M	Must contain the same value from the original advice message.
99	Settlement Institution Identification Code	•	M	M	Must contain the same value from the original advice message.
100	Receiving Institution Identification Code	•	M	M	Must contain the same value from the original advice message.

Issuer Reconciliation Advice/0522 Message Format

Table 3.22—Issuer Reconciliation Advice/0522 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Value must be 0522 for all Issuer Reconciliation Advice/0522 messages.
-	Bit Map, Primary	•	M	M	Mandatory.
1	Bit Map, Secondary	•	M	M	Mandatory.
7	Transmission Date and Time	•	M	M	The system initiates the date and time (in UTC format) of the message.
11	System Trace Audit Number	•	M	M	Transaction trace number. Contents of this data element must be unique for each transaction initiated by a message originator on any single UTC date.
15	Date, Settlement	•	M	M	Contains the effective settlement date for all financial information provided in this message.
33	Forwarding Institution Identification Code	•	M	M	Contains the processor ID of the CPS receiving this advice message.
50	Currency Code, Settlement	•	M	M	Contains a numeric code to indicate the currency of settlement. The system provides all "amount" data elements in this message in this currency.
60	Advice Reason Code (ARC)	•	M	M	The ARC indicates the specific reason for this advice.
63	Network Data	•	M	M	Contains financial network ID and the network reference number assigned to this message.
74	Credits, Number	•	C	C	Number of issuer credit transactions.
75	Credits, Reversal Number	•	C	C	Number of issuer reversal credit transactions.
76	Debits, Number	•	C	C	Number of issuer debit transactions.
77	Debits, Reversal Number	•	C	C	Number of issuer reversal debit transactions.
78	Transfers, Number	•	C	C	Contains all zeros.
79	Transfers, Reversal Number	•	C	C	Contains all zeros.
80	Inquiries, Number	•	C	C	Contains all zeros.
81	Authorizations, Number	•	C	C	Contains all zeros.
82	Credits, Processing Fee Amount	•	C	C	Contains all zeros.

No.	Data Element Name	Org	Sys	Dst	Comments
83	Credits, Transaction Fee Amount	•	C	C	Contains all zeros.
84	Debits, Processing Fee Amount	•	C	C	Contains all zeros.
85	Debits, Transaction Fee Amount	•	C	C	Contains all zeros.
86	Credits, Amount	•	C	C	Total amount of all issuer credit transactions due to issuer.
87	Credits, Reversal Amount	•	C	C	Total amount of all issuer reversal credit transactions due to issuer.
88	Debits, Amount	•	C	C	Total amount of all issuer debit transactions due to issuer.
89	Debits, Reversal Amount	•	C	C	Total amount of all issuer reversal debit transactions due to issuer.
97	Amount, Net Settlement	•	M	M	Total Issuer Net Settlement amount. The system calculates this as the sum of all amounts specified in the message and includes all transaction and processing fees. If this computation results in a positive number, the “x” portion of this data element is set to “C” to indicate that the issuer has a net credit position with respect to the financial network entity (such as funds are due to the issuer). Otherwise, the issuer has a net debit position and the “x” portion of this data element is set to “D” (funds are due from the issuer).
99	Settlement Institution Identification Code	•	M	M	Processor ID of the issuer settlement agent contains the value 1072000326.
100	Receiving Institution Identification Code	•	M	M	Processor ID of the system to which the MDS transmits this settlement message.

Issuer Reconciliation Advice Response/0532 Message Format

Table 3.23—Issuer Reconciliation Advice Response/0532 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Value must be 0532.
-	Bit Map, Primary	•	M	M	Mandatory.
1	Bit Map, Secondary	•	M	M	Mandatory.
7	Transmission Date and Time	•	M	M	Must contain the same value from the original advice message.
11	System Trace Audit Number	•	M	M	Must contain the same value from the original advice message.
15	Date, Settlement	•	M	M	Must contain the same value from the original advice message.
33	Forwarding Institution Identification Code	•	M	M	Must contain the same value from the original advice message.
39	Response Code	•	M	M	Response Code for this message.
44	Additional Response Data	•	C	C	Used to indicate the data element location where the message format error occurred.
50	Currency Code, Settlement	•	M	M	Must contain the same value from the original advice message.
63	Network Data	•	M	M	Must contain the same value from the original advice message.
66	Settlement Code	•	O	O	Optional DE to indicate receipt of reconciliation message without performance of balancing (value of “9”).
97	Amount, Net Settlement	•	M	M	Must contain the same value from the original advice message.
99	Settlement Institution Identification Code	•	M	M	Must contain the same value from the original advice message.
100	Receiving Institution Identification Code	•	M	M	Must contain the same value from the original advice message.

Administrative Advice/0620 Message Format

Table 3.24—Administrative Advice/0620 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Value must be 0620.
-	Bit Map, Primary	•	M	M	Mandatory.
1	Bit Map, Secondary	•	M	M	Mandatory.
7	Transmission Date and Time	•	M	M	The system transmits the date and time, in UTC format of this message.
11	System Trace Audit Number	•	M	M	Must contain the same value from the original message.
33	Forwarding Institution Identification Code	•	M	M	Must contain the same value from the original message.
60	Advice Reason Code	•	M	M	Indicates the specific reason for this message. 600 = rejected message.
63	Network Data	•	X	M	Provided by the MDS. Contains the MDS reference number for this transaction.
100	Receiving Institution Identification Code	•	M	M	Must contain the Processor ID of the CPS or Network destination for this message.
120	Record Data	•	C	C	If ARC = 600, this data element may be used to contain the original (rejected) message.
127	Private Data	•	PR	PR	Available for private use by the message originator. Does not pass through the MDS.

Administrative Advice Response/0630 Message Format

Table 3.25—Administrative Advice Response/0630 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Value must be 0630.
-	Bit Map, Primary	•	M	M	Mandatory.
1	Bit Map, Secondary	•	M	M	Mandatory.
7	Transmission Date and Time	•	M	M	Must contain the same value from the original advice message.
11	System Trace Audit Number	•	M	M	Must contain the same value from the original advice message.
33	Forwarding Institution Identification Code	•	M	M	Must contain the same value from the original advice message.
39	Response Code	•	M	M	Response code for this message.
44	Additional Response Data	•	C	C	May contain data element number where edit error occurred in a rejected message.
63	Network Data	•	M	M	Must contain the same value from the original advice message.
100	Receiving Institution Identification Code	•	M	M	Must contain the same value from the original advice message.
127	Private Data	•	PR	PR	Available for private use by the message originator. This data does not pass through the MDS.

Administrative Advice/0644 Message Format

Table 3.26—Administrative Advice/0644 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Value must be 0644.
-	Bit Map, Primary	•	M	M	Mandatory.
1	Bit Map, Secondary	•	M	M	Mandatory.
7	Transmission Date and Time	•	M	M	The system transmits the date and time, in UTC format of this message.
11	System Trace Audit Number	•	M	M	Must contain the same value from the original message.
33	Forwarding Institution Identification Code	•	M	M	Must contain the same value from the original message.
60	Advice Reason Code	•	M	M	Indicates the specific reason for this message. 690x = rejected message.
63	Network Data	•	X	M	Contains the Banknet reference number. The interface replaces this number with the MDS switch serial number.
100	Receiving Institution Identification Code	•	•	•	Must contain the Processor ID of the CPS or Network destination for this message.
120	Record Data	•	C	C	If ARC = 690x, this data element may be used to contain the original (rejected) message.
127	Private Data	•	PR	PR	Available for private use by the message originator. Does not pass through the MDS.

Network Management Request/0800 Message Format

Table 3.27—Network Management Request/0800 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Value must be 0800 for all Network Management Request/0800 messages.
-	Bit Map, Primary	•	M	M	Mandatory.
1	Bit Map, Secondary	•	M	M	Mandatory.
2	Primary Account Number	•	M	M	This data element must contain the originating processor ID for sign–on/sign-off by group NCID network management codes (see DE 70, below).
7	Transmission Date and Time	•	M	M	The system transmits the date and time (in UTC format) of the message.
11	System Trace Audit Number	•	M	M	Transaction trace number; must be unique value for transaction initiator within each UTC day.
33	Forwarding Institution Identification Code	•	M	M	Contains the processor ID of the CPS or NCID originating this message.
48	Additional Data	•	X	C	Key data for PIN encryption key exchange messages only, where DE 70 = 161
63	Network Data	•	X	M	Provided by the MDS. Includes a network reference number for this transaction.
70	Network Management Information Code	•	M	M	Indicates the specific purpose of this 0800 message. Valid values are as follows: **060** Store and Forward session request **061** Sign-on by processor to the MDS **062** Sign-off by processor from the MDS **065** Sign-off by processor from the MDS, MDS to begin Stand-In processing **066** Sign-on by processor to the MDS, MDS to cease Stand-In processing **161** PIN encryption key change **270** Echo test
96	Message Security Code	•	C	C	Contains a MDS Password security code to verify that the originator of the Sign-on Request is allowed access to the requested functions.
127	Private Data	•	PR	PR	Available for private use by the message originator. Does not pass through the MDS.

Network Management Request Response/0810 Message Format

Table 3.28—Network Management Request Response/0810 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	M	M	•	Value must be 0810 for all Network Management Request/0810 messages.
-	Bit Map, Primary	M	M	•	Mandatory.
1	Bit Map, Secondary	M	M	•	Mandatory.
7	Transmission Date and Time	M	M	•	Must contain the same value from the original request message.
11	System Trace Audit Number	M	M	•	Must contain the same value from the original request message.
33	Forwarding Institution Identification Code	M	M	•	Must contain the same value from the original request message.
39	Response Code	M	M	•	Response code for this message.
44	Additional Response Data	C	C	•	May contain additional response or diagnostic information when Response Code (DE 39) is 30.
48	Additional Data	C	C	•	For key change messages (DE 70 = 161), this field may contain the PIN encryption key.
63	Network Data	•	M	•	Must contain the same value from the original request message.
70	Network Management Information Code	M	M	•	Must contain the same value from the original request message.
127	Private Data	PR	PR	•	Available for private use by the message originator. Does not pass through the MDS.

Network Management Advice/0820 Message Format

Note **Network Management Advice/0820 messages DO NOT have corresponding Network Management Advice Acknowledgment/0830 messages**

Table 3.29—Network Management Advice/0820 Message Format

No.	Data Element Name	Org	Sys	Dst	Comments
-	Message Type Identifier (MTI)	•	M	M	Value must be 0820 for all Network Management Advice /0820 messages.
-	Bit Map, Primary	•	M	M	Mandatory.
1	Bit Map, Secondary	•	M	M	Mandatory.
7	Transmission Date and Time	•	M	M	The system transmits the date and time (in UTC format) of the message.
11	System Trace Audit Number	•	M	M	Transaction trace number. Must be unique value for transaction initiator within each UTC day.
33	Forwarding Institution Identification Code	•	M	M	Contains the processor ID of the network facility originating this message.
48	Additional Data	C	C	•	For key change messages (DE 70 = 161), this field will be present. Will not be present when the 0810 message is denied.
63	Network Data	•	X	M	Provided by the MDS. Includes a network reference number for this transaction.
70	Network Management Information Code	•	C	C	Indicates the specific purpose of this 0820 message. Will not be present when the 0810 message is denied. Valid values are as follows: **161** Encryption key change confirmed **363** Store and Forward complete
127	Private Data	•	PR	PR	Available for private use by the message originator. Does not pass through the MDS.

4 ISO 8583–1987 Data Element Definitions

This chapter provides a detailed definition of all data elements used within ISO 8583–1987 online financial institution card (bank card) messages.

Overview

This chapter provides a detailed definition of all data elements that are used within ISO 8583–1987 bank card message types. Information is presented in the following order:

1. Explanation of the notation used throughout this chapter to describe all data element attributes.
2. Summary list of all ISO 8583–1987 data elements in the order of their ISO-assigned bit map numbers, including annotation of those data elements which are currently **not implemented** within the ISO 8583–1987 specification.
3. Detailed definition of each message data element, presented in the order of the data element bit map number. Information provided for each data element includes the following:
   - Data element definition
   - Data element usage
   - Data element formats, including data representation attributes, data field format, and data field length
   - Data element values or field edits indicating the specific value(s) or permissible range of values that may be present within the data element
   - Product application notes, where applicable, that detail unique, product-specific, or message-specific usage of the data element

Annotation Conventions for Data Element Attributes

The following notation conventions are used throughout this chapter to describe the attributes of ISO 8583–1987 message data elements:

Conventions for Data Representation

The data encoding conventions listed below have been adapted for all ISO 8583–1987 messages:

- The system aligns all message data element fields on byte boundaries; for example; a data field cannot begin with the low order "nibble" or any bit other than the high-order bit of any byte.
- All of the data types listed in Table 4.1, are encoded for transmission between the MDS and processor systems using EBCDIC display character representation:
- All numeric (attribute **n**) data elements are **right justified** with **leading zeroes** unless otherwise specified in the individual data element definitions. All other data elements are **left justified** with **trailing blanks** unless otherwise specified.
- All binary (attribute **b**) data elements are constructed of bit-strings which have lengths that are an integral number of 8-bit bytes. No binary data element has a length of less than eight bits (one byte).
- All track-2 or track-3 (attribute **z**) data elements are encoded as EBCDIC representations of the hexadecimal data specified in ISO specification 7811 and 7812. Thus, a hex "D" (binary "1101") is encoded as an EBCDIC "D" character, and so on. The LLVAR or LLLVAR length specification associated with these data elements specifies the field length in number of bytes.
- The system encodes all length subfields as numeric EBCDIC, right justified with leading zeroes.
  - Fields designated LL are 2-character numeric fields with values from "01" to "99".
  - Fields designated LLL are 3-character numeric fields with values from "001" to "999".

General Representation

Table 4.1—Data Types

Notation	Description
a	alphabetic characters only (MUST USE UPPERCASE LETTERS) [a]
n	numeric characters only
s	special characters only
an	alpha and numeric characters
as	alpha and special characters
ns	numeric and special characters
ans	alpha, numeric, and special characters
b	binary data
z	magnetic stripe track-2 or track-3 data
x	character "C" or "D" to indicate "credit" or "debit" value of a dollar amount

[a] The MDS application requires use of uppercase letters in data specifying state and country codes. Thus, safe, general practice is to code alpha data as uppercase.

Length Attributes

Table 4.2—Data Length Attributes

Notation	Description
-digit(s)	Fixed length in number of positions. Example: "n-3" indicates a 3-position numeric field. Example: "an-10" indicates a 10-position alphanumeric field.
…digit(s)	Variable length field, with maximum number of positions specified. Example: "n...11" indicates a variable length numeric field of up to 11 digits. Example: "an...25" indicates a variable length alphanumeric field of up to 25 characters.
LLVAR	Present with a variable-length data element attribute, indicates that the data element contains two fields: "LL" is the length field and represents the number of positions in the variable-length data field that follows. The length field contains a value in the range of 01-99. "VAR" is the variable-length data field. Example: "an...25; LLVAR" represents a variable-length alphanumeric data element with a length of 1-25 positions.
LLLVAR	Present with a variable-length data element attribute, indicates that the data element contains two fields: "LLL" is the length field and represents the number of positions in the variable-length data field that follows. The length field contains a value in the range 001-999. "VAR" is the variable-length data field. Example: "an...500; LLLVAR" indicates a variable-length alphanumeric data element having a length of 1-500 positions.

Field Content Attributes

Table 4.3—Data and Time Attributes

Notation	Description
MM	month (two digits, 01–12)
DD	day (two digits, 01–31)
YY	year (last two digits of calendar year, 00–99)
hh	hour (two digits, 00–23)
mm	minute (two digits, 00–59)
ss	second (two digits, 00–59)

Message Data Elements

Table 4.4 lists all data elements implemented within the ISO 8583–1987 message standard. Where indicated, some data elements are not currently used. **ISO 8583–1987 messages should not contain these data elements.**

Table 4.4—ISO 8583–1987 Message Standard Data Elements

Number	Name	Attributes
1	Bit map, Secondary	b-64
2	Primary Account Number	n...19; LLVAR
3	Processing Code	n-6
4	Amount, Transaction	n-12
5	Amount, Settlement	n-12
6	Amount, Cardholder Billing	n-12
7	Transmission Date and Time	n-10; MMDDhhmmss
8	Amount, Cardholder Billing Fee (not currently used)	n-8
9	Conversion Rate, Settlement	n-8
10	Conversion Rate, Cardholder Billing	n-8
11	System Trace Audit Number	n-6
12	Time, Local Transaction	n-6; hhmmss
13	Date, Local Transaction	n-4; MMDD
14	Date, Expiration	n-4; YYMM
15	Date, Settlement	n-4; MMDD
16	Date, Conversion	n-4; MMDD
17	Date, Capture (not currently used)	n-4; MMDD
18	Merchant Type (MCC)	n-4
19	Acquiring Institution Country Code (not currently used)	n-3
20	Primary Account Number, Extended, Country Code (not currently used)	n-3
21	Forwarding Institution Country Code (not currently used)	n-3
22	Point of Service Entry Mode	n-3

Number	Name	Attributes
23	Card Sequence Number	n-3
24	Network International Identifier (not currently used)	n-3
25	Point of Service Condition Code (not currently used)	n-2
26	Point of Service PIN Capture Code	n-2
27	Authorization Identification Response Length (not currently used)	n-1
28	Amount, Transaction Fee	x+n-8
29	Amount, Settlement Fee	x+n-8
30	Amount, Transaction Processing Fee (not currently used)	x+n-8
31	Amount, Settlement Processing Fee	x+n-8
32	Acquiring Institution Identification Code	n...11; LLVAR
33	Forwarding Institution Identification Code	n...11; LLVAR
34	Primary Account Number, Extended (not currently used)	ns...28; LLVAR
35	Track-2 Data	z...37; LLVAR
36	Track-3 Data (not currently used)	z...104; LLLVAR
37	Retrieval Reference Number	an-12
38	Authorization Identification Response	an-6
39	Response Code	an-2
40	Service Restriction Code (not currently used)	an-3
41	Card Acceptor Terminal Identification	ans-8
42	Card Acceptor Identification Code	ans-15
43	Card Acceptor Name/Location	ans-40
44	Additional Response Data	ans...25; LLVAR
45	Track-1 Data	ans...79; LLVAR
46	Additional Data (ISO) (not currently used)	ans...999; LLLVAR
47	Additional Data (National) (not currently used)	ans...999; LLLVAR
48	Additional Data (Private/ISO 8583–1987)	ans...999; LLLVAR
49	Currency Code, Transaction	n-3

Number	Name	Attributes
50	Currency Code, Settlement	n-3
51	Currency Code, Cardholder Billing	n-3
52	Personal Identification Number (PIN) Data	b-64
53	Security Related Control Information (not currently used)	n-16
54	Additional Amounts	an...120; LLLVAR
55	Integrated Circuit Card (ICC) System Related Data	b...255; LLLVAR
56	Reserved (ISO) (not currently used)	ans...999; LLLVAR
57–59	Reserved (National) (not currently used)	ans...999; LLLVAR
60	Advice Reason Code	ans...060; LLLVAR
61	Point of Service (POS) Data	ans...026; LLLVAR
62	Intermediate Network Facility (INF) Data	ans...050; LLLVAR
63	Network Data	ans...044; LLLVAR
64	Message Authentication Code (MAC) (not currently used)	b-64
65	Bit map, Extended (not currently used)	b-64
66	Settlement Code	n-1
67	Extended Payment Code (not currently used)	n-2
68	Receiving Institution Country Code (not currently used)	n-3
69	Settlement Institution Country Code (not currently used)	n-3
70	Network Management Information Code	n-3
71	Message Number (not currently used)	n-4
72	Message Number Last (not currently used)	n-4
73	Date, Action (not currently used)	n-6; YYMMDD
74	Credits, Number	n-10
75	Credits, Reversal Number	n-10
76	Debits, Number	n-10
77	Debits, Reversal Number	n-10
78	Transfers, Number	n-10
79	Transfers, Reversal Number	n-10

Number	Name	Attributes
80	Inquiries, Number	n-10
81	Authorizations, Number	n-10
82	Credits, Processing Fee Amount	n-12
83	Credits, Transaction Fee Amount	n-12
84	Debits, Processing Fee Amount	n-12
85	Debits, Transaction Fee Amount	n-12
86	Credits, Amount	n-16
87	Credits, Reversal Amount	n-16
88	Debits, Amount	n-16
89	Debits, Reversal Amount	n-16
90	Original Data Elements	n-42
91	File Update Code	an-1
92	File Security Code (not currently used)	an-2
93	Response Indicator (not currently used)	an-5
94	Service Indicator (not currently used)	an-7
95	Replacement Amount	n-42
96	Message Security Code (not currently used)	b-64
97	Amount, Net Settlement	x+n-16
98	Payee (not currently used)	ans-25
99	Settlement Institution Identification Code	n...11; LLVAR
100	Receiving Institution Identification Code	n...11; LLVAR
101	File Name	ans...17; LLVAR
102	Account Identification-1	ans...28; LLVAR
103	Account Identification-2	ans...28; LLVAR
104	Transaction Description (not currently used)	ans...100; LLLVAR
105–111	Reserved for ISO use (not currently used)	ans...999; LLLVAR
112	Parcelas Data	ans...248; LLLVAR
113–119	Reserved for National use (not currently used)	ans...999; LLLVAR
120	Record Data	ans...999; LLLVAR
121	Authorizing Agent Identification Code (not currently used)	ans...011; LLLVAR

Number	Name	Attributes
122	Additional Record Data	ans…100; LLLVAR
123–125	Reserved for future definition and use by MasterCard (not currently used)	ans...999; LLLVAR
126	Reserved (Private/ISO 8583–1987) (not currently used)	ans...999; LLLVAR
127	Private Data	ans...050; LLLVAR
128	Message Authentication Code (not currently used)	b-64

Data Element Definitions

The remainder of this chapter contains detailed definitions of all the ISO 8583–1987 message data elements.

Message Type Identifier (MTI)

The Message Type Identifier is a four-digit numeric field describing the type of interchange message.

Attribute

n-4

Usage

This data element must be present as the first field of each ISO 8583–1987 message.

Values

Table 4.5 lists the valid MTIs for the MDS.

Table 4.5—ISO 8583–1987 Message Types

Code	Description
Financial Transaction (02xx) Messages	
0200	Financial Transaction Request
0210	Financial Transaction Request Response
0220	Financial Transaction Advice
0230	Financial Transaction Advice Response
0280	Financial Transaction Acknowledgment
0290	Financial Transaction Negative Acknowledgment
File Update (03xx) Messages	
0302	File Update Request
0312	File Update Request Response

Code	Description
Reversal Advice (04xx) Messages	
0420	Acquirer Reversal Advice
0430	Acquirer Reversal Advice Response
0422	Issuer Reversal Advice
0432	Issuer Reversal Advice Response
Reconciliation Advice (05xx) Messages	
0520	Acquirer Reconciliation Advice
0530	Acquirer Reconciliation Advice Response
0522	Issuer Reconciliation Advice
0532	Issuer Reconciliation Advice Response
Administrative Advice (06xx) Messages	
0620	Administrative Advice
0630	Administrative Advice Response
0644	Administrative Advice (Banknet connected members only)
Network Management (08xx) Messages	
0800	Network Management Request
0810	Network Management Request Response
0820	Network Management Advice

Note **The MDS-generated Network Management Advice/0820 messages do not require a subsequent response message.**

Primary and Secondary Bit Maps

A bit map is a series of 64 bits used to identify the presence or absence (denoted by "1" or "0", respectively) of each data element. The MDS interprets the bit map from left to right. The leftmost bit represents DE 1 in the Primary Bit Map and DE 65 in the Secondary Bit Map. The rightmost bit represents DE 64 in the Primary Bit Map and DE 128 in the Secondary Bit Map.

Attribute

b-64 (for each bit map)

Note **If both bit maps are present, the total length of the bit map field is 128 bits (16 bytes).**

Usage

All bit positions are interpreted from left to right within each bit map; such as within the Primary Bit map the leftmost bit is "DE 1", and the rightmost bit is "DE 64".

- ISO 8583–1987 messages are variable length, with a bit map scheme that indicates the presence or absence of additional fields in the message
- Each bit map is a 64-bit string contained within an 8-byte field.
- The first bit in each bit map is set to "1" indicating the presence or "0" indicating the absence of an additional 64-bit bit map field which immediately follows the preceding bit map field.
- ISO 8583–1987 message format uses a maximum of 2 bit maps: a "Primary" and a "Secondary" bit map.
- Bits set to "1" or "0" in the Primary Bit map indicate the presence or absence of DE 2 through DE 64.
- Bits set to "1" or "0" in the Secondary Bit map indicate the presence or absence of DE 66 through DE 128.

- Bit No. 1 in the Primary Bit Map and DE 65 in the Secondary Bit Map (such as the first bit in each bit map) do not have corresponding data elements. These bits indicate the presence or absence of additional bit map fields in the message.
  - If bit No. 1 is set to "1", it indicates that the Secondary Bit Map is present, and selected data elements in the DE 66 through DE 128 range are also present in the message (as indicated by bit positions in the Secondary Bit Map.)
  - Bit No. 65 **must always be set to "0"**, because there are no additional bit maps defined beyond the Secondary Bit Map in the ISO 8583–1987 message specification.
- All ISO 8583–1987 messages must contain a Primary Bit Map.
  - The Secondary Bit Map is only included in a message when data elements in the range DE 66 through DE 128 are present in the message.
- Although additional bit maps are accommodated in ISO Standard 8583 (such as setting the first bit in any bit map to "1" to indicate the presence of a following extended bit map), the ISO 8583–1987 implementation uses a **maximum of two bit maps** (Primary and Secondary), with a maximum number of message data elements in the range DE 1 through DE 128. Consequently, DE 65 (the first bit in Bit Map, Secondary) **must always be set to zero**.
- Bits corresponding to **mandatory** data elements for a specific message type must be set to "1" to indicate the presence of the data element in the message. Otherwise, the message will be rejected by the MDS through the appropriate response message or through an Administrative Advice (Reject)/0620 message.

DE 1—Bit Map, Secondary

The Bit Map, Secondary (DE 1) is a series of 64 bits used to identify the presence (“1”) or absence (“0”) of each data element in the second segment of a message (for example, data elements in the range Settlement Code (DE 66) through Message Authentication Code (DE 128]).

Attribute

b-64

DE 2—Primary Account Number (PAN)

Primary Account Number (PAN) (DE 2) is a series of digits used to identify a customer account or relationship.

Attribute

n...19; LLVAR

Usage

This data element has two primary uses:

1. It contains the Primary Account Numbers (PAN) in authorization (01xx), financial transaction (02xx) and reversal advice (04xx) messages.

   The MDS uses this data element for all PANs up to 19 digits in length, in authorization (01xx), financial transaction (02xx), and reversal advice (04xx) messages.

   PAN data consists of three primary components:

   – Issuer identification number (IIN)

   – Individual account identification number

   – PAN check digit

   ISO specification 7812 and 7813 details the specific requirements for PAN composition. All PANs used in ISO 8583–1987 messages must conform to the ISO PAN encoding requirements as specified in these documents.

   The (class 0) network management (08xx) messages may use the data element to contain a valid processor ID number or a variable-length issuer card prefix.

   The PAN field may contain only an Issuer Identification Number or card prefix sequence identified by a card-issuing institution. It may also contain a valid processor ID for certain 08xx-series messages. The individual Message Format Specification charts detail the specific usage requirements for each message.

2. The MDS will accommodate variable-length prefix sequences from four to eleven digits, where card prefix information is required. Processor IDs used in this data element must be valid values assigned by MasterCard.

Note **The processor ID is a ten-digit number of the form: "9000000xxx" where "xxx" is the 3-digit MDS–assigned processor ID.**

DE 3—Processing Code

The Processing Code (DE 3) is a series of digits used to describe the effect of a transaction on the customer account and the type of accounts affected.

Attribute

n-6

Usage

Table 4.6 describes the subfields in DE 3.

Table 4.6—Processing Code Subfields

Subfield	Position	Attribute	Value
Transaction Type	1-2	n	Describes the transaction being performed. **00** Purchase (00 code also used for cash back and scrip transactions) **01** Withdrawal **02** Debit Adjustment **17** Cash Disbursement (can be sent to 0100 issuers) **20** Refund/Correction **23** Credit Adjustment **28** Payment **30** Balance Inquiry **40** Account Transfer **50** Bill Payment **90** Electronic Commerce (Set Certificate and Cardholder Certificate information)
Account Type (From)	3-4	n	Describes the cardholder account type affected for cardholder account debits and inquiries, and the “from” account type for cardholder account transfer transactions. **00** No account specified (NAS)/Default Account **10** Savings Account **20** Checking Account **30** Credit Card Account

Subfield	Position	Attribute	Value
Account Type (To)	5-6	n	Describes the cardholder account type affected for cardholder account credits and the "to" account type for cardholder account transfer transactions. **00** No account specified (NAS)/Default Account **10** Savings Account **20** Checking Account **30** Credit Card Account

Table 4.7 lists all valid combinations of subfields supported by the MasterCard® Debit Switch (MDS) as Processing Codes.

Table 4.7—MasterCard® Debit Switch Processing Code Values

Code	Description
000000	Purchase; no account specified *Not valid for ATM Gateway transactions (for example, Plus, or Visa).*
001000	Purchase from savings account
002000	Purchase from checking account
010000	Withdrawal; no account specified
011000	Withdrawal from savings account
012000	Withdrawal from checking account
013000	Withdrawal from credit card account. *Acquirer Processing Systems (APS) connected to the MDS must use this processing code for cash advance transactions. The MDS translates this processing code as an Authorization Request/0100 message with a processing code of 173000.* *NOTE: The MDS performs the above processing for any cash withdrawal request.*
200000	Online refund; no account specified *Codes valid for debit MasterCard and Maestro transactions only.*
201000	Online refund to savings account
202000	Online refund to checking account
230000	Credit adjustment; no account specified *Codes valid for debit MasterCard only. Debit MasterCard adjustments must be 230000.*
231000	Credit adjustment to savings account
232000	Credit adjustment to checking account

Code	Description
280000	Payment to NAS
280010	Payment to Savings
280020	Payment to Checking
280030	Payment to Credit Account
300000	Balance inquiry; no account specified *When no account is specified on a balance inquiry transaction, the issuer may return both checking and savings account balances if applicable.*
301000	Balance inquiry on savings account
302000	Balance inquiry on checking
303000	Balance inquiry on credit card (credit line)
401020	Transfer from savings account to checking account
402010	Transfer from checking account to savings account
500000	Bill Payment, no account specified
501000	Bill Payment, savings
502000	Bill Payment, checking
503000	Bill Payment, credit card
900000	Electronic Commerce, no account specified, certificate request
901000	Electronic Commerce, savings, certificate request
902000	Electronic Commerce, checking, certificate request
903000	Electronic Commerce, credit card, certificate request

Note **The MasterCard® Debit Switch (MDS) only supports the specific Processing Code subfield combinations listed in Table 4.7.**

When the Account Type value processing code in the Financial Transaction Request/0200 message indicates "no account specified", the issuer may specify an Account Type in the Financial Transaction Request Response/0210 message. For example, when an acquirer sends processing code 010000 (withdrawal, no account specified), the issuer may send a Financial Transaction Request Response/0210 message containing processing code 012000 (withdrawal, checking account).

DE 4—Amount, Transaction

Amount, Transaction (DE 4) is the amount of funds requested by the cardholder in the local currency of the acquirer or source location of the transaction. The Transaction Fee Amount (DE 28) must be included in DE 4. DE 4 may include ATM Access Charges if DE 28 is present.

Attribute

n-12

Usage

The local currency of the card acceptor (currency used by the cardholder and merchant at the point of service) must always be specified using the Currency Code, Transaction (DE 49). The MDS refers to this currency as the "currency of the acquirer" or the "currency of the transaction at the point of service".

Amounts are expressed without a decimal separator. For currencies that support exponents, users and systems are responsible for placing the decimal separator appropriately. Refer to appendix B for a listing of currencies and their exponents. See Table 4.8 for examples of decimal separator placement.

Table 4.8—Decimal Separator Example (DE 4)

Amount, Transaction DE 4	Currency Code DE 49	Currency Exponent	Currency Name	Actual Monetary Value of DE 4
000000001500	792	0	Turkish Lira	1500 Lira
000000001500	124	2	Canadian Dollar	15.00 Dollars
000000001500	788	3	Tunisian Dinar	1.500 Dinars

Values

For debit MasterCard clearings (Financial Transaction Advice/0220 message), this data element will contain the completed amount.

DE 5—Amount, Settlement

Amount, Settlement (DE 5) is the amount of funds to be transferred between the acquirer and the issuer (exclusive of fees) equal to the Amount, Transaction (DE 4) in the currency of settlement, U.S. dollars.

Attribute

n-12

Usage

The currency of this data element must always be specified using the Currency Code, Settlement (DE 50). The MDS automatically inserts this data element into all originating Financial Transaction (02xx) messages as a currency conversion service when the currency of transaction differs from the currency of settlement or currency of the issuer.

This data element will generally be present whenever the currency of Amount, Transaction, (DE 4) is not equal to U.S. dollars. Exceptions to this are:

- ISIS transactions (currency of settlement)
- International transactions acquired in the United States

Amounts are expressed without a decimal separator. For currencies that support exponents, users and systems are responsible for placing the decimal separator appropriately. Refer to appendix B for a listing of currencies and their exponents. See Table 4.9 for examples of decimal separator placement.

Table 4.9—Decimal Separator Example (DE 5)

Amount, Settlement DE 5	Currency Code DE 50	Currency Exponent	Currency Name	Actual Monetary Value of DE 5	Comments
000000001500	840	2	United States Dollar	15.00 Dollars	
000000001500	792	0	Turkish Lira	1500 Lira	ISIS only
000000001500	124	2	Canadian Dollar	15.00 Dollars	ISIS only
000000001500	788	3	Tunisian Dinar	1.500 Dinars	ISIS only

When this field is present in a message, Conversion Rate, Settlement (DE 9), Date, Conversion (DE 16), and Currency Code, Settlement (DE 50) must also be present.

DE 6—Amount, Cardholder Billing

Amount, Cardholder Billing (DE 6) is the amount billed to the cardholder in the currency of the cardholder account, exclusive of cardholder billing fees.

Note **This data element is used only when an 0200 issuer (Maestro and/or Cirrus) is eligible for the MasterCard multiple currency conversion rate (MCCR) service and participates in the MasterCard issuer currency conversion rate (ICCR) service.**

Attribute

n-12

Usage

The currency of this data element must always be specified using the Currency Code, Cardholder Billing (DE 51). The MDS calculates the Amount, Cardholder Billing (DE 6) for issuers that participate in the ICCR service. The Cardholder Billing amount is in U.S. dollars. This data element is not returned to the acquirer in the 0210 message. The ICCR amount is not included in the MDS settlement amounts.

The MDS performs the calculation to adjust the Amount, Settlement (DE 5) to obtain the Amount, Cardholder Billing (DE 6), which appears in the Financial Transaction Request/0200 message sent by the MDS to the issuer. The issuer has the option to post this adjusted value to the cardholder account. Issuers may calculate the ICCR amount by subtracting Amount, Settlement (DE 5) from Amount, Cardholder Billing (DE 6).

When this field is present in a message, Conversion Rate, Cardholder Billing (DE 10), Date, Conversion (DE 16), and Currency Code, Cardholder Billing (DE 51) must also be present.

DE 7—Transmission Date and Time

Transmission Date and Time (DE 7) is the date and time a message was transmitted by a processing entity, to be expressed in coordinated universal time (UTC) time units.

The UTC timestamp is the date and time that a processor, including the MDS, transmits any message (as opposed to the initiation of an entire transaction), containing this data element, to another processor.

MasterCard recommends that processors do not include the values of this data element as part of their message key if they expect this data element to contain the original acquirer transmission timestamp.

If processors want to use the original acquirer timestamp as part of their message key, MasterCard recommends that they use Time, Local Transaction (DE 12) and Date, Local Transaction (DE 13).

Attribute

n-10; MMDDhhmmss

Usage

Upon receipt of the Financial Transaction Request/200 message, with limited exceptions, the MDS updates Data Element 7 with its internal time stamp before sending the message to the issuer. The issuer may return this time to the MDS in the Financial Transaction Request Response/0210 message, or the issuer may send its own transmission time in the data element.

When sending the Financial Transaction Request Response/0210 message to the acquirer, the MDS re-inserts the acquirer's time stamp from the request message in this data element.

Values

This field must contain a valid date and time.

- **MM** must be in the range 01 – 12
- **DD** must be in the range 01 – 31
- **hh** must be in the range 00 – 23
- **mm** must be in the range 00 – 59
- **ss** must be in the range 00 – 59

DE 8—Amount, Cardholder Billing Fee

Amount, Cardholder Billing Fee (DE 8) is the fee the issuer will bill to the cardholder in the same currency as Amount, Cardholder Billing (DE 6).

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-8

DE 9—Conversion Rate, Settlement

Conversion Rate, Settlement (DE 9) is the factor used in the conversion from transaction to settlement amount. The MDS multiplies the Amount, Transaction (DE 4) by DE 9 to determine the Amount, Settlement (DE 5).

Attribute

n-8

Usage

The MDS provides automatic currency conversion as a service for customers that participate in international interchange and will supply the conversion rate in this field.

When this data element is present in a message, Amount, Settlement (DE 5), Date, Conversion (DE 16), and Currency Code, Settlement (DE 50) must also be present.

Values

The format is left justified with trailing zeroes. The left-most digit denotes the number of positions that the MDS moves the decimal separator **from the right**. The leftmost digit must be in the range 0 to 7. For example, a field value of 76887050 is interpreted as a conversion rate of 0.6887050.

DE 10—Conversion Rate, Cardholder Billing

Conversion Rate, Cardholder Billing (DE 10) is the factor used in the conversion from transaction to cardholder billing amount. The Amount, Transaction (DE 4) is multiplied by DE 10 to determine Amount, Cardholder Billing (DE 6).

Attribute

n-8

Usage

When this data element is present in a message, Amount, Cardholder Billing (DE 6), Date, Conversion (DE 16), and Currency Code, Cardholder Billing (DE 51) must also be present.

This data element should only be present in an Issuers Currency Conversion Rate (ICCR) transaction.

Values

The format is left justified with trailing zeroes. The left-most digit denotes the number of positions that the MDS moves the decimal separator **from the right**. The leftmost digit must be in the range zero to seven. For example, a field value of 69972522 is interpreted as a conversion rate of 9.972522.

DE 11—System Trace Audit Number

The System Trace Audit Number (STAN) is the unique identifier assigned to each transaction by the originator of the message.

Attribute

n-6

Usage

The message originator assigns DE 11. This identifier must be unique for each transaction that occurs within an originator's day. Each originator's day must be based on coordinated universal time (UTC).

DE 11 must remain constant throughout the life of the transaction including responses and acknowledgments (see the following note) that relate to the original request or advice message.

Note **In the Acquirer Reversal Advice/0420 message, an acquirer may submit a new trace number in DE 11; however, that is not recommended practice. The MDS will always send the trace number from the original Financial Transaction Request/0200 message to the issuer. The Acquirer Reversal Advice Response/0430 message to the acquirer will contain the same trace number as the Acquirer Reversal Advice/0420.**

DE 11 from the original transaction request is located in DE 90, subfield 2 of a reversal message.

Enhanced Delivery Option

In debit MasterCard Financial Advice/0220 clearing messages, the MDS distinguishes messages depending on whether they originate from the store-and-forward (SAF) file through the SAF process or directly from the main processing module, through the Enhanced Delivery option. The values for the STAN in these cases are:

- 999999 Store and Forward
- 999998 Enhanced Delivery

In the Financial Advice Response/0230 message, issuers must return the value received in DE 11 of the Financial Advice/0220 message.

DE 12—Time, Local Transaction

Time, Local Transaction (DE 12) is the local time at which the transaction takes place at the point of service.

Attribute

n-6; hhmmss

Usage

DE 12 is the local time that a cardholder transaction takes place, and should be the same value that is printed on the cardholder receipt, if possible. This time must be specified in local time zone units, and **not** in coordinated universal time (UTC) units.

For debit MasterCard clearings (Financial Transaction Advice/0220 messages), this will contain the MDS time that the store and forward message is sent, **not** the value set in the original pre-authorization.

Values

The value in this field must be a valid time.

DE 13—Date, Local Transaction

Date, Local Transaction (DE 13) is the local month and day on which the transaction takes place at the point of service.

Attribute

n-4; MMDD

Usage

DE 13 is the local date that a cardholder transaction takes place, and should be the same value printed on the cardholder receipt, if possible. This time must be specified in local time zone units, and **not** in coordinated universal time (UTC) units.

Values

The value in this field must be a valid date.

DE 14—Date, Expiration

The Date, Expiration (DE 14) specifies the year and month that a cardholder's bank card expires.

Attribute

n-4; YYMM

Usage

This data element may be present in manually-keyed debit MasterCard transactions, where allowed.

Values

The value in this field must be a valid date.

Note **For debit MasterCard, the MDS uses a default date of 4912 if not present in track data.**

DE 15—Date, Settlement

Date, Settlement (DE 15) is the date (month and day) that funds will be transferred between an acquirer and an issuer by the MDS.

Attribute

n-4; MMDD

Usage

This data element is present in Financial Transaction (02xx) and Reversal (04xx) messages that convey a settlement value. The exception is a Timeout-Induced Reversal/0420 message; which does not contain DE 15. It contains the calendar date on which funds for a transaction will be settled.

A Timeout Induced Reversal/0420 message does not contain DE 15, since the acquirer never received the Financial Transaction Request Response/0210 message providing the settlement date.

For settlement reconciliation (05xx) messages, DE 15 identifies the financial network calendar date for which the reconciliation count and amount totals are applicable.

Values

This data element must contain a valid date.

DE 16—Date, Conversion

The Date, Conversion (DE 16) is the month and day that the conversion rate is effective to convert the transaction amount from the original currency into the currency of settlement.

Attribute

n-4; MMDD

Usage

DE 16 indicates the effective date (month and day) of the Conversion Rate, Settlement (DE 9) whenever that data element is present within a message.

DE 16 also indicates the effective date (month and day) of the conversion rate and must be present whenever the Currency Code, Transaction (DE 49) does not equal 840.

When this field is present in a message, Amount, Settlement (DE 5), Conversion Rate, Settlement (DE 9), and Currency Code, Settlement (DE 50) must also be present. Cardholder Billing (DE 6), Conversion Rate, Cardholder Billing (DE 10), and Currency Code, Cardholder Billing (DE 51) may also be present.

Values

The value in this field must be a valid date.

DE 17—Date, Capture

Date, Capture (DE 17) is the month and day the acquirer processed the transaction data.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-4; MMDD

Usage

The MasterCard® Debit Switch does not currently use this data element.

Values

The value in this field must be a valid date.

DE 18—Merchant Type

The Merchant Type (DE 18) code is the classification of the merchant's type of business or service.

Attribute

n-4

Usage

DE 18 code is a four-digit indicator used to classify a merchant's product or service, selected from a standard list of classification codes referred to as merchant category codes (MCCs). The MCC is included in Financial Transaction Request/0200 messages and Financial Transaction Advice/0220 completion messages.

The issuer has the option of whether or not to receive DE 18 in ATM cash transactions.

Values

Refer to the *Quick Reference Booklet* for a list of MCCs and transaction category codes (TCCs).

Processors should use the following codes for cash transactions:

- 6010—Bank teller over-the-counter (OTC) cash transaction
- 6011—ATM cash transaction - Processing Code (DE3) = 01, 30, or 40
- 6012—ATM purchase transaction

For Cirrus Purchase transactions (DE 3 = "00xx00"), the Merchant Type Code data element must contain a value of 6012. The value 6012 is not valid for Maestro POS transactions.

For ATM cash disbursements, this data element must contain a value of 6011. In these instances, the MDS, unless otherwise directed by the issuer's configuration file, removes DE 18 from the message before sending the message to the issuer.

MasterCard reserves the right to reject online financial request messages when the merchant category code (MCC) value in the Merchant Type field (DE 18) equals 0000. Acquirers should anticipate receiving a format error when an 0200/0210, 0220/0230 message contains a value of 0000 in DE 18.

For debit MasterCard and Maestro POS transactions, appendix C, provides a list of permissible values that members must use in the Merchant Type Code data element.

Note **The Merchant Type Code of 6012 is not valid for Maestro point of sale transactions.**

DE 19—Acquiring Institution Country Code

Acquiring Institution Country Code (DE 19) is the code of the country where the acquirer is located. Refer to the ISO 3166 specification for more information.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-3

Usage

The MasterCard® Debit Switch does not currently use this data element.

DE 20—Primary Account Number (PAN) Country Code

The PAN Country Code (DE 20) is a code identifying the country where the card issuer is located.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-3

Usage

DE 20 is required to be included within any message whenever the associated PAN (in Primary Account Number [DE 2] or Primary Account Number Extended [DE 34]) is present and begins with a "59" prefix. PANs beginning with a "59" prefix are **not** guaranteed to be unique without the use of this associated Country Code.

Values

Country codes must be selected from the numeric ISO Standard Country Codes listed in ISO 8583–1987 Appendix 2, ISO Country and Currency Codes.

DE 21—Forwarding Institution Country Code

Forwarding Institution Country Code (DE 21) is the code of the country where the forwarding institution is located.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-3

DE 22—Point of Service Entry Mode

The Point of Service Entry Mode (DE 22) consists of numeric codes that indicate the method used to enter the PAN into the terminal device and the PIN entry capability of that device.

Attribute

n-3

Usage

Table 4.10 describes the subfields in DE 22.

Table 4.10—Point of Service Entry Mode Subfields

Subfield	Position	Attribute	Value
POS Terminal PAN Entry Mode	1-2	n	Describes the method used for PAN entry to initiate a transaction. **00** PAN entry mode unknown **01** PAN manual entry **02** PAN auto-entry via magnetic stripe, full unaltered **03** PAN auto-entry via bar code reader **04** PAN auto-entry via optical character reader (OCR) **05** PAN auto-entry via integrated circuit card **06** PAN key entry **79** Chip card or chip-capable terminal was unable to process the transaction using the data on the chip or magnetic stripe, the PAN was entered manually *Or,* Acquirer is not certified to process the value 80, the MDS replaces 80 with 79 **80** Chip card or chip-capable terminal was unable to process the transaction using the data on the chip, the PAN was entered via magnetic stripe. The full track data was read from the data encoded on the card and transmitted within the authorization request on Track-2 Data (DE 35 [a]) or Track-1 Data (DE 45) without alteration or truncation. **81** PAN manual entry via electronic commerce. **90** PAN auto-entry via magnetic stripe—the full track data has been read from the data encoded on the card and transmitted within the authorization request on DE 35 [a] or DE 45 without alteration or truncation.

Subfield	Position	Attribute	Value
POS Terminal PIN Entry Mode	3	n	Describes the capability of the terminal device to support/accept PIN entry. **0** Unspecified or unknown. **1** Terminal has PIN entry capability. **2** Terminal *does not* have PIN entry capability. **8** Terminal has PIN entry capability, but PIN pad is out of service. **9** PIN verified by terminal device.

[a] If the MDS creates any track 2 Data using track 1 Data, the processors must be prepared to accept any character that would have been present in the track-1.

For ATM transactions, the POS Terminal Entry Mode (position 1-2) should contain either 02 or 90. The MDS will overwrite any other value received from an acquirer with 02. Issuers will therefore receive only values 02x or 90x or in this data element.

Note **All debit MasterCard "force post" Financial Transaction Advice/0220 messages contain the value "000" in DE 22. MasterCard sends the value "00" (PAN entry mode unknown) in the POS Terminal PAN Entry Mode subfield, and the value "0" (Unspecified or unknown) in the POS Terminal PIN Entry Mode subfield because the member does not send this information in the GCMS message that generates the 0220 force post message.**

If acquirers submit Integrated Circuit Card (ICC) System-Related Data (DE 55) in the message, then the first two digits of DE 22 must be 05 or 81.

DE 23—Card Sequence Number

The Card Sequence Number (DE 23) is used to distinguish among individual cards having the same Primary Account Number (DE 2) or Primary Account Number, Extended (DE 34).

Attribute

n-3

Usage

For ICC transactions (DE 22 = 05x), if DE 55 is present, then DE 23 is mandatory and must contain the Application PAN Sequence Number (EMV tag 5F34) if present on the ICC. If this information is not available, then DE 23 must contain the Card Sequence Number data taken from the Track 2 Equivalent Data (EMV tag 57) on the ICC and the Card Sequence Number is present in this data object.

If DE 55 is not present, then DE 23 is optional for ICC transactions. Valid values for Card Sequence Number are 000-099.

DE 24—Network International Identifier

Network International Identifier identifies a single international network of card issuers.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-3

DE 25—Point of Service Condition Code (ISO)

Point of Service Condition Code (ISO) (DE 25) is an identification of the condition under which the transaction takes place at the point of service.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-2

Usage

All programs and services that the MasterCard® Debit Switch supports use Point of Service (POS) Data (DE 61) as MasterCard defined and implemented for use by all customers in all countries to specify the applicable conditions at the point of service.

DE 26—Point of Service (POS) PIN Capture Code

The POS PIN Capture Code (DE 26) is a code indicating the technique, maximum number of PIN characters, or both, that can be accepted by the point of service device used to construct the personal identification number (PIN) data.

Attribute

n-2

Usage

The point of service PIN capture code must be used to indicate the maximum number of PIN characters that the acquiring terminal device (ATM, POS terminal, etc.) is **capable** of accepting.

Note **If this field is not present in debit MasterCard transactions the MDS defaults to a value of 12.**

The MDS does not use this data element to specify the number of PIN characters actually accepted by a point of service terminal device.

The MDS requires that this data element be included in 0200 Financial Transaction messages **only** when PIN Data (DE 52) is present **and** the maximum PIN character acceptance capability of the terminal is known to be **other than 12 digits**.

Values

Table 4.11 describes the subfields in DE 26.

Table 4.11—Point of Service (POS) PIN Capture Code Values

Code	Description
00–03	Invalid.
04–12	Indicates the maximum number of PIN characters that the terminal can accept.
13–99	Reserved.

DE 27—Authorization Identification Response Length

The Authorization Identification Response Length (DE 27) is the maximum length of the authorization response that the acquirer can accommodate. MasterCard expects the issuer, or its agent, to limit response to this length.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-1

DE 28—Amount, Transaction Fee

Amount, Transaction Fee (DE 28) is the fee charged (for example, by the acquirer) for transaction activity in the currency of the Amount, Transaction (DE 4).

Attribute

x+n-8

Usage

This data element may be present in a message whenever an online transaction fee is permitted by the operating rules of a bank card product.

The credit or debit indicator (the first position of the data element) applies to the message recipient. Within acquirer-generated message types, a D (debit) fee amount indicates that the fee is to be applied as a debit to the message recipient, the issuer (and therefore as a credit to the message originator, the acquirer).

Edits

This data element must contain valid numeric data with an appropriate indicator (C or D) in the first character position.

Note **For acquirers that are approved to levy ATM Access Charges, this data element must contain the access fee amount and this amount must also be added to the requested amount contained in Transaction Amount (DE 4).**

Acquirers must test with the MDS before implementation of this data element.

DE 29—Amount, Settlement Fee

Amount, Settlement Fee (DE 29) is the transferred fee between the acquirer and the issuer in the currency of Amount, Settlement (DE 5).

Attribute

x+n-8

Usage

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

DE 30—Amount, Transaction Processing Fee

In some transaction processing systems, Amount, Transaction Processing Fee (DE 30) can represent the switch fee for the handling and routing of messages in the currency of Amount, Transaction (DE 4).

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

x+n-8

DE 31—Amount, Settlement Processing Fee

Amount, Settlement Processing Fee (DE 31) is the fee charged by the MDS for handling and routing messages in U.S. dollars.

Attribute

x+n-8

Usage

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

DE 32—Acquiring Institution Identification Code

The Acquiring Institution Identification Code (DE 32) identifies the acquirer (for example, merchant bank) or its agent.

Attribute

n...11, LLVAR.

Usage

DE 32 identifies the institution acting as the acquiring bank or merchant bank for a transaction. The length value ll is always 09, followed by nine digits. For CPS systems connected to the MDS, this data element must contain either:

- The 9-digit Federal Reserve Routing and Transit (R & T) number of the institution that owns the terminal device.

  Or,

- The 9-digit ID pseudo number assigned by MasterCard in accordance with applicable product rules.

For debit MasterCard, this will contain 7084xxxxC where:

- **7084** Literal text
- **xxxx** Four digits of the acquiring/forwarding institution code
- **C** Check digit

For the EPSS acquirer transactions, this will contain 99xxxxxxC, where:

- **99** Literal text
- **xxxxxx** Six digits of the acquiring/forwarding institution code
- **C** Check digit

For the MasterCard PIN for Credit request, this will contain:

- **786** Literal text
- **xxxxx** Five digits for the group signin ID (GSI)
- **C** Check digit

Note **For 0200 requests received by the MDS destined for a 0100 issuer, the MDS inserts the five-digit value 02202 in this data element (where ll = 05). The value 02202 is the Banknet ICA identifier for the MDS.**

DE 33—Forwarding Institution Identification Code

The Forwarding Institution Identification Code (DE 33) identifies the institution forwarding a Request or Advice message in an interchange system if not the same institution as specified in the Acquiring Institution Identification Code (DE 32).

Attribute

n...11; LLVAR.

Usage

This data element **must** be present in all authorization (01xx), financial transaction (02xx), and reversal (04xx) messages. Routing of all these messages throughout the MDS is based upon the Acquirer ID Code (DE 32), the DE 33, and PAN (DE 2) information. It is important that all of these data elements are properly encoded to ensure accurate routing for both the original transaction and any subsequent reversals, chargebacks, adjustments, or representments.

Values

When present in a message, this data element must contain the processor ID.

Note **The processor ID is a ten-digit number of the form:**

"9000000xxx"

where "xxx" is the 3-digit MDS Processor ID assigned by MasterCard.

This data element must always contain a length value of "10" followed by ten characters of numeric data in the associated variable-length data field.

DE 34—Primary Account Number, Extended

The Primary Account Number, Extended (DE 34) identifies a customer account or relationship. It is used only when a PAN is longer than 19 digits in length or contains special characters, and therefore cannot be placed into Primary Account Number (DE 2).

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

ns...28; LLVAR.

DE 35—Track 2 Data

Track 2 Data (DE 35) is the information encoded on track 2 of the card magnetic stripe as defined in ISO 7813, **including field separators**, but excluding beginning and ending sentinels and Longitudinal Redundancy Check (LRC) characters as defined therein.

If any track 2 data must be created by the MDS using track 1 data, the processors must be prepared to accept any character that would have been present in the track 1.

Attribute

z...37; LLVAR.

Usage

Whenever track 2 data is captured automatically at the point of service, this field must contain whatever is encoded on the magnetic stripe (track 2) of the card (regardless of whether or not the card has been properly encoded with information in accordance with ISO specifications).

The member's system must encode the following minimum data on track 2 (Table 4.12):

Table 4.12—Track-2 Minimum Data Subfields

Subfield	Attribute	Value
Start Sentinel	n-1	binary "1011" *(not transmitted)*
PAN	n…19	
Field Separator	ans-1	binary "1101" (see note)
Expiration Date	ans-4	"YYMM" format
Service Code	ans-3	
Discretionary Data	ans…9	optional by issuer
End Sentinel	n-1	binary "1111" *(not transmitted)*
LRC	n-1	*(not transmitted)*

Total maximum characters transmitted = 37

For manually keyed Maestro and debit MasterCard transactions, the MDS builds track 2 using the PAN (DE 2) and the Expiration Date (DE 14). If the Expiration Date is not available, the MDS uses 4912 as the default.

For electronic commerce (e-commerce) requests that do not contain track 2 data, the MDS builds track 2 using the PAN (DE 3), expiration date (DE 14) if available, and a Service Code subelement value of "101". E-commerce requests contain a POS entry mode (DE 22) value of "81x" or a value of "05x" with DE 48 subelements 40, 42, and 43 present.

Values

This data element must contain the hexadecimal digits "0" through "9" and "D" or "=" (the "equal sign").

Note **The field separator character (binary "1101") is represented as the EBCDIC character "D." However, because many ATM and POS devices perform non-standard character translation while reading binary coded decimal (BCD)-encoded magnetic stripe data, the EBCDIC character "=" may also be used to represent the field separator character in magnetic stripe data forwarded to the MDS.**

If the MDS must create track 2 data from track 1 information, the issuer must be prepared to accept ANY character sent from the acquirer in track 1 data.

Track 2 data is not present in the 0220 messages for Cirrus, Maestro, and debit MasterCard transactions.

DE 36—Track 3 Data

Track 3 Data (DE 36), as represented in the ISO 8583 specification, is the information encoded on track 3 of the card magnetic stripe as defined in ISO 4909-1986. This includes field separators, but excludes beginning and ending sentinels and LRC characters as defined therein.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

z...104; LLLVAR

DE 37—Retrieval Reference Number

The Retrieval Reference Number (DE 37) is a document reference number supplied by the system retaining the original source document of the transaction. It is used to assist in locating that source document or a copy thereof.

Attribute

an-12

Usage

This data element is reserved for use by the acquiring institution (or an affiliated merchant organization) for the purpose of recording a document retrieval reference number. The number can be used to locate original cardholder transaction information in subsequent retrieval request or any subsequent chargeback action.

The issuer in all corresponding response messages and in any subsequent chargeback action must return DE 37. The retrieval reference number should be printed on a customer's ATM or POS receipt.

This data element is left-justified and blank-filled.

DE 38—Authorization Identification Response

The Authorization Identification Response (DE 38) is a transaction response identification code assigned by the authorizing institution.

Attribute

an-6

Usage

The issuer processing system may use this data element for authorization tracking information. It is not mandatory for use in the MDS; however, MasterCard credit card issuers participating in the program via the Banknet network will provide this data element in approved MasterCard ATM transactions that are forwarded to the APS via the MDS.

This data element is required in "approved" debit MasterCard Financial Transaction Request Response/0210 messages.

Note **The Authorization Code may also be present in any "denied" Financial Transaction Request Response/0210 message.**

Values

Any valid alphanumeric character sequence may be used.

Debit MasterCard clearing 0220 messages will contain all six positions from the original authorization response.

The MDS Stand-In service will set this data element to "**STND**" followed by two (2) blank characters. The MDS will change this data element only when the MDS 0220 Stand-In advice is invoked.

Note **The debit MasterCard program does not support the MDS Stand-In service. Banknet performs MasterCard authorization Stand-In support to the debit MasterCard program. For more information regarding Stand-In Processing refer to the *MDS Programs and Services* manual.**

DE 39—Response Code

The Response Code (DE 39) is a code that defines the disposition of a message.

Attribute

an-2

Usage

Response codes indicate the disposition of a previous message or indicate approval or denial of a transaction. When an authorization is declined, the response code will indicate the reason for rejection and may indicate an action to be taken by the card acceptor or POS terminal device (for example, to capture the card).

This data element must be present in all response messages. In addition, it will also be present in Financial Transaction Advice (Stand-In)/0220 messages to indicate that DE 39 was utilized in the Financial Transaction Request Response (Stand-In)/0210 message response to the original Financial Transaction Request/0200 message.

For File Update Request Response/0312 messages, the response code will indicate whether the account record was successfully updated or the messages resulted in an error.

Values

Table 4.13 provides a list of valid values for DE 39.

Table 4.13—Response Code Values

Code	Response	Description
Valid Response Codes for 0210 and 0220 messages		
00	Approve	Approved or completed successfully
01	Decline	Refer to card issuer
04	Capture	Capture Card
05	Decline	MDS use only

Code	Response	Description
12	Decline	Invalid transaction *For MDS use only*
13	Decline	Invalid amount
14	Decline	Invalid PAN
15	Decline	Invalid issuer *For MDS use only*
30	Decline	Message format error *Debit MasterCard issuers may only use this code if the MDS sends a message indicating a format error*
41	Capture	Lost Card
43	Capture	Stolen Card
51	Decline	Non-sufficient funds
54	Decline	Expired card
55	Decline	Invalid PIN
57	Decline	Transaction not permitted to issuer or cardholder
58	Decline	Transaction not permitted to acquirer or cardholder
61	Decline	Exceeds withdrawal limit
62	Decline	Restricted Card
63	Decline	Error in decryption of PIN block
65	Decline	Exceeds withdrawal count limits
75	Decline	Allowable number of PIN tries exceeded
76	Decline	Invalid “To” account specified
77	Decline	Invalid “From” account specified
78	Decline	Invalid account specified *For MDS use only*
79	Decline	Key exchange validation failed
80	Decline	System not available
85	NA	Valid for AVS only requests
91	Decline	Destination processor (CPS or INF) not available
92	Decline	Unable to route transaction
94	Decline	Duplicate transmission detected
96	Decline	System error

Code	Response	Description
Valid Response Codes for 0280 messages		
00	—	Completed successfully
Valid Response Codes for 0290 messages		
30	—	Format error
68	—	Response received late *The MDS forwards a response code "68" in the Financial Transaction Negative Acknowledgment/0290 message in reply to a late 0210 message (or in reply to a late 0280 message) or other unsolicited response message.* *The Financial Transaction Acknowledgment/0280 timer is only relevant for those acquirers that support the Financial Transaction Acknowledgment/0280 message.*
80	—	System not available *Response codes from the acquirer Financial Transaction Advice/0220 and Acquirer reversal Advice/0420 messages are stored temporarily in a queue. When the MDS formulates the Financial Transaction Advice Response/0230 or Acquirer Reversal Advice Response/0430 message, the MDS returns the response code from the queue in the online message. If the acquirer sends a response code 71, 72, 80, 96, the MDS echoes it back to the acquirer.*
96	—	System error or system timer expired on expected CPS Message *Response codes from the acquirer's 0220 and 0420 messages are stored temporarily in a queue.* *The MDS forwards a response code "96" in the Financial Transaction Negative Acknowledgment/0290 message to the issuer when the transaction timer expires before a valid Financial Transaction Request Response/0210 transaction response is received.* *The MDS forwards a response code "96" in the Financial Transaction Negative Acknowledgment/0290 message to the acquirer when the acquirer Financial Transaction Acknowledgment/0280 timer expires before a valid Financial Transaction Acknowledgment/0280 message is received.* *When the MDS formulates the 0230 or 0430 response message, the MDS returns the response code from the queue in the online message.* *If the acquirer sends a response code 71, 72, 80, 96, the MDS echoes it back to the acquirer.*
Valid Response Codes for 0230 messages		
00	—	Approved or completed successfully
30	—	Format error

Code	Response	Description
80	—	System not available *Response codes from the acquirer's 0220 and 0420 messages are stored temporarily in a queue.* *When the MDS formulates the 0230 or 0430 response message, the MDS returns the response code from the queue in the online message.* *If the acquirer were to send a response code 71, 72, 80, 96, the MDS echoes it back to the acquirer.*
96	—	System error or system timer expired on expected CPS Message *Response codes from the acquirer's 0220 and 0420 messages are stored temporarily in a queue.* *When the MDS formulates the 0230 or 0430 response message, the MDS returns the response code from the queue in the online message.* *If the acquirer sends a response code 71, 72, 80, 96, the MDS echoes it back to the acquirer.*
Valid Response Codes for 0312 messages		
00	—	File update action completed successfully.
25	—	Unable to locate record on file (no action taken). *Valid for debit MasterCard only.*
27	—	File update field edit error. *Valid for debit MasterCard only.*
30	—	Format error
40	—	Requested function not supported. *Valid for debit MasterCard only.*
63	—	Security violation. *Valid for debit MasterCard only.*
80	—	Duplicate add; action not performed. *Valid for debit MasterCard only.*
96	—	System error
Valid Response Codes for Timeout-Induced Reversal/0420 messages		
00	—	Required value

Code	Response	Description
Valid Response Codes for 0430, 432 messages		
00	—	Approved or completed successfully
30	—	Format error
80	—	System not available *Response codes from the acquirer's 0220 and 0420 messages are stored temporarily in a queue.* *When the MDS formulates the 0230 or 0430 response message, the MDS returns the response code from the queue in the online message.* *If the acquirer sends a response code 71, 72, 80, 96 the MDS echoes it back to the acquirer.*
96	—	System error or system timer expired on expected CPS Message *Response codes from the acquirer's 0220 and 0420 messages are stored temporarily in a queue.* *When the MDS formulates the 0230 or 0430 response message, the MDS returns the response code from the queue in the online message.* *Therefore, if the acquirer sends a response code 71, 72, 80, 96, the MDS echoes it back to the acquirer.*
Valid Response Codes for 0530, 0532 and 0630 messages (for reference only)		
00	—	Approved or completed successfully
30	—	Format error
80	—	System not available
96	—	System error or system timer expired on expected CPS Message
Valid Response Codes for 0810 messages		
00	—	Approved or completed successfully
96	—	System error or system timer expired on expected CPS Message

DE 40—Service Restriction Code

Service Restriction Code (DE 40) identifies geographic or service availability.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

an-3

DE 41—Card Acceptor Terminal Identification

The Card Acceptor Terminal Identification (DE 41) is a unique code identifying the terminal at the Card Acceptor Location. It is mandatory for initial requests, and must be returned, unchanged, in subsequent response message.

Attribute

ans-8

Usage

The MDS uses this data element to identify specific terminal devices of acquiring institutions or merchant point of service (POS) systems. The terminal owner assigns each terminal ID. It must be unique within the terminal-owning organization.

When this data element is included within an originating financial transaction (02xx) or reversal (04xx) message, it must be returned in the corresponding response message.

Values

The MDS does not perform edits on this data element.

DE 42—Card Acceptor Identification Code

The Card Acceptor Identification Code (DE 42) identifies the card acceptor, which defines the point of the transaction in both local and interchange environments.

Attribute

ans-15

Usage

The MDS uses DE 42 as an "merchant ID" to uniquely identify the merchant in a POS transaction. For a Maestro or debit MasterCard transaction, this data element must contain an alphanumeric merchant identifier.

Note **This data element is required in Maestro (U.S.) and debit MasterCard transactions and is forwarded to the issuer in Financial Transaction Request (Pre-Authorization)/0200 and Financial Transaction Advice/0220 messages. The MDS does not require Maestro International acquirers to use this data element.**

Values

The MDS does not perform edits on this data element.

DE 43—Card Acceptor Name and Location

The Card Acceptor Name and Location (DE 43) field contains the name and location of the card acceptor, which defines the point of service in both local and interchange environments.

Attribute

ans-40

Usage

The MDS uses this data element to satisfy national regulatory requirements concerning merchant identification within financial transaction (02xx) messages. It is a required data element within all financial transaction (02xx) request and advice messages.

Table 4.14 describes the subfields in DE 43.

Table 4.14—Card Acceptor Name and Location Subfields

Subfield	Position	Attribute	Value
1	1–22	ans-22	ATM owning institution and/or Terminal/Merchant address
2	23	ans-1	Delimiter (space)
3	24–36	ans-13	ATM or Merchant location city
4	37	ans-1	Delimiter (space)
5	38–40	a-3	For U.S.A and U.S. territories: ATM or Merchant location state code. This data must be right-justified, blank-filled, and in upper case. Or, For Canada and Canadian territories: ATM or Merchant location province code. This data must be right-justified, blank-filled, and in upper case. Or, For all other countries: ATM or Merchant location country code. This data must be right-justified, blank-filled, and in upper case.

Members must select all State, Province, and Country Codes from appendix A or appendix B. If a country code is used, it must be the ISO 3-character alphabetic (not numeric) Country Code. If used, a State or Province Code should be right justified in this subfield with one leading blank space.

Delimiter fields must be the **blank** character. This is required because most cardholder statement rendering systems in operation today are not designed to perform printer output editing or formatting of the acquirer-supplied data contained within DE 43. The acquirer must pre-format DE 43 exactly as they want it printed on the cardholder's statement.

DE 44—Additional Response Data

The MDS uses the Additional Response Data (DE 44) field to provide other supplemental data (for example, a telephone number) that may be required in response to an authorization or other type of transaction request.

Attribute

ans...25; LLVAR.

Usage

This data element may be present in any response message when the Response Code (DE 39) is set to "30", indicating that a Format Error condition was detected in the preceding message. The first three bytes of DE 44, if present, will contain a 3-digit numeric value indicating the ISO data element number where the MDS encountered the format error.

When DE 44 exists in a File Update Request Response/0312, it contains the data element (in the File Update Request/0302 message) in which the error exists. If the error is in DE 120 of the 0302 message, the next three digits provide more specific information to identify the error condition. Valid values are shown in Table 4.15.

Table 4.15—Additional Response Data Subfields

Value	Description
120001	The code has one of the following meanings: PAN is not numeric BIN in PAN is not numeric BIN does not belong to message initiator Check digit of PAN is incorrect PAN is not on the Account file
120002	Entry reason is not one of the following: P, L, S, X, O, F, V, G, C, U
120005	PIN length not numeric or spaces
120006	Entry reason V and VIP limit is not numeric

For electronic commerce cardholder certificate requests, the issuer must send a telephone number in DE 44 of the 0210 message only when the response code is equal to "01". If the response code is equal to "85" then send delay date and time in DE 44 of the 0210 message. Table 4.16 describes these subfields in DE 44.

Table 4.16—DE 44 Values for 0210 Responses

Subfield	Position	Attribute	Value
telephone number	ans…25	25	Contains the telephone number for "call issuer" response codes.
delay date and time	ans…10	10	Contains the date and time after which a cardholder may reapply for a certificate.

Note **When DE 39 = "30", DE 44 is optional; DE 44 will not always be present to indicate the source location of a format error. When it is provided in the response, the MDS will pass DE 44 to the requestor.**

DE 45—Track 1 Data

Track 1 Data (DE 45) is the information encoded on track 1 of a bankcard magnetic stripe as defined in ISO 7813, including field separators. However, this excludes beginning and ending sentinels and LRC characters, as defined therein.

Attribute

ans...79; LLVAR.

Usage

DE 45 is used in point of service (POS) applications where the POS terminal equipment reads and transmits track 1 data in lieu of or in addition to the track 2 information encoded on the card magnetic stripe.

When it is received, as a part of a financial authorization request (debit MasterCard **only**), the MDS will build track 2 data (DE 35) from this data element to forward to the issuer. DE 45 will also be sent, when present.

If any track 2 data must be created by the MDS using track 1 data, the processors must be prepared to accept any character that would have been present in the track 1.

The acquirer must encode the following minimum data on DE 45 (Table 4.17):

Table 4.17—Track-1 Data Subfields

Subfield	Attribute	Value
Start Sentinel	n-1	*(not transmitted)*
Format Code	an-1	Literal character “B”
PAN	n…19	
Field Separator	ans-1	Binary 1101
Cardholder Name	ans…26	
Field Separator	ans-1	Binary 1101
Expiration Date	ans-4	“YYMM” format
Service Code	ans-3	
Discretionary Data	ans…24	optional by issuer
End Sentinel	n-1	*(not transmitted)*

Subfield	Attribute	Value
LRC	n-1	*(not transmitted)*

Maximum record length = 79

Note **The field separator character (binary "1101") is represented as the EBCDIC character "D." However, because many ATM and POS devices perform non-standard character translation while reading binary coded decimal (BCD)-encoded magnetic stripe data, the EBCDIC character "=" may also be used to represent the field separator character in magnetic stripe data forwarded to the MDS.**

If the MDS must create track 2 data from track 1 information, the issuer must be prepared to accept ANY character sent from the acquirer in track 1 data.

DE 46—Additional Data (ISO)

Additional Data (ISO) (DE 46) provides data supplemental to that already conveyed in the specific data elements in the message.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

ans...999; LLLVAR

Usage

ISO reserves this data element for future definition and use.

DE 47—Additional Data (National)

Additional Data (National) (DE 47) is reserved for national organizations to define data unique to country applications

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

ans...999; LLLVAR

Usage

This data element is reserved for future definition and use by appropriate national standards organizations.

DE 48—Additional Data

Additional Data (DE 48) is reserved for use based on product type.

Attribute

ans...100; LLLVAR

Usage

Table 4.19 provides formats and descriptions for the subelements (SE) in DE 48. The subelement sequence does not have to be in the order of tag value. For example, subelement 11 does not have to precede subelement 40, which does not have to precede subelement 41, and so on.

DE 48 provides other supplemental data in a message when a specific ISO-designated data element is not available. It is a free-format, variable-length alphanumeric data element that may be used for multiple purposes. This data element's content may vary by program and service.

Note **The length of this data element has been limited to 100 bytes for practical operational and system constraints.**

Transaction Category Code (TCC)

If the fourth character in DE 48 (the first character following the length value) is an *alphabetic* character, the MDS reads that character as the transaction category code (TCC). Refer to appendix C or the *Quick Reference Booklet* for TCC values.

Inclusion of the TCC received from the acquirer in DE 48 of the 0200 request, into the outbound 0200 message sent by the MDS to the issuer, adheres to the following rules:

- For Cirrus ATM and Maestro ATM requests, any acquirer-supplied TCC is never passed to the issuer in the outbound message.
- For Maestro POS requests, if the TCC is the **only** subelement contained in DE 48 from the acquirer, the MDS **will not** send the TCC to the issuer.
- For Maestro POS requests, if the TCC is **not** the only subelement in DE 48, the MDS **will** send the TCC to the issuer.

- For Cirrus purchase requests, if the TCC is the **only** subelement contained in DE 48 from the acquirer, the MDS **will not** send the TCC to the issuer. If the TCC is accompanied by only Implied Decimal (SE 70), the MDS **will not** send the TCC to the issuer. If the TCC is accompanied by only Implied Decimal (SE 70) and Nation's ID (SE 41, code 11), the MDS **will** send the TCC to the issuer.
- For debit MasterCard transactions, regardless of whether the TCC is alone or with other subelements, the MDS **will** send the TCC to the issuer.

Table 4.18 illustrates the TCC message inclusion rules.

Table 4.18—TCC Message Inclusion Rules

Transaction Type	TCC	Nation's ID (SE 41)	Implied Dec (SE 70)	Other SEs	To Issuer
MS ATM	•	NA	NA	Any	No
MS ATM	•	NA	NA	None	No
CI ATM	•	NA	NA	Any	No
CI ATM	•	NA	NA	None	No
MS POS	•	NA	NA	Any	Yes
MS POS	•	NA	NA	None	No
CI Purchase	•	No	No	NA	No
CI Purchase	•	No	•	NA	No
CI Purchase	•	•	•	NA	Yes
CI Purchase	•	•	No	NA	Yes
MD	•	NA	NA	Any	Yes
MD	•	NA	NA	None	Yes

Subelement Encoding Scheme

Note **This is the encoding scheme if subelements exist in transactions.**

DE 48 consists of encoded subelements. Except for the TCC, each subelement begins with a two-byte tag **and** an associated two-byte length indicator. The subelements do not need to be in any particular order or sequence within DE 48. Members should be able to send and receive all subelements available within DE 48.

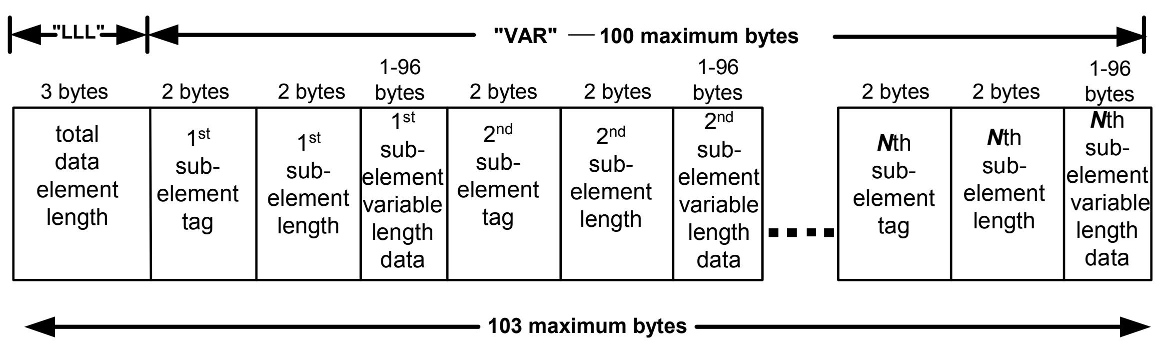
The first two bytes of each subelement must contain a tag in the range 00–99 to specify the type of DE 48 subelement. MDS universally defines values 00–69 for use by all programs and services. Values 70–99 are defined for use within individual programs and services only; individual program and service requirements dictate the use and content of the DE 48 subelement.

The second two bytes of each subelement must contain a length indicator in the range 00–99.

The overall length of the DE 48 is specified in its first three bytes (the “LLL” portion of the data element). The overall length of DE 48 is restricted to 100 bytes to accommodate practical operational limitations.

Figure 4.1 illustrates the construction of the entire DE 48 as well as subelements that may exist within it:

Figure 4.1—Subelements of DE 48



Subelement Descriptions

Table 4.19 provides formats and descriptions for the SEs in DE 48.

Table 4.19—Additional Data Subelements

Subelement	Value		
11	Processors must use the following subfield formatting for Network Management/0800 and 0810 messages used in Key Exchange Data Block sequences. During a key exchange, the key length being exchanged will be verified against the set-up for that particular encryption zone. The database will have an indicator for each zone (based on processor number, group sign-in (GSI)) to represent the key length required for that zone. For example, if the link is set-up of double-length keys and DE 48 subelement 11 contains single- or a triple-length key, the key exchange will be denied.		
	Subfield	**Attribute**	**Value**
	Single DES Prefix	an-4	1138 (Constant value for MDS; indicates that this is a key exchange data block)
	Triple DES Double Length Prefix	an-4	1154 (Constant value for MDS; indicates that this is a key exchange data block)
	Triple DES Triple Length Prefix	an-4	1170
	Key Class Identifier	an-2	PK (Pin Key Change)
	Key Index Number	n-2	00 (Constant value for MDS)
	Key Cycle Number	n-2	00 ... 99
	Encrypted Key – Single DES	an-16	Hex characters 0 ... 9 and A ... F. Contains the hexadecimal representation of the 64 bits of the new encryption key, encrypted under the current communications key.
	Encrypted Key – Triple DES Double Length	an-32	Hex characters 0 ... 9 and A ... F. Contains the hexadecimal representation of the 64 bits of the new encryption key, encrypted under the current communications key.
	Encrypted Key – Triple DES Triple Length	an-48	Hex characters 0 ... 9 and A ... F. Contains the hexadecimal representation of the 64 bits of the new encryption key, encrypted under the current communications key.
	Key Check Value	an-16	Hex characters 0 ... 9 and A ... F Contains the hexadecimal representation of the first four check digits (binary zeroes encrypted under the new key).

Subelement	Value		
11	Processors must expect the following subfield formatting for Network Management/0820 message used in Key Exchange Data Block sequences. During a key exchange, the key length being exchanged will be verified against the set-up for that particular encryption zone. The database will have an indicator for each zone (based on processor number, group sign-in (GSI)) to represent the key length required for that zone. For example, if the link is set-up of double-length keys and DE 48 subelement 11 contains single- or a triple-length key, the key exchange will be denied.		
	Subfield	**Attribute**	**Value**
	Single DES Prefix	an-4	1138 (Constant value for MDS; indicates that this is a key exchange data block)
	Key Class Identifier	an-2	PK (Pin Key Change)
	Key Index Number	n-2	00 (Constant value for MDS)
	Key Cycle Number	n-2	00 ... 99
	Encrypted DES Key	an-16	Blank-filled
	Key Check Value	an-16	Blank-filled
40	For electronic commerce transactions, the acquirer can send the merchant certificate serial number and/or cardholder certificate serial number in subelement 40 of DE 48 of the 0100 and 0200 messages. The subelement designation has the format 40xx, where xx is the length of the data in the subelement. Should only be used for purchases and cardholder certificate information.		
	Subfield	**Attribute**	**Value**
	Merchant/Cardholder Certificate Serial Number	2	40 contains SET compliant information.
	Subelement length	2	05 … 40
	Contains one or both of the following subfields:		
	01-Merchant Certificate Serial Number	2	01
	Subfield 01 length	2	Length of Merchant Certificate Serial Number
	Merchant Certificate Serial Number, if present	1-16	Must be binary data
	02- Cardholder Certificate Serial Number	2	02
	Subfield 02 length	2	Length of Cardholder Certificate Serial Number
	Cardholder Certificate Serial Number, if present	1-16	Must be binary data

Subelement	Value
41	Contains 10 potential subfields. The prefix of this subelement is 41xx, where xx is the length of the sublelement. These subfields contain authentication data used in the 0100 and 0200 electronic commerce cardholder certificate message. A few examples of the data stored in these subfields are password, date of birth, mother's maiden name, social security number, etc. NOTE: not all of the data shown can be sent at once because it would exceed the maximum length of DE 48.

Subfield		Attribute
01	Password	1-26
02	Date of Birth (YYMMDD)	6
03	Card Validation Code 2	3
04	Cardholder's Name (as it appears on the card)	1-22
05	Street Address	1-20
06	Cardholder's City of Residence	1-13
07	Cardholder's State/ Country Code	3
08	Cardholder's Postal Code	1-10
09	Mother's Maiden Name	1-22
10	Social Security No.	9
11	National ID	1-20
12	Home Phone Number	1-20
13	Work Phone Number	1-20
14	Passport Number	1-20
15	Birth Date (YYYY/MM/DD)	10
16	Member Since (YY)	2
17	Photo Card Indicator(Y/N)	1
18	Country	1-20
19	Miscellaneous Authorization 1	1-30
20	Miscellaneous Authorization 2	1-30
21	Miscellaneous Authorization 3	1-30

Subelement	Value		
42	For electronic commerce purchases, the acquirer can send a level of security in subelement 42 of DE 48 of the 0200 message. The prefix is 42xx, where xx has a value of "07".		
	Subfield	**Length**	**Value**
	Subelement identifier	2	42
	Subelement data length	2	07
	Subfield number (currently only one value = 01)	2	01
	Subfield data length	2	03
	Security level code	2	**11** SET encryption; cardholder certificate not used **12** SET encryption; cardholder certificate used **13** SET encryption; chip cryptogram used, cardholder certificate not used **14** SET encryption; chip cryptogram used, cardholder certificate used **21** channel encryption; cardholder certificate not used **23** channel encryption; chip cryptogram used, cardholder certificate not used **91** no security protocol; cardholder certificate not used
	Universal Cardholder Authentication Field (UCAF) data status	1	**0** UCAF data collection is not supported at the merchant's Web site. **1** UCAF data collection is supported by the merchant but the UCAF data is not populated. (DE 48, SE 43 is not present) **2** UCAF data collection is supported by the merchant and the UCAF data is populated. (DE 48, SE 43 must be present)

Subelement	Value
43	For electronic commerce purchases, this subelement can carry Universal Cardholder Authentication Field (UCAF) data. This subelement is only present when a UCAF-enabled merchant has collected authentication data from the cardholder and passed it to the acquirer for inclusion in the 0200 financial request or 0200 preauthorization request. The format of the tag and length is 43xx, where xx is the length of the data that follows. The structure of these subfields is as follows: **Subfield** / **Length** / **Value** Subelement identifier / 2 / 43 Subelement data length / 2 / Variable up to 40 bytes UCAF data / 32 / Accountholder Authentication Value (AAV)
70	MDS support for the implied decimal will be applicable for Cirrus, Maestro, and MasterCard® debit card-branded transactions. The MDS supports implied decimal exponent values for all currencies maintained at the switch. **Subfield** / **Attribute** / **Value** Subelement Tag / 2 / 70 Subelement Length / n-2 / 01 Subelement Value / n-1 / 0 through 3
82	Contains the AVS address verification request option code. **Subfield** / **Attribute** / **Value** Transaction Subelement Identifier / n-2 / 82 Subelement Length / n-2 / 02 AVS Option Code / n-2 / **51** AVS only; **52** AVS and Authorization Request/0100

Subelement	Value			
83	Contains the AVS address verification response.			
	Subfield	**Attribute**	**Value**	
	Transaction Subelement Identifier	n-2	83	
	Subelement Length	n-2	01	
	AVS Result Code	an-1	**X**	for U.S. addresses, all digits match, nine-digit ZIP code; for addresses outside the U.S., the postal code matches
			Y	yes, all digits match, five-digit ZIP code
			A	address matches, postal/ZIP code does not
			W	for U.S. addresses, nine-digit ZIP code matches, address does not; for address outside the U.S., the postal code matches, address does not
			Z	five-digit ZIP code matches, address does not
			N	nothing matches
			U	no data from issuer/Authorization System
			R	retry, system unable to process
			S	AVS currently not supported
87	Contains the magnetic stripe/CVC error tag, provided by the issuer when applicable.			
	Subfield	**Attribute**	**Value**	
	Magnetic strip/CVC error tag	n-2	Issuer provides this subelement when applicable. Value = 87.	
	Subelement Length	n-2	Value is 01; the subelement designation will be 87xx, where xx = 01.	
	Code value	an-1	**M**	CVC2 match (in issuer response)
			N	CVC2 non-match (in issuer response)
			P	CVC2 not processed (may be in acquirer request)
			U	CVC2 issuer not registered (may be in acquirer request)
			X	Invalid Magneprint
			Y	CVC1 is invalid (in issuer response)
			Z	CVC1 and Magneprint are invalid (in issuer response)

Subelement	Value		
88	Contains the magnetic stripe/CVC error tag, provided by the issuer when applicable.		
	Subfield	**Attribute**	**Value**
	Magnetic stripe/CVC error tag	n-2	Authorization System provides this subelement when applicable. Value = 88.
	Subelement Length	n-2	Value is 01
	Monitoring Status	an-1	**Y** indicates that the Authorization System replaced to 90 in DE 22 with 02, meaning that the acquirer submitting the transaction is in monitoring status for CVC processing.
89	Contains the magnetic stripe/CVC error tag provided by the issuer when applicable.		
	Subfield	**Attribute**	**Value**
	Magnetic stripe/CVC error tag	n-2	Authorization System provides this subelement when applicable. Value = 89.
	Subelement Length	n-2	Value is 01
	Data/Code Indicators	an-1	The following codes indicate track data, POS data, or TCC errors: **A** track 1 or track 2 not present in the message **B** track 1 and track 2 present in the message **C** PAN (DE 2) not equal in track data **D** Expiration Date (DE 14) not equal in track data **E** service code invalid in track data **F** field separator(s) invalid in track data **G** a field within the track data exceeds maximum length **H** TCC (in DE 48) is T **I** POS customer presence indicator (DE 61, position 4) is 1, 2, 3, 4, or 5 **J** POS card presence indicator (DE 61, position 5) is 1

Subelement	Value		
90	Contains an indication of cardholder participation in the MasterCard Travel Industries Premier Service (TIPS).		
	Subfield	**Attribute**	**Value**
	TIPS Tag	n-2	90
	Subelement Length	n-2	Value is 01
	Enrolled Program	an-1	**P** Indicates the request is from a cardholder enrolled in a merchant preferred customer program, and magnetic stripe data may be absent.
92	Contains the CVC2 value indent printed on the card signature panel for MO/TO and e-commerce transactions.		
	Subfield	**Attribute**	**Value**
	CVC2 Tag	n-2	Value = 92
	Subelement Length	n-2	Value = 03
	CVC2 value	n-3	The value for CVC2 sent in the request
93	Contains the airline ticket number information.		
	Subfield	**Attribute**	**Value**
	T&E Tag	n-2	Value = 93
	Subelement Length	n-2	Value is variable up to 15. Current value is fixed at 15.
	Ticket no.	ans-15	Ticket number
95	This subelement indicates participation in a particular program or service established between issuers and merchants.		
	Subfield	**Attribute**	**Value**
	Promotion Tag	n-2	95
	Subelement Length	n-2	Value is 06
	Promotion Code	an-6	Six-character alphanumeric code identifying a unique use of a MasterCard card for a specific program or service established between issuers and merchants.

Note **Registered issuers should note that if an acquirer transmits both CVC1 and CVC2 data, CVC1 processing takes precedence over CVC2 processing.**

If the CVC1 value is incorrect, registered issuers should respond with a value of Y (incorrect CVC1) in subelement 87 without validating the CVC2 value. However, if the CVC1 value is correct, then registered issuers must validate the CVC2 and send the appropriate response to the acquirer.

DE 49—Currency Code, Transaction

The Currency Code, Transaction (DE 49) is the local currency of the acquirer or source location of the transaction. The MDS uses it to specify the currency used in Amount, Transaction (DE 4).

Attribute

n-3

Usage

This data element is mandatory whenever DE 4 is present in a message.

Values

Acquirers must select all currency codes from the numeric ISO Standard Currency Codes provided in appendix B of this document. Members must not use alpha currency codes.

DE 50—Currency Code, Settlement

The Currency Code, Settlement (DE 50) is the code defining the currency of Amount, Settlement (DE 5), and Amount, Settlement Fee (DE 29).

Attribute

n-3

Usage

This data element is mandatory whenever DE 5 is present in a message. For transactions where the MDS performs automatic currency conversion, the MDS automatically inserts this data element into the message.

When this field is present in a message, Conversion Rate, Settlement (DE 9) and Date, Conversion (DE 16) must also be present.

Values

The data element value of 840 is populated by the MDS except for an ISIS transaction where the value will equal the ISIS intra-country currency code.

DE 51—Currency Code, Cardholder Billing

The Currency Code, Cardholder Billing (DE 51) is the code defining the currency of Amount, Cardholder Billing (DE 6), and Amount, Cardholder Billing Fee (DE 8).

Attribute

n-3

Usage

This data element is mandatory whenever Amount, Cardholder Billing (DE 6) is present in a message. For transactions where the MDS performs the ICCR service for the issuer, the MDS automatically inserts this data element into the message.

When this field is present in a message, Conversion Rate, Cardholder Billing (DE10) and Date, Conversion (DE16) must also be present.

This data element should only be present in an ICCR transaction.

DE 52—Personal Identification Number (PIN) Data

The Personal Identification Number (PIN) Data (DE 52) contains a number assigned to a cardholder intended to uniquely identify that cardholder at the point of interaction (POI).

Attribute

b-64

Usage

The MDS uses this data element to transmit a cardholder's PIN, in **encrypted form** for issuer verification or validation. It is required in all ATM Financial Transaction Request/0200 messages and in some POS Financial Transaction Request/0200 messages.

Acquirers must encrypt all PINs using the procedures identified in Chapter 6 of this manual.

The MasterCard® Debit Switch (MDS) permits PINs from 4 to 12 characters in length. Regardless of the original PIN length, the encrypted PIN block is always 64 bits (8 bytes) in length.

Strict security requirements implemented within data communications with the MDS, mandate that PINs are never transmitted in the clear as character data. PINs must always be encrypted into a 64-bit Encrypted PIN block. PIN Data is never included in online 0220 store-and-forward or other bank card transaction messages.

If the MDS performs PIN validation or verification on behalf of an issuer, this data element will be turned off, unless the transaction is a balance inquiry, in which case the MDS will transmit the following binary value:

0000 0001 0000 0001 0000 0001 0000 0001 0000 0001 0000 0001 0000 0001 0000 0001

Note **A Banknet® telecommunications-connected 0100 member may elect not to receive this data element in balance inquiry messages. Contact your Implementation Representative for additional information.**

DE 53—Security Related Control Information

As of the publication date of this document, the ISO 8583 organization has not determined the specific definition and usage requirements for Security Related Control Information (DE 53).

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-16

DE 54—Additional Amounts

Additional Amounts (DE 54) provides information for a maximum of six amounts and related account data for which specific data elements have not been defined.

Attribute

an...120; LLLVAR

Usage

DE 54 can return account balance information in a balance inquiry (0210) requests. It can also provide cash back information on a purchase with a cash back transaction. Table 4.20 describes the subelements of DE 54.

Account Balance Information

Use of DE 54 subfields is optional within other Authorization Response/0110 and Financial Transaction Response/0210 messages. When present, they can contain account balance information that the acquirer may print on transaction receipts for the benefit of the cardholder.

The terminal screen displays the account balance information on the receipt in the currency of the acquirer.

The MDS provides the account balance information in the Financial Transaction Response/0210 message in the issuer's currency. The MDS performs a currency conversion from the issuer's currency to the acquirer's currency. The MDS returns this in the Financial Transaction Response/0210 message to the acquirer.

Cash Back Processing

A "purchase with cash back" Financial Transaction Request/0200 is indicated by Processing Code Transaction Type (DE 3, Positions 1 and 2) of "00" (POS purchase) and the presence of DE 54. DE 54 has no account type specified (Account Type = 00) and a value of "40" for Amount Type. The format for DE 54 is shown in Table 4.20. The MDS will not return DE 54 to the acquirer if the content is determined to be cash back information.

Table 4.20—Additional Amounts Subfields

Subfield	Position	Attribute	Value
Account Type	1–2	n-2	**00** no account specified **10** savings account **20** checking account **30** credit card account
Amount Type	3–4	n-2	**01** Ledger Balance **02** Available Balance **40** Cash Back **90** Available Credit **91** Credit Limit
Currency Code	5–7	n-3	Valid numeric Currency Code selected from appendix B.
Debit or Credit Indicator	8	a-1	**C** credit amount or positive balance **D** debit amount or negative balance
Amount	9–20	n-12	12 digits, right-justified with leading zeroes

Note

The MasterCard® Debit Switch (MDS) only uses two Additional Amount data subfields.

Acquirers not capable of printing/displaying negative balances should print/display a zero balance value.

DE 55—Integrated Circuit Card (ICC) System-Related Data

Integrated Circuit Card (ICC) System-Related Data (DE 55) contains chip data formatted in accordance with the Europay-MasterCard-Visa (EMV) 2000 specifications. EMV uses Basic Encoding Rules (BER). (Reference the EMV 2000 specifications for further details regarding the coding of BER-TLV (Tag, Length, Value) data objects.)

Note **This data element is used only when the financial transaction card is equipped with an integrated circuit and when that mode is activated and selected by the cardholder for the transaction.**

Attribute

b...255; LLLVAR

Usage

For integrated circuit card transactions, an issuer and the smart card use DE 55 to communicate with each other during a Financial Transaction Request/0200 or a Financial Transaction Response/0210. The data is encrypted and only the issuer or issuer's agent is able to properly use it.

Note **If DE 55 is present in the message, DE 22 (POS Entry Mode) must contain "05x" or "81x".**

Note **The MDS performs no edits on the data in DE 55; it passes the data from the acquirer to the issuer and from the issuer to the acquirer. The description of this data is intended to show relevant information from the EMV 2000 specification.**

The provided functionality is known as Online Mutual Authentication (OMA). The chip is able to authenticate itself to the issuer in the Financial Transaction Request/0200 and the issuer is able to authenticate itself to the chip in the Financial Transaction Response/0210 message.

After initial designation of the overall data-element length (LLL), the remaining binary data consists of a series of subelement tag-length-value (TLV) segments up to a total of 255 bytes. The subelement tags can be from one to two bytes long, which are followed by a one-byte designation of the length, and the subelement data, respectively.

OMA-Card Application Data is Financial Transaction Request/0200 messages inbound to an issuer and Financial Transaction Response/0210 messages outbound to an acquirer. DE 55 contains binary data that only the issuer or the issuer agent can process. The chip in a smart card uses it locally at a chip-capable terminal. The MDS does not edit the contents of this data element.

If DE 55 is present in a Financial Transaction Request/0200 message, the POS Entry Mode Code (DE 22) must equal 05x or 81x. Otherwise, the MDS rejects the message with a format error.

DE 55 is not mandatory in Financial Transaction Response/0210 messages if it was present in the Financial Transaction Request/0200 message. Issuers are expected to return a 0210 response to the acquirer with a different value in DE 55 than was received by the issuer in the 0200 request.

For debit MasterCard issuers, if the contents of DE 55 in the Financial Transaction Request Response/0210 message are the same as the contents of DE 55 in the Financial Transaction Request/0200 message, then the MDS will not pass DE 55 back to the acquirer in the Financial Transaction Request Response/0210 message. Also for debit MasterCard transactions, if the MDS sends DE 55 in the Financial Transaction Response/0210 message, DE 55 must have been present in the corresponding Financial Transaction Request/0200 message.

Required Subelements for DE 55 in a Financial Transaction Request/0200

Table 4.21 conveys current chip specification requirements for subelements in DE 55 for a Financial Transaction Request/0200. These subelements are considered mandatory.

Table 4.21—Required DE 55 Subelements in 0200 Message

Subelement Name	Tag Value [a]	Length [b]
Application Cryptogram (AC)	9F26	8
Cryptogram Information Data	9F27	1
Issuer Application Data (IAD)	9F10	7–32
Unpredictable Number	9F37	4

Subelement Name	Tag Value [a]	Length [b]
Application Transaction Counter	9F36	2
Terminal Verification Result (TVR)	95	5
Transaction Date	9A	3
Transaction Type	9C	1
Transaction Amount or Amount Authorized	9F02	6
Transaction Currency Code	5F2A	2
Application Interchange Profile	82	2
Terminal Country Code	9F1A	2
Amount Other [c]	9F03	6

a Hexadecimal representation: two characters = one byte binary.

b This column shows the actual character length of the data for the subelement. The actual length designator in the TLV is the one-byte binary designation of the data that follows.

c If the card populates the six bytes of data in Amount, Other, the acquirer must pass all six bytes.

Optional Subelements for DE 55 in a Financial Transaction Request/0200

Table 4.22 conveys current smart specification requirements for subelements in DE 55 for a Financial Transaction Request/0200. These subelements are considered optional.

Table 4.22—Optional DE 55 Subelements in 0200 Message

Subelement Name	Tag Value	Length
Cardholder Verification Method (CVM) Results	9F34	3
Terminal Capabilities	9F33	3
Terminal Type	9F35	1
Interface Device (IFD) Serial Number	9F1E	8
Transaction Category Code	9F53	1
Dedicated File Name	84	5–16
Terminal Application Version Number	9F09	2
Transaction Sequence Counter	9F41	2–4

Required Subelement for DE 55 in a Financial Transaction Request Response/0210

Table 4.23 conveys current chip specification requirements for subelements in DE 55 for a Financial Transaction Request Response/0210. This subelement is considered mandatory.

Table 4.23—Required DE 55 Subelement in 0210 Message

Subelement Description	Tag Value	Length
Issuer Authentication Data (Provides data to be transmitted to the card for issuer authentication.)	91	8–16

Optional Subelements for DE 55 in a Financial Transaction Request Response/0210

Table 4.24 conveys current chip specification requirements for subelements in DE 55 for a Financial Transaction Request Response/0210. This subelement is considered optional.

Table 4.24—Optional DE 55 Subelement in 0210 Message

Subelement Description	Tag Value	Length
Issuer Script Template 1 and 2 (Allows the issuer to provide a command for transmission to the card; present if issuer sends commands to ICC; acquirer network must support a subfield length of 128 bytes.)	71 and 72	1–128

Required Subelements for DE 55 in a Debit MasterCard Financial Transaction Advice/0220

For debit MasterCard completion messages (stemming from chip-based authorizations) acquired from the batch system and sent to the issuer in the form of Financial Transaction Advice/0220 messages, the MDS uses DE 55 for supplying transaction certificate chip data to the issuer.

Note **It is assumed that track 2 equivalent data (PAN, PAN Sequence Number, and Expiration Date) are already present in the clearing message.**

Transaction certificate data is defined in the EMV 2000 specification, and consists of the mandatory subelements defined for the Financial Transaction Request/0200 in chapter 2.

DE 56—Reserved for ISO Use

Reserved for ISO Use (DE 56) is reserved for future definition and use.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

ans...999; LLLVAR

DE 57—Reserved for National Use

ISO reserved this data element for future definition and use.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

ans…999; LLLVAR

DE 58—Authorizing Agent Institution ID

This data element is the institution identifier of the card issuer.

Note **The MasterCard® Debit Switch (MDS) does not currently use this data element.**

Attribute

n...11; LLVAR.

Usage

The MDS does not currently use this data element.

However, for the initial release of 2003 (03.1), the MDS anticipates conditional receipt of this data from the IPS in the 0210 message. The data is defined as either a routing number or a product-based pseudo number assigned by MasterCard. For additional reference see DE 32.

DE 59—Reserved for National Use

ISO reserved this data element for future definition and use.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

ans...999; LLLVAR

DE 60—Advice Reason Code

The MDS uses the Advice Reason Code (DE 60) to indicate the specific purpose of an advice message.

Attribute

ans...060; LLLVAR

Usage

DE 60 is present in all advice messages, with the exception of network management advices (08xx message types). The data element has one or more of the following subelements:

Table 4.25—Data Element 60 Advice Reason Structure

Subelement	Position	Description	Value
1	1-3	Advice Reason Code	This subfield is mandatory for all advice messages and indicates the general purpose of the advice message.
2	4-7	Advice Reason Detail Code	This subfield usage may be conditionally required; it provides additional, specific information as to the exact nature of the advice message.
3	8-60	Advice Reason Detail Text	This optional subfield contains textual information supplementary to the Advice Detail Code.

The value and meaning of the contents of the Advice Reason data element vary according to the advice message type (i.e. 02xx, 04xx, 05xx, or 06xx), card product type, and whether the advice message is inbound to or outbound from the MDS.

Table 4.26 displays the general values for subelements 1 and 2 based on message type for all products, though each code does not pertain to all products. To determine whether the processor or the MDS will populate DE 60, refer to the message flow descriptions and diagrams for advice messages in chapter 2.

Table 4.26—Data Element 60 Advice Reason Subelements 1 and 2

Message Type	Advice Reason Code	Advice Detail Code	Description
0220	200	–	Network Stand-In: issuer selected option
	201	–	Network Stand-In: IPS signed out
	202	–	Network Stand-In: IPS timed out
	203	–	Network Stand-In: IPS unavailable
	251	0010	APS approved transaction; card returned; partial dispense
	251	1010	APS approved transaction; card retained; partial dispense
	260	0091	Clear Chip
	280	–	APS approved transaction; Late Acquirer Financial Acknowledgment/0280
	280	0000	Outbound from MDS APS approved transaction; Late Acquirer Financial Acknowledgment/0280
	290	–	APS approved transaction; pre-authorized by issuer
	291	–	APS approved transaction; network timeout
	293	–	APS approved transaction; APS system error
	Any of the advice reason codes for a 0220 message.	x2yy	Network Advice: Possible Duplicate As indicated by the 2 in the second position of the detail code, this 0220 message has been determined by the MDS to be a possible duplicate of a previous 0220 message. The reason code and the other positions of the detail code are from the original advice message: **x** card disposition of previous message, 1 = retain, 0 =return **yy** detail code (position 3, 4) of previous message If the Advice Detail Code of the original advice message is not present or contains zeroes, the Advice Detail Code of this possible duplicate message is “0200”.
0420	400	–	Network advice: APS error; unable to deliver response
	401	0080	Network advice: APS error; no APS 0180/0280 Acknowledgment
	402	0090	Network advice: IPS timeout error not acceptable from acquirer
	450	0011	Zero dispense: card returned; no receipt issued
	450	1011	Zero dispense: card retained; no receipt issued
	450	0018	Zero dispense/Over dispense card returned; POI failure. For the Timeout-Induced Reversal/0420 message, this code indicates an 0210 timeout at the acquirer.

Message Type	Advice Reason Code	Advice Detail Code	Description
	450	1018	Zero dispense: card retained; POI failure
	450	0019	Zero dispense: card returned, POI timeout
	450	1019	Zero dispense: card retained; POI timeout
	450	0040	Zero dispense: card returned; cardholder timeout
	450	1040	Zero dispense: card retained; cardholder timeout
	451	0010	Partial dispense: card returned
	451	1010	Partial dispense: card retained
	453	0041	Financial transaction cancellation: card returned
	453	1041	Financial transaction cancellation: card retained
	454	–	APS unable to deliver response
	455	0090	APS timeout; card returned
	455	1090	APS timeout; card retained
	487	0005	Retrieval request: cardholder does not agree
	487	0021	Retrieval request: transaction not recognized
	487	0023	Retrieval request: need for personal records
	487	0041	Retrieval request: fraud investigation
	487	0042	Retrieval request: potential chargeback
	488	–	Fulfillment
	489	0017	Chargeback - Cardholder dispute (ATM only)
	489	0071	Chargeback - Disputed Amount (POS only)
	489	0072	Chargeback - Credit Posted as Debit (POS only)
	489	0073	Chargeback - Duplicate Transaction (POS and ATM)
	489	0074	Chargeback - Missing or Illegible Signature (POS only)
	489	0075	Chargeback - Credit not Received (POS only)
	489	0076	Chargeback – Documentation not received on retrieval request (POS only)
	489	0077	Chargeback - Cardholder Denies Transaction Finalized (POS only)
	489	0078	Chargeback – Documentation not legible on retrieval request (POS only)
	489	0079	Chargeback – Goods or services not delivered (e-commerce only)

Message Type	Advice Reason Code	Advice Detail Code	Description
	490	0004	Duplicate Transaction – indicates two copies of a transaction were posted and this adjustment is backing out one of the transactions.
	490	0007	Correction processed by the MDS to reverse adjustment, chargeback or representment
	490	0010	Adjustment
	490	0019	Reversal of a representment - no documentation fulfillment
	491	0013	Representment
	491	0086	Invalid chargeback for IPM, dollar amount does not match original transaction
	491	0088	Invalid chargeback for IPM, rejection of adjustment due to a duplicate request
	491	0089	Invalid chargeback for IPM, adjustment over 180 days
	491	0099	Invalid chargeback for IPM, unable to locate transaction in Cirrus file
	Any of the advice reason codes for a 0420 message.	x2yy	Network Advice: Possible Duplicate As indicated by the 2 in the second position of the detail code, this 0420 message has been determined by the MDS to be a possible duplicate of a previous 0420 message. The reason code and the other positions of the detail code are from the original advice message: **x** card disposition of previous message, 1 = retain, 0 =return **yy** detail code (position 3, 4) of previous message If the Advice Detail Code of the original advice message is not present or contains zeroes, the Advice Detail Code of this possible duplicate message is “0200”.
0422	487	0005	Retrieval request: cardholder does not agree
	487	0021	Retrieval request: transaction not recognized
	487	0023	Retrieval request: need for personal records
	487	0041	Retrieval request: fraud investigation
	487	0042	Retrieval request: potential chargeback
	488	–	Fulfillment
	489	0017	Chargeback - Cardholder dispute (ATM only)
	489	0071	Chargeback - Disputed Amount (POS only)
	489	0072	Chargeback - Credit Posted as Debit (POS only)
	489	0073	Chargeback - Duplicate Transaction (POS and ATM)

Message Type	Advice Reason Code	Advice Detail Code	Description
	489	0074	Chargeback - Missing or Illegible Signature (POS only)
	489	0075	Chargeback - Credit not Received (POS only)
	489	0076	Chargeback – Documentation not received on retrieval request (POS only) Chargeback
	489	0077	Chargeback – Goods or services not delivered (e-commerce only)
	489	0078	Chargeback – Documentation not legible on retrieval request (POS only)
	489	0079	Chargeback – Goods or services not delivered (e-commerce only)
	490	0004	Duplicate Transaction
	490	0007	Correction processed by the MDS to reverse adjustment, chargeback or representment
	490	0010	Adjustment
	490	0019	Reversal of a representment - no documentation fulfillment
	491	0013	Representment
	491	0086	Invalid chargeback for IPM, dollar amount does not match original transaction
	491	0088	Invalid chargeback for IPM, rejection of adjustment due to a duplicate request
	491	0089	Invalid chargeback for IPM, adjustment over 180 days
	491	0099	Invalid chargeback for IPM, unable to locate transaction in Cirrus file
	Any of the advice reason codes for a 0422 message.	x2yy	Network Advice: Possible Duplicate As indicated by the 2 in the second position of the detail code, this 0422 message has been determined by the MDS to be a possible duplicate of a previous 0422 message. The reason code and the other positions of the detail code are from the original advice message: **x** disposition of previous message, 1 = retain, 0 =return **yy** detail code (position 3, 4) of previous message If the Advice Detail Code of the original advice message is not present or contains zeroes, the Advice Detail Code of this possible duplicate message is “0200”.
0520	500	–	Automatic reconciliation/totals complete
	501	–	Reconciliation request response/totals complete
	502	–	Reconciliation request response/totals incomplete
0522	500	–	Automatic reconciliation/totals complete

Message Type	Advice Reason Code	Advice Detail Code	Description
	501	–	Reconciliation request response/totals complete
	502	–	Reconciliation request response/totals incomplete
0620	600	–	Message unreadable/indecipherable/contains invalid data; (Advice Detail Code field MAY contain bit map number of data element where message scanning was aborted)
	601	–	Retrieval Request (Not used by MDS)
	602	–	Fulfillment notification (Not used by MDS)
0644	650	6904	Message not dispatched from remote MIP
	650	6905	Message not dispatched to remote MIP
	650	6906	Message not delivered to remote MIP
	650	6907	Message not delivered from remote MIP
	650	6908	No confirmation 0210 message was delivered to the remote MIP

Note **For debit MasterCard transactions, 04xx message types, a number of additional advice detail codes are valid. Please refer to the *Chargeback Guide* for a listing of valid values.**

Subelement 2 Usage Notes

The advice detail code (SE 2) description above has certain special meanings and the code has some special applications as indicated below:

In the column for Advice Detail Code, a hyphen indicates no data will be present in subelement 2.

The first two digits of subelement 2 Advice Detail Code are normally “00”; common practice for acquirers designating “card retained” in an inbound 0420 reversal is to use “10” in these positions.

The MDS has a facility for determining whether an inbound exception advice is a possible duplicate. When positions 4 and 5 of DE 60 (positions 1 and 2 of SE 2) in an exception message outbound from the MDS have the value “20”, this indicates the advice is a possible duplicate.

Subelement 3 Usage Notes

A debit MasterCard issuer must provide subelement 3, as defined in Table 4.27, when sending an online exception (chargeback) Issuer Reversal Advice/0422 message to the MDS through the issuer's online facility.

Table 4.27—DE 60, SE 3 for Debit MasterCard 0422 from Issuer

DE 60 Position	Subfield Name and Values
8	Usage Code **1** Issuer disputing initial presentment **2** Acquirer sending a second presentment **3** Arbitration chargeback
9	Documentation Indicator **Blank** No documentation **1** Documentation will follow **2** Invalid ARN in prior chargeback, no documentation required or received **3** Invalid ARN in prior chargeback, documentation received **4** Non-receipt of required documentation
10-11	Condition Code, reference only, generated by the MDS to batch
12-49	Message Block, optional message text
50-51	Chargeback Flag, reference only, generated by the MDS to batch
52-53	Future use – space fill
54	Reject Code – space fill
55-60	Future use – space fill

Maestro and Cirrus processors can provide the contact information (Table 4.28) to the MDS in the online exception 042x advice DE 60 SE 3. Similar information may be sent from the processor's terminal entry on a NICS exception advice.

Table 4.28—DE 60 SE 3, Advice Reason Text (Maestro and Cirrus)

Subelement	Position	Description
3	8-28	Contact name
	29-44	Contact phone
	45-60	Contact fax

DE 61—Point of Service (POS) Data

The MDS uses the Point of Service (POS) Data (DE 61) to indicate the specific conditions present at the point of service (POS) at the time that a transaction takes place.

Attribute

ans...026; LLLVAR

Usage

Issuers of debit MasterCard require the 26 characters of DE 61 in the request message, and acquirers must supply this information in the request message in accordance with Table 4.29. Maestro and Cirrus issuers must be prepared to accept the full 26 characters of DE 61 in the request message. Acquirers should supply at least the first 11 bytes of this information in the request message in accordance with Table 4.29.

Note **The MDS does not perform edits on the contents of this data element in the Financial Transaction Request/0200 message it receives from the acquirer. The MDS passes the data element contents to the issuer in the outbound request message to the issuer.**

Debit MasterCard force post messages will only contain the first ten positions described below. The MDS populates the first nine positions with 010001000. The 10th position (CAT level) value is a one-digit translation of the three-character code obtained from the POS Data Terminal Type field of the batch GCMS integrated product message (IPM) record.

Table 4.29 describes the subfields in DE 61.

Table 4.29—Point of Service (POS) Data Subfields

Subfield	Position	Attribute	Value	
POS Terminal Attendance Indicator	1	n-1	**0**	Attended terminal (Valid for debit MasterCard and Maestro transactions only)
			1	Unattended terminal
			2	No terminal used (voice/audio response unit [ARU] authorization) (Valid for debit MasterCard transactions only)

Subfield	Position	Attribute	Value
POS Terminal Operator Indicator	2	n-1	No longer used—zero filled
POS Terminal Location Indicator	3	n-1	**0** On premises of card acceptor facility **1** Off premises of card acceptor facility (remote location) **2** On premises of cardholder (home PC) (valid for debit MasterCard transactions only) **3** No terminal used (valid for debit MasterCard transactions only)
POS Customer Presence Indicator	4	n-1	**0** Customer present **1** Cardholder not present, unspecified (valid for debit MasterCard transactions only) **2** Cardholder not present, mail/facsimile order (valid for debit MasterCard transactions only) **3** Cardholder not present, phone/ARU order (valid for debit MasterCard transactions only) **4** Cardholder not present, recurring transaction (valid for debit MasterCard transactions only) **5** Electronic order (home PC, Internet)
POS Card Presence Indicator	5	n-1	**0** Card present **1** Card not present (valid for debit MasterCard transactions only)
POS Card Retention Indicator	6	n-1	**0** Terminal/operator does not have card capture capability **1** Terminal/operator has card capture capability
POS Transaction Status Indicator	7	n-1	**0** Normal request (original presentment) **1** Merchant authorized (valid for Maestro POS transactions only) **3** CDC inquiry request **4** Pre-authorization request (valid for debit MasterCard and Maestro transactions only) **5** Stand-In **6** Address verification request; verify shipping address (valid for debit MasterCard transactions only) **7** Cash back (valid for Maestro POS transactions only) **8** Downtime submission request (valid for debit MasterCard transactions only)
POS Transaction Security Indicator	8	n-1	**0** No security concern **1** Suspected fraud **2** Identification verified

Subfield	Position	Attribute	Value
POS Transaction Routing Indicator	9	n-1	**0** Zero fill; field no longer used
Cardholder-Activated Terminal Level Indicator	10	n-1	**0** Not a CAT transaction **1** Authorized Level 1 CAT: automated dispensing machine with PIN **2** Authorized Level 2 CAT: self service terminal **3** Authorized Level 3 CAT: limited amount terminal (valid for debit MasterCard transactions only) **4** Authorized Level 4 CAT: In-flight Commerce (valid for debit MasterCard transactions only) **5** Scrip device (valid for Maestro POS transactions only) **6** Electronic Commerce Transactions **7** Authorized Level 7 CAT: transponder (valid for debit MasterCard transactions only).
POS Card Data Terminal Input Capability Indicator	11	n-1	**0** Unknown **1** No terminal used **2** Magnetic stripe reader **3** Bar code (reserved for future use) **4** Optical Character Recognition (reserved for future use) **5** Magnetic stripe reader and EMV specification compatible integrated circuit card (ICC) reader **6** Key entry only **7** Magnetic stripe reader and key entry **8** Magnetic stripe reader and key entry and EMV-compatible ICC reader **9** EMV compatible ICC reader
POS Authorization Life Cycle Indicator	12-13	n-2	Indicates the number of days pre-authorization stays in effect (ATM and Maestro POS transactions should use 01.)
POS Country Code Indicator	14-16	n-3	Indicates the country of the terminal location (use valid three digit ISO numeric country code.)
POS Postal Code Indicator	17-26	ans-10	Indicates the geographic code of the terminal location (if data is unknown or unavailable, zero fill.)

DE 62—Intermediate Network Facility (INF) Data

Intermediate Network Facility (INF) Data (DE 62) is provided for use by acquiring network processors (CPS or INF) to contain acquiring network trace information that is useful for routing chargeback or adjustment transactions to the original acquiring institution.

Attribute

ans...50; LLLVAR

Usage

INF data is an optional data element within any originating Financial Transaction Request/0200 or Financial Transaction Advice/0220 message originated by an acquiring CPS or INF. Subsequently, this data element (if present within an original transaction) is returned without alteration in any chargeback or adjustment related to the original transaction.

This data element is provided to assist acquiring processor facilities directly-connected to the MDS. It allows these processors to maintain sufficient data within a message to facilitate online routing of chargebacks and adjustment messages without maintaining an online database of original transaction routing data.

Values

INF data is a free-format, variable length alphanumeric field that may be used to store unique acquiring processor ID codes, acquiring network linking data, or other information useful to processors in routing online chargeback and adjustment messages. The MDS does not edit or modify the field.

DE 63—Network Data

Network Data (DE 63) is a mandatory switch-generated data element composed of subelements that contain various descriptive and identifying attributes of the transaction.

Attribute

ans...044; LLLVAR

Usage

The MDS generates this data element for each originating message routed to the MDS. The receiver must retain and use this data element in any response or acknowledgment message associated with the originating request or advice message. The exception is a Timeout-Induced Reversal/0420 message; which does not contain DE 63.

The MDS determines the appropriate financial network code for all transactions routed through the MDS, based upon customer-established product configuration tables, customer parameter tables, and MDS routing priority tables.

Table 4.30 describes the subfields in DE 63.

Table 4.30—Network Data Subfields

Subfield	Position	Attribute	Value
Financial network code	1-2	a-2	Identifies the financial bank card product associated with the transaction. **MC** MasterCard **CI** Cirrus® **MS** Maestro® **MD** MasterCard® debit card **MX** Mondex™ **PL** Plus® **VI** VISA

Subfield	Position	Attribute	Value
Interchange rate indicator	3	n-1	Identifies the transaction as domestic (within the U.S.A. and Canada), International (Asia Pacific, Europe, Latin America and the Caribbean and Middle East/Africa), or Intra-country (within a country where an ISIS agreement is in effect). **0** U.S. and Canada Regions **1** Asia Pacific, Europe, Latin America and the Caribbean, and Middle East/Africa Regions **2** Intracountry (ISIS)
Network reference number	4-12	n-9	A unique transaction identification number (switch serial number) generated (or assigned) by the MDS.
Banknet reference number	13-21	an-9	A unique identifier assigned to debit MasterCard authorizations and is present in both Financial Transaction Request/0200 Authorization and Financial Transaction Advice/0220 Clearing messages. Only present in debit MasterCard transactions.
Acquirer's reference number	22-44	n-23	A unique identifier assigned by the acquirer of debit MasterCard transactions. Only present in debit MasterCard 0220 clearing messages.

Note **The issuer processor must use the switch serial number contained in this data element to match an online same day Acquirer Reversal Advice/0420 message and a Financial Transaction Acknowledgment/0290 message for issuer late response to the original Financial Transaction Request/0220 message.**

Online same day reversals contain the original switch serial number of the 0200 message. MDS generated (such as NICS™) exception items will be assigned a unique switch serial number.

The batch record will contain both switch serial numbers (the original switch serial number of the 0200 and the switch serial number of the NICS™ processed exception item).

The online reversal (04xx) message will only contain the new switch serial number.

For debit MasterCard file updates, the Financial Request Response/0312 messages returning to the MDS from the Banknet Account Management Service (AMS) contain a unique Banknet identifier in DE 63.

For 0312 messages returned to the issuer by the MDS, the data conforms to Table 4.30. In those 0312 messages, the following values are used:

Financial network code = CI, MS, or MD

Interchange rate indicator = 0 (U.S. and Canadian regions)

Network reference identifier = Nine digits, generated by the internal MDS file update process for Cirrus or Maestro.

Banknet reference number = Nine alphanumeric characters, received from AMS by the MDS for debit MasterCard updates

The acquirer reference number is not applicable to 0312 messages.

There are some messages (for example, network management [08xx] messages) where financial network code data is not required; however, the subfield must always be present. In these cases, the subfield must be blank-filled.

DE 64—Message Authentication Code (MAC)

Message Authentication Code (MAC) (DE 64) validates the source and the text of the message between the sender and the receiver.

The MDS reserves the last bit position within any bit map for DE 64. If the member uses authentication on a message, the final bit of the final bit map of that message indicates the MAC information. The final bit of all preceding bit maps shall contain 0; for example, there shall be only one DE 64 per message and that DE 64 must be the last data element of the message.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

b-64

DE 65—Bit Map, Extended

Bit map, Extended (DE 65) is a series of 64 bits used to identify the presence ("1") or absence ("0") of each data element in an extended [third] message segment.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

b-64

Usage

The MasterCard® Debit Switch defines only two message segments, the presence or absence of which is indicated by Primary and Secondary bit maps. DE 65 would indicate the presence of a "third" message segment, and must never be present in an MasterCard® Debit Switch message. The corresponding bit (number 65) must always be "0" in the Secondary Bit Map.

Refer to the Primary and Secondary Bit Maps subsection.

DE 66—Settlement Code

The Settlement Code (DE 66) is a code indicating the result of a reconciliation request.

Attribute

n-1

Usage

The MDS does not use this data element. However, if an 05xx message arrives at the MDS interface with this data element present, the MDS will accept the data; however, it will ignore the data.

Processors can insert this data element into Acquirer Reconciliation Advice Response/0530 messages or into Issuer Reconciliation Response/0532 messages to indicate the disposition of a previous Acquirer Reconciliation Advice/0520 or Issuer Reconciliation Advice/0522 message.

Values

Table 4.31 lists design values for the Settlement Code data element.

Table 4.31—Settlement Code Values

Code	Description
1	Reconciliation message totals balance.
2	Reconciliation message totals do not balance.
9	Reconciliation message received; no balancing performed.

DE 67—Extended Payment Code

Extended Payment Code (DE 67) indicates the number of months that the cardholder prefers to pay for an item (the item purchased during the course of this transaction), if permitted by the card issuer.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-2

DE 68—Receiving Institution Country Code

Receiving Institution Country Code (DE 68) is the code of the country where the receiving institution is located.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-3

DE 69—Settlement Institution Country Code

The Settlement Institution Country Code (DE 69) is the code of the country where the settlement institution is located.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-3

DE 70—Network Management Information Code

The MDS uses the Network Management Information Code (DE 70) to identify network status. The Customer Processing System may use Additional Data (DE 48) in conjunction with DE 70 to provide network status or control information.

Attribute

n-3

Usage

This data element indicates the specific classification and purpose of network management (08xx) messages. It must be present in all network management (08xx) messages.

Values

Table 4.32 lists all Network Management Information Codes valid on the MDS.

Table 4.32—Network Management Information Codes

Code	Description
060	Processor-initiated Store-and-Forward (SAF) session request
061	General sign-on by the processor to the MDS
062	General sign-off by the processor from the MDS
065	Issuer sign-off, directing the MDS to begin Stand-In processing for the issuer
066	Issuer sign-on, directing the MDS to cease Stand-In processing for the issuer
161	Encryption key change
162	Initiate Encryption Key Change (by processor)
270	Echo test
363	End-of-file (EOF) encountered for SAF traffic. SAF complete.

DE 71—Message Number

Message Number (DE 71) is a sequential, cyclic number the message initiator assigns to a message. The Message Number monitors the integrity of interchange.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-4

DE 72—Message Number Last

Message Number Last (DE 72) is a sequential, cyclic number the message initiator assigns to a message. The Message Number monitors the integrity of interchange.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-4

DE 73—Date, Action

Date, Action (DE 73) specifies the date (year, month, and day) of a future action. In addition, the member may use it as a static time such as a birthdate.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-6; YYMMDD.

DE 74—Credits, Number

Credits, Number (DE 74) is the number of transactions the MDS processes as credits (to the CPS) during the daily settlement reporting period.

Attribute

n-10

Usage

When present in Acquirer Reconciliation Advice/0520 messages, this data element contains the sum number of all cardholder credit transactions (for example, transfer credits) processed during the specified settlement reporting period.

When present in Issuer Reconciliation Advice/0522 messages, this data element contains the sum number of all cardholder credit transactions processed during the specified settlement reporting period. These transactions include all cardholder account credit transactions performed in interchange by the issuer's cardholders on another (acquiring) institution's terminal network.

Values

This data element must contain valid numeric data.

Note **The MDS only populates this data element for transfer and refund transactions.**

DE 75—Credits, Reversal Number

Credits, Reversal Number (DE 75) is the number of transactions the MDS processes as credit reversals (to the CPS) during the daily settlement reporting period.

Attribute

n-10

Usage

When present in Acquirer Reconciliation Advice/0520 messages, this data element contains the sum of all acquirer reversal credit transactions processed during the specified settlement reporting period. These transactions include all reversals of cardholder account credit transactions performed in interchange by **not** On-Us cardholders on the acquirer's terminal network.

When used in Issuer Reconciliation Advice/0522 messages, this data element contains the sum number of all issuer reversal credit transactions processed during the specified settlement reporting period. These transactions include all reversals of cardholder account debit transactions (such as reversals of transactions) performed in interchange by the issuer's cardholders on another institution's terminal network.

Values

This data element must contain valid numeric data.

DE 76—Debits, Number

Debits, Number (DE 76) is the number of transactions the MDS processes as debits (to the CPS) during the daily settlement reporting period.

Attribute

n-10

Usage

When present in Acquirer Reconciliation Advice/0520 messages, this data element contains the sum number of all acquirer debit transactions processed during the specified settlement reporting period. These transactions include all cardholder account credit transactions performed in interchange by **not** On-Us cardholders on the acquirer's terminal network.

When present in Issuer Reconciliation Advice/0522 messages, this data element contains the sum number of all issuer debit transactions processed during the specified settlement reporting period. These transactions include all cardholder account debit transactions performed in interchange by the issuer's cardholders on another institution's terminal network.

Values

This data field must contain valid numeric data.

DE 77—Debits, Reversal Number

Debits, Reversal Number (DE 77) is the number of transactions the MDS processes as reversal debits (to the CPS) during the daily settlement reporting period.

Attribute

n-10

Usage

When present in Acquirer Reconciliation Advice/0520 messages, this data element contains the sum number of all acquirer reversal debit transactions processed during the specified settlement reporting period. These include all reversals of cardholder account debit transactions (such as reversals of transactions) performed in interchange by **not** On-Us cardholders on the acquirer's terminal network.

When present in Issuer Reconciliation Advice/0522 messages, this data element contains the sum number of all issuer reversal debit transactions processed during the specified settlement reporting period. These include all reversals of cardholder account credit transactions performed in interchange by the issuer's cardholders on another institution's terminal network.

Values

This data field must contain valid numeric data.

DE 78—Transfers, Number

Transfers, Number (DE 78) is the number of transactions the MDS processes as transfer transactions (to the CPS) during the daily settlement reporting period.

Attribute

n-10

Usage

The MDS currently generates no functional data for this data element. The MDS fills DE 78 with zeroes in the outbound 0520 or 0522 message.

Values

This data element will contain zeroes.

DE 79—Transfers, Reversal Number

Transfers, Reversal Number (DE 79) is the number of transactions the MDS processes as transfer reversals (to the CPS) during the daily settlement reporting period.

Attribute

n-10

Usage

The MDS currently generates no functional data for this data element. The MDS fills DE 79 with zeros in the outbound 0520 or 0522 message.

Values

This data element will contain numeric zeroes.

DE 80—Inquiries, Number

Inquiries, Number (DE 80) is the number of transactions the MDS processes as inquiries (to the CPS) during the daily settlement reporting period.

Attribute

n-10

Usage

The MDS currently generates no functional data for this data element. The MDS fills DE 80 with zeroes in the outbound 0520 or 0522 message.

Values

This data field will contain numeric zeroes.

DE 81—Authorizations, Number

Authorizations, Number (DE 81) is the number of transactions the MDS processes as Authorization Request/100 and Authorization Advice/0120 messages (to the CPS) during the daily settlement reporting period.

Attribute

n-10

Usage

The MDS currently generates no functional data for this data element. The MDS fills DE 81 with zeroes in the outbound 0520 or 0522 message.

Values

This data field will contain numeric zeroes.

DE 82—Credits, Processing Fee Amount

Credits, Processing Fee Amount (DE 82) is the amount the MDS processes as processing fees (to the CPS) during the daily settlement reporting period.

Attribute

n-12

Usage

The MDS currently generates no functional data for this data element. The MDS fills DE 82 with zeroes in the outbound 0520 or 0522 message.

Values

This data field will contain numeric zeroes.

DE 83—Credits, Transaction Fee Amount

Credits, Transaction Fee Amount (DE 83) is the amount the MDS processes as interchange transactions (to the CPS) during the daily settlement reporting period.

Attribute

n-12

Usage

The MDS currently generates no functional data for this data element. The MDS fills DE 83 with zeroes in the outbound 0520 or 0522 message.

Values

This data field will contain numeric zeroes.

DE 84—Debits, Processing Fee Amount

Debits, Processing Fee Amount (DE 84) is the amount the MDS processes as processing fees dealing with handling and routing (to the CPS) during the daily settlement reporting period.

Attribute

n-12

Usage

The MDS currently generates no functional data for this data element. The MDS fills DE 84 with zeroes in the outbound 0520 or 0522 message.

Values

This data field will contain numeric zeroes.

DE 85—Debits, Transaction Fee Amount

Debits, Transaction Fee Amount (DE 85) is the amount the MDS processes as processing fees of interchange transactions (to the CPS) during the daily settlement reporting period.

Attribute

n-12

Usage

The MDS currently generates no functional data for this data element. The MDS fills DE 85 with zeroes in the outbound 0520 or 0522 message.

Values

This data field will contain numeric zeroes.

DE 86—Credits, Amount

Credits, Amount (DE 86) is the amount the MDS processes as cardholder credits (to the CPS) during the daily settlement reporting period.

Attribute

n-16

Usage

Within an Acquirer Reconciliation Advice/0520 message, this data element contains the total amount for cardholder transactions credited to the acquirer processor. These include all cardholder account transactions (such as transfer credits) performed by the processor's cardholders.

Within an Issuer Reconciliation Advice/0522 message, this data element contains the total amount for cardholder transactions credited to the issuer processor. These include all cardholder account credit transactions performed via the MDS by the issuer's cardholders on another (acquiring) institution's terminal network.

The Acquirer Reconciliation Advice/0520 message and the issuer Reconciliation Advice/0522 message are denominated in U.S. dollars.

Values

This data field must contain valid numeric data.

Amounts are expressed without a decimal separator. Users and systems are responsible for placing the decimal separator appropriately for U.S. dollars. For example, 0000000000000150 represents $1.50 in U.S. currency.

DE 87—Credits, Reversal Amount

The Credits, Reversal Amount (DE 87) is the amount the MDS processes as reversal credits (to the CPS) during the daily settlement reporting period.

Attribute

n-16

Usage

Within an Acquirer Reconciliation Advice/0520 message, this data element contains the sum amount of all acquirer credit-reversal transactions due to the acquirer processor. These include reversals of cardholder account credit transactions performed via the MDS by **not** On-Us cardholders on the acquirer's terminal network.

Within an Issuer Reconciliation Advice/0522 message, this data element contains the sum amount of all issuer reversal credit transactions due to the institution or customer. These include reversals performed via the MDS by the issuer's cardholders on another (acquiring) institution's terminal network.

The Acquirer Reconciliation Advice/0520 message and the issuer Reconciliation Advice/0522 message are denominated in U.S. dollars.

Values

This data field must contain valid numeric data.

Amounts are expressed without a decimal separator. Users and systems are responsible for placing the decimal separator appropriately for U.S. dollars. For example, 0000000000000150 represents $1.50 in U.S. currency.

DE 88—Debits, Amount

Debits, Amount (DE 88) is the amount the MDS processes as debits (to the CPS) during the daily settlement reporting period.

Attribute

n-16

Usage

Within an Acquirer Reconciliation Advice/0520 message, this data element contains the sum amount of all cardholder transactions the MDS registers as debits of the acquirer. These include all cardholder account debit transactions performed via the MDS by **not** On-Us cardholders on the acquirer's terminal network.

Within an Issuer Reconciliation Advice/0522 message, this data element contains the sum amount of cardholder transactions the MDS registers as debits of the issuer. These include cardholder account debit transactions performed via the MDS by the issuer's cardholders on another (acquiring) institution's terminal network.

The Acquirer Reconciliation Advice/0520 message and the issuer Reconciliation Advice/0522 message are denominated in U.S. dollars.

Values

This data field must contain valid numeric data.

Amounts are expressed without a decimal separator. Users and systems are responsible for placing the decimal separator appropriately for U.S. dollars. For example, 0000000000000150 represents $1.50 in U.S. currency.

DE 89—Debits, Reversal Amount

Debits, Reversal Amount (DE 89) is the amount the MDS processes as reversal debits (to the CPS) during the daily settlement reporting period.

Attribute

n-16

Usage

Within an Acquirer Reconciliation Advice/0520 messages, this data element contains the sum amount of all debit reversal debit transactions due **from** the acquirer. These include reversals of cardholder account debit transactions performed via the MDS by **not** On-Us cardholders on the acquirer's terminal network.

Within an Issuer Reconciliation Advice/0522 messages, this data element contains the sum amount of all debit reversal transactions due **from** the issuer. These include all debit reversals performed by the issuer's cardholders on another (acquiring) institution's terminal network.

The Acquirer Reconciliation Advice/0520 message and the issuer Reconciliation Advice/0522 message are denominated in U.S. dollars.

Values

This data field must contain valid numeric data.

Amounts are expressed without a decimal separator. Users and systems are responsible for placing the decimal separator appropriately for U.S. dollars. For example, 0000000000000150 represents $1.50 in U.S. currency.

DE 90—Original Data Elements

Original Data Elements (DE 90) are data elements contained in an original message that may identify a transaction for correction or reversal.

Attribute

n-42

Usage

DE 90 **must** be present in the following messages as a reference to an original transaction being affected by a new message or transaction:

- Acquirer Reversal Advices/0420
- Issuer Reversal Advices/0422
- Financial Transaction Advices/0220

Note **For transactions processed by the MDS, please refer to data element (DE 63) subfield 3—switch serial number, as the key data element to match online same day acquirer and issuer reversal advices.**

Please note that this data element is not present in the Financial Transaction Negative Acknowledgment/0290 messages.

Values

This data element is composed of 5 fixed-length subfields. Each subfield is encoded as alphanumeric, right-justified with leading zeroes, as described in Table 4.33.

Table 4.33—Original Data Elements Subfields

Subfield	Position	Attribute	Value
1		n-4	Original MTI, Message Type Identifier
2		n-6	Original DE 11, System Trace Audit Number
3		n-10	Original DE 7, Transmission Date and Time
4		n-11	Original DE 32, Acquiring Institution ID Code
5		n-11	Original DE 33, Forwarding Institution ID Code

DE 91—File Update Code

The File Update Code (DE 91) is used in File Update Request/0302 messages. It indicates, to the MDS or to the MasterCard Account Management System (AMS), the action to perform to the file named in File Name (DE 101).

Note **The MasterCard® Debit Switch (MDS) does not use this data element for the Financial Transaction/02xx messages.**

Attribute

an-1

Usage

For File Update Request/0302 messages, the value of this data element indicates the execution of a specific file update action. The File Update Request Response/0312 message must return the same value as was sent in the 0302 message.

Values

Table 4.34 describes the File Update Code values for DE 91.

Table 4.34—File Update Code Values

Code	Description
1	Add record
2	Change record
3	Delete record
5	Inquiry

Table 4.35 lists valid update codes for each of the files identified in DE 101.

Table 4.35—Valid Updates by File Type

Filename (DE 101)	Add (DE 91 = 1)	Change (DE 91 = 2)	Delete (DE 91 = 3)	Inquiry (DE 91 = 5)
MCC102	•	•	•	•
MCC103	•		•	
MCCNEG	•	•	•	•

DE 92—File Security Code

File Security Code (DE 92) is a file update security code that indicates a message originator is authorized to update a file.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n-2

DE 93—Response Indicator

Response Indicator (DE 93) indicates the update action a POS system takes.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

an-5

DE 94—Service Indicator

Service Indicator (DE 94), in some systems, is an indication of the type of support service required by the recipient of a File Update Request/0302 message.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

an-7

DE 95—Replacement Amounts

Replacement Amounts (DE 95) are the new actual amount data elements necessary to perform a partial or full reversal of a financial transaction or a partial completion amount. This data element can also be used for the completed amount in a Maestro Financial Transaction Advice/0220 completion message for automated fuel transactions.

Attribute

n-42

Usage

The Customer Processing System must use DE 95 in the following messages when the original transaction amounts are being modified:

- Financial Transaction Advice/0220 (Acquirer-generated only) except debit MasterCard
- Acquirer Reversal Advice/0420 message
- Issuer Reversal Advice/0422 message

This data element will not be present in debit MasterCard transaction clearing messages.

Values

This data element is composed of 4 fixed-length subfields. Each subfield is encoded as alphanumeric, right justified with leading zeroes, as described below.

Subfield No. 1 must contain valid numeric data. The message initiator must zero-fill all other subfields. Currency conversion of actual Amount, Transaction (DE 4) into actual Amount, Settlement (DE 5) will be performed by the MDS, when required.

Table 4.36 describes the subfields in DE 95.

Table 4.36—Replacement Amounts Subfields

Subfield	Position	Attribute	Value
1		n-12	Actual Amount, Transaction. For the acquirer Timeout-Induced Reversal Advice/0420 message, this field contains all zeros.
2		n-12	Actual Amount, Settlement (provided by the MDS). For the acquirer Timeout-Induced Reversal Advice/0420 message, this field contains all zeros.
3		n-12	Actual Amount, Cardholder Billing (provided by the MDS). For the acquirer Timeout-Induced Reversal Advice/0420 message, this field contains all zeros.
4		n-6	zero fill

DE 96—Message Security Code

Message Security Code (DE 96) contains an MDS "password" security code to verify that the originator of the sign-on request is allowed access to the requested functions.

Attribute

b-64

Usage

This data element is used in Network Management/0800 messages.

DE 97—Amount, Net Settlement

Amount Net Settlement (DE 97) is the net value of all gross settlement amounts including fees.

Attribute

X+n-16

Usage

Within an Acquirer Reconciliation Advice/0520 message, this data element contains the calculated sum of all credit and debit amounts for the acquiring processor for the settlement period.

The credit or debit indicator (the first position of the data element) applies to the message recipient. Within acquirer-generated message types, a D (debit) fee amount indicates that the fee is to be applied as a debit to the message recipient, the issuer (and therefore as a credit to the message originator, the acquirer).

DE 97 indicates the processor's net settlement position as an acquirer. A credit value in this data field indicates monies are due to the acquiring institution or processor. A debit value in this field indicates monies are due from the acquiring institution or processor.

In Issuer Reconciliation Advice/0522 messages, this data element contains the calculated sum of all credit and debit amounts for the issuing processor for the settlement period.

DE 97 indicates the processor's net settlement position as an issuer. A credit value in this data field indicates monies are due to the issuing institution or processor. A debit value in this field indicates monies are due from the issuing institution or processor.

If an institution or processor participates as both an acquirer and an issuer within MDS, the DE 97 from both the Acquirer Reconciliation Advice/0520 and the Issuer Reconciliation Advice/0522 are added together to obtain the net settlement position.

If the sum of these data elements results in a credit value, the institution or processor has a net credit position for the applicable program. If the sum of these data elements results in a debit value, the institution or processor has a net debit position for the applicable program.

The Acquirer Reconciliation Advice/0520 message and the issuer Reconciliation Advice/0522 message are denominated in U.S. dollars.

Values

This data field must contain valid numeric data.

Amounts are expressed without a decimal separator. Users and systems are responsible for placing the decimal separator appropriately for U.S. dollars. For example, 0000000000000150 represents $1.50 in U.S. currency.

DE 98—Payee

Payee (DE 98) is the third party beneficiary in a payment transaction.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

ans-25

DE 99—Settlement Institution Identification Code

The Settlement Institution Identification Code (DE 99) is a code identifying a settlement institution or its agent.

Attribute

n...11; LLVAR.

Usage

This data element identifies the settlement agent associated with an acquirer, issuer, intermediate network facility, or other entity that participates on the MDS. The settlement institution or agent is responsible for the actual transfer of funds resulting from reconciliation and settlement processing.

Values

This data element will contain the Routing and Transit number assigned to the bank selected by MDS to act as the settlement agent for all processors connected to the MDS.

DE 100—Receiving Institution Identification Code

The MDS uses the Receiving Institution Identification Code (DE 100) to determine the destination routing of administrative (06xx) messages. For these messages, the Forwarding Institution ID (DE 33) identifies the sender of the message; the receiver of the message is identified by the Receiving Institution ID (DE 100).

In an Acquirer Reconciliation Advice/0520 message, this field contains the APS Processor ID. In an Issuer Reconciliation Advice/0522 message, this field contains the IPS Processor ID.

Attribute

n...11; LLVAR.

Usage

This data element must contain a valid Processor ID number assigned by the MDS.

Values

This data element must contain a valid Processor ID.

Note **Processing systems must not exchange the contents of the Forwarding and Receiving Institution ID Code data elements in response messages; the contents must remain the same for accurate response message routing.**

For the initial release of 2003 (03.1), the MDS may require the IPS to return (in the 0210 message and 04xx reversal responses) the processor ID of the issuer. The processor ID is a ten-digit number of the form "9000000xxx" where xxx is a three-digit number assigned by MasterCard. Please refer to DE 33 description.

DE 101—File Name

File Name (DE 101) is the actual or abbreviated name of a referenced file that is updated in accordance with the File Update Code (DE 91) of a File Update Request/0302 message.

Note **The MasterCard® Debit Switch (MDS) does not use this data element in the Financial Transaction/02xx messages.**

Attribute

ans...17; LLVAR.

Usage

This data element is used to identify the specific name of a Network data file, product parameter table, or Stand-In processing database that is being updated via a File Update Request/0302 message. The File Update Request Response/0312 message contains the same value in DE 101 that was sent in the 0302 message.

Values

Table 4.37 shows the valid values and the allowable file update actions for the file name.

Table 4.37—Valid Filenames and Allowable Updates

File Name (DE 101)	Description	Add (DE 91 = 1)	Change (DE 91 = 2)	Delete (DE 91 = 3)	Inquiry (DE 91 = 5)
MCC102	Account File	•	•	•	•
MCC103	Account Management File	•		•	
MCCNEG	MDS Stand-In Negative File	•	•	•	•
MCCVIP	MDS Stand-In VIP File	•	•	•	•

DE 102—Account Identification-1

Account Identification-1 (DE 102) is a series of digits used to identify a customer account or relationship. It is primarily used to identify the "**from account**" in a transaction.

Attribute

n...28; LLVAR.

Usage

Issuers may use DE 102 in an Authorization Response/0110 or a Financial Transaction Response/0210 messages to identify the specific cardholder "from" account number affected by a transaction. Acquirers may use DE 102 for printing on cardholder transaction receipts.

The "from" account is the account specified by the third and fourth digits of the Processing Code (DE 3).

Values

The MDS restricts the values of this data element to be numeric only.

DE 103—Account Identification-2

Account Identification-2 (DE 103) is a series of digits used to identify a customer account or relationship. It is primarily used to identify the "**to" account** in a transaction.

Attribute

n...28; LLVAR.

Usage

Issuers may use DE 103 in an Authorization Response/0110 or a Financial Transaction Response/0210 messages to identify the specific cardholder to account number affected by a transaction. Acquirers may use DE 103 for printing on cardholder transaction receipts. The "to" account is the account specified by the fifth and sixth digits of the Processing Code (DE 3).

Values

The MDS restricts the values of this data element to be numeric only.

DE 104—Transaction Description

Transaction Description (DE 104) can be used to describe additional characteristics of the transaction for billing purposes.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

ans...100; LLLVAR

DE 105–DE 111—Reserved for ISO Use

ISO reserves these data elements for future definition and use.

Note **The MasterCard® Debit Switch (MDS) does not use these data elements.**

Attribute

ans...999; LLLVAR

DE 112—Additional Data (National Use)

Additional Data (National Use) (DE 112) is reserved for national organizations to define data unique to specific networks or specific programs and services.

Attribute

ans...248; LLLVAR

Usage

The MDS uses this data element to support the Parcelas product. Parcelas provides acquirers and issuers with the ability to support a recurring payment option at the point of service.

Note **Parcelas and CDC support are presently limited to Maestro® ISIS transactions in Brazil.**

The MDS supports Parcelas and Consumer Direct Credit (CDC) options.

- **CDC.** Consumers at the point of interaction can make an "inquiry" requesting time pay options for a major purchase. The issuer can respond, providing up to six financing plans for the consumer to choose. The consumer can then select one of the options and the merchant submits the "purchase" request in another transaction. The consumer is then billed monthly for the installment payment.
- **Parcelas with interest.** Under this option, the length of repayment installments range from 2 to 12 months. The transaction is billed monthly to the cardholder. The authorization returned to the acquirer includes the amount of interest over the total installment value and taxes on interest.

Note **CDC Inquiry transactions must include Point of Service (POS) Data (DE 61) where position 7 equals 3.**

Values

The first three bytes (the "LLL" length field of LLLVAR) specify the overall length of DE 112. The overall length of DE 112 is restricted to 251 bytes.

The MDS organizes DE 112 into a group of encoded subelements. A three-byte ID and an associated three-byte length indicator identify each subelement.

The first three bytes of each subelement must contain an ID in the range 000–999 to specify the type of DE 112 subelement. Individual requirements define the use and content of the DE 112 subelement.

The second three bytes of each subelement must contain a length indicator in the range 000–999.

Figure 4.2 reflects the construction of DE 112 as well as the subelements it may contain.

Figure 4.2—DE 112 Subelement Contents

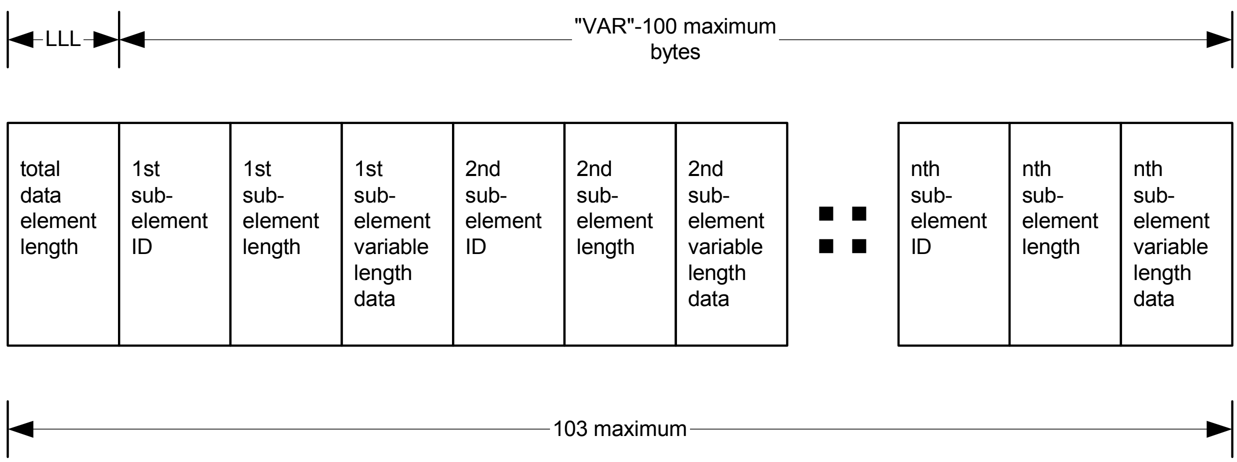


Table 4.38—Parcelas and CDC Transactions Subelement Requirements

Message Type	Subelements								
	1	2	11	12	13	14	15	16	17
Parcelas 0200	M	•	•	•	•	•	•	•	•
Parcelas 0210	M	M	•	•	•	•	•	•	•
CDC Inquiry 0200	M	•	•	•	•	•	•	•	•
CDC Inquiry 0210	M	•	M	O	O	O	O	O	O
CDC Purchase 0200	M	•	•	•	•	•	•	•	•
CDC Purchase 0210	M	•	M	O	•	•	•	•	•

Table 4.39—Additional Data (National Use) Subelements

Subelement	Position	Attribute	Value		
1			For Financial Transaction Request/0200 messages and Financial Transaction Request Response/0210 messages.		
			Subfield	**Attribute**	**Value**
	1–3		Subfield tag	n-3	**001** 1st subfield
	4–6		Subfield length	n-3	**004** Length of the subfield
	7–8		Subfield data	an-2	Transaction type: **21** Parcelas transaction **20** CDC transaction
	9–10			n-2	Number of installments
2			For Financial Transaction Request Response/0210 messages.		
			Subfield	**Attribute**	**Value**
	1–3		Subfield tag	n-3	**002** 2nd subfield
	4–6		Subfield length	n-3	**032** Length of the subfield
	7–8		Subfield data	an-2	Transaction type: **21** Parcelas transaction
	9–10			n-2	Number of installments
	11–22			n-12	Installment Interest
	23–34			n-12	Purchase plus interest amount
	35–38			n-4	Annual interest rate
11			For CDC Inquiry and Purchase Financial Transaction Request/0200 and Response/0210 messages		
			Subfield	**Attribute**	**Value**
	1-3		Subfield tag	n-3	**011**
	4-6		Subfield length	n-3	**029** Length of the subfield
	7-18		Subfield data	n-12	Estimated installment amount for YY installment
	19-30		Subfield data	n-12	Total amount of transaction with interest
	31-35		Subfield data	n-5	Monthly interest rate xxx.xx

Subelement	Position	Attribute	Value
12			For CDC **Inquiry** Financial Transaction Request Response/0210 messages. Items below are optional at the issuer's discretion.
			Subfield / **Attribute** / **Value**
	1-3		Subfield tag / n-3 / **012**
	4-6		Subfield length / n-3 / **002** Length of the subfield
	7-8		Subfield data / n-2 / Value = nn nn = number of additional installment options the issuer is offering (detailed in subtags 13 through 17, if applicable)
12			For CDC **Purchase** Financial Transaction Request Response/0210 messages.
			Subfield / **Attribute** / **Value**
	1-3		Subfield tag / n-3 / **012** This subtag is optional
	4-6		Subfield length / n-3 / **197** Length of the subfield
	7-11		Subfield data / n-5 2 decimal positions / Fee for CDC transaction
	12-15		Subfield data / n-4 2 decimal positions / Percentage penalty if not paid on specific date
	16-19		Subfield data / n-4 2 decimal positions / Percentage of interest on deferred payment
	20-203		Subfield data / an-184 / Free text available to issuer to print marketing or agreement data messages to appear on terminal receipt

Subelement	Position	Attribute	Value		
13-17			For CDC **Inquiry** Financial Transaction Request Response/0210 messages.		
			Subfield	**Attribute**	**Value**
	1-3		Subfield tag	n-3	**013**
	4-6		Subfield length	n-3	**033** Length of the subfield
	7-10		Subfield data	an-4	Value = XXYY XX = 20 installment with interest YY = option 2 number of installments
	11-22		Subfield data	n-12 2 decimal positions	Estimated installment amount for YY installment
	23-34		Subfield data	n-12 2 decimal positions	Total amount of transaction with interest
	35-39		Subfield data	n-5 2 decimal positions	Monthly interest rate xxx.xx

DE 113–DE 119—Reserved for National Use

The National Standards Organization uses and defines these data elements.

Note **The MasterCard® Debit Switch (MDS) does not use these data elements.**

Attribute

ans...999; LLLVAR

DE 120—Record Data

Record Data (DE 120) is a free-format variable-length data field used for transmitting file record data or textual character string data in various message types.

Attribute

ans...999; LLLVAR

Usage

When used in Administrative Advice/0620 messages having an Advice Reason Code set to “600” (Invalid message; rejected by Network), this data element contains the original (rejected) message.

When used in the File Update Request/0302 message, DE 120 contains the new, actual file record data used in “add” or “change” file-update actions. Table 4.40 illustrates the subelement structure of DE 120 in the 0302 message for each file that the 0302 message updates.

Table 4.40—Message Type 0302 Data Element 120 Structure

Subelement	Position	Attribute	Description
Account File MCC102 DE 120 [a]			
PAN	1-19	n-19	Primary account number to be updated by this request.
Issuer ICA	20-24	n-5	MasterCard assigned member ID.
Entry Reason	25	an-1	Must be one of the following codes: P Capture card S Stolen card O Other V VIP C Credit L Lost X Counterfeit F Fraud G Gold (only for a BIN with a product code of MCC) U Unauthorized use
Date Last Update	26-31	mmddyy	Returned in inquiry, ignored on add, update, and delete.

Subelement	Position	Attribute	Description
Time Last Update	32-35	hhmm	Retained in inquiry, ignored on add, update, and delete.
PIN Length	36, 37	n-2	Always has a value of “00”.
VIP Limit	38-49	n-12	Amount, whole dollars. Zero except for Entry Reason = V.
VIP Currency Code	50-52	n-3	Currency code for VIP, only valid for Entry Reason = V.
Account Management File MCC103 DE 120 [b]			
PAN	1-19	n-19	Primary account number to be updated by this request.
Issuer ICA	20-24	n-5	MasterCard assigned member ID.
Card Program	25-27	an-3	Must be one of the following codes: **MCB** MasterCard Corporate card **MCC** Mixed BIN **MCD** Debit MasterCard **MCF** MasterCard Corporate Fleet card **MCG** Gold MasterCard Corporate Purchasing card **MCP** MasterCard Corporate Purchasing card **MCS** MasterCard Standard card **MCW** World MasterCard card **MNS** Non-standard **MPL** Platinum MasterCard card **OTH** Other
Response Code	28, 29	n-2	Value is “04”, capture card.
Entry Reason	30	an-1	Must be one of the following codes: **C** Credit **X** Counterfeit **O** Other **F** Fraud
Filler	31-56	an-25	Reserved for future AMS enhancements.
Regional Information			Nonpositional, may occur up to six times in ascending order.

Subelement	Position	Attribute	Description
Indicator	57	an-1	Valid regions are: **1** United States **A** Canada **B** Caribbean, Central America, Mexico and South America **C** Asia/Pacific **D** Europe **E** Middle East/Africa
Purge Date	58-63	yymmdd	The member-requested purge date.
MDS Stand-In Negative File MCCNEG DE 120 [a]			
PAN	1-19	n-19	Primary account number to be updated by this request.
Issuer ICA	20-24	n-5	MasterCard assigned member ID.
Capture Code	25	an-1	Y(es) or N(o), indicates whether to capture the card.
Entry Reason	26	an-1	Must be one of the following codes: **P** Capture card **S** Stolen card **L** Lost **X** Counterfeit **F** Fraud **U** Unauthorized use
Purge Date	27-32	yymmdd	The member-requested purge date. If not included, the MDS calculates a purge date 180 days from the date of the account listing.
MDS Stand-In VIP File MCCVIP DE 120			
Record Data	1-3	n-3	Length of DE 120
PAN	4-22	n-19	Primary account number to be updated via this request.
Issuer ICA	23-27	n-5	MasterCard assigned member id
Entry Reason	28	an-1	Must be one of the following codes: **P** Capture card **S** Stolen card **O** Other **L** Lost **X** Counterfeit **F** Fraud **U** Unauthorized use
Purge Date	29-34	yymmdd	The member requested purge date

Subelement	Position	Attribute	Description
Total Usage Count	35-38	n-4	The total number of ATM and POS debits
Total Usage Amount	39-46	n-8	The total amount of ATM and POS debits
ATM Usage Total Count	47-50	n-4	The total number of ATM debits from all related accounts
ATM Usage Total Amount	51-58	n-8	The total amount of ATM debits from all related accounts
ATM Usage Savings Count	59-62	n-4	The total number of ATM debits from savings
ATM Usage Savings Amount	63-70	n-8	The total amount of ATM debits from savings
ATM Usage DDA Count	71-74	n-4	The total number of ATM debits from checking
ATM Usage DDA Amount	75-82	n-8	The total amount of ATM debits from checking
ATM Usage Credit Card Count	83-86	n-4	The total number of ATM debits from credit card
ATM Usage Credit Card Amount	87-94	n-8	The total amount of ATM debits from credit card
ATM Usage NAS Count	95-98	n-4	The total number of ATM debits from no account specified
ATM Usage NAS Amount	99-106	n-8	The total amount of ATM debits from no account specified
POS Usage Total Count	107-110	n-4	The total number of POS debits
POS Usage Total Amount	111-118	n-8	The total amount of POS debits

[a] Only PAN and Issuer ICA are required for delete or inquiry requests.

[b] Only PAN and Issuer ICA are required for delete requests.

For MCC102 adds, changes, and deletes, the File Update Request Response/0312 echoes back in Record Data (DE 120) the same information that was received in DE 120 of the File Update Request/0302. For MCC102 inquiries, the 0312 response contains in DE 120 the full record of the database that was requested.

For MCC103 adds, changes, and deletes, the File Update Request Response/0312 echoes back in Record Data (DE 120) the same information that was received in DE 120 of the File Update Request/0302. Corresponding to the contents of DE 120 in the MCC103 add/change 0312 message responses, DE 122 will contain the effective dates and purge dates for each region on file for the account.

For MCCNEG adds, changes, and deletes, the File Update Request Response/0312 echoes back in Record Data (DE 120) the same information that was received in DE 120 of the File Update Request/0302.

Note **If a BIN is flagged as debit MasterCard and is eligible for the MDS Stand-In processing, then the issuer will only need to send one Online File Update Message/03XX to update the MCC103 file on the Account Management System (AMS). This message will update the MCCNEG file on the MDS Stand-In system and the MCC103 file on the AMS.**

Alternatively, this data element is used in 0100 requests to contain billing address data for the MasterCard Address Verification Service (AVS). When the MDS receives an 0100 message with this data present, the MDS submits the data within DE 120 within an 0200 message to the issuing processor and the issuer must respond with the same data in DE 120 that it received from the MDS. The MDS then returns the DE 120 to the acquirer.

The AVS condensed format begins with the left-most position and uses up to five numeric values that appear before the first alphabetic character or space. Once AVS finds a space or an alphabetic character, it stops interrogating the cardholder billing address, and constructs the condensed AVS key.

For example: 223 NW 31st Street, Apartment #3, in this case the match is 223.

Table 4.41 describes the Record Data values for DE 120 for the AVS usage.

Table 4.41—DE 120 Subfields for AVS Usage

Subfield	Position	Attribute	Value
Subelement tag identifier	1–2	n-2	03
Length of subelement	3–4	n-2	14
Cardholder postal/zip code	5–13	an	Cardholder postal zip code (left justified, blank filled)

Subfield	Position	Attribute	Value
Cardholder address	14–18	an	Cardholder billing address (left justified, blank filled)

Note **To receive a complete match, the issuer must base its keys using the AVS condensed algorithm logic.**

DE 121—Authorizing Agent Identification Code

This data element identifies the actual processing facility that approved or denied a Transaction Request message.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

n...011; LLLVAR

Usage

When a Stand-In or alternate authorizer processing facility performs an authorization or a financial transaction on behalf of a card issuer, it must insert this data element into the response message and any advice message transmitted to the actual card issuer. This procedure ensures that a transaction audit trail clearly identifies the authorizing agent that actually approved the transaction.

DE 122—Additional Record Data

By provision of the ISO 8583–1987 specification, MasterCard redefined this data element for use as "Additional Record Data."

Attribute

ans...100; LLLVAR

Usage

Additional Record Data (DE 122) is a "free-format" variable length data element used for transmitting file record data in various message types. In a File Update Request Response/0312 message, this data element is available to pass data sent to the issuer by the AMS in response to a File Update Request/0302 inquiry.

Table 4.42 indicates the data returned in DE 122 for accepted File Update Request/0302 updates.

Table 4.42—Message Type 0312 Data Element 122 Structure

Subelement	Position	Attribute	Description
Regional data on file			Nonpositional, may occur up to six times in ascending order.
Indicator	1	an-1	Valid regions are: **1** United States **A** Canada **B** Caribbean, Central America, Mexico and South America **C** Asia/Pacific **D** Europe **E** Middle East/Africa
Effective Date	2-7	yymmdd	Effective date of the listing within this region.
Purge Date	8-13	yymmdd	Purge date of the regional listing.

DE 123–DE 125—Reserved for Future Use and Definition by MasterCard

These data elements are for future definition and use by private organizations.

Note **The MasterCard® Debit Switch (MDS) does not use these data elements.**

Attribute

ans...999; LLLVAR

DE 126— Switch Private Data

The MDS generates this information to facilitate its own message processing.

Note **The MasterCard® Debit Switch (MDS) does not currently use this data element.**

Attribute

ans...50; LLVAR.

Usage

For the initial release of 2003 (03.1), the MDS anticipates using this data element for placement of internal data to enhance message tracking and/or routing capabilities. It will be required for processors to accept and return this information when they receive it.

DE 127—Processor Private Data

The MDS reserves this data element for the proprietary use of Customer Processing Systems (CPS) that connect directly to the MDS.

Attribute

ans...50; LLVAR.

Usage

Any message originator (for example, any CPS or INF facility communicating directly with the MDS) may use this data element to contain private-use data up to a maximum length of 50 characters. Data placed in this field is not passed through to the message receiver, but is stored temporarily by the MDS and returned to the message originator in any subsequent response or acknowledgment message.

Typically, this data element is used by CPS or INF processors to contain online transaction matching or queuing data that can be accessed readily upon receipt of the corresponding response to any originating request or advice message.

Values

The MDS does not perform edits on this data field.

DE 128—Message Authentication Code (MAC)

Message Authentication Code (MAC) (DE 128) validates the source and the text of the message between the sender and the receiver.

The last bit position within any bit map is reserved for DE 128. If a member is using authentication on a message, the final bit of the final bit map of that message indicates the MAC information. The final bit of all preceding bit maps must contain "0." For example, there must be only one DE 128 per message and that DE 128 must be the last data element of the message.

Note **The MasterCard® Debit Switch (MDS) does not use this data element.**

Attribute

b-64

5 Communication Protocols

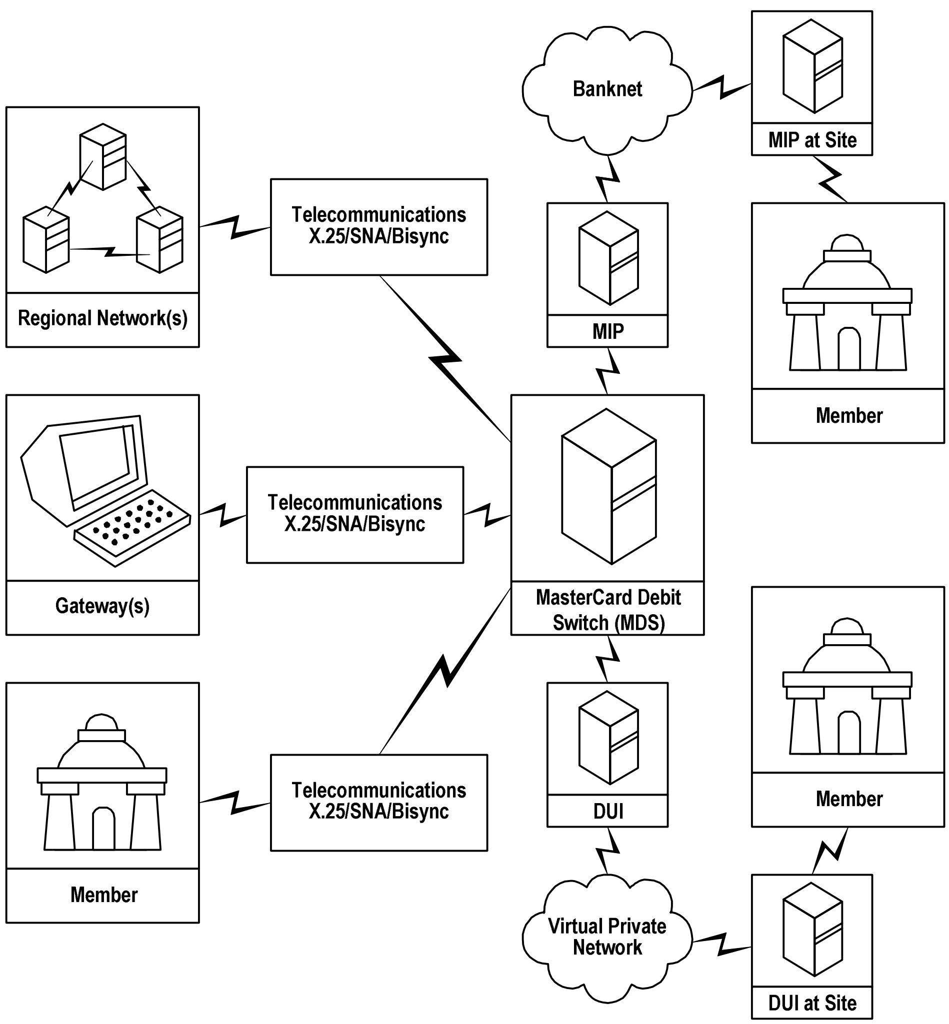
This chapter describes the linking methods that customer processing systems (CPSs) use to connect to the MDS.

Overview

All customer processing systems (CPSs) participating in services offered by the MDS must link to the MDS in one of three ways:

1. Connect through MDS Virtual Private Network (VPN)
2. Direct leased-line connection (this method will be phased out during 2003).
3. Connect through the Banknet® telecommunications network. In this case, the CPS is connected to a MIP (MasterCard interface processor) operated by MasterCard from the Banknet Operations Center in St. Louis, Missouri, USA.

Figure 5.1—MasterCard® Debit Switch Communications Network



MIP (Banknet) Connect to MDS

MasterCard transmits data through the Banknet telecommunications network. The Banknet network uses a peer-to-peer architecture and a mesh configuration. A peer-to-peer architecture allows data to follow the most efficient route to its destination and flows through distributed intelligence. Therefore, multiple locations are available for processing information, even during peak times. Another important feature of the architecture of the Banknet network is its on-demand dial back-up equipment.

The Banknet network uses X.25 or packet protocol. X.25 is a global standard for data networks with universal connectivity to 130 public data networks worldwide.

MasterCard Interface Processor (MIP)

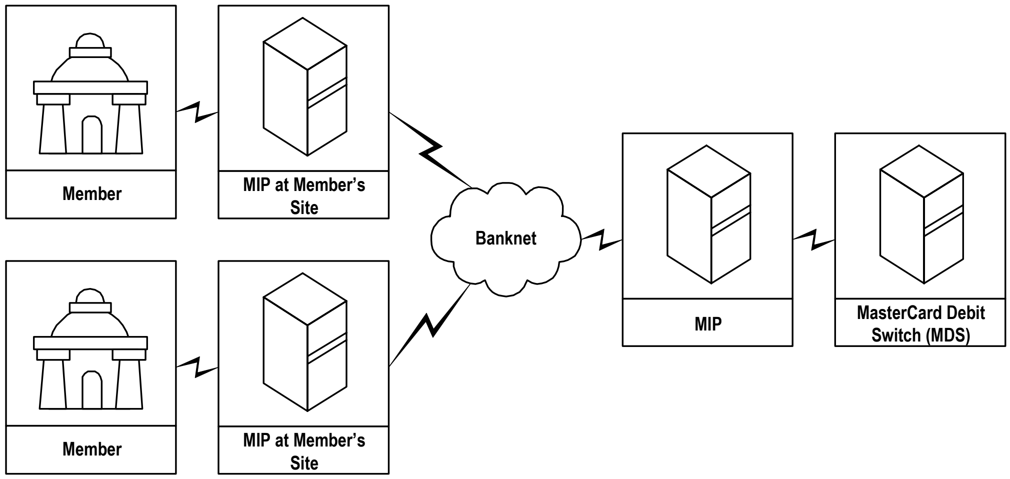
The MIP provides the hardware and software at the member site that allows the member host system to interface with the Banknet network in a standard, efficient manner. The MIP also permits distributed data processing functions to support the financial service applications of MasterCard.

Members may already have a MIP on-site or may share a MIP with other financial institutions in the area. To accommodate Maestro and Cirrus processing, MasterCard will upgrade the MIP, and route the online real time financial transactions between the member and the MDS.

MasterCard needs to know the member's communication protocol required for online debit processing. The precise configuration at a given member site is engineered jointly by the member and the MasterCard Network Engineering Group. Historical or anticipated transaction volume is the basis for the MIP configuration.

Figure 5.2 shows the member host system's connection to the MDS through the Banknet network.

Figure 5.2—Connecting to the MasterCard® Debit Switch



Refer to the *Banknet Systems Manual* for additional descriptive technical information on the Banknet network and MIP options.

Note **If a member selects the Banknet/MIP connectivity option, they must contact their MasterCard Regional Office and/or Debit Payment Systems—Implementation Support Manager for more detailed requirements.**

Virtual Private Network

This communications solution, MasterCard's Virtual Private Network (VPN), is available for North American-based debit processors.

The VPN consists of a Frame Relay service infrastructure that uses Transport Control Protocol/Internet Protocol (TCP/IP). In order to access the network, each member site will be equipped (at a minimum) by MasterCard with a Service Delivery Point (SDP) that consists of a Debit Interface Unit (DIU), dual state-of–the-art Cisco routers and dual Ethernet hubs.

Online Transaction Communications

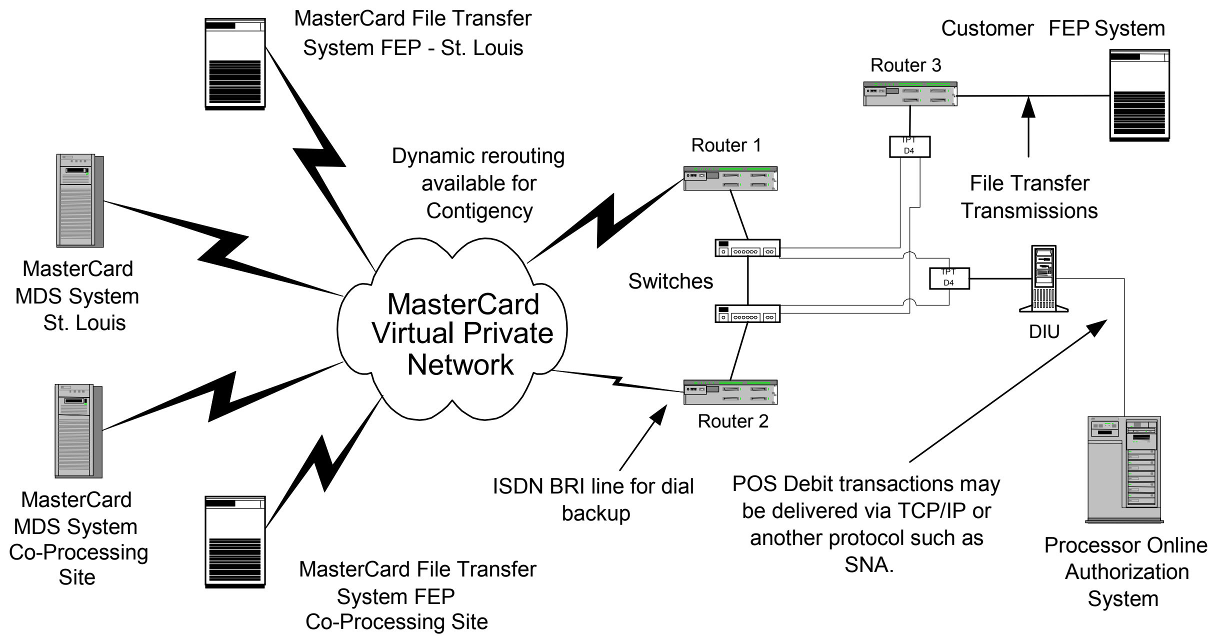
For online transaction communications, customers can choose to continue to use their current data communications protocol rather than convert applications to TCP/IP. The Debit Interface Unit (DIU) will convert the Bisync or SNA protocol to TCP/IP for transport through the network. New processors connecting to the MDS network should use either TCP/IP or SNA protocols.

Batch File Transmission

Members using batch file transmissions (Complex to Complex–CTC) in conjunction with VPN can choose between SNA Token Ring or TCP/IP protocol.

In the event that a problem occurs with the VPN, the ISDN service provides backup. The routers immediately initiate the ISDN service upon detecting a problem.

Figure 5.3—Typical VPN Configuration



VPN Infrastructure

There are two primary VPN components that are used for both Online and File Transfer communications: frame relay technology and the Service Delivery Point (SDP).

Frame Relay

Frame relay is a high-speed communications technology for sending information over a wide area network (WAN) that divides the information into frames, or packets. Each frame has an address that the network uses to determine the destination of the frame. The frames travel through a series of switches within the frame relay network and arrive at their destination. Frame relay supports the MasterCard Virtual Private Network, and provides the foundation for connectivity for all MasterCard members.

The frame relay network is not a single physical connection between one endpoint and the other. Instead, a logical path, or virtual circuit, is defined within the network. This logical path is called a virtual circuit. The routers do not allocate bandwidth to the path until the transmission of the actual data. At that time, the routers allocate bandwidth within the network on a packet-by-packet basis.

MasterCard will typically have a single Data Link Channel Identifier (DLCI) that links from a member location to the nearest Frame Relay switch. Across this DLCI, there will be two Permanent Virtual Circuits (PVCs), one connects to MasterCard's St. Louis Global Hub and the other to MasterCard's New York Global Hub.

Service Interruption

If there is an interruption in the frame relay network there will be little, if any, service interruption unless both PVCs are impacted. If the customer access circuit (DLCI) is impacted by a Local Exchange Carrier (LEC) outage, both PVCs will be impacted and the processor connectivity will be restored through Integrated Services Digital Network (ISDN).

Service Delivery Points (SDP)

A typical MasterCard VPN Service Delivery Point (SDP), that encompasses the MasterCard equipment placed at a customer site, includes:

- MasterCard equipment cabinet
- Debit Interface Unit (DIU)

- Drop Ethernet Splitter
- Two Ethernet Switches
- Two Cisco 26XX series routers
- 8PK-CAS switch & analog modem
- Frame Relay Circuit
- ISDN Circuit
- A third Cisco 26xx router for Data Transfer via IP

MasterCard has worked very closely with AT&T to develop a highly reliable, highly redundant SDP (with regards to the communications equipment.) The SDP has automatic failover if any of the following components become inoperable:

- Primary Cisco Router
- Primary Ethernet Switch
- Frame Relay Circuit

Hot-Standby Routing Protocol (HSRP)

MasterCard VPN routers use Enhanced Internet Gateway Routing Protocol (EIGRP) to distribute updated network routing information between routers. EIGRP allows MasterCard to take advantage of a feature called Hot-Standby Routing Protocol (HSRP).

This feature equips the SDP with the ability to restore service automatically using ISDN in case of a failure of one or more of the above listed components. With HSRP, the two Cisco routers at the customer site are in constant communication with each other so that the secondary router will know if it is necessary to establish an ISDN connection to restore service.

If the primary router senses an outage on the Frame Relay circuit (impacting both PVCs) it will notify the secondary router to "take over." At that point, the secondary router will initiate an ISDN call to either St. Louis or New York to reestablished connectivity to the VPN. In the event of a primary router failure or primary Ethernet hub failure, communication between the primary and secondary routers is lost and the secondary router will restore the site on ISDN.

The Debit Interface Unit (DIU) connects to the Ethernet hubs via a dual output Ethernet transceiver. The transceiver takes the single Ethernet output from the DIU and splits it, connecting to both Ethernet hubs. With this design, a single Ethernet hub failure will not isolate an SDP from the VPN.

The 8PK-CAS and analog modem are strictly used for out-of-band access into the routers and hubs for troubleshooting purposes in the event that in-band management becomes inoperable.

The SDP can also be configured to support direct connect data transfer customers using either TCP/IP or Token Ring.

- **TCP/IP**: To support data transfer via TCP/IP MasterCard will have a third Cisco 26xx router installed as part of the SDP. This router will connect to the Ethernet hubs via a transceiver for redundancy. The primary purpose of this router is to provide a firewall between a customer's LAN and MasterCard's VPN. This router also performs Network Address Translation (NAT) functionality converting customer host IP addresses to MasterCard recognized IP addresses for transport across the VPN.
- **Token Ring**: To support data transfer via Token Ring MasterCard will utilize Cisco 26xx routers as the primary and secondary routers at the customer site SDP. These routers, via their built in Token Ring cards, will be connected to a customer's Token Ring through either MasterCard-provided Multistation Access Units (MAUs) or customer provided MAUs. The customer's Token Ring network is connected to MasterCard's Mainframe via DLSw+ switching.

Online Communication Using VPN

A Debit Interface Unit (DIU) facilitates online communications between a member and the MasterCard® Debit Switch (MDS). The DIU is a Netfinity computer connected to the SDP at the customer site. It allows a customer to send debit transactions to MasterCard across the VPN.

The DIU supports legacy protocols (SNA and Bisync) as well as IP. The customer will connect their host equipment to the DIU via a serial connection (for SNA or Bisync) to a DB25 (RS232 or V.35) interface on the DIU's multi-interface card. A DIU will typically have one or two of these serial interface cards. Each card can support up to eight serial connections ranging in speeds from 4800bps to 56,000bps.

If the member's preferred method of connectivity is IP, their host system will connect to a secondary Ethernet card in the DIU (the network interface Ethernet card is commonly referred to as the **primary** Ethernet card). The DIU receives transactions from the customer, performs certain logging and statistical functions (i.e., transaction and program counters), and encapsulates the transactions into TCP/IP format for transport across the VPN.

The MasterCard Network Command Center (NCC) has remote access to each DIU for troubleshooting purposes. Using this remote console access, the NCC is able to perform various monitoring and troubleshooting functions, which assist in problem determination and resolution.

Note **MasterCard does not support new member installations using the bisync protocol.**

Effective 1 July 2005, MasterCard will no longer support the bisync protocol. All existing members must complete migration to TCP/IP or SNA over token ring.

File Transfer Using VPN

There are two options available that use the VPN for support of settlement detail and report file transmissions:

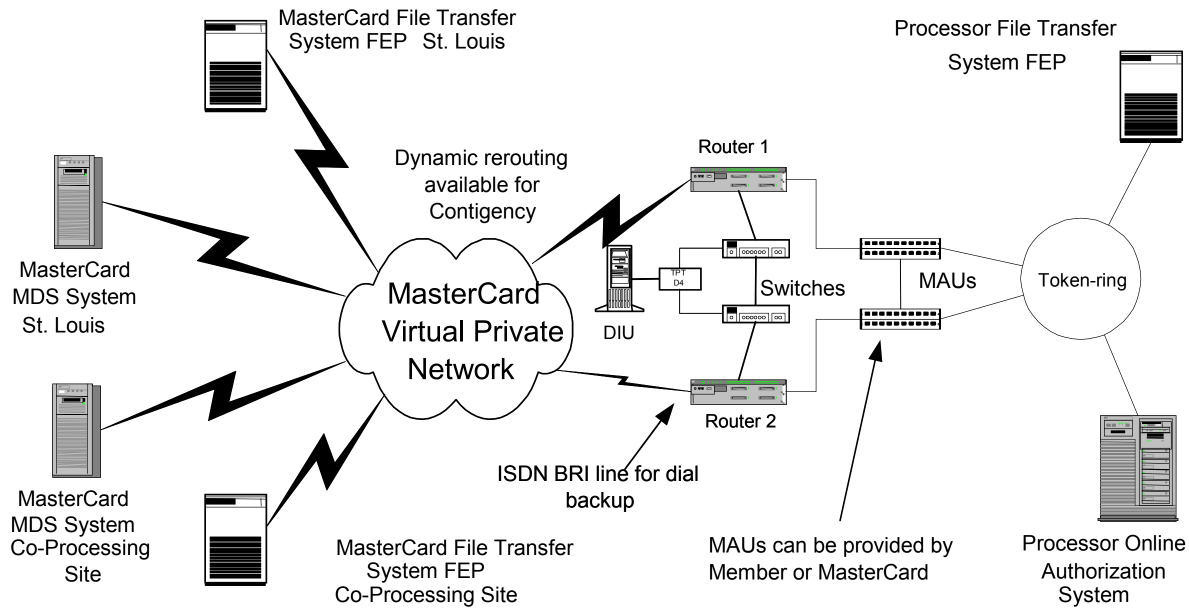
- Token Ring
- TCP/IP

The following subsections describe both options.

VPN File Transfer Using Token Ring

Figure 5.4 illustrates a VPN File Transfer configuration using a Token Ring network. VPN uses a sub-area-only Token Ring Interface Card (TIC) on the processor's Front End Processor (processor provided). MasterCard provides the Multistation Access Units (MAUs) that interface to the Cisco routers. If the processor prefers, they may provide their own MAUs.

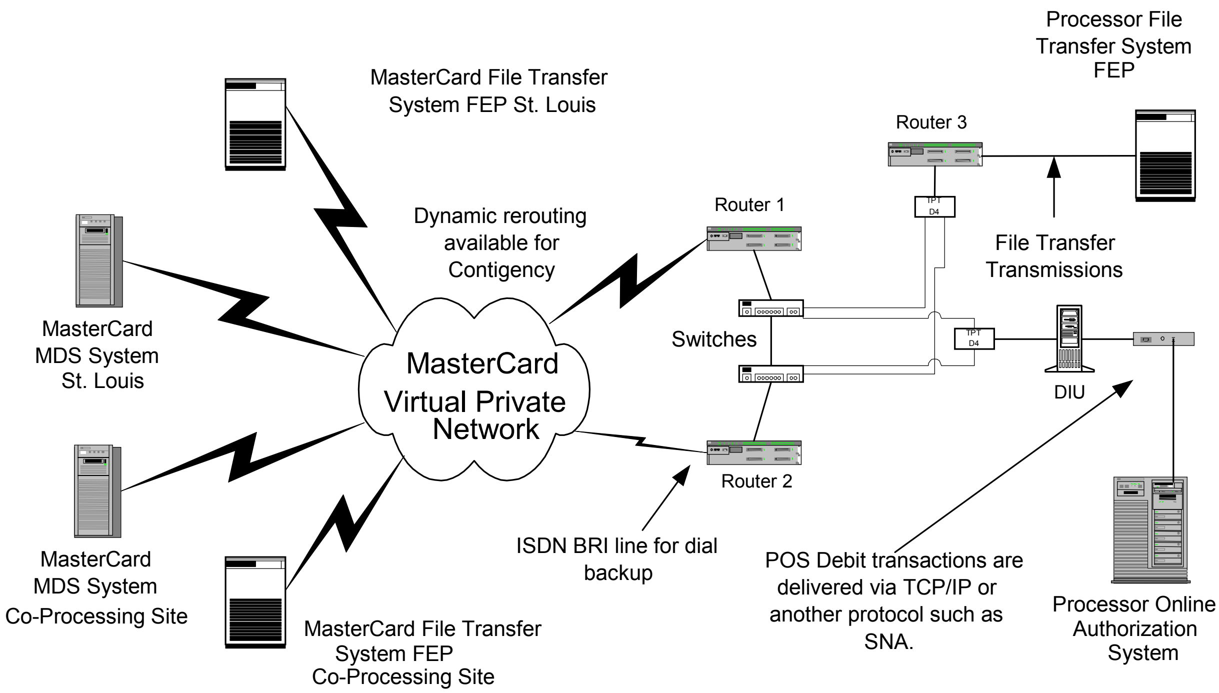
Figure 5.4—VPN Token Ring File Transfer Configuration



VPN File Transfer Using TCP/IP

MasterCard also offers File Transfer support using SDP integrated TCP/IP. File Transfer traffic will be on a dedicated segment by providing a second Ethernet connection on router 1 & 2.

Figure 5.5—Typical TCP/IP Configuration



Direct Connect to MDS

All CPS processors that connect directly to the MDS have the option of using bisynchronous point-to-point connection or SNA leased-line(s). Either the EBCDIC or the ASCII character set is available for any processor.

MDS Connectivity

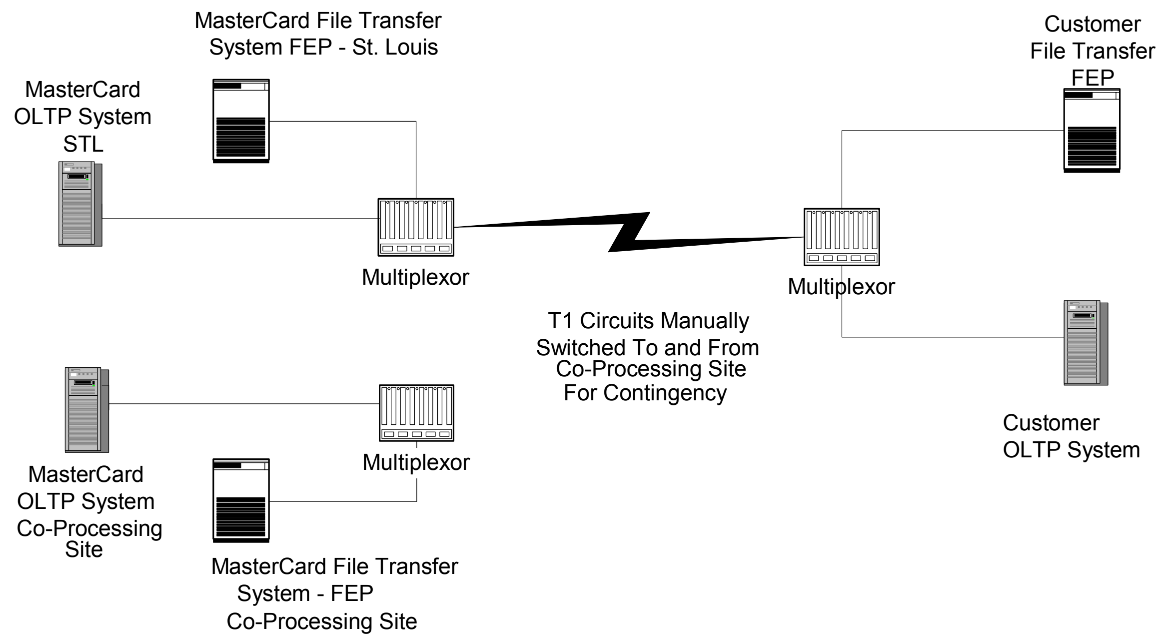
Members use high speed 56 Kbps line sets for connectivity between member and MasterCard systems for debit online authorization traffic, and for file transfer of batch settlement and reporting files.

The diagram below illustrates the typical telecommunications configuration. Following is a list of abbreviations for all diagrams in this chapter:

- **FEP**—Front-end processor
- **OLTP**—Online transaction processor
- **LKS**—Lake Success, NY
- **DIU**—Debit interface unit
- **SDP**—Service delivery point
- **MAU**—Multistation access unit

Note **Connection to the MDS through a leased line is no longer an option for processors joining the MDS network.**

Figure 5.6—Direct Connect Communication Configuration



Bisynchronous Point-to-Point Protocol Support

Bisync point-to-point contention protocol makes it possible for an application process to act as a station in a binary synchronous point-to-point contention network. Point-to-point contention bisync lines operate in a mode where neither station has total control of the line. A station that sends data normally obtains control of the line by issuing a line bid and by receiving a positive response from the other station.

The bisynchronous protocol software subsystem employed by the MDS processor includes the following capabilities:

- Support for MDS or CIS (ISO 8583-1987) message formats.
- Automatic character-check generation and verification, performed by the synchronous controller hardware.
- Automatic retry of various communication I/O operations.
- Protocol timing in accordance with established (IBM) protocol standards.

Note **MasterCard does not support new member installations using the bisync protocol.**

Effective 1 July 2005, MasterCard will no longer support the bisync protocol. All existing members must complete migration to TCP/IP or SNA over token ring.

Data Communication Lines and Modems

Backup lines are available at the MDS in the event of leased line failure. All DCE used at the MDS facility for backup functions make the backup lines appear as leased lines to each processor.

Dial-up modems that use full duplex bisynchronous point-to-point contention protocols are also available for ongoing testing with the MDS Test and Certification Subsystem (such as test network). The member's host system should generate all CPS processor lines with the CPS as primary.

The MDS Test and Certification Subsystem is currently configured using Bell 208B or compatible modems which can be used for both batch transmission and interactive online testing.

When ordering Bell 208B modems, specify full duplex 4800 bps.

The point-to-point bisync contention protocol software used by the MDS conforms to the IBM Binary Synchronous Communications Standard as documented in *IBM General Information Manual GA27-3004-2*, the following exceptions exists.

- The (WACK) control character in response to a text transmission forces the MDS to resolicit (ENQ) the last transmission. The MDS system will do this for a preset number of times and then report an error if it did not receive an (ACK).
- The MDS does not support ETBs.

Bisync Message Formats

The MDS allows the following general types of message formats and classifies them as **received** messages and **transmitted** messages.

Received Messages

Normal text message format:

Message:	[STX]- - - - text - - - -[ETX]
Message Length:	2 + text length.

Transmitted Messages

Normal text message format:

Message:	[STX] - - - - text - - - - [ETX]
Message Length:	2 + text length.

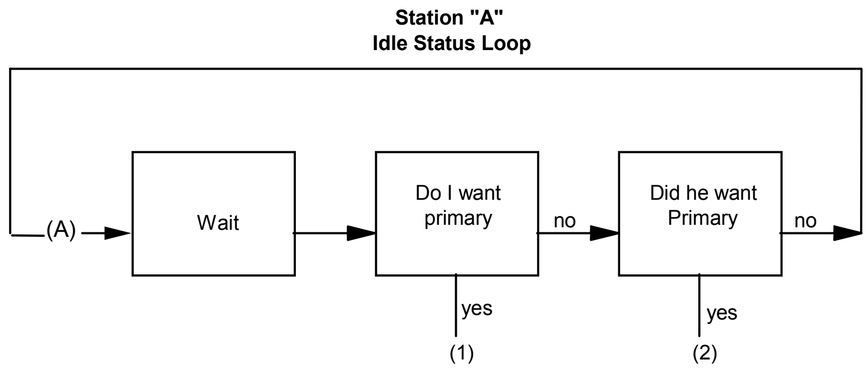
The following figures detail the protocol logic diagrams for the transmission and reception of electronic file transfer (EFT) messages using the Binary Synchronous protocol. In the diagrams, standard bisync terminology and character symbols are used. Table 5.1 defines these standard terms and symbols.

Table 5.1—Data Communication Control Characters

Name	Symbol	Function
Start of heading	SOH	Resets control mode. BCC accumulation starts with the first character after the first SOH or STX character is transmitted/received.
Start of text	STX	Resets control mode. BCC accumulation starts with the first character after the first SOH or STX character is transmitted/received.
End of transmission block	ETB	Resets text mode with BCC comparison.
End of text	ETX	Resets text mode with BCC comparison.
End of transmission	EOT	End of transmission.
Enquiry	ENQ	Reset text mode without BCC transmission and comparison. ENQ received in response to a transmission means a solicitation of the last transmission.
Negative acknowledge	NAK	Negative response to a request for a reply, or to a block of heading or a block of text in error.
Synchronous idle	SYN	Transmitted automatically to establish and maintain synchronization.
Data link escape	DLE	DLE is always the first byte in the following two-byte control characters: ACK 0, ACK 1, WACK, and RVI. The MDS uses DLE as the first character in several two-byte sequences in transparent mode.
Intermediate text block (ITB)	IUS	Included in the BCC; it causes the BCC to be sent.
Even acknowledge (two-bytes)	DLE 0	Indicates affirmative acknowledgment of even blocks.
Odd acknowledge (two-bytes)	DLE 1	Indicates affirmative acknowledgment of odd blocks.
Wait before transmit positive acknowledge (two-bytes)	DLE ;	Indicates a temporary not ready to continue condition.
Mandatory disconnect (two-bytes)	DLE EOT	Used only on switched communication to initiate a disconnect.
Reverse interrupt (two-bytes)	DLE @	Reverse direction on data transfer.
Temporary text delay (two-bytes)	STX ENQ	Alerts the receiving station of a temporary text delay.

Name	Symbol	Function
Transparent start of text (two-bytes)	DLE STX	Turns off control mode and set the adapter to transparent text mode.
Transparent intermediate text block (ITB) (two-bytes)	DLE IUS	Same as ITB but turns off transparent text mode.
Transparent end of text (two-bytes)	DLE ETX	Same as ETX but turns off transparent text mode.
Transparent end of text block (two-bytes)	DLE ETB	Same as ETB but turns off transparent text mode.
Transparent block cancel (two-bytes)	DLE ENQ	Turn off transparent text mode and cancel the current block of data.
Transparent temporary text delay (four-bytes)	DLE STX DLE ENQ	Alerts the receiving station to a temporary text delay in transparent text mode
Transparent data link escape (two-bytes)	DLE DLE	In transparent text mode, the transmitter adds a second DLE after each data DLE. At the receiver removes the first DLE and does not enter storage of the BCC.

Figure 5.7—Bisync Point-to-Point Contention Idle Status Loop



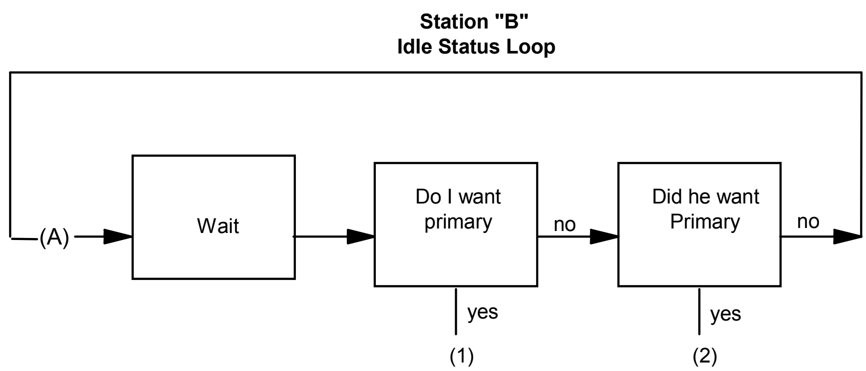


Figure 5.8—Bisync Point-To-Point Contention Diagram

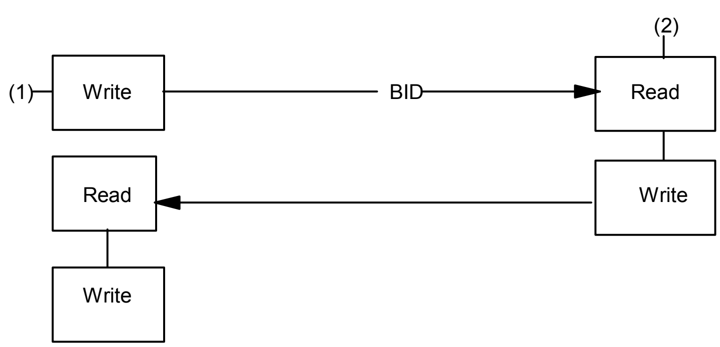


Figure 5.9—Initial Read (Wait For Bid) Error Handling

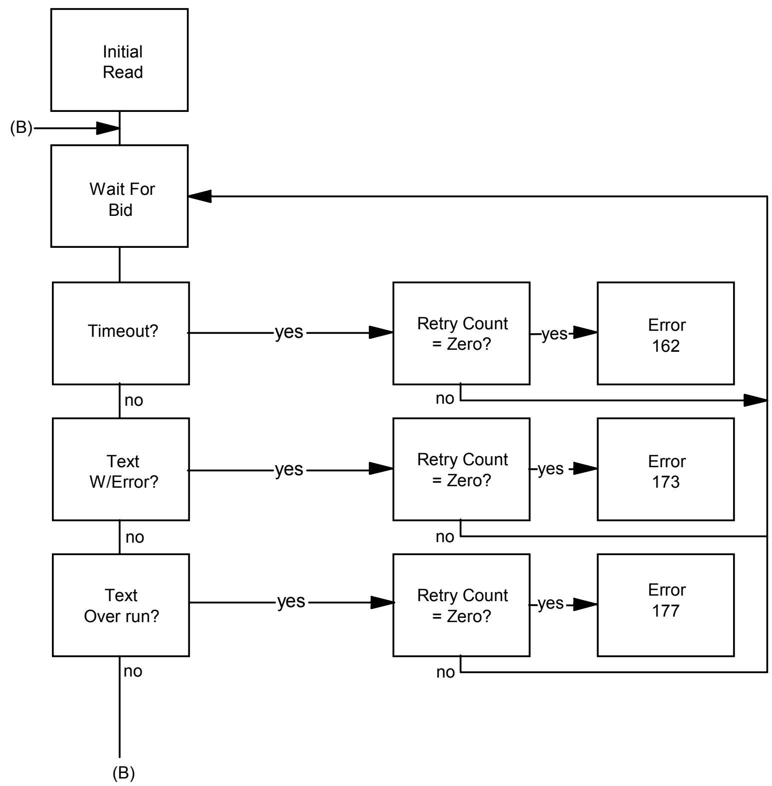


Figure 5.10—Initial Read (Wait for Bid) Error Handling

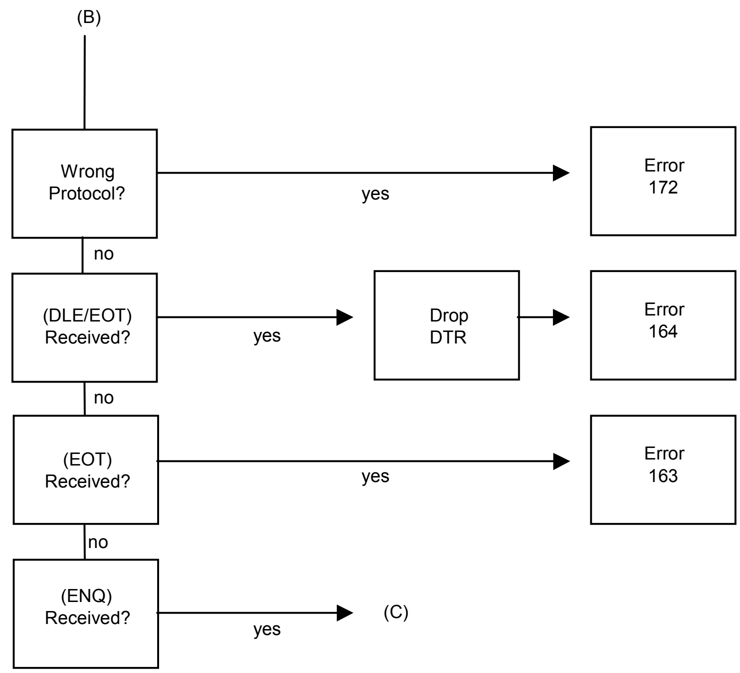


Figure 5.11—Continue Read (ACK Message) Error Handling

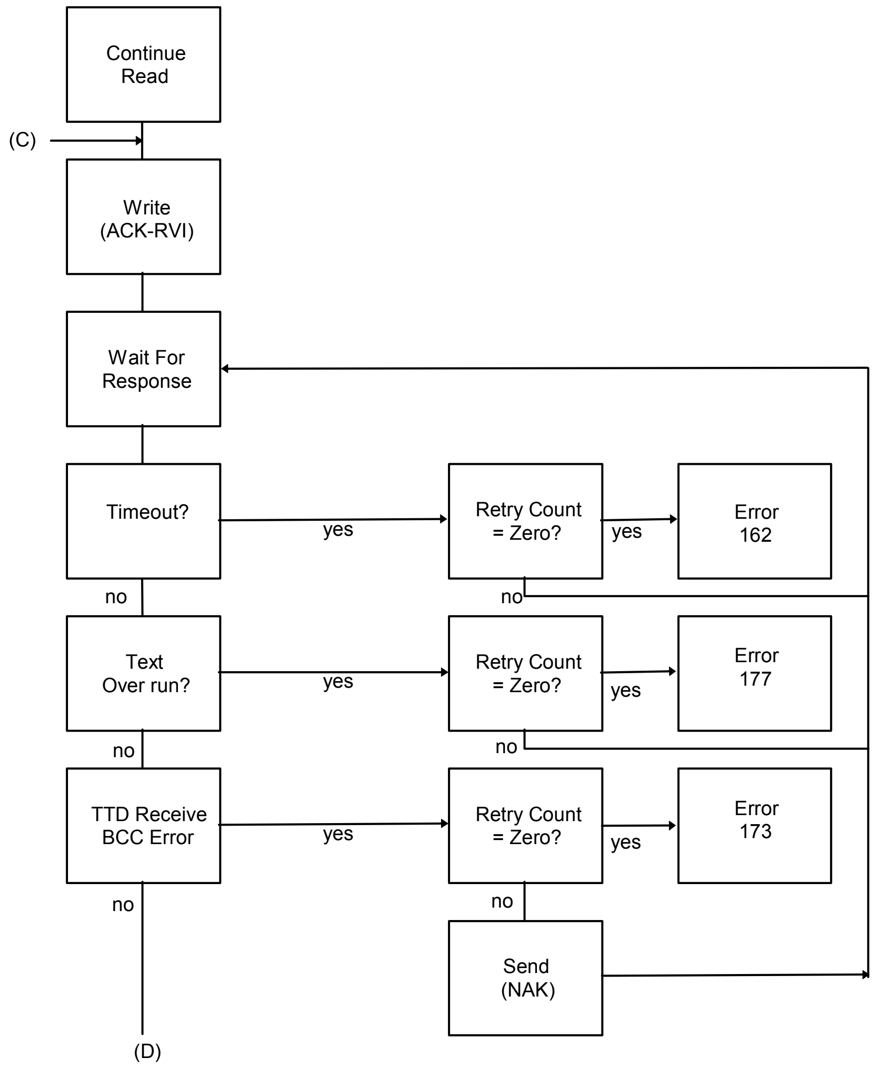


Figure 5.12—Continue Read (ACK Message) Error Handling

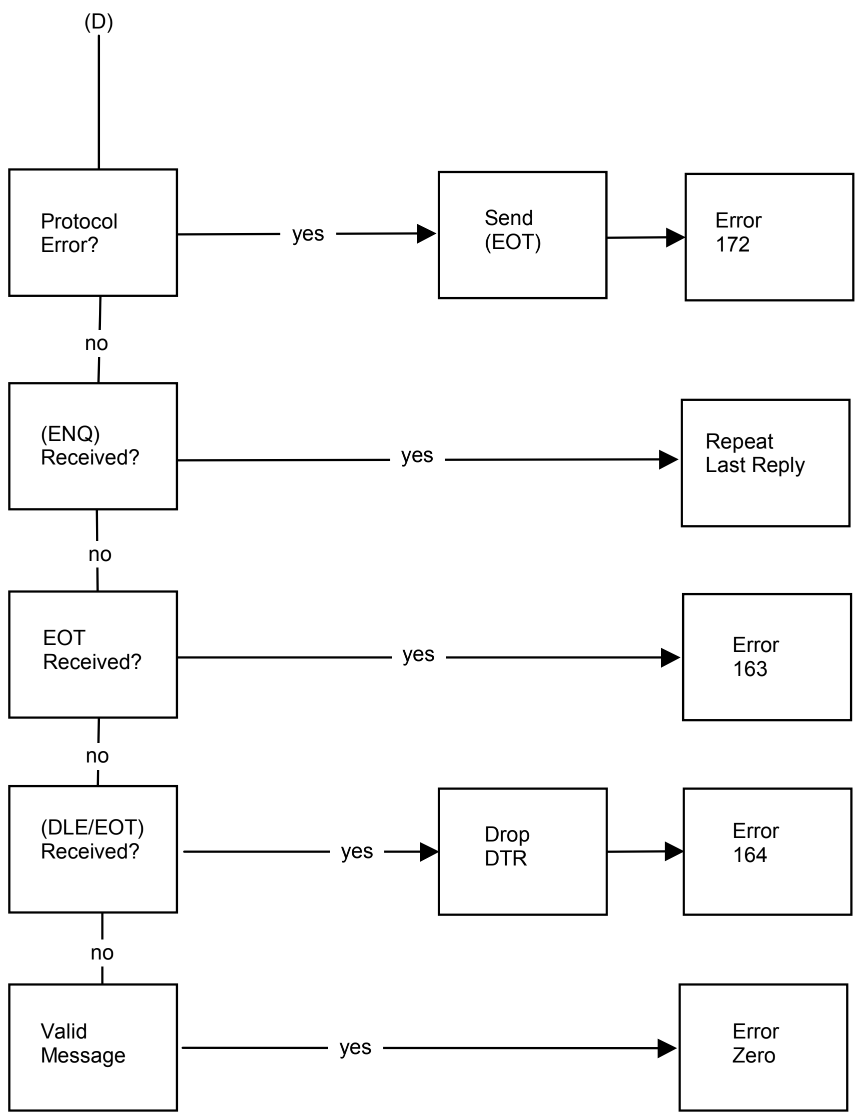


Figure 5.13—Initial Write (Bid for Line) Error Handling

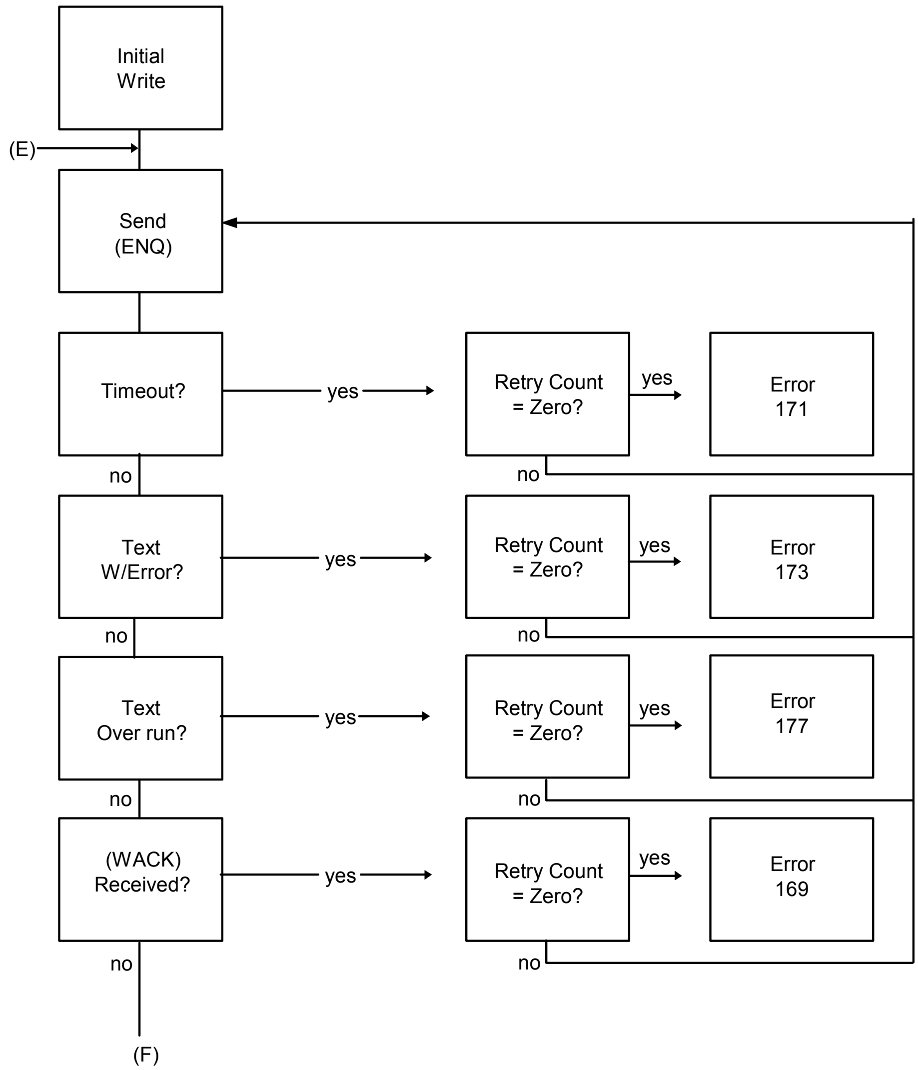


Figure 5.14—Initial Write (Bid for Line) Error Handling

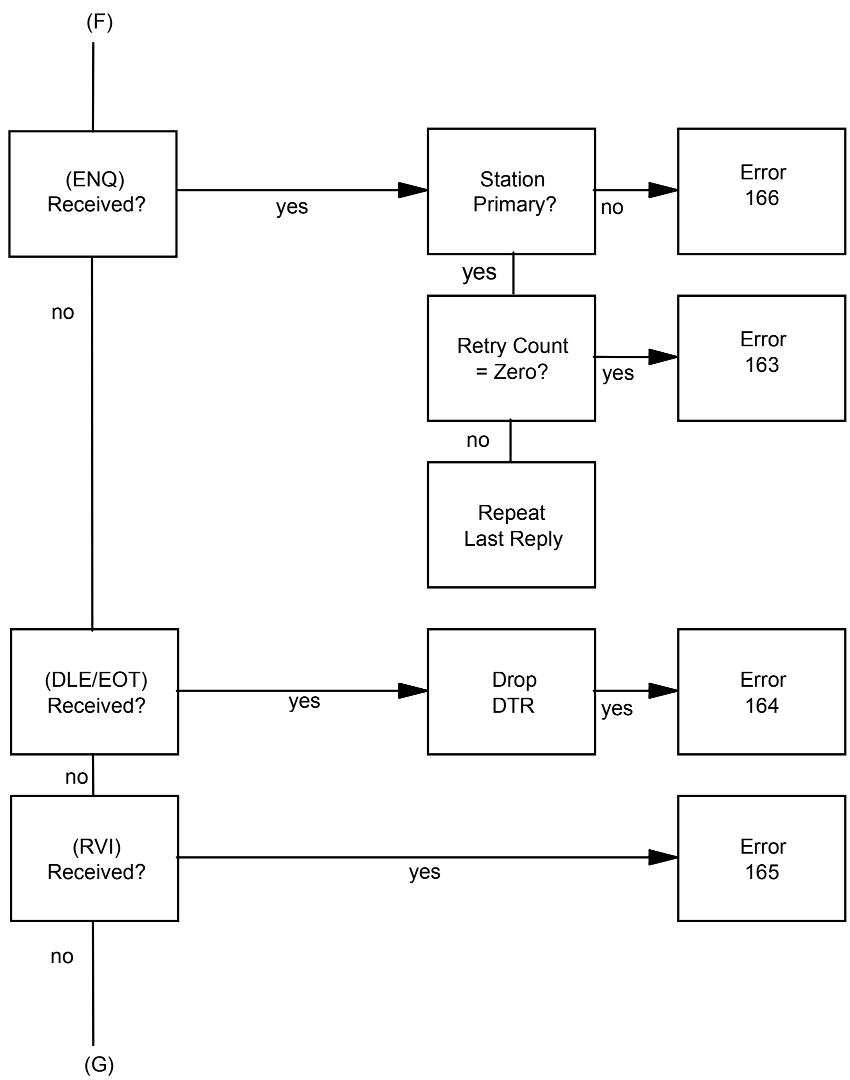


Figure 5.15—Initial Write (Bid for Line) Error Handling

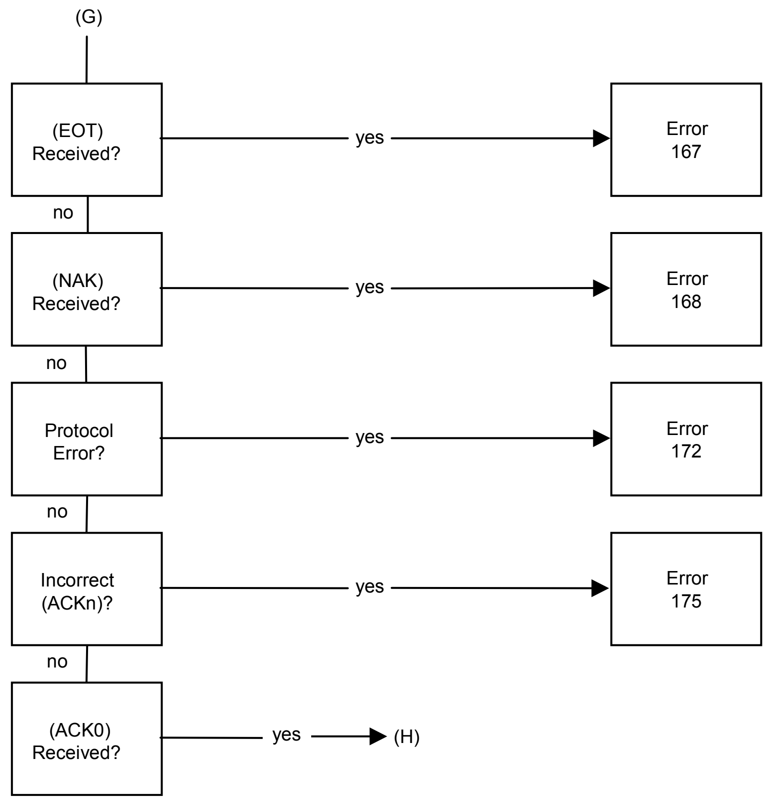


Figure 5.16—Continue Write (Write Message) Error Handling

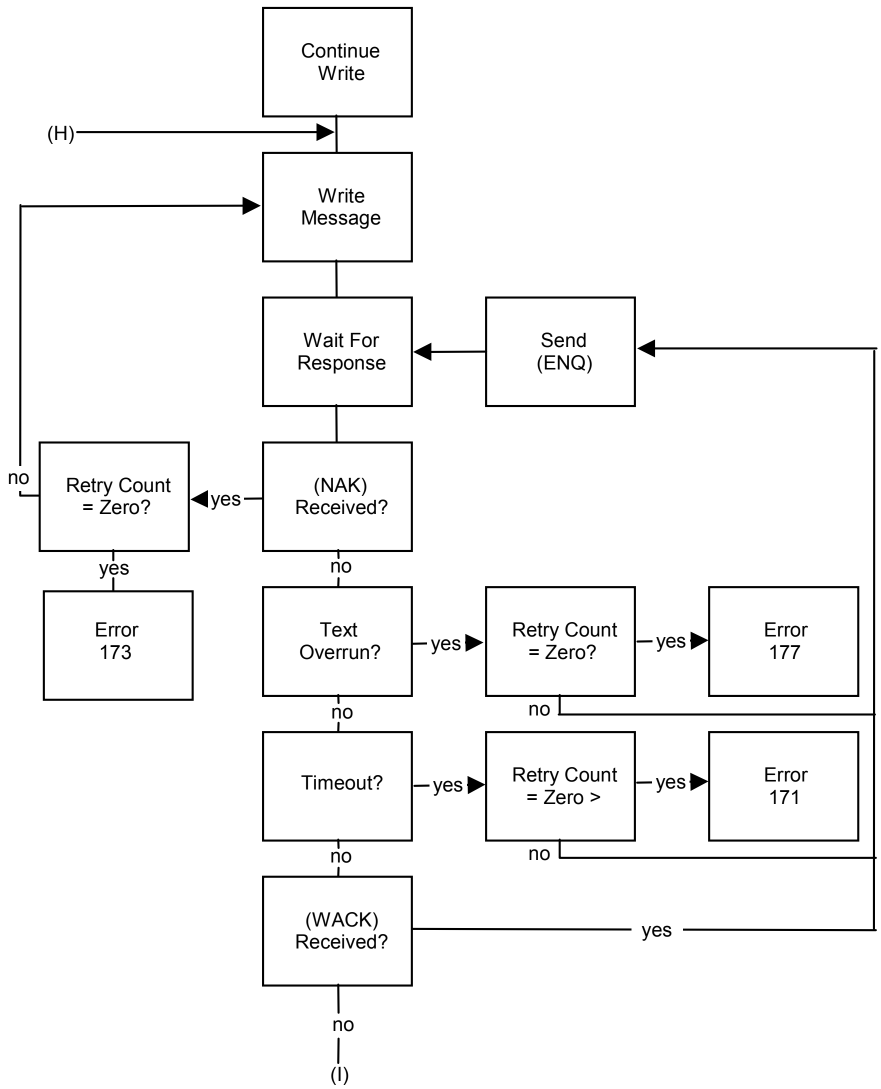
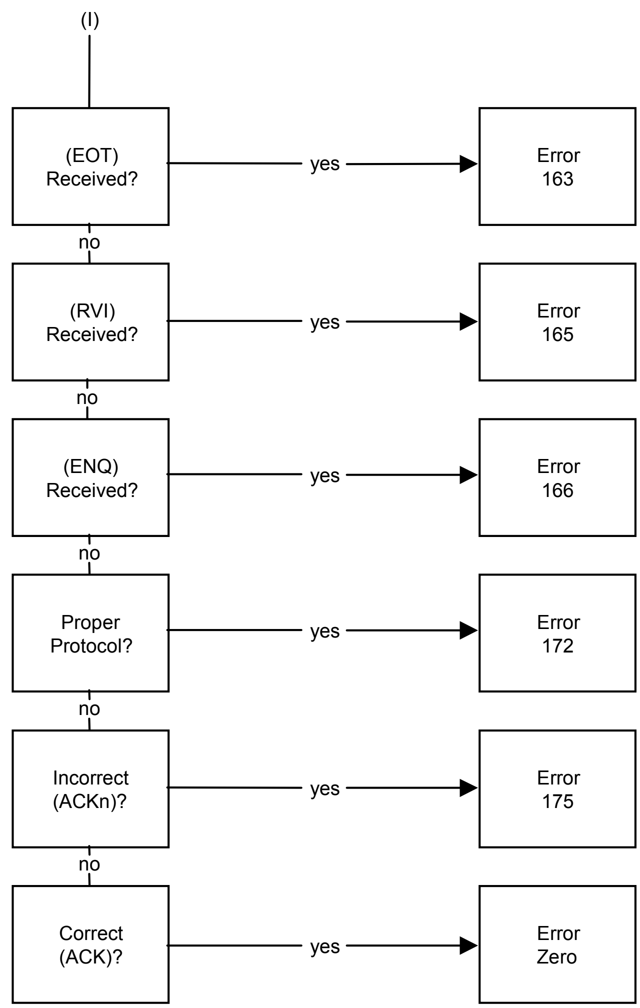


Figure 5.17—Continue Write (Write Message) Error Handling



Communication Protocol Logic

The following communications protocol tables detail the events, actions, and error codes associated with the protocol diagrams. These logic tables explain the protocol procedures used by the MDS.

Note **The entry "Line error" under the action category will not mark down the line. The entry "Line close" under the action category will mark down the line.**

- The annotation "L" with an error code indicates a line error.
- The annotation "S" with an error code indicates a station error.

Standard Bisync Processing Logic Table

Table 5.2 describes the logic and error codes for standard bisync processing.

Table 5.2—Logic and Error Codes for Bisync

State	Error Code	Action	Next State
Evaluate line (Idle status)	n/a	If want to read?	Initial read
		If want to write?	Initial write
		If neither?	Exit line handler
Initial read (Wait for bid)	000	Process message	Continue read
	S 172	Error action	Reply with EOT
	S 173	Error action	Reply with EOT
	S 177	Error action	Reply with EOT
	162	n/a	Evaluate line
	163	n/a	Reply with EOT
	140	n/a	Evaluate line
	164	n/a	Evaluate line
	L 200	Line error	Reply with EOT
	L 201	Line error	Reply with EOT
	L 210	Line error	Reply with EOT
	L 211	Line error	Reply with EOT
	L all other	Line close	Reply with EOT
Continue read (ACK and read)	000	Process message	Continue read
	S 162	Error action	Reply with EOT
	S 172	Error action	Reply with EOT
	S 173	Error action	Reply with EOT
	S 177	Error action	Reply with EOT
	163	n/a	Evaluate line
	140	n/a	Evaluate line
	164	n/a	Evaluate line
	L 200	Line error	Reply with EOT
	L 201	Line error	Reply with EOT
	L 210	Line error	Reply with EOT
	L 211	Line error	Reply with EOT
	L all other	Line close	Reply with EOT

State	Error Code	Action	Next State
Reply with EOT	000	n/a	Evaluate line
	L 200	Line error	Reply with EOT
	L 201	Line error	Reply with EOT
	L 210	Line error	Reply with EOT
	L 211	Line error	Reply with EOT
	L all other	Line close	Evaluate line
Initial write (Bid for line)	000	Data waiting	Continue Write
		Data not waiting	Reply with EOT
	S 163	Error action	Reply with EOT
	S 167	Error action	Reply with EOT
	S 168	Error action	Reply with EOT
	S 169	Error action	Reply with EOT
	S 171	Error action	Reply with EOT
	S 172	Error action	Reply with EOT
	S 173	Error action	Reply with EOT
	S 175	Error action	Reply with EOT
	S 177	Error action	Reply with EOT
	165	n/a	Continue write
	140	n/a	Evaluate line
	164	n/a	Evaluate line
	L 200	Line error	Reply with EOT
	L 201	Line error	Reply with EOT
	L 210	Line error	Reply with EOT
	L 211	Line error	Reply with EOT
	L all other	Line close	Evaluate line
	166	If line primary?	Reply with EOT
	166	If line secondary and want to read?	Initial read
	166	If line secondary/no read	Reply with EOT

State	Error Code	Action	Next State
Continue Write (Write Msg)	000	Data waiting	Continue write
		Data not waiting	Reply with EOT
	S 163	Error action	Reply with EOT
	S 166	Error action	Reply with EOT
	S 171	Error action	Reply with EOT
	S 172	Error action	Reply with EOT
	S 173	Error action	Reply with EOT
	S 177	Error action	Reply with EOT
	140	n/a	Evaluate line
	164	n/a	Evaluate line
	165	n/a	Continue write
	L 200	Line error	Reply with EOT
	L 201	Line error	Reply with EOT
	L 210	Line error	Reply with EOT
	L 211	Line error	Reply with EOT
	L all other	Line close	Evaluate line
Reply with EOT	000	n/a	Evaluate line
	L 200	Line error	Reply with EOT
	L 201	Line error	Reply with EOT
	L 210	Line error	Reply with EOT
	L 211	Line error	Reply with EOT
	L all other	Line close	Evaluate line

Read Initial Error Codes

The following data communication file management errors may occur on an initial READ. An initial READ causes the MDS to wait for a line bid from the remote station. If a line bid is received, the MDS will do the following:

- Acknowledge the line bid
- Receive the initial message
- Return the message to the application process buffer

A condition code of CCE upon completion of the read indicates the successful receipt of the initial message. Following receipt of the initial message, the line is in the read state.

Table 5.3—Read Initial Error Codes

Error Number	Description	Resultant Line State
160	Request is invalid for line state.	
162	Operation timed out. A line bid was not received within the timeout/retry limits.	Control
163	EOT received. An EOT was received while waiting for the line bid.	Control
164	A disconnect sequence (such as DLE-EOT) was received. For non-switched lines: For switched lines; data terminal ready is cleared.	 Control Disconnect
172	Reply not proper for protocol. The data received while waiting for a line bid or for text was not proper for the BISYNC protocol.	Control
173	Transmission error. The message received was in error and a NAK sequence was sent the number of times specified by the retry value.	Control
177	Text overrun. The length of the message received exceeds the length specified in the 'read count' parameter. The READ has been attempted the number of times specified by the retry value.	Control

READ Continue Error Codes

The following data communication file management errors may occur on a continuation READ. All reads following the initial read are continuation reads. A continuation READ causes the MDS to send the proper acknowledgment for the preceding message, read the next message, and return the message to the application process buffer. A condition code of CCE upon completion of the read indicates the successful receipt of a message. Following receipt of an initial message, the line is in the read state.

Table 5.4—Read Continue Error Codes

Error Number	Description	Resultant Line State
160	Request is invalid for line state.	
162	Operation timed out. The MDS sent the acknowledgment but the member host system did not receive a message within the timeout/retry limits.	Control

Error Number	Description	Resultant Line State
163	EOT received.	Control
164	The member host system received a disconnect sequence (such as DLE-EOT.) For non-switched lines: For switched lines; data terminal ready is cleared.	Control Disconnect
172	Reply not proper for protocol. The data received while waiting for a message was not proper for the BISYNC protocol.	Control
173	Transmission error. The message received was in error and a NAK sequence has been sent the number of times specified by the retry value.	Control
177	Text overrun. The length of the message received exceeds the length specified in the 'read count' parameter. The READ has been attempted the number of times specified by the retry value.	Control

WRITE Initial Error Codes

The following data communication file management errors may occur on an initial WRITE. An initial WRITE causes the MDS to bid for the line. If the remote station acknowledges the line bid, the MDS transmits the message in the application process buffer to the remote station. The call to the WRITE procedure is complete when the remote station acknowledges the message. A condition code of CCE, upon completion of the write indicates the successful transmission of the initial message. Following transmission of the initial message, the line is in the write state.

Table 5.5—Write Initial Error Codes

Error Number	Description	Resultant Line State
160	Request is invalid for line state.	
163	EOT received. The MDS received an EOT while waiting for an acknowledgment.	Control
164	A disconnect sequence (such as DLE-EOT) was received.	
	For non-switched lines.	Control
	For switched lines; data terminal ready is cleared.	Disconnect

Error Number	Description	Resultant Line State
165	RVI received. The remote station transmitted a reverse interrupt sequence as a positive acknowledgment to the message.	Write
166	ENQ in response to the line bid. The remote station is also bidding for the line. If the local station is a primary station, the bid was attempted the number of times specified by the retry value.	
	If station is primary.	Control
	If station is secondary.	Read
167	EOT in response to line bid. The remote station rejected the line bid.	Control
168	NAK in response to line bid.	Control
169	WACK in response to line bid.	Control
171	No response to line bid. The line bid was attempted the number of time specified by the retry value.	Control
172	Reply not proper for protocol. The data received when bidding for the line or a message transmitted was not proper for the BISYNC protocol.	Control
173	Transmission error. A NAK sequence in response to the text has been received the number of times specified by the retry value.	Control
175	Incorrect ACK received. The MDS received an ACK1 on the line bid.	Control
177	Text overrun. The length of the response received exceeds two characters. The line bid or transmission has been attempted the number of times specified by the retry value.	
	If overrun is on line bid.	Control
	If overrun is on message.	Write

WRITE Continue Error Codes

The following data communication file management errors may occur on a continuation WRITE. All writes following the initial write are continuation writes. A continuation WRITE causes the MDS to send the message in the application process buffer to the remote station. The call to WRITE procedure is complete when the remote station acknowledges the message. A condition code of CCE, upon completion of the write, indicates the successful transmission of a continuation message. Following transmission of a continuation message, the line is in the write state.

Table 5.6—Write Continue Error Codes

Error Number	Description	Resultant Line State
160	Request is invalid for line state.	
163	EOT received. The MDS received an EOT while waiting for an acknowledgment.	Control
165	RVI received. The remote station transmitted a reverse interrupt sequence as a positive acknowledgment to the message.	Write
166	ENQ in response to the message.	Write
171	No response to the message. A response was requested from the remote station the number of times specified by the retry value.	Write
172	Reply not proper for protocol. The data received as a response in the transmitted message was not proper for the BISYNC protocol.	Write
173	Transmission error. A NAK sequence in response to the text has been received the number of times specified by the retry value.	Write
177	Text overrun. The length of the response received exceeds two characters. The transmission has been attempted the number of times specified by the retry value.	Write

SNA Protocol Support

In addition to the BSC data communication protocols described in the previous subsection(s), the MDS also supports SNA/SDLC data communications protocols.

Protocol Functionality

MDS SNA Protocol emulates an IBM SNA and SDLC Protocol. Two one-way directional Logical Units (LU), or bi-directional LUs facilitate SNA. One LU is for messages that originate from the MDS, and the other LU is for messages originating from the processor.

A processor is designated as the primary station and session acquirer when using the MDS SNA Protocol. The MDS is configured as a secondary station and acts as the session acceptor.

Session Initiator Requirements

Action Required—The processor or the session acquirer, must send a bind frame to initiate the session between the MDS and the processor. If the line of communication between the processor and the MDS is interrupted, and later resumed, the processor must re-initiate the bind frame without operator intervention.

Communication Outages

Action Required—When the MDS is experiencing system outages, processors using SNA-protocol must continue sending receive ready frames for a minimum of thirty minutes.

Communication Sequences

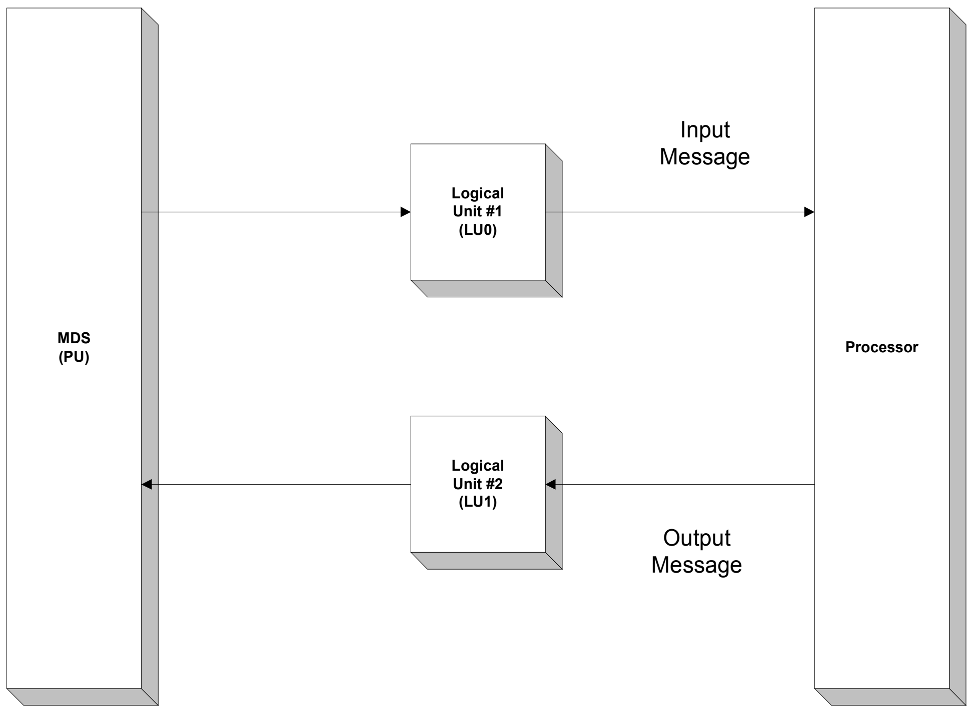
Action Required—The processor **must** define the MDS as: one Physical Controller (PU), with two Logical Terminals (LUs).

Logical Unit 1 (LU0)—the MDS uses the LU to write messages to the processor. The processor accepts input messages from the MDS on LU0, (Address 2).

Logical Unit 2 (LU1)—the processor **must** use LU1 for all outbound messages to the MDS. The MDS accepts input messages from the processor on LU1, (Address 3).

Note **The MDS has a POLL address of %HC1.**

Figure 5.18—Communication Sequences



Expedited/Normal Flow

The logical progression of requests are processed as received, on a first-in/first-out basis. There will be no preferential treatment. The internal network control program processing for each type does not differ in request priority.

Bind Frames

The following sequences of OCTETS defines a suggested bind sequence. The Y constant indicates four bytes of hexadecimal data. In the bind message, the member host system must specify exception responses and turn off pacing.

Y '31010404'	**Y 'B0B00000'**	**Y '00008686'**
Y '80000000'	**Y '00000000'**	**Y '00000000'**
Y '00000005'	**Y 'NNNNNNNN'**	**Y 'NNNNNNNN'**

NNNNNNNN = 8 characters of virtual device name

Device Emulation/Emulation Exceptions

The MDS emulates the following IBM devices: **3601/02**, **3274**, and **3276**.

The following is a list of emulation exceptions:

1. The MDS does not support the bracketing of related SNA commands. Bracketing is supported only if requested during session initiation bind operation, beginning and ending brackets will then be set on all outbound blocks.
2. Segmenting is not supported.
3. Only the FID-2 Transmission Header is supported.
4. All data blocks must be presented only in a chain.
5. Only the ASCII character string is supported.
6. Maximum RU size is 1024 bytes. Maxdata equals 1030.
7. Definite Response is not recommended.
8. SDLC must be full-duplex, non-nrzi (direct).
9. Two logical units (LU), LU0 and LU1 must be used for data transfer; these are Type 2 and 3 Lus.
10. MDS does not support BID SNA commands.
11. MasterCard does not recommend pacing.

IBM Line Set Up

Table 5.7—SLVNCP Slave SDLC Parameters for Vtam NCP Generations

	Space 5			
LINX	Line	ADDRESS = (028,029),	The Switch Line	x
		SPEED = 9600,		x
		CLOCKING = EXT,		x
		DATARATE = LOW,		x
		DUPLEX = FULL,		x
		INTPRI = 3,		x
		NEWSYNC = NO,		x
		RETRIES = (5,1,3),		x
		NRZI = NO,		x
		SERVLIM = 4,		x
		SPDSEL = NO,		x
		TRANSFER = 4,		x
	Space 5			
SERVX	Service	ORDER = (SWITCH)		
	Space 5			
SWITCH	PU	ADDR = 80,		x
		MAXDATA = 489,		x
		MAXOUT = 7,		x
		PASSLIM = 7,		x
		RETRIES = (1,3),		x
		ANS = CONTINUE,		x
		ISTATUS = INACTIVE,		x
		PUTYPE2		x
	Space 5			
SWWS01	LU	LOCADDR = 2,		x x
		ISTATUS = INACTIVE,		x
		VPACING = 3,		
	Space 5			
SWWS02	LU	LOCADDR = 3,		x
		ISTATUS = INACTIVE,		x
		VPACING = 3,		
	Space 5			

Note **Some of the parameters previously listed are site dependent.**

MDS Standard Configuration

Table 5.8—MDS Profile—Secondary Accept Mode

Category	Value
init-name	IMDSSACC
open-mode	ACCEPT
hs-polarity	SECONDARY
want-sscp-text	FALSE
response-type	1
pipeline-lu-ind	FALSE
translate	FALSE
Imo	FALSE
max-request	2
want-bind-control	NONE
user-responses	FALSE
want-flow-control	FALSE
encode-eri	FALSE
single-fmh	FALSE
omit-modem-wait	FALSE

Table 5.9—MDS Line/Unit Parameters

Category	Value
Line	
maxpus	1
maxlus	2
recsize	1030
character set	ASCII
speed	9600
duplex	FULL
station	SECONDARY

Category	Value
PU	
address	%HC1
recsize	1024
maxlus	2
LU	
address	2 or 3
character set	ASCII
puname	#PU1
recsize	1024
protocol	SNALU

Related Information

- Advanced communications function for network control and system support programs for the IBM 3705 (Release 3) SC30-3154-0
- Advance communications function for VTAM planning and installation reference (Version 1, Release 3) SC27-0584-1

SNA/SDLC Supported Formats/Structures/Codes

Table 5.10—Unnumbered Format

Code	Frame	Description
93	SNRM	Set normal response mode command
73	UA	Unnumbered acknowledged response
53	DISC	Disconnect command
	IFDM	Disconnect mode response
97	FRMR	Frame reject response

Table 5.11—Supervisory Format

Bits 0-2	Bit 3	Bits 4-7	Frame	Description	BSC Equivalent
NR	P/F	0001 (1)	RR	Receive ready	ACK
NR	P/F	0101 (8)	RNR	Receive not ready	WACK
NR	P/F	1001 (9)	REJ	Reject	NAK
NR	P/F	NS 0	I-FRM	Information frame	DATA

NR ACK of all frames (Receive sequence number) up to NR–1
NR NAK of any others (Receive sequence number)
P/F Poll/final bit
NS Send sequence number (other sides NR)

Supported Requests/Response Units

X'0X' Function management data—user data
X'6X' Session control—SNA command data
X'4X' Data flow control—SNA command data

SNA Command Code Set Units

Table 5.12—SNA Command Codes

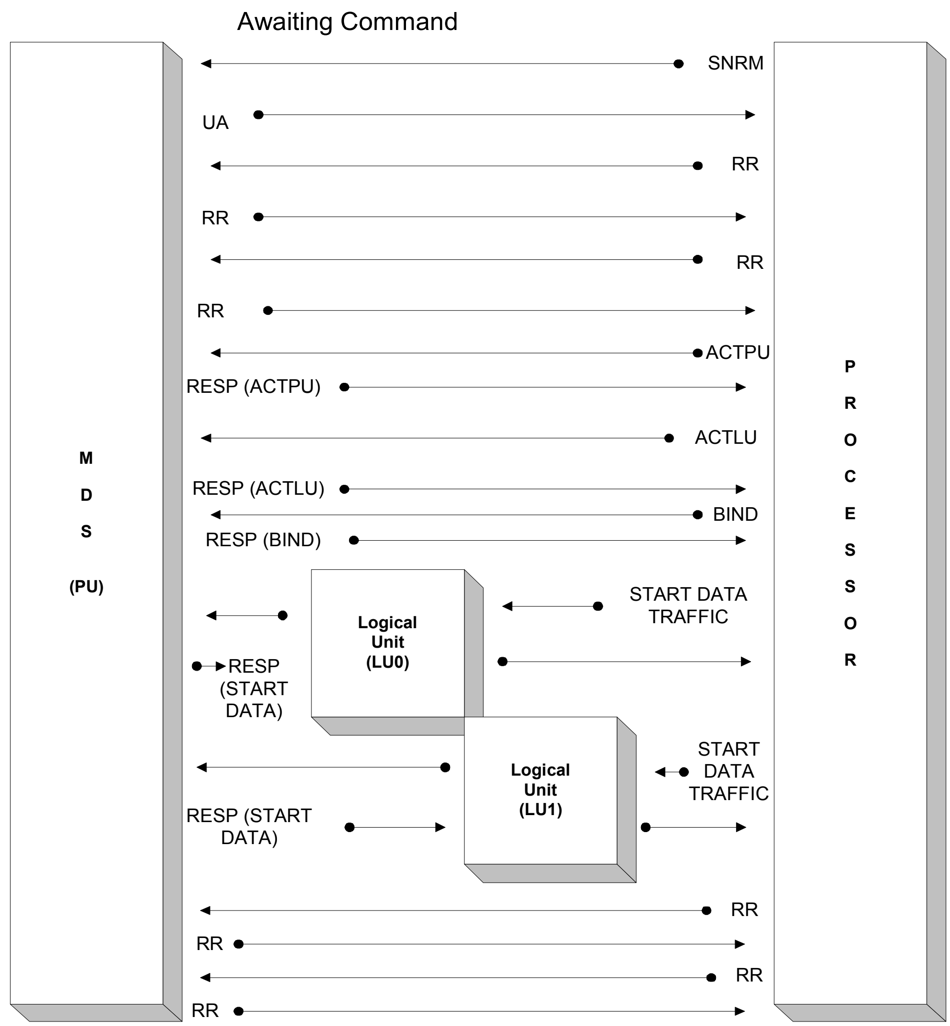
Command Codes	Description	Command Codes	Description
X'11'	Activate physical unit	X'C0'	Shutdown request
X'0D'	Activate logical unit	X'C1'	Shutdown complete
X'31'	Bind Into session	X'83'	Cancel
X'12'	Deactivate physical unit	X'84'	Chase
X'0E'	Deactivate logical unit	X'C9'	signal
X'32'	Unbind	X'O4'	LU status command
X'A0'	Start data traffic	X'A2'	Set and test sequence NBRs command
X'A1'	Clear/stop traffic	X'83010000'	Isolate pacing response-really RM

SNA Sense/Error Codes

(Sent to Host in Exception Responses)

X'2005' Data traffic state not active
X'2007' Data traffic state active
X'1003' Invalid command code
X'1007' Invalid request/response category
X'8005' Session not active
X'8009' LU not active
X'8008' PU not active
X'0815' LU already bound in session-function active

Figure 5.19—SNA Protocol Flow



SNRM = Set Normal Response Mode
UA = Unnumbered Ackowledge Response
RR = Ready to Receive
ACTPU = Activate Physical Unit
ACTLU = Activate Logical Unit LU0 and LU1
BIND = Bind Into Session LU0 and LU1

6

Encryption

This chapter describes network key management, the exchange of encryption keys, and the maintenance of security in the MDS online environment.

Overview

Network key management involves the exchange and security of encryption keys in the MDS online environment.

The MasterCard® Debit Switch supports two options for Key Management:

- Dynamic
- Static

Note **MasterCard does not permit software encryption of PIN data within the MasterCard ATM network or the Maestro and Cirrus Programs. All processors participating in the MasterCard ATM network or the Cirrus/Maestro Programs must use hardware encryption devices only.**

Dynamic Key Encryption—Working Key

The online working key exchange uses the MDS ISO 8583-1987 Network Management/0800 message. The MDS hardware security module (HSM) randomly generates the online working key exchange. Only the MDS can send the key exchange, however, the member may request a key exchange at any time.

The working key is changed online every 12 hours or every 2,500 transactions, whichever occurs first. It is used by processors connected to the MDS via an ISO 8583-1987 interface.

The member and the MDS jointly establish and use the communication key to encrypt the new working key and check value in the online MDS ISO 8583-1987 Network Management/0800 message.

The member validates the check value and loads the new working key. The processor uses the check values to ensure that the MDS generated the new working key from the same unique communication key established between the processor and the MDS.

The member has a time limit of five minutes to implement the new working key; during that period, the MDS will attempt to send the encrypted message using the old and new working key.

The working key encrypts and decrypts the Personal Identification Number (PIN) Data (DE 52) of the ISO 8583-1987 Financial Transaction Request/0200 message, the Financial Authorization Request/0100 message that the member and the MDS sends. The member uses the same working key for issuing and acquiring transactions. The working key supports both ATM and POS products.

MasterCard requests that processors submit new communication key components, used to encrypt the dynamically exchanged working keys, on an annual basis. MasterCard debit staff will contact the member, provide a copy of the MDS Production Communications Key Exchange form and schedule the exchange and loading of the new communication key at the MDS and member site.

Static Key Encryption—Working Key

The working key, used to encrypt the PIN in DE 52 of the Authorization Request/0100 authorization request message, is not exchanged online; it is entered manually (offline) and is used by processors who perform their own PIN verification. Members connected to the MDS via the Banknet® telecommunications network Authorization Request/0100 message interface use this encryption method. Only ATM activity can employ a static encryption key.

The processor creates the working key; the MDS does not generate any component of the key. However, the working key is a shared key between the processor and the MDS. The MDS uses the working key to encrypt the PIN for safe transmission to the issuer and the issuer uses it to decrypt the PIN before performing PIN Validation. The working key or KPE (PIN Encryption Key) is changed annually.

The Authorization Request/0100 message using the Banknet network interface supports credit card, issuer only, transactions; the Authorization Request/0100 message interface does not process debit cards.

MDS PIN Verification Services

The MDS provides PIN verification services for the following:

Credit card issuers using the Authorization Request/0100 Banknet network connection to interface with the MDS for ATM cash advances. Issuers connected to the MDS via a Financial Transaction/0200 message interface who have opted for MDS Stand-In processing. If the issuer wants the MDS to perform PIN verification, they must provide the PIN Verification Key(s) and PIN processing parameters to the MDS. Please contact your MasterCard Regional Office representative for the required forms.

The MDS supports the following PIN Verification methods:

- IBM 3624
- ABA

MDS Key Management

The MDS key management scheme is a method of exchanging encryption keys and maintaining encryption key integrity in an online environment. The MDS employs the following hierarchy of encryption keys:

- **Master file key**: Encrypts working keys and communication keys for safe storage on a database.
- **Communication key**: Encrypts a working key for transmission between the MDS and the processor during the online key exchange.
- **Working key**: Encrypts a PIN for transmission between the processors and the MDS.

Master File Keys

Master file keys are unique to the MasterCard® Debit Switch and each processor that connects to the MDS. Their function is to protect the communication keys and working keys for storage at each site (processor and MDS).

It is the joint responsibility of the MDS and each processor to generate and securely maintain a proprietary master file key.

Communication Keys

Both the MDS and the processor share communication keys. These keys encrypt the new working keys during the dynamic online key exchange.

It is the joint responsibility of the MDS and each processor to generate the unique communication key used to exchange/encrypt working keys.

MasterCard recommends that the communication key for each processor be generated and exchanged at least once every twelve-month period. Members should follow these procedures:

1. Members must use the "MDS Communications Key Part Exchange Form" to request a new communication key exchange. The processor can obtain this form from their MasterCard Regional Office representative or their Debit Services Implementation Manager.

2. Processor Sending Key Officer generates and records the processor clear key component, making an original and a copy. The original is stored in a sealed envelope. The MDS Receiving Key Officer receives the copy in a sealed envelope via courier. Members must use the “MDS Communications Key Part Exchange Form” to accomplish this task. The processor can obtain this form from their MasterCard Regional Office representative or their Debit Services Implementation Manager.
3. MDS Sending Key Officer generates and records the MDS clear key component and a check value. This key part consists of either a sixteen (16) digits or twenty four (24) hexadecimal digits with odd parity on each pair of digits. The MDS Receiving Key Officer makes an original and a copy; storing the original in a sealed envelope. The Processor Receiving Key Officer receives the copy in a sealed envelope via courier.
4. At each site, each key officer enters the key part in his custody into the host security module to be used to generate the communication key (for example, an Atalla security device). A dual control environment handles the management of key components. The system performs a binary “Exclusive-Or” function on the two key parts, thus generating the communication key.

The processor’s master key encrypts this communication key and the resulting cryptogram is stored in the database for use during the online key exchange of working keys.

1. The key parts at each site are stored under dual custody in sealed envelopes for thirty days then destroyed after that time.
2. The generation of new key parts occurs at least once a year on the anniversary of first use.

Working Key

PIN encryption keys are working keys generated by the MDS for each direct connect online processor. These keys encrypt the PIN in Data Element 52 of the online authorization request message.

It is the responsibility of the MDS to generate and distribute the working key.

New working keys are changed dynamically and have the following life cycle:

- Used for no more than twenty four hours.
- Used for no more than 2,500 authorization requests.
- Changed after five consecutive “sanity check” errors.
- Changed upon request by the intermediate network facility (INF, acquirer processor or issuer processor).

It is the responsibility of the processor to safely store this working key by encrypting it under a proprietary master key using hardware security procedures. The processor must use this working key to encrypt all PINs (using ANSI PIN Block formatting) sent to the MDS as well as to decrypt all PINs received from the MDS.

MDS Security Requirements

Within the MDS environment, security considerations include measures for ensuring message security and integrity as well as protection against cardholder personal identification number (PIN) disclosure. The MDS uses secure PIN encryption to protect all PINs. This chapter describes the key management implementation scheme within the MDS, which provides an enhanced degree of protection against PIN disclosure.

The MDS employs PIN encryption using the Data Encryption Standard (DES) algorithm for network security management. Security under DES is dependent on the secrecy of the keys used, and therefore on the management of those keys. The MDS implements the "zone" approach to key management with dynamic keys. MasterCard chose this approach instead of an "end-to-end" approach for the following reasons:

1. Key exchange is required only between connected intermediate network facilities (INF), acquirer processor systems (APS), and issuer processing systems (IPS).
2. Keys are not required to be loaded at the terminal for every issuer participating in the ATM or POS programs.

Figure 6.1 and Figure 6.2 outline the "zone" approach and the flow of key exchange and PIN validation.

In ATM and POS programs, all PINs must be encrypted at the point of entry (the terminal) using the DES algorithm and the approved ANSI PIN block format. The PIN will remain encrypted until the issuer receives it for verification. It will be translated from one zone's working key to another zone's working key as it is passed from one processor to another through the MDS. The MDS must receive the PIN encrypted using the ANSI PIN Block Format.

Members must execute all PIN encryption, translation, and decryption for the ATM or POS programs using hardware encryption through physically secure devices. Both the host and the point of entry, such as the ATM or POS terminal, must use physically secure hardware.

Physically Secure Device (PSD)

A physically secure device (PSD) is a hardware device that cannot be successfully penetrated to disclose all or part of any key or PIN resident within such a device. Penetration of a PSD shall cause the automatic and immediate erasure of all PINs, keys, and all useful residue of PINs and keys contained within the device.

The member's host computer system must use a hardware security module (HSM) to ensure that the cardholder PIN and the PIN keys used to encrypt the PIN do not reside within the processor's host system.

The ATM must use a PSD such as an encryption board or a keyboard encryption controller to encrypt the PIN before it leaves the terminal and is sent to the acquirer's host.

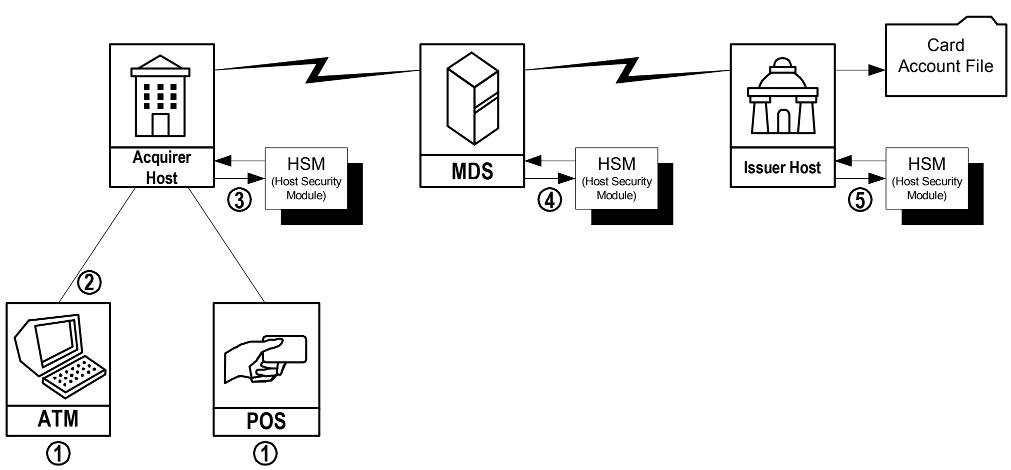
PIN Encryption/Decryption Process

PIN encryption, translation, or decryption must not be performed using software DES routines. Use of DES software in acquirer processing systems (APS), issuer processing systems (IPS), or intermediate network facilities (INF) is a violation of the rules. Following are the PIN encryption/decryption steps:

1. Cardholder enters PIN at point of entry.
2. The terminal encrypts the PIN in hardware under a PIN encryption key and sends it to the acquirer's host.
3. The acquirer's host receives the encrypted PIN, which is then decrypted in hardware using the terminal working key. The host system then encrypts it in hardware under a different key that the acquirer and the MDS share. The MDS then receives the newly encrypted PIN.
4. The MDS decrypts the PIN in hardware. It re-encrypts the PIN using a different key that the MDS and the issuer share. The MDS sends the newly encrypted PIN in hardware to the issuer for verification.

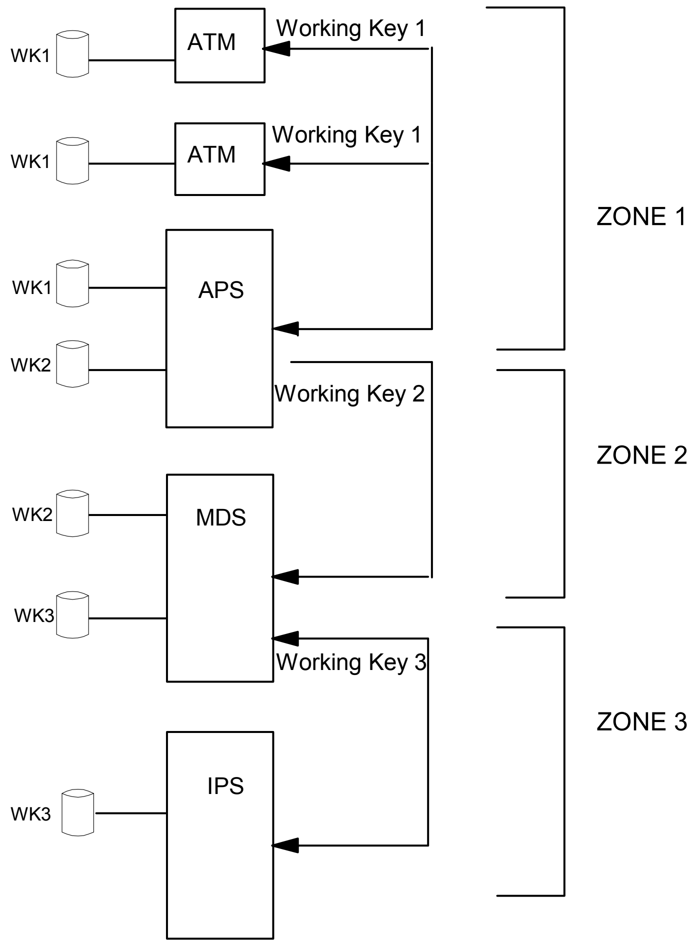
The issuer decrypts the PIN using the key it shares with the MDS and verifies that the PIN is valid.

Figure 6.1—PIN Encryption/Decryption Process



Zone Key Management

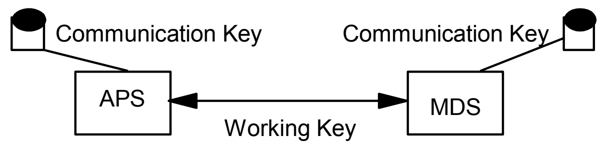
Figure 6.2—Zone Key Management



Key Exchange and PIN Validation Data Flows

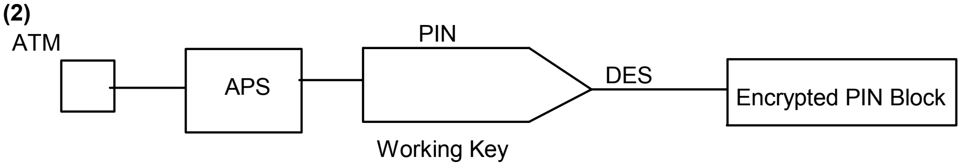
Figure 6.3—Key Exchange and PIN Validation Data Flows

(1)



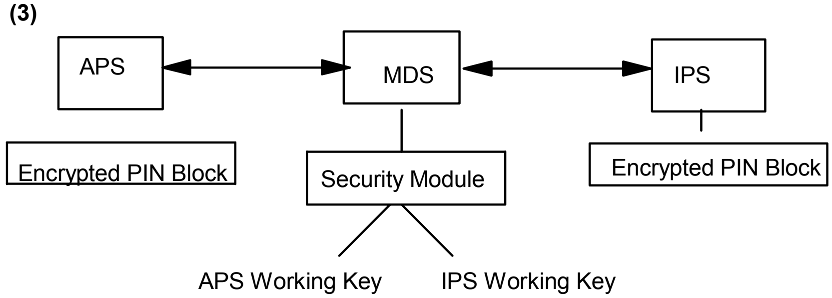
The working key is encrypted under the communication key by the MDS and transmitted to the APS. The APS decrypts the working key using the same communication key, and then re-encrypts it with their master key.

(2)



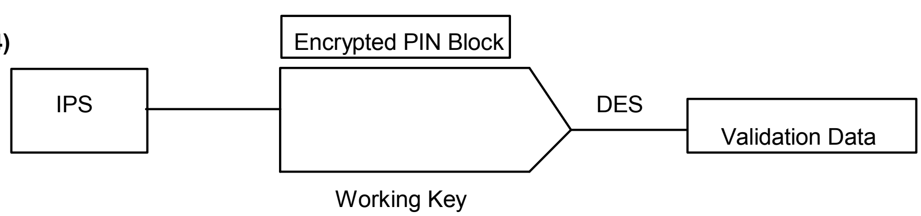
The cardholder-entered PIN is padded to 16 positions with the PIN PAD character. The 16 digit PIN block and the working key are input to DES and the resultant block is the encrypted PIN block.

(3)



The encrypted PIN block is routed to the MDS, where it is translated from the APS working key to the IPS working key. The MDS then routes the encrypted PIN block to the IPS for verification.

(4)



The encrypted PIN block and the IPS working key are then input to DES, with an offset if the issuer uses customer selected PINs. The resultant is then verified.

Triple DES

In February 2000, the International Security Committee (ISC) endorsed the migration of members and processors to the triple DES standard. MDS encryption support currently uses the 3DES encryption method to effectively counter sophisticated "brute force" key attacks.

Member Requirements

Beginning 6 January 2001, the MDS testing environment will be available to generate key exchange (08xx) message sequences using the expanded length key.

MDS processors will need to establish a new communications key with the MDS as part of the processor's conversion effort to triple DES support. Implementation of the new communications key is a manual process managed by the MasterCard Debit Product Support group.

The MasterCard encryption mode for the encryption of multiple-length keys in an online key exchange is electronic codebook (ECB). MasterCard will not support cipher block chaining (CBC) mode in the initial triple DES implementation.

MasterCard will require that all keys for a defined key zone be of the same length. For example, both the zone master key (KEK) and the zone working key (KPE) for a processor link must be double length, or both must be triple length.

Single Key Length

Currently, processors use the ISO 0800/0810/0820 message sequence to complete encryption key exchanges with the MDS. The key information is stored in data element (DE) 48, subtag 11 in the following format:

Table 6.1—Key Information Format

Subfield	Attribute	Value
Subtag ID	n-2	11
Subtag Length	n-2	38
Key Class Identifier	an-2	PK (PIN key exchange)
Key Index Number	n-2	00 (constant
Key Cycle Number	n-2	00-99
Encrypted Key	an-16	Hexadecimal characters 0-9, A-F; **single key size**
Key Check Value	an-16	Hexadecimal characters 0-9, A-F

Double Key Length

Processors that use double length keys under triple DES will follow the key exchange process below in ISO 0800/0810/0820 message sequences to complete encryption key exchanges with the MDS. The key information is stored in DE 48, subtag 11 in the following format:

Table 6.2—Key Information Format

Subfield	Attribute	Value
Subtag ID	n-2	11
Subtag Length	n-2	54
Key Class Identifier	an-2	PK (Pin key exchange
Key Index Number	n-2	00 (constant)
Key Cycle Number	n-2	00-99
Encrypted Key	an-32	Hexadecimal characters 0-9, A-F; **double key length**
Key Check Value	an-16	Hexadecimal characters 0-9, A-F

Triple Key Length

Processors using triple length keys under triple DES will follow the key exchange process below in ISO 0800/0810/0820 message sequences to complete encryption key exchanges with the MDS. The key information is stored in DE 48, subtag 11 in the following format:

Table 6.3—Key Information Format

Subfield	Attribute	Value
Subtag ID	n-2	11
Subtag Length	n-2	70
Key Class Identifier	an-2	PK (Pin key exchange
Key Index Number	n-2	00 (constant)
Key Cycle Number	n-2	00-99
Encrypted Key	an-48	Hexadecimal characters 0-9, A-F; **triple key length**
Key Check Value	an-16	Hexadecimal characters 0-9, A-F

Member Testing

The MasterCard Test Simulator-Debit version 1.5 will include enhancements to support testing of triple DES encryption. During testing, MasterCard will require acquirers and issuers to re-certify with the MDS, verifying that they are able to support triple DES encryption. The MDS will track the encryption method used at the issuer and processor levels and generate the appropriate key exchanges and PIN translations.

For more information about the conversion to triple DES encryption, please contact Debit Product Support:

Telephone: 1-914-249-5620

Fax: 1-914-249-4301

Network Key Management Responsibilities

MasterCard® Debit Switch

1. Identify, authorize, and brief the appropriate staff for management of the master key and the communications keys.
2. Appoint communication key-part holders.
3. Keep the key parts in dual custody.
4. Coordinate the exchange of new communication keys with each processor on an annual basis.
5. Initiate dynamic online key exchanges as required.

Processors

1. Select and purchase a hardware security module.
2. Demonstrate ability to receive and process working key exchange requests.
3. Demonstrate ability to translate PINs from one key to another in hardware.
4. Develop procedure for managing the communication key.
5. Identify, authorize, and brief the appropriate staff for management of the master key and the communications keys.
6. Appoint communications key-part holders.
7. Keep the logical key parts under dual control.

Note **It is recommended that security officers selected for key management not have extensive technical backgrounds.**

All PINs must be encrypted at the point of entry using the DES algorithm and the approved ANSI PIN Block Format. The ANSI PIN block is the only format supported by the MDS. Below is the description of the PIN block creation.

ANSI PIN Block Format

A technical staff member builds the ANSI PIN block by performing a binary "Exclusive OR" of the two sixteen-hexadecimal digit data elements together.

1. The first hexadecimal data element contains cardholder PIN information.
   a. The first digit is zero.
   b. The second digit is the length of the PIN (such as 4-9, A (10), B (11), or C (12). The maximum length is twelve digits.
   c. The third digit is the start of the cardholder PIN; twelve is the maximum length.
   d. The PIN is padded on the right with hexadecimal "F"s to complete the 16-digit data element.
2. The second hexadecimal data element contains Primary Account Number (PAN) information.
   a. The first four digits are set to zero.
   b. The next 12 digits of the data element contain the right-most 12 digits of the Primary Account Number (PAN), excluding the check digit. If the PAN contains 12 or less digits, then the entire PAN excluding the check digit is used. The field is padded on the left with zeroes to complete the 16-digit data element.

The two hexadecimal data elements are "Exclusive OR'ed" to obtain the ANSI PIN Block result.

PIN Encryption

The acquirer must send the entered PIN to the MasterCard® Debit Switch, encrypted in an ANSI block format (Figure 9.4). The acquirer must meet the following requirements when encrypting a PIN:

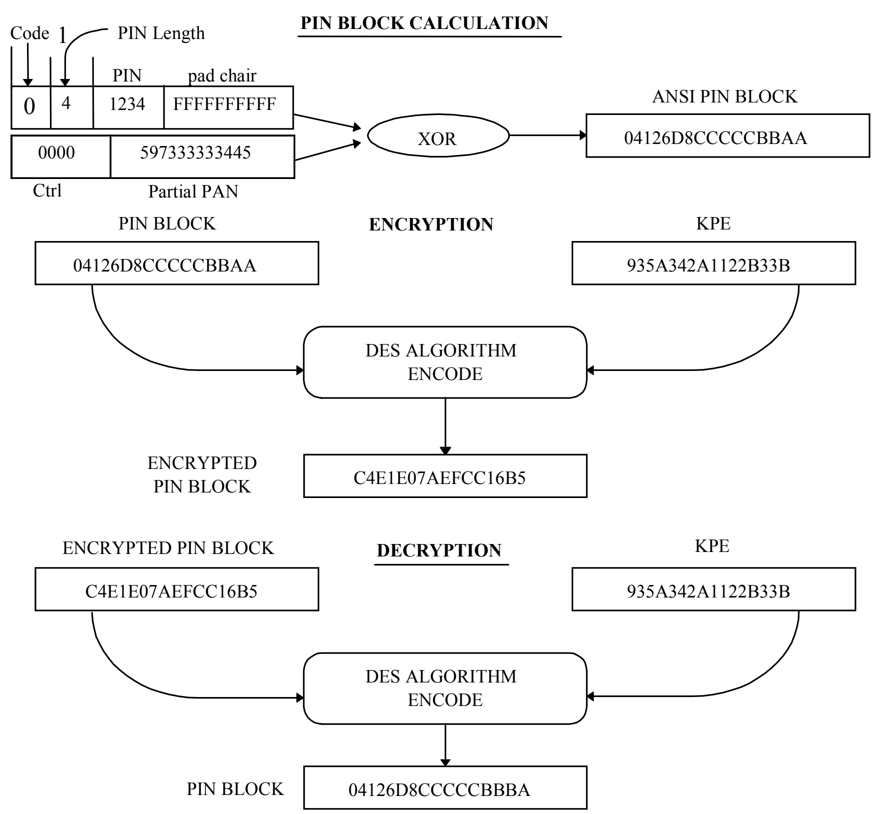
1. The first digit of the first block will contain the control character 0, followed by a number representing the length of the PIN, and then the PIN itself. The remaining digits of the block are filled with the pad character "F".
2. The first four characters of the second block will contain 0000, followed by the 12-rightmost digits of the PAN, excluding the check digit.
3. In formatting an ANSI block, the acquirer will "Exclusive-OR" (XOR) the two 16-digit blocks.

4. After creation of the PIN block, it is sent through the DES algorithm along with the 16-digit key (KPE), producing the encrypted PIN block, which is sent to the MDS.
5. The MDS will decrypt the encrypted PIN block with the KPE it shares with the acquirer.

The MDS will then re-encrypt the PIN block with the KPE that the MDS shares with the issuer and then forward it in the authorization request message sent to the issuer.

Figure 6.4—ANSI PIN Block

ENTERED PIN "1234"
PAD "F"
KPE "935A342A1122B33B"
ACCOUNT# "5415973333334456"
PARTIAL PAN "597333333445"



Sanity Checks

All physically-secure devices (PSD) must be able to detect possible working key corruption by verifying that the clear text PIN block is in the expected format. Failure of this “Sanity Check” should result in a denied transaction and the initiation of the key exchange sequence between the processor and MDS.

Security Provisions

The security provisions at the MasterCard® Debit Switch require adherence to the following:

- All terminals must encrypt PINS in a physically secure device using the ISO approved algorithm(s) for PIN encipherment listed in ISO 9564–2 (DES algorithm) and a working key that is used at the terminal (ATM or POS). Working keys that are loaded manually must be loaded in a dual control environment.
- MasterCard recommends that the same working key not be assigned to any two terminals driven by the same hardware, software and/or both in a predictable manner.
- All keys must be stored encrypted using the ISO approved algorithm(s) for PIN encipherment listed in ISO 9564-2 (DES algorithm) and a proprietary master key. Alternately, keys may be stored within a physically secure device. All encryption/decryption processing must occur within a physically secure device.
- All intermediate network facilities (INFs) between the terminal and the cardholder processor must receive and send customer entered PINs in the form of a cryptogram to other INFs using the ISO approved algorithm(s) for PIN encipherment listed in ISO 9564-2 (DES algorithm) and working keys statically or dynamically maintained by the INFs processing MDS activity.
- PINs may be decrypted and re-encrypted, during INF transmission processing, to change the format of the PIN block or the working key used to protect the PIN. It must be translated in a physically secure device.
- Working keys that are dynamically maintained must have the following life cycle:
  - Used for no more than 24 hours.
  - Used for no more than 2,500 transactions.

PIN Generation Verification

The MDS supports two methods of PIN generation verification. Below is a description of each methodology.

IBM 3624

To generate PINs using the IBM 3624 method (Figure 6.5), the institution must determine whether it will support customer-selected PINs and establish the following:

- Validation data
- PAN pad character
- Decimalization table
- DES Key (KPV)

The validation data is a portion of the PAN used in constructing the first 16-digit block. If the portion of the PAN being used is less than 16 digits, this data is left- or right-justified, and padded with the PAN pad character. This block, along with the 16-digit KPV, is sent through the DES algorithm to produce a 16-digit generated PIN block.

The decimalization table is then applied to the generated PIN block to map all alphabetic characters to numeric digits. The resulting block is called the natural PIN block.

The MDS security hardware requires that the member use at least eight distinct digits to create the decimalization table. The member may not use any digit more than four times.

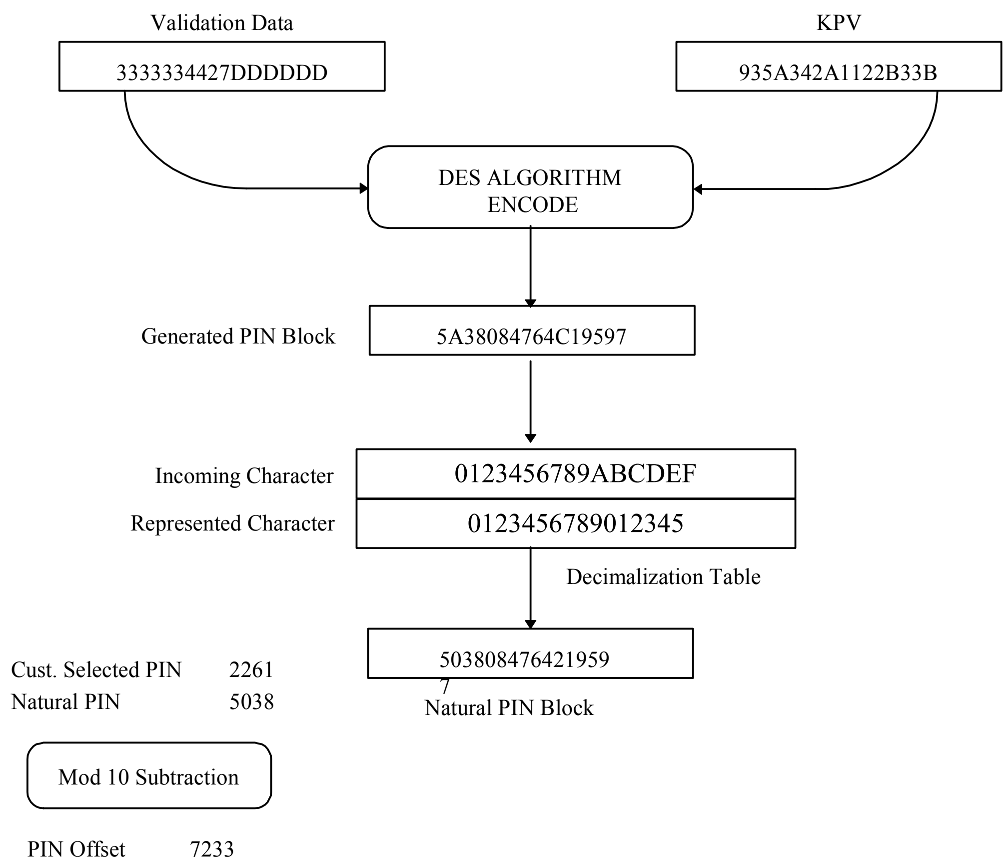
If customer-selected PINs are supported, then the MDS will need the location of the PIN offset that is produced during generation; this is encoded on Track-2 of the card's magnetic stripe.

If the institution does not support customer-selected PINs, the first 4 through 12 digits are the PIN assigned to the customer. This PIN, called the natural PIN explicitly implies that the PIN offset is all zeros.

The customer-selected PIN is module 10 subtracted from the natural PIN block to produce the PIN offset.

Figure 6.5—IBM 3624 PIN Generation

PAN “5415973333334427”
VALIDATION DATA “3333334427”
PAN PAD “D”
DECTABLE “0123456789012345”
KPV “935A342A1122B33B”



ABA

To generate PINs using the ABA method (Figure 6.6) the institution must determine the following:

- Validation data
- PVKI (PIN verification key index)
- Key left
- Key right
- PIN (customer-selected)

This generation process will produce a PVV (PIN verification value), which is encoded on track 2 of the card's magnetic stripe.

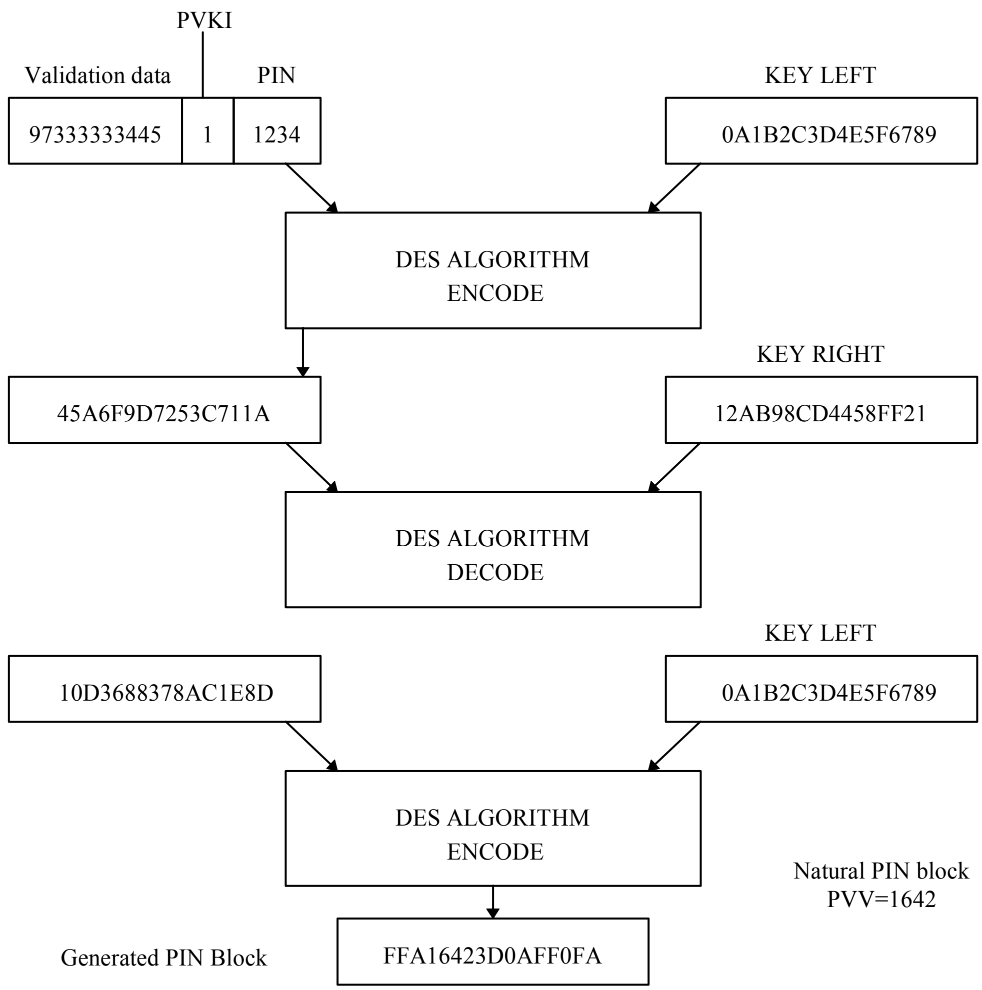
The validation data is the last 11 digits of the PAN, excluding the check digit. The PVKI is appended to the validation data, which is in turn appended by the PIN to complete the 16-digit block.

This block along with the 16-digit key left is sent through the DES algorithm to produce another 16-digit block. The resultant block is sent through DES using the 16-digit key right, producing a new 16-digit resulting block. This new block is sent through DES with the key left to produce the final block called the generated PIN block.

The PVV is determined by taking the first four numeric digits of the natural PIN block. If there are not four numeric digits, the A-F are mapped to 0-5, and the remaining digits of the PVV are completed.

Figure 6.6—ABA PIN Generation

ASSIGNED PIN	"1234"
KEY LEFT	"0A1B2C3D4E5F6789"
KEY RIGHT	"12AB98CD4458FF21"
PAN	"5415973333333445"
VAL. DATA:	"97333333445"



Any intermediate network facility (INF) that is not a processor and does not translate, use, or verify PIN and/or key data for any transaction processed by such INF is not subject to any of the provisions listed above.

Required Functionality

Using a physically secure device (PSD) only, a processor must be able to support the functionality indicated in Table 6.4 and Table 6.5.

Note **Software emulation of these DES security functions** are not permitted **under MasterCard/ Cirrus® and Maestro® Operating Rules.**

Table 6.4—Key Management Functions

Functionality	MDS	Processors	Others [a]	Terminals
Store Master Key	Required	Required	Required	N/A
Establish/reset Communication Key	Required	Required	Required	Required
Generate working key	Required	Optional	Optional	N/A
Receive working key	N/A	Required	Optional	Required

[a] Processors or networks not directly connected to the MDS.

Table 6.5—PIN Processing Functions

Functionality	MDS	Processors	Others [a]	Terminals
Encrypt PIN	Required	Required	Required	Required
Verify PIN	N/A	Required (IPS)	Optional	Optional (IPS)
Translate from ANSI to ANSI	Required	Optional	Optional	N/A
Translate from PIN PAD to ANSI	N/A	Required	Optional	N/A
Translate from ANSI to PIN PAD	N/A	Optional	Optional	N/A
Translate from Clear to ANSI	Required	Required	Required	N/A
Translate from ANSI to Clear	Required	Required	Required	N/A

[a] Processors or networks not directly connected to the MDS.

Detection of Working Key Corruption

All PSDs must be able to detect possible working key corruption by verifying that the clear text PIN block is in the expected format. Failure to perform this sanity check should result in a denied transaction and the initiation of a key exchange sequence between the INFs where the corruption is detected.

After there has been a successful working key exchange, it is the responsibility of the processor and of the MDS to preserve the old working key for a period of five minutes. If the system receives a sanity check error during this five-minute period, the system should try the old working key before returning an error. If the stored key receives a sanity check, the system should return a response code appropriate to the type of message format used.

Fallback to Clear Text

In the event of a major problem with security equipment (for example, a faulty PSD or DES circuit board), the MDS will have no choice but to suspend all transaction processing with the processor.

Note **Use of clear text processing of transactions is expressly prohibited by the Operating Rules.**

Emergency Communication Key Procedures

In the event that a successful working key exchange cannot be performed, the MDS will invoke the emergency communication key procedure using the following procedures:

1. The MDS marks the faulty processor as down.
2. Authorized MDS personnel randomly generate an emergency communication key (generating both parts of the key).
3. MDS personnel call the security or operations staff at the processor. The emergency communication key is given verbally to the processor.
4. Both the MDS and the processor insert the new emergency communication key in their security modules.
5. The MDS initiates key exchange and log-on, using the new emergency communication key.

The emergency communication key procedure is only to be used as an interim measure to enable a processor to resume transaction processing with the MDS as quickly as possible following a key exchange failure. The personnel responsible for key management must be notified immediately of the security failure situation and must conduct a secure key exchange at the earliest possible time.

Use of the emergency communication key in any one occurrence is limited to six business days. After such time, the processor must have reestablished the jointly established communication key, in accordance with the provisions outlined in this chapter.

Key Naming Convention

The master key encrypts the working key and communication key for storage.

The communication key is used to encrypt and decrypt the working key in the MDS or CIS ISO 8583-1987 Network Management/0800.

The working key is used to encrypt the PIN block in DE 52 of the Financial Transaction Request (Pre-Authorization)/0200 message.

Table 6.6—Functional versus Vendor-Specific

FUNCTIONAL	ATALLA	RACAL
Master	MFK (Master File Key)	LMK (Local Master Key)
Communication	KEK (Key Exchange Key)	ZMK (Zone Master Key)
Working	KPE (Key PIN Encryption)	ZPK (Zone PIN Key)

7 Database Forms

This chapter describes the procedures for completing the forms to establish the member participation database on the MDS.

Overview

This chapter describes the procedures for completing the forms to establish the member participation database on the MDS. There are two types of forms required, an Institution Definition File (IDF) and an Institution Routing Table (IRT). The IDF defines the Institution and the IRT establishes the card prefix for routing purposes. The following pages outline instructions for completing each form.

Please make copies of the forms in this chapter for database maintenance. Submit the forms to Debit Product Support for production processing:

Address: MasterCard International
Debit Product Support
2000 Purchase Street
Purchase NY 10577-2509

Members and processors can receive the electronic version of the MDS database forms by emailing a request to Debit Product Support:

email: MDS_Database_Forms@mastercard.com

Institution Definition File (IDF)

The IDF (Institution Definition File) identifies each financial institution participating in the MDS network. It defines the products supported by the member and the processor that is providing the authorization services for the institution.

Procedures to Complete an IDF Form

Following are the list of fields and descriptions found on the IDF form. Any **Add**, **Modify** (note exceptions **), and **Transfer** fields **must** have contracts. The following fields are required for all actions taken on the IDF:

Field Descriptions

- **Add**
- **Modify**
  - A member cannot modify a Routing and Transfer (R&T) number using an IDF form. The member must submit a request to delete the old R&T with a request to Add the new R&T.
  - The following changes require a new Cirrus contract:
    Name change
    City and/or State change
    Network participation change (Cirrus issuer and acquirer to acquirer-only member or acquirer-only member to Cirrus issuer and acquirer)
  - Change in acquirer supported products (adding Plus and Visa)
  - **Comments**—Use this space to indicate what is being modified (for example, for name change, indicate old name of institution).
- **Delete**
  - The Operating Rules require six months notification to leave the network. Please submit the letter of termination from the institution along with the IDF, six months prior to deletion.
  - For issuers, please submit an Institution Routing Table (IRT) form indicating the necessary action for each prefix.

- **Transfer**
  - For a transfer to be effective, the current principal member must sign the IDF (Current Principal Member Approval).
  - For issuers, submit an IRT form with the proper signature from the current principal member.
  - For any prefix that will not be transferred, an IRT must be submitted from the current principal member indicating the necessary action to be taken.
  - Processor-only changes do not require a new contract or sign-offs, **however**, the principal member must submit the paperwork.

Effective Date

Allow ten business days from the date received.

Institution (R&T) Number

This is the Federal Reserve routing and transit number of the institution or a nine digit pseudo number assigned by the MDS to identify the financial institution as a member of the network.

Institution Name

This is the name of the participating institution (a maximum of 25 alphanumeric characters).

Member ID/ICA Number

Refer to the "Exceptions" subsection later in this chapter.

City

The city where the main office of the institution is located (maximum of 12 alphanumeric characters).

State

State where the main office of the institution is located. This field must contain the ISO defined state code (see appendix A).

Numeric Currency Code

This is the local currency of the institution. Members must use the ISO-defined numeric currency code in the field (see appendix B).

Numeric Country Code

The ISO-defined Country Code of the institution (see appendix B).

Processor Number

The MDS assigns the processor number.

Processor Name

The name of the processor providing the network services (maximum of 25 alphanumeric characters).

Principal Member Name

This is the principal member who sponsors the institution into the network.

Acquirer-Only Participation

Indicate with **Y** or **N** if the member is an acquirer-only participant.

Acquirer Product Participation

Identify the product(s) supported by indicating **Y** or **N** in the boxes next to the specific products.

Signatures

Sign and date the form providing a contact telephone number to contact the originator of the form, if necessary.

Institution Routing Table (IRT)

The IRT (Institution Routing Table) form contains information regarding the card prefix identifying the institution participating in the network and the products supported. The IRT also determines the routing of transactions to the appropriate processor.

Procedures to Fill out an IRT Form

Members must submit a contract and an IDF for Adds (new sponsored members) and Transfers. For pre-existing members, submit only an IRT form for the new prefix.

Following are the list of fields and descriptions required for Adds and Transfers.

Field Descriptions

- **Add**
- **Modify**
  - A member cannot modify a prefix number using an IRT form. The member must submit a request to delete the old prefix with a request to Add the new prefix.
  - You **do not** need an IRT for a name change, use the IDF form to change this information.
  - Use the IRT form to **modify** any of the following fields:
    Change in R&T number
    Product participation change
- **Delete**
  - The Operating Rules require six months notification to leave the network. Please submit the letter of termination from institution along with the IDF six months prior to deletion.
  - Complete an IRT indicting the effective date of termination.

- **Transfer**
  - For a transfer to be effective, the current principal member must sign the IRT (current principal member approval).
  - For issuing members, submit the IDF form with the proper signature from the current principal member.
  - For any prefix that will not be transferred, the member must submit an IRT from the current principal member indicating the action necessary.
  - Processor-only changes do not require a new contract, or signature, **however**, the principal member must submit the paperwork.

Effective Date

Allow ten business days from the date received.

Prefix Number (BIN)

The ISO assigned number, MasterCard or Visa assigned BIN number encoded on the card uniquely identifying the owner of the prefix. The number must be 5–11 positions in length and is used for routing the transactions.

Institution (R&T Number)

This is the Federal Reserve routing and transit number of the institution or a nine digit pseudo number assigned by the MDS to identify the financial institution as a member of the Network.

Institution Name

This is the name of the participating institution (maximum of 25 alphanumeric characters).

Member ID/ICA Number

See subsection “Exceptions” later in this chapter.

State

State where the main office of the institution is located. Members must use the ISO defined state code in the field (see appendix A).

Card Type

Indicate whether the prefix is a debit or credit card BIN.

Numeric Currency Code

This is the local currency of the institution. Members must use the ISO defined numeric currency code in the field (see appendix B).

Numeric Country Code

The ISO defined Country Code of the institution (see appendix B).

Processor Number

The MDS assigns the processor number.

Processor Name

The name of the processor providing the network services (maximum of 25 alphanumeric characters).

Principal Member Name

Name of the principal member who sponsors the institution.

Encrypted PIN Key

For MasterCard use only.

Intracountry Settlement

For MasterCard use only.

Product Participation

Identify the ATM product(s) supported by indicating **Y** or **N** in the boxes next to the specific products. For MasterCard, the BIN range must be 51–55.

Credit Processor Number

If the processor is different for credit card cash advances, indicate the processor number providing the credit authorization services.

POS Product Participation

Identify the POS product(s) supported by indicating **Y** or **N** in the boxes next to the specific products.

POS Debit Processor Number

If the POS processor is different from the ATM processor, include the POS processor number on the form.

Debit MasterCard Processor Number

If the debit MasterCard processor is different from the ATM or POS processor, include the debit MasterCard processor number on the form.

Signatures

Sign and date the form. Provide a contact telephone number to contact the originator of the form, if necessary.

Exceptions

Pseudo Numbers

A pseudo number is used under the following conditions and must be assigned by MasterCard:

1. To set up an IDF or IRT before a transfer that will temporarily identify a member or processor (until an actual R&T is ready to convert over to the new principal member or the new processor).
2. To assist processors' internal settlement.
3. To establish Non-Member Terminal Agreement and Service Mark License members.

Please contact MasterCard for pseudo number assignments.

Banknet Members

For some credit card processors, their connection to the MDS is through a Banknet network interface.

Note **A Member ID/ICA Number is required for these members.**

Member ID/ICA Number—This field contains a four-digit identification number assigned by MasterCard International to an institution, third-party processor, or other type of customer for the purpose of uniquely identifying that customer as a member of MasterCard. If a Member ID/ICA Number is needed, contact the Franchise Management Department in New York. Complete this field with the assigned identification number.

Note **The MasterCard Franchise Management Department assigns Member ID/ICA Numbers (numbers require some time to obtain).**

A North American State, Province and Territory Codes

U.S. State and Territory Codes

Table A.1—State Codes

State or Territory	Code	State or Territory	Code
Alabama	AL	Mississippi	MS
Alaska	AK	Missouri	MO
Arizona	AZ	Montana	MT
Arkansas	AR	Nebraska	NE
California	CA	Nevada	NV
Colorado	CO	New Hampshire	NH
Connecticut	CT	New Jersey	NJ
Delaware	DE	New Mexico	NM
District of Columbia	DC	New York	NY
Florida	FL	North Carolina	NC
Georgia	GA	North Dakota	ND
Guam	GU[a]	Ohio	OH
Hawaii	HI	Oklahoma	OK
Idaho	ID	Oregon	OR
Illinois	IL	Pennsylvania	PA
Indiana	IN	Puerto Rico	PR[b]
Iowa	IA	Rhode Island	RI
Kansas	KS	South Carolina	SC
Kentucky	KY	South Dakota	SD
Louisiana	LA	Tennessee	TN
Maine	ME	Texas	TX
Maryland	MD	Utah	UT
Massachusetts	MA	Vermont	VT
Michigan	MI	Virginia	VA
Minnesota	MN	Virgin Islands	VI[b]

State or Territory	Code	State or Territory	Code
Washington	WA	Wisconsin	WI
West Virginia	WV	Wyoming	WY

[a] Warning Bulletin region C

[b] Warning Bulletin region B

Canadian Province and Territory Codes

Table A.2—Province Codes

Province or Territory	Code
Alberta	AB
British Columbia	BC
Manitoba	MB
New Brunswick	NB
Newfoundland	NL
Northwest Territories	NT
Nova Scotia	NS
Nunavut	NU
Ontario	ON
Prince Edward Island	PE
Quebec	QC
Saskatchewan	SK
Yukon	YT

B Country and Currency Codes

Country and Currency Codes

Table B.1—Country and Currency Codes (Sorted by Name of Country or Entity)

Country Codes				Currency Codes			
Name of Country or Entity	Numeric	Alpha	Warning Bulletin Region	Currency	Numeric	Alpha	Currency Exponent
Afghanistan	004	AFG	C	Afghani	004	AFA	2
Albania	008	ALB	D	Lek	008	ALL	2
Algeria	012	DZA	E	Algerian Dinar	012	DZD	2
American Samoa	016	ASM	C	Spanish Peseta U.S. Dollar	724 840	ESP USD	0 2
Andorra	020	AND	D	Euro Andorran Peseta French Franc	978 020 250	EUR ADP FRF	2 0 2
Angola	024	AGO	E	New Kwanza Kwanza Reajustado	024 982	AON AOR	2 2
Anguilla	660	AIA	B	East Caribbean Dollar	951	XCD	2
Antarctica	010	ATA	D	No universal currency			
Antigua and Barbuda	028	ATG	B	East Caribbean Dollar	951	XCD	2
Argentina	032	ARG	B	Argentine Peso	032	ARS	2
Armenia	051	ARM	D	Armenian Dram	051	AMD	2
Aruba	533	ABW	B	Aruban Guilder	533	AWG	2
Australia	036	AUS	C	Australian Dollar	036	AUD	2
Austria	040	AUT	D	Euro Schilling	978 040	EUR ATS	2 2
Azerbaijan	031	AZE	D	Azerbaijanian Manat	031	AZM	2
Bahamas	044	BHS	B	Bahamian Dollar	044	BSD	2
Bahrain	048	BHR	E	Bahraini Dinar	048	BHD	3
Bangladesh	050	BGD	C	Taka	050	BDT	2
Barbados	052	BRB	B	Barbados Dollar	052	BBD	2
Belarus	112	BLR	D	Belarussian Ruble	974	BYR	0

Country Codes				Currency Codes			
Name of Country or Entity	Numeric	Alpha	Warning Bulletin Region	Currency	Numeric	Alpha	Currency Exponent
Belgium	056	BEL	D	Euro Belgian Franc	978 056	EUR BEF	2 0
Belize	084	BLZ	B	Belize Dollar	084	BZD	2
Benin	204	BEN	E	CFA Franc BCEAO✪	952	XOF	0
Bermuda	060	BMU	B	Bermudian Dollar	060	BMD	2
Bhutan	064	BTN	C	Indian Rupee Ngultrum	356 064	INR BTN	2 2
Bolivia	068	BOL	B	Boliviano	068	BOB	2
Bosnia and Herzegovina	070	BIH	D	Convertible Mark	977	BAM	2
Botswana	072	BWA	E	Pula	072	BWP	2
Bouvet Island	074	BVT	E	Norwegian Krone	578	NOK	2
Brazil	076	BRA	B	Brazilian Real	986	BRL	2
British Indian Ocean Territory	086	IOT	C	U.S. Dollar	840	USD	2
Brunei Darussalam	096	BRN	C	Brunei Dollar	096	BND	2
Bulgaria	100	BGR	D	Lev	975	BGN	2
Burkina Faso	854	BFA	E	CFA Franc BCEAO✪	952	XOF	0
Burundi	108	BDI	E	Burundi Franc	108	BIF	0
Cambodia	116	KHM	C	Riel	116	KHR	2
Cameroon	120	CMR	E	CFA Franc BEAC✸	950	XAF	0
Canada	124	CAN	A	Canadian Dollar	124	CAD	2
Cape Verde	132	CPV	E	Cape Verde Escudo	132	CVE	2
Cayman Islands	136	CYM	B	Cayman Islands Dollar	136	KYD	2
Central African Republic	140	CAF	E	CFA Franc BEAC✸	950	XAF	0
Chad	148	TCD	E	CFA Franc BEAC✸	950	XAF	0
Chile	152	CHL	B	Chilean Peso	152	CLP	2

Country Codes				Currency Codes			
Name of Country or Entity	Numeric	Alpha	Warning Bulletin Region	Currency	Numeric	Alpha	Currency Exponent
China, People's Republic of	156	CHN	C	Yuan Renminbi	156	CNY	2
Christmas Island	162	CXR	C	Australian Dollar	036	AUD	2
Cocos (Keeling) Islands	166	CCK	C	Australian Dollar	036	AUD	2
Colombia	170	COL	B	Colombian Peso	170	COP	2
Comoros	174	COM	E	Comoro Franc	174	KMF	0
Congo	178	COG	E	CFA Franc BEAC✸	950	XAF	0
Cook Islands	184	COK	C	New Zealand Dollar	554	NZD	2
Costa Rica	188	CRI	B	Costa Rican Colon	188	CRC	2
Côte D'Ivoire	384	CIV	E	CFA Franc BCEAO✪	952	XOF	0
Croatia, Republic of	191	HRV	D	Kuna	191	HRK	2
Cyprus	196	CYP	D	Cyprus Pound	196	CYP	2
Czech Republic	203	CZE	D	Czech Koruna	203	CZK	2
Democratic Republic of the Congo	180	COD	E	Franc	180	CDF	2
Denmark	208	DNK	D	Danish Krone	208	DKK	2
Djibouti	262	DJI	E	Djibouti Franc	262	DJF	0
Dominica	212	DMA	B	East Caribbean Dollar	951	XCD	2
Dominican Republic	214	DOM	B	Dominican Peso	214	DOP	2
East Timor	626	TMP	C	Rupiah	360	IDR	0
Ecuador	218	ECU	B	Sucre Unidad de Valor U.S. Dollar Constante	218 840 983	ECS USD ECV	2 2 2
Egypt	818	EGY	E	Egyptian Pound	818	EGP	2

Country Codes				Currency Codes			
Name of Country or Entity	Numeric	Alpha	Warning Bulletin Region	Currency	Numeric	Alpha	Currency Exponent
El Salvador	222	SLV	B	El Salvador Colon U. S. Dollar	222 840	SVC USD	2 2
Equatorial Guinea	226	GNQ	E	CFA Franc BEAC✸	950	XAF	0
Eritrea	232	ERI	E	Nafka Ethiopian Birr	232 230	ERN ETB	2 2
Estonia	233	EST	D	Kroon	233	EEK	2
Ethiopia	230	ETH	E	Ethiopian Birr	230	ETB	2
Euro				Euro	978	EUR	2
Faeroe Islands	234	FRO	D	Danish Krone	208	DKK	2
Falkland Islands (Malvinas)	238	FLK	D	Falkland Islands Pound	238	FKP	2
Fiji	242	FJI	C	Fiji Dollar	242	FJD	2
Finland	246	FIN	D	Euro Markka	978 246	EUR FIM	2 2
France	250	FRA	D	Euro French Franc	978 250	EUR FRF	2 2
French Guiana	254	GUF	B	Euro French Franc	978 250	EUR FRF	2 2
French Polynesia	258	PYF	C	CFP Franc	953	XPF	0
French Southern Territories	260	ATF	E	Euro French Franc	978 250	EUR FRF	2 2
Gabon	266	GAB	E	CFA Franc BEAC✸	950	XAF	0
Gambia	270	GMB	E	Dalasi	270	GMD	2
Georgia	268	GEO	D	Lari	981	GEL	2
Germany, Republic of	280	DEU	D	Euro Deutsche Mark	978 280	EUR DEM	2 2
Ghana	288	GHA	E	Cedi	288	GHC	2
Gibraltar	292	GIB	D	Gibraltar Pound	292	GIP	2
Greece	300	GRC	D	Euro Drachma	978 300	EUR GRD	2 0

Country Codes				Currency Codes			
Name of Country or Entity	Numeric	Alpha	Warning Bulletin Region	Currency	Numeric	Alpha	Currency Exponent
Greenland	304	GRL	D	Danish Krone	208	DKK	2
Grenada	308	GRD	B	East Caribbean Dollar	951	XCD	2
Guadeloupe	312	GLP	B	Euro French Franc	978 250	EUR FRF	2 2
Guam	316	GUM	C	U.S. Dollar	840	USD	2
Guatemala	320	GTM	B	Quetzal	320	GTQ	2
Guinea	324	GIN	E	Guinea Franc	324	GNF	0
Guinea-Bissau	624	GNB	E	Guinea-Bissau Peso	624	GWP	2
Guyana	328	GUY	B	Guyana Dollar	328	GYD	2
Haiti	332	HTI	B	Gourde U.S. Dollar	332 840	HTG USD	2 2
Heard and McDonald Islands	334	HMD	C	Australian Dollar	036	AUD	2
Honduras	340	HND	B	Lempira	340	HNL	2
Hong Kong	344	HKG	C	Hong Kong Dollar	344	HKD	2
Hungary	348	HUN	D	Forint	348	HUF	2
Iceland	352	ISL	D	Iceland Krona	352	ISK	2
India	356	IND	C	Indian Rupee	356	INR	2
Indonesia	360	IDN	C	Rupiah	360	IDR	0
Iran, Islamic Republic of	364	IRN	E	Iranian Rial	364	IRR	2
Iraq	368	IRQ	E	Iraqi Dinar	368	IQD	3
Ireland	372	IRL	D	Euro Irish Pound	978 372	EUR IEP	2 2
Israel	376	ISR	D	New Israeli Shekel	376	ILS	2
Italy	380	ITA	D	Euro Italian Lira	978 380	EUR ITL	2 0
Jamaica	388	JAM	B	Jamaican Dollar	388	JMD	2
Japan	392	JPN	C	Yen	392	JPY	0

Country Codes				Currency Codes			
Name of Country or Entity	Numeric	Alpha	Warning Bulletin Region	Currency	Numeric	Alpha	Currency Exponent
Jordan	400	JOR	E	Jordanian Dinar	400	JOD	3
Kazakhstan	398	KAZ	D	Tenge	398	KZT	2
Kenya	404	KEN	E	Kenyan Shilling	404	KES	2
Kiribati	296	KIR	C	Australian Dollar	036	AUD	2
Korea, Democratic People's Republic of	408	PRK	C	North Korean Won	408	KPW	2
Korea, Republic of	410	KOR	C	Won	410	KRW	0
Kuwait	414	KWT	E	Kuwaiti Dinar	414	KWD	3
Kyrgyzstan	417	KGZ	D	Som	417	KGS	2
Lao People's Democratic Republic	418	LAO	C	Kip	418	LAK	2
Latvia	428	LVA	D	Latvian Lats	428	LVL	2
Lebanon	422	LBN	E	Lebanese Pound	422	LBP	2
Lesotho	426	LSO	E	Loti Rand	426 710	LSL ZAR	2 2
Liberia	430	LBR	E	Liberian Dollar	430	LRD	2
Libyan Arab Jamahiriya	434	LBY	E	Libyan Dinar	434	LYD	3
Liechtenstein	438	LIE	D	Swiss Franc	756	CHF	2
Lithuania	440	LTU	D	Lithuanian Litas	440	LTL	2
Luxembourg	442	LUX	D	Euro Luxembourg Franc	978 442	EUR LUF	2 0
Macau	446	MAC	C	Pataca	446	MOP	2
Macedonia, the former Yugoslav Republic of	807	MKD	D	Denar	807	MKD	2
Madagascar	450	MDG	E	Malagasy Franc	450	MGF	0
Malawi	454	MWI	E	Kwacha	454	MWK	2

Country Codes				Currency Codes			
Name of Country or Entity	Numeric	Alpha	Warning Bulletin Region	Currency	Numeric	Alpha	Currency Exponent
Malaysia	458	MYS	C	Malaysian Ringgit	458	MYR	2
Maldives	462	MDV	C	Rufiyaa	462	MVR	2
Mali	466	MLI	E	CFA Franc BCEAO✪	952	XOF	0
Malta	470	MLT	D	Maltese Lira	470	MTL	2
Marshall Islands	584	MHL	C	U.S. Dollar	840	USD	2
Martinique	474	MTQ	B	Euro French Franc	978 250	EUR FRF	2 2
Mauritania	478	MRT	E	Ouguiya	478	MRO	2
Mauritius	480	MUS	E	Mauritius Rupee	480	MUR	2
Mexico	484	MEX	B	Mexican Peso	484	MXN	2
Micronesia	583	FSM	C	U.S. Dollar	840	USD	2
Moldova, Republic of	498	MDA	D	Moldovan Leu	498	MDL	2
Monaco	492	MCO	D	Euro French Franc	978 250	EUR FRF	2 2
Mongolia	496	MNG	C	Tugrik	496	MNT	2
Montserrat	500	MSR	B	East Caribbean Dollar	951	XCD	2
Morocco	504	MAR	E	Moroccan Dirham	504	MAD	2
Mozambique	508	MOZ	E	Metical	508	MZM	2
Myanmar	104	MMR	C	Kyat	104	MMK	2
Namibia	516	NAM	E	Namibia Dollar Rand	516 710	NAD ZAR	2 2
Nauru	520	NRU	C	Australian Dollar	036	AUD	2
Nepal	524	NPL	C	Nepalese Rupee	524	NPR	2
Netherlands	528	NLD	D	Euro Netherlands Guilder	978 528	EUR NLG	2 2
Netherlands Antilles	530	ANT	B	Netherlands Antillian Guilder	532	ANG	2
New Caledonia	540	NCL	C	CFP Franc	953	XPF	0

Country Codes				Currency Codes			
Name of Country or Entity	Numeric	Alpha	Warning Bulletin Region	Currency	Numeric	Alpha	Currency Exponent
New Zealand	554	NZL	C	New Zealand Dollar	554	NZD	2
Nicaragua	558	NIC	B	Cordoba Oro	558	NIO	2
Niger	562	NER	E	CFA Franc BCEAO✪	952	XOF	0
Nigeria	566	NGA	E	Naira	566	NGN	2
Niue	570	NIU	C	New Zealand Dollar	554	NZD	2
Norfolk Island	574	NFK	C	Australian Dollar	036	AUD	2
Northern Mariana Islands	580	MNP	C	U.S. Dollar	840	USD	2
Norway	578	NOR	D	Norwegian Krone	578	NOK	2
Oman	512	OMN	E	Rial Omani	512	OMR	3
Pakistan	586	PAK	C	Pakistan Rupee	586	PKR	2
Palau	585	PLW	C	U.S. Dollar	840	USD	2
Palestinian Territory, Occupied	275	PSE	E	U.S. Dollar	840	USD	2
Panama	591	PAN	B	Balboa U.S. Dollar	590 840	PAB USD	2 2
Papua New Guinea	598	PNG	C	Kina	598	PGK	2
Paraguay	600	PRY	B	Guarani	600	PYG	0
Peru	604	PER	B	Nuevo Sol	604	PEN	2
Philippines	608	PHL	C	Philippine Peso	608	PHP	2
Pitcairn	612	PCN	C	New Zealand Dollar	554	NZD	2
Poland	616	POL	D	Zloty	985	PLN	2
Portugal	620	PRT	D	Euro Portuguese Escudo	978 620	EUR PTE	2 0
Puerto Rico	630	PRI	B	U.S. Dollar	840	USD	2
Qatar	634	QAT	E	Qatari Rial	634	QAR	2

Country Codes				Currency Codes			
Name of Country or Entity	Numeric	Alpha	Warning Bulletin Region	Currency	Numeric	Alpha	Currency Exponent
Reunion	638	REU	D	Euro French Franc	978 250	EUR FRF	2 2
Romania	642	ROM	D	Leu	642	ROL	0
Russian Federation	643	RUS	D	Russian Ruble New Ruble	810 643	RUR RUB	2 2
Rwanda	646	RWA	E	Rwanda Franc	646	RWF	0
Samoa	882	WSM	C	Tala	882	WST	2
San Marino	674	SMR	D	Euro Italian Lira	978 380	EUR ITL	2 0
Sao Tome and Principe	678	STP	E	Dobra	678	STD	2
Saudi Arabia	682	SAU	E	Saudi Riyal	682	SAR	2
Senegal	686	SEN	E	CFA Franc BCEAO✪	952	XOF	0
Seychelles	690	SYC	E	Seychelles Rupee	690	SCR	2
Sierra Leone	694	SLE	E	Leone	694	SLL	2
Singapore	702	SGP	C	Singapore Dollar	702	SGD	2
Slovakia	703	SVK	D	Slovak Koruna	703	SKK	2
Slovenia	705	SVN	D	Tolar	705	SIT	2
Solomon Islands	090	SLB	C	Solomon Islands Dollar	090	SBD	2
Somalia	706	SOM	E	Somali Shilling	706	SOS	2
South Africa	710	ZAF	E	Rand	710	ZAR	2
Spain	724	ESP	D	Euro Spanish Peseta	978 724	EUR ESP	2 0
Sri Lanka	144	LKA	C	Sri Lanka Rupee	144	LKR	2
St. Helena	654	SHN	D	St. Helena Pound	654	SHP	2
St. Kitts-Nevis	659	KNA	B	East Caribbean Dollar	951	XCD	2
St. Lucia	662	LCA	B	East Caribbean Dollar	951	XCD	2

Country Codes				Currency Codes			
Name of Country or Entity	Numeric	Alpha	Warning Bulletin Region	Currency	Numeric	Alpha	Currency Exponent
St. Pierre and Miquelon	666	SPM	A	Euro French Franc	978 250	EUR FRF	2 2
St. Vincent and the Grenandines	670	VCT	B	East Caribbean Dollar	951	XCD	2
Sudan	736	SDN	E	Sudanese Dinar	736	SDD	2
Suriname	740	SUR	B	Surinam Guilder	740	SRG	2
Svalbard and Jan Mayen Islands	744	SJM	D	Norwegian Krone	578	NOK	2
Swaziland	748	SWZ	E	Lilangeni	748	SZL	2
Sweden	752	SWE	D	Swedish Krona	752	SEK	2
Switzerland	756	CHE	D	Swiss Franc	756	CHF	2
Syrian Arab Republic	760	SYR	E	Syrian Pound	760	SYP	2
Taiwan	158	TWN	C	New Taiwan Dollar	901	TWD	2
Tajikistan	762	TJK	D	Somoni	972	TJS	2
Tanzania, United Republic of	834	TZA	E	Tanzanian Shilling	834	TZS	2
Thailand	764	THA	C	Baht	764	THB	2
Togo	768	TGO	E	CFA Franc BCEAO✪	952	XOF	0
Tokelau	772	TKL	C	New Zealand Dollar	554	NZD	2
Tonga	776	TON	C	Pa’anga	776	TOP	2
Trinidad and Tobago	780	TTO	B	Trinidad and Tobago Dollar	780	TTD	2
Tunisia	788	TUN	E	Tunisian Dinar	788	TND	3
Turkey	792	TUR	D	Turkish Lira	792	TRL	0
Turkmenistan	795	TKM	D	Manat	795	TMM	2
Turks and Caicos Islands	796	TCA	B	U.S. Dollar	840	USD	2
Tuvalu	798	TUV	C	Australian Dollar	036	AUD	2

Country Codes				Currency Codes			
Name of Country or Entity	Numeric	Alpha	Warning Bulletin Region	Currency	Numeric	Alpha	Currency Exponent
U.S. Minor Outlying Islands	581	UMI	C	U.S. Dollar	840	USD	2
Uganda	800	UGA	E	Uganda Shilling	800	UGX	0
Ukraine	804	UKR	D	Hryvnia	980	UAH	2
United Arab Emirates	784	ARE	E	UAE Dirham	784	AED	2
United Kingdom	826	GBR	D	Pound Sterling	826	GBP	2
United States	840	USA	1	U.S. Dollar	840	USD	2
Uruguay	858	URY	B	Peso Uruguayo	858	UYU	2
Uzbekistan	860	UZB	D	Uzbekistan Sum	860	UZS	2
Vanuatu	548	VUT	C	Vatu	548	VUV	0
Vatican City State (Holy See)	336	VAT	D	Euro Italian Lira	978 380	EUR ITL	2 0
Venezuela	862	VEN	B	Bolivar	862	VEB	2
Vietnam	704	VNM	C	Dong	704	VND	2
Virgin Islands, British	092	VGB	B	U.S. Dollar	840	USD	2
Virgin Islands, U.S.	850	VIR	B	U.S. Dollar	840	USD	2
Wallis and Futuna Islands	876	WLF	C	CFP Franc	953	XPF	0
Western Sahara	732	ESH	E	Moroccan Dirham	504	MAD	2
Yemen	887	YEM	E	Yemeni Rial	886	YER	2
Yugoslavia	891	YUG	D	Yugoslavian Dinar New Dinar	890 891	YUN YUM	2 2
Zambia	894	ZMB	E	Kwacha	894	ZMK	2
Zimbabwe	716	ZWE	E	Zimbabwe Dollar	716	ZWD	2

✪ CFA Franc BCEAO; Responsible authority: Banque Centrale des Etats de l'Afrique de l'Ouest
✸ CFA Franc BEAC; Responsible authority: Banque des Etats de l'Afrique Centrale

C Merchant Category Codes

Merchant Category Codes

Table C.1—All Transaction Category Codes

MCC	TCC	Description
-A-		
3351	A	Affiliated Auto Rental
3352	A	American International
3353	A	Brooks Rent A Car
3354	A	Action Auto Rental
3357	A	Hertz
3359	A	Payless Car Rental
3360	A	Snappy Car Rental
3361	A	Airways Rent-A-Car
3362	A	Altra Auto Rental
3364	A	Agency Rent-A-Car
3366	A	Budget Rent-A-Car
3368	A	Holiday Rent-A-Car
3370	A	Rent A Wreck
3374	A	Accent Rent-A-Car
3376	A	Ajax Rent-A-Car
3380	A	Triangle Rent A Car
3381	A	Europ Car
3385	A	Tropical Rent-A-Car
3386	A	Showcase Rental Cars
3387	A	Alamo Rent-A-Car
3389	A	Avis Rent A Car
3390	A	Dollar Rent A Car
3391	A	Europe By Car
3393	A	National Car Rental
3394	A	Kemwell Group Rent-A-Car
3395	A	Thrifty Car Rental

MCC	TCC	Description
3396	A	Tilden Rent-A-Car
3398	A	Econo Car Rent-A-Car
3400	A	Auto Host Car Rental
3405	A	Enterprise Rent-A-Car
3409	A	General Rent-A-Car
3412	A	A1 Rent-A-Car
3414	A	Godfrey National Rent-A-Car
3420	A	ANSA International
3421	A	Allstate Rent-A-Car
3423	A	Avcar Rent-A-Car
3425	A	Automate Rent-A-Car
3427	A	Avon Rent-A-Car
3428	A	Carey Rent-A-Car
3429	A	Insurance Rent-A-Car
3430	A	Major Rent-A-Car
3431	A	Replacement Rent-A-Car
3432	A	Reserve Rent-A-Car
3433	A	Ugly Duckling Rent-A-Car
3434	A	USA Rent-A-Car
3435	A	Value Rent-A-Car
3436	A	Autohansa Rent-A-Car
3437	A	Cite Rent-A-Car
3438	A	Interent Rent-A-Car
3439	A	Milleville Rent-A-Car
3441	A	Advantage Rent A Car
7512	A	Automobile Rental Agency—not elsewhere classified
7513	A	Truck Rental
7519	A	Motor Home and Recreational Vehicle Rental
-C-		
6010	C	Member Financial Institution—Manual Cash Disbursements

MCC	TCC	Description
-F-		
5811	F	Caterers
5812	F	Eating Places, Restaurants
5813	F	Bars, Cocktail Lounges, Discotheques, Nightclubs, and Taverns—Drinking Places (Alcoholic Beverages)
5814	F	Fast Food Restaurants
-H-		
3501	H	Holiday Inns
3502	H	Best Western Hotels
3503	H	Sheraton Hotels
3504	H	Hilton Hotels
3505	H	Forte Hotels
3506	H	Golden Tulip Hotels
3507	H	Friendship Inns
3508	H	Quality Inns
3509	H	Marriott
3510	H	Days Inns
3511	H	Arabella Hotels
3512	H	Intercontinental Hotels
3513	H	Westin Hotels
3514	H	Amerisuites
3515	H	Rodeway Inns
3516	H	LaQuinta Inns
3517	H	Americana Hotels
3518	H	Sol Hotels
3519	H	Pullman International Hotels
3520	H	Meridien Hotels
3522	H	Tokyo Hotel
3523	H	Peninsula Hotels
3524	H	WelcomGroup Hotels
3525	H	Dunfey Hotels

MCC	TCC	Description
3526	H	Prince Hotels
3527	H	Downtowner Passport
3528	H	Red Lion Inns
3529	H	CP (Canadian Pacific) Hotels
3530	H	Renaissance Hotels
3533	H	Hotel Ibis
3534	H	Southern Pacific Hotels
3535	H	Hilton International
3536	H	AMFAC Hotels
3537	H	ANA Hotels
3538	H	Concorde Hotels
3539	H	Summerfield Suites Hotel
3540	H	Iberotel Hotels
3541	H	Hotel Okura
3542	H	Royal Hotels
3543	H	Four Seasons Hotels
3544	H	Ciga Hotels
3545	H	Shangri-La International
3546	H	Sierra Suites Hotel
3548	H	Hotels Melia
3549	H	Auberge des Governeures
3550	H	Regal 8 Inns
3551	H	Mirage Hotel and Casino
3552	H	Coast Hotel
3553	H	Park Inns International
3555	H	Treasure Island Hotel and Casino
3558	H	Jolly Hotels
3561	H	Golden Nugget
3562	H	Comfort Inns
3563	H	Journey’s End Motels
3564	H	Sam’s Town Hotel and Casino

MCC	TCC	Description
3565	H	Relax Inns
3568	H	Ladbroke Hotels
3570	H	Forum Hotels
3572	H	Miyako Hotel
3573	H	Sandman Hotels
3574	H	Venture Inn
3575	H	Vagabond Hotels
3577	H	Mandarin Oriental Hotels
3579	H	Hotel Mercure
3581	H	Delta Hotels
3582	H	California Hotel and Casino
3583	H	SAS Hotels
3584	H	Princess Hotels International
3585	H	Hungar Hotels
3586	H	Sokos Hotel
3587	H	Doral Hotels
3588	H	Helmsley Hotels
3590	H	Fairmont Hotels
3591	H	Sonesta Hotels
3592	H	Omni Hotels
3593	H	Cunard Hotels
3595	H	Hospitality Inns
3598	H	Regent International Hotel
3599	H	Pannonia Hotels
3603	H	Noah's Hotel
3612	H	Movenpick Hotels
3615	H	Travelodge
3620	H	Binion's Horseshoe Club
3622	H	Merlin Hotel Group
3623	H	Dorint Hotels
3624	H	Lady Luck Hotel and Casino

MCC	TCC	Description
3625	H	Hotel Universale
3628	H	Excalibur Hotel and Casino
3629	H	Dan Hotels
3631	H	Sleep Inns
3632	H	Phoenician, The
3633	H	Rank Hotels
3634	H	Swissotel
3635	H	Reso Hotel
3636	H	Sarova Hotels
3637	H	Ramada Inns
3638	H	Howard Johnson
3639	H	Mount Charlotte Thistle
3640	H	Hyatt Hotels
3641	H	Sofitel Hotels
3642	H	Novotel Hotels
3643	H	Steigenberger Hotels
3644	H	EconoLodges
3645	H	Queens Moat Houses
3646	H	Swallow Hotels
3647	H	Husa Hotels
3648	H	De Vera Hotels
3649	H	Radisson Hotels
3650	H	Red Roof Inns
3651	H	Imperial London Hotels
3652	H	Embassy Hotels
3653	H	Penta Hotels
3654	H	Loews Hotels
3655	H	Scandic Hotels
3656	H	Sara Hotels
3657	H	Oberoi Hotels
3658	H	Otani Hotels

MCC	TCC	Description
3659	H	Taj Hotels International
3660	H	Knights Inn
3661	H	Metropole Hotels
3662	H	Circus Circus Hotel and Casino
3663	H	Hoteles El Presidente
3664	H	Flag Inns
3665	H	Hampton Inn Hotels
3666	H	Stakis Hotels
3667	H	Luxor Hotel and Casino
3668	H	Maritim Hotels
3669	H	Eldorado Hotel and Casino
3670	H	Arcade Hotels
3671	H	Arctia Hotels
3672	H	Campanile Hotels
3673	H	IBUSZ Hotels
3674	H	Rantasipi Hotels
3675	H	Interhotel CEDOK
3676	H	Monte Carlo Hotel and Casino
3677	H	Climat de France Hotels
3678	H	Cumulus Hotels
3679	H	Silver Legacy Hotel and Casino
3680	H	Hoteis Othan
3681	H	Adams Mark Hotels
3682	H	Sahara Hotel and Casino
3683	H	Bradbury Suites
3684	H	Budget Hosts Inns
3685	H	Budgetel Inns
3686	H	Susse Chalet
3687	H	Clarion Hotels
3688	H	Compri Hotels
3689	H	Consort Hotels

MCC	TCC	Description
3690	H	Courtyard Inns
3691	H	Dillon Inn
3692	H	Doubletree Hotels
3693	H	Drury Inn
3694	H	Economy Inns of America
3695	H	Embassy Suites
3696	H	Excel Inn
3697	H	Fairfield Hotels
3698	H	Harley Hotels
3699	H	Midway Motor Lodge
3700	H	Motel 6
3701	H	La Mansion Del Rio
3702	H	The Registry Hotels
3703	H	Residence Inn
3704	H	Royce Hotels
3705	H	Sandman Inn
3706	H	Shilo Inn
3707	H	Shoney's Inn
3708	H	Virgin River Hotel and Casino
3709	H	Super 8 Motels
3710	H	The Ritz Carlton
3711	H	Flag Inns (Australia)
3712	H	Buffalo Bill's Hotel and Casino
3713	H	Quality Pacific Hotel
3714	H	Four Seasons (Australia) Hotels
3715	H	Fairfield Inn
3716	H	Carlton Hotels
3717	H	City Lodge Hotels
3718	H	Karos Hotels
3719	H	Protea Hotels
3720	H	Southern Sun Hotels

MCC	TCC	Description
3721	H	Conrad Hotels
3722	H	Wyndham Hotels
3723	H	Rica Hotels
3724	H	Inter Nor Hotels
3725	H	Sea Pines Resort
3726	H	Rio Suites
3727	H	Broadmoor Hotel
3728	H	Ballys Hotel and Casino
3729	H	John Ascuagas Nugget
3730	H	MGM Grand Hotel
3731	H	Harrahs Hotels and Casinos
3732	H	Opryland Hotel
3733	H	Boca Raton Resort
3734	H	Harvey Bristol Hotels
3735	H	Masters Economy Inns
3736	H	Colorado Belle Edgewater Resort
3737	H	Riviera Hotel and Casino
3738	H	Tropicana Resort and Casino
3739	H	Woodside Hotels and Resorts
3740	H	TownePlace Suites
3741	H	Millennium Broadway Hotel
3742	H	Club Med
3743	H	Biltmore Hotel and Suites
3744	H	Carefree Resorts
3745	H	St. Regis Hotel
3746	H	Eliot Hotels
3747	H	Club Corp/Club Resorts
3748	H	Wellesley Inns
3749	H	The Beverly Hills Hotel
3750	H	Crowne Plaza Hotels
3751	H	Homewood Suites

MCC	TCC	Description
3752	H	Peabody Hotels
3753	H	Greenbriar Resorts
3754	H	Amelia Island Plantation
3755	H	The Homestead
3756	H	South Seas Resorts
3757	H	Canyon Ranch
3758	H	Kahala Mandarin Oriental Hotel
3759	H	The Orchid at Mauna Lani
3760	H	Halekulani Hotel/Waikiki Parc
3761	H	Primadonna Hotel and Casino
3762	H	Whiskey Pete's Hotel and Casino
3763	H	Chateau Elan Winery and Resort
3764	H	Beau Rivage Hotel and Casino
3765	H	Bellagio Hotel and Casino
3766	H	Fremont Hotel and Casino
3767	H	Main Street Hotel and Casino
3768	H	Silver Star Hotel and Casino
3769	H	Stratosphere Hotel and Casino
3770	H	SpringHill Suites
3771	H	Caesars Hotel and Casino
3772	H	Nemacolin Woodlands
3773	H	Venetian Resort Hotel and Casino, The
3774	H	New York, New York Hotel and Casino
3775	H	Sands Resort
3776	H	Nevele Grande Resort and Country Club
3777	H	Mandalay Bay Resort
3778	H	Four Points Hotels
3779	H	W Hotels
3780	H	Disneyland Hotels
3781	H	The Patricia Grand Resort Hotels
3782	H	Rosen Hotels and Resorts

MCC	TCC	Description
3783	H	Town and Country Resort & Convention Center
3784	H	First Hospitality Hotels
3785	H	Outrigger Hotels & Resorts
3786	H	Ohana Hotels of Hawaii
3787	H	Caribe Royale Resort Suites & Villas
3788	H	Ala Moana Hotel
4411	H	Cruise Lines
7011	H	Lodging—Hotels, Motels, Resorts—not elsewhere classified
-O-		
8050	O	Nursing and Personal Care Facilities
8062	O	Hospitals
8220	O	Colleges, Universities, Professional Schools, and Junior Colleges
-P-		
6532	P	Payment Service Provider—Member Financial Institution—Payment Transaction
6533	P	Payment Service Provider—Merchant—Payment Transaction
-R-		
0742	R	Veterinary Services
0763	R	Agricultural Cooperatives
0780	R	Horticultural and Landscaping Services
1520	R	General Contractors—Residential and Commercial
1711	R	Air Conditioning, Heating, and Plumbing Contractors
1731	R	Electrical Contractors
1740	R	Insulation, Masonry, Plastering, Stonework, and Tile Setting Contractors
1750	R	Carpentry Contractors
1761	R	Roofing and Siding, Sheet Metal Work Contractors
1771	R	Concrete Work Contractors
1799	R	Contractors, Special Trade—not elsewhere classified
4119	R	Ambulance Services
4121	R	Limousines and Taxicabs

MCC	TCC	Description
4214	R	Motor Freight Carriers, Trucking—Local/Long Distance, Moving and Storage Companies, Local Delivery
4215	R	Courier Services—Air and Ground, Freight Forwarders
4225	R	Public Warehousing—Farm Products, Refrigerated Goods, Household Goods Storage
4457	R	Boat Leases and Boat Rentals
4468	R	Marinas, Marine Service/Supplies
4582	R	Airports, Airport Terminals, Flying Fields
4812	R	Telecommunication Equipment Including Telephone Sales
4814	R	Calls Through Use of Magnetic Stripe–Reading Telephones
4821	R	Telegraph Services
4899	R	Cable and other Pay Television Services
4900	R	Utilities—Electric, Gas, Sanitary, Water
5200	R	Home Supply Warehouse Stores
5211	R	Building Materials, Lumber Stores
5231	R	Glass, Paint, Wallpaper Stores
5251	R	Hardware Stores
5261	R	Lawn and Garden Supply Stores
5271	R	Mobile Home Dealers
5300	R	Wholesale Clubs
5309	R	Duty Free Stores
5310	R	Discount Stores
5311	R	Department Stores
5331	R	Variety Stores
5399	R	Miscellaneous General Merchandise
5411	R	Grocery Stores, Supermarkets
5422	R	Freezer, Locker Meat Provisioners
5441	R	Candy, Nut, Confectionery Stores
5451	R	Dairy Products Stores
5462	R	Bakeries
5499	R	Miscellaneous Food Stores—Convenience Stores, Markets, Specialty Stores, and Vending Machines

MCC	TCC	Description
5511	R	Automobile and Truck Dealers—Sales, Service, Repairs, Parts, and Leasing
5521	R	Automobile and Truck Dealers—(Used Only)—Sales
5531	R	Auto Store, Home Supply Stores
5532	R	Automotive Tire Stores
5533	R	Automotive Parts, Accessories Stores
5541	R	Service Stations (with or without Ancillary Services)
5542	R	Fuel Dispenser, Automated
5551	R	Boat Dealers
5561	R	Camper Dealers, Recreational and Utility Trailers
5571	R	Motorcycle Shops and Dealers
5592	R	Motor Home Dealers
5598	R	Snowmobile Dealers
5599	R	Miscellaneous Automotive, Aircraft, and Farm Equipment Dealers—not elsewhere classified
5611	R	Men’s and Boys’ Clothing and Accessories Stores
5621	R	Women’s Ready to Wear Stores
5631	R	Women’s Accessory and Specialty Stores
5641	R	Children’s and Infants’ Wear Stores
5651	R	Family Clothing Stores
5655	R	Sports Apparel, Riding Apparel Stores
5661	R	Shoe Stores
5681	R	Furriers and Fur Shops
5691	R	Men’s and Women’s Clothing Stores
5697	R	Alterations, Mending, Seamstresses, Tailors
5698	R	Wig and Toupee Shops
5699	R	Accessory and Apparel Stores—Miscellaneous
5712	R	Equipment, Furniture, and Home Furnishings Stores (except Appliances)
5713	R	Floor Covering Stores
5714	R	Drapery, Upholstery, and Window Coverings Stores
5718	R	Fireplace, Fireplace Screens and Accessories Stores

MCC	TCC	Description
5719	R	Miscellaneous House Furnishing Specialty Shops
5722	R	Household Appliance Stores
5732	R	Electronic Sales
5733	R	Music Stores—Musical Instruments, Pianos, Sheet Music
5734	R	Computer Software Stores
5735	R	Record Shops
5912	R	Drug Stores, Pharmacies
5921	R	Package Stores, Beer, Wine, Liquor
5931	R	Second Hand Stores, Used Merchandise Stores
5932	R	Antique Shops—Sales, Repairs, and Restoration Services
5933	R	Pawn Shops
5935	R	Salvage and Wrecking Yards
5937	R	Antique Reproduction Stores
5940	R	Bicycle Shops—Sales and Service
5941	R	Sporting Goods Stores
5942	R	Book Stores
5943	R	Office, School Supply, and Stationery Stores
5944	R	Clock, Jewelry, Watch, and Silverware Store
5945	R	Game, Toy, and Hobby Shops
5946	R	Camera and Photographic Supply Stores
5947	R	Card, Gift, Novelty, and Souvenir Shops
5948	R	Leather Goods and Luggage Stores
5949	R	Fabric, Needlework, Piece Goods, and Sewing Stores
5950	R	Crystal and Glassware Stores
5963	R	Door-to-Door Sales
5970	R	Artist Supply Stores, Craft Shops
5971	R	Art Dealers and Galleries
5972	R	Stamp and Coin Stores—Philatelic and Numismatic Supplies
5973	R	Religious Goods Stores
5975	R	Hearing Aids—Sales, Service, Supply Stores
5976	R	Orthopedic Goods—Artificial Limb Stores

MCC	TCC	Description
5977	R	Cosmetic Stores
5978	R	Typewriter Stores—Rentals, Sales, Service
5983	R	Fuel Dealers—Coal, Fuel Oil, Liquefied Petroleum, Wood
5992	R	Florists
5993	R	Cigar Stores and Stands
5994	R	News Dealers and Newsstands
5995	R	Pet Shops—Pet Food and Supplies
5996	R	Swimming Pools—Sales and Supplies
5997	R	Electric Razor Stores—Sales and Service
5998	R	Tent and Awning Shops
5999	R	Miscellaneous and Specialty Retail Stores
6012	R	Financial Institutions—Merchandise and Services
6211	R	Securities—Brokers/Dealers
6300	R	Insurance Sales, Underwriting, and Premiums
7012	R	Timeshares
7032	R	Recreational and Sporting Camps
7033	R	Campgrounds and Trailer Parks
7210	R	Cleaning, Garment, and Laundry Services
7211	R	Laundry Services—Family and Commercial
7216	R	Dry Cleaners
7217	R	Carpet and Upholstery Cleaning
7221	R	Photographic Studios
7230	R	Barber and Beauty Shops
7251	R	Hat Cleaning Shops, Shoe Repair Shops, Shoe Shine Parlors
7261	R	Funeral Service and Crematories
7273	R	Dating and Escort Services
7276	R	Tax Preparation Service
7277	R	Debt, Marriage, Personal—Counseling Service
7278	R	Buying/Shopping Clubs, Services
7296	R	Clothing Rental—Costumes, Uniforms, and Formal Wear
7297	R	Massage Parlors

MCC	TCC	Description
7298	R	Health and Beauty Spas
7299	R	Other Services—not elsewhere classified
7311	R	Advertising Services
7321	R	Consumer Credit Reporting Agencies
7333	R	Commercial Art, Graphics, Photography
7338	R	Quick Copy, Reproduction, and Blueprinting Services
7339	R	Stenographic and Secretarial Support Services
7342	R	Exterminating and Disinfecting Services
7349	R	Cleaning and Maintenance, Janitorial Services
7361	R	Employment Agencies, Temporary Help Services
7372	R	Computer Programming, Data Processing, and Integrated Systems Design Services
7392	R	Consulting, Management, and Public Relations Services
7393	R	Detective Agencies, Protective Agencies, Security Services including Armored Cars, Guard Dogs
7394	R	Equipment Rental and Leasing Services, Furniture Rental, Tool Rental
7395	R	Photo Developing, Photofinishing Laboratories
7399	R	Business Services—not elsewhere classified
7523	R	Automobile Parking Lots and Garages
7531	R	Automotive Body Repair Shops
7534	R	Tire Retreading and Repair Shops
7535	R	Automotive Paint Shops
7538	R	Automotive Service Shops
7542	R	Car Washes
7549	R	Towing Services
7622	R	Electronic Repair Shops
7623	R	Air Conditioning and Refrigeration Repair Shops
7629	R	Appliance Repair Shops, Electrical and Small
7631	R	Clock, Jewelry, and Watch Repair Shops
7641	R	Furniture—Reupholstery and Repair, Refinishing
7692	R	Welding Repair

MCC	TCC	Description
7699	R	Miscellaneous Repair Shops and Related Services
7832	R	Motion Picture Theaters
7841	R	Video Entertainment Rental Stores
7911	R	Dance Halls, Schools, and Studios
7922	R	Theatrical Producers (except Motion Pictures), Ticket Agencies
7929	R	Bands, Orchestras, and Miscellaneous Entertainers—not elsewhere classified
7932	R	Pool and Billiard Establishments
7933	R	Bowling Alleys
7941	R	Athletic Fields, Commercial Sports, Professional Sports Clubs, Sports Promoters
7991	R	Tourist Attractions and Exhibits
7992	R	Golf Courses, Public
7993	R	Video Amusement Game Supplies
7994	R	Video Game Arcades/Establishments
7996	R	Amusement Parks, Carnivals, Circuses, Fortune Tellers
7997	R	Clubs—Country Clubs, Membership (Athletic, Recreation, Sports), Private Golf Courses
7998	R	Aquariums, Dolphinariums, and Seaquariums
7999	R	Recreation Services—not elsewhere classified
8011	R	Doctors—not elsewhere classified
8021	R	Dentists, Orthodontists
8031	R	Osteopathic Physicians
8041	R	Chiropractors
8042	R	Optometrists, Ophthalmologists
8043	R	Opticians, Optical Goods, and Eyeglasses
8049	R	Chiropodists, Podiatrists
8071	R	Dental and Medical Laboratories
8099	R	Health Practitioners, Medical Services—not elsewhere classified
8111	R	Attorneys, Legal Services
8211	R	Schools, Elementary and Secondary
8241	R	Schools, Correspondence

MCC	TCC	Description
8244	R	Schools, Business and Secretarial
8249	R	Schools, Trade and Vocational
8299	R	Schools and Educational Services—not elsewhere classified
8351	R	Child Care Services
8398	R	Organizations, Charitable and Social Service
8641	R	Associations—Civic, Social, and Fraternal
8651	R	Organizations, Political
8661	R	Organizations, Religious
8675	R	Automobile Associations
8699	R	Organizations, Membership—not elsewhere classified
8911	R	Architectural, Engineering, and Surveying Services
8931	R	Accounting, Auditing, and Bookkeeping Services
8999	R	Professional Services—not elsewhere classified
9211	R	Court Costs including Alimony and Child Support
9222	R	Fines
9223	R	Bail and Bond Payments
9311	R	Tax Payments
9399	R	Government Services—not elsewhere classified
9402	R	Postal Services—Government Only
9751	R	U.K. Supermarkets, Electronic Hot File
9752	R	U.K. Petrol Stations, Electronic Hot File
-T-		
2741	T	Miscellaneous Publishing and Printing*
2791	T	Typesetting, Plate Making, and Related Services*
2842	T	Sanitation, Polishing, and Specialty Cleaning Preparations
4813	T	Key-entry Telecom Merchant providing single local and long-distance phone calls using a central access number in a non–face-to-face environment using key entry
4814	T	Telecommunication Services including but not limited to prepaid phone services and recurring phone services
4815	T	MasterPhone™ Telephone Service—Monthly Summary Telephone Charges

Wholesale Distributors and Manufacturers are noted by an asterisk.

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MCC	TCC	Description
4816	T	Computer Network/Information Services
5013	T	Motor Vehicle Supplies and New Parts*
5021	T	Office and Commercial Furniture*
5039	T	Construction Materials—not elsewhere classified*
5044	T	Office, Photographic, Photocopy, and Microfilm Equipment*
5045	T	Computers, Computer Peripheral Equipment, Software*
5046	T	Commercial Equipment—not elsewhere classified*
5047	T	Dental/Laboratory/Medical/Ophthalmic Hospital Equipment and Supplies*
5051	T	Metal Service Centers and Offices*
5065	T	Electrical Parts and Equipment*
5072	T	Hardware Equipment and Supplies*
5074	T	Plumbing and Heating Equipment*
5085	T	Industrial Supplies—not elsewhere classified*
5094	T	Precious Stones and Metals, Watches and Jewelry*
5099	T	Durable Goods—not elsewhere classified*
5111	T	Stationery, Office Supplies, Printing and Writing Paper*
5122	T	Drugs, Drug Proprietaries, and Druggists Sundries*
5131	T	Piece Goods, Notions, and Other Dry Goods*
5137	T	Men’s, Women’s, and Children’s Uniforms and Commercial Clothing*
5139	T	Commercial Footwear*
5169	T	Chemicals and Allied Products—not elsewhere classified*
5172	T	Petroleum and Petroleum Products*
5192	T	Books, Periodicals and Newspapers*
5193	T	Florists Supplies, Nursery Stock, and Flowers*
5198	T	Paints, Varnishes, and Supplies*
5199	T	Nondurable Goods—not elsewhere classified*
5960	T	Direct Marketing—Insurance Services
5962	T	Direct Marketing—Travel-Related Arrangement Services
5964	T	Direct Marketing—Catalog Merchants

The Abbreviated Airline Name is in bold following the actual airline name.

MCC	TCC	Description
5965	T	Direct Marketing—Combination Catalog and Retail Merchants
5966	T	Direct Marketing—Outbound Telemarketing Merchants
5967	T	Direct Marketing—Inbound Telemarketing Merchants
5968	T	Direct Marketing—Continuity/Subscription Merchants
5969	T	Direct Marketing—Other Direct Marketers—not elsewhere classified
6531	T	Payment Service Provider—Payment Transfer for a Purchase
7375	T	Information Retrieval Services*
7379	T	Computer Maintenance, Repair, and Services—not elsewhere classified*
7829	T	Motion Picture and Video Tape Production and Distribution*
8734	T	Testing Laboratories (Non-Medical)*
9405	T	Intra-Government Purchases—Government Only
-U-		
4829	U	Money Transfer—Merchant
6050	U	Quasi Cash—Member Financial Institution
6051	U	Quasi Cash—Merchant
6529	U	Remote Stored Value Load—Member Financial Institution
6530	U	Remote Stored Value Load—Merchant
6534	U	Money Transfer—Member Financial Institution
7511	U	Truck Stop Transactions
7995	U	Gambling Transactions
-X-		
3000	X	United Airlines—**UNITED**
3001	X	American Airlines—**AMERICAN**
3002	X	Pan American—**PAN AM AIR**
3004	X	Trans World Airlines—**TWA**
3005	X	British Airways—**BRITISH AWYS**
3006	X	Japan Air Lines—**JAL**
3007	X	Air France—**AIR FRAN**
3008	X	Lufthansa—**LUFTHAN**
3009	X	Air Canada—**AIR CANA**

MCC	TCC	Description
3010	X	Royal Dutch Airlines (KLM)—**KLM**
3011	X	Aeroflot—**AEROFLOT**
3012	X	Qantas—**QANTAS**
3013	X	Alitalia—**ALITALIA**
3014	X	Saudi Arabian Airlines—**SAUDI AI**
3015	X	Swissair—**SWISSAIR**
3016	X	Scandinavian Airline System (SAS)—**SAS**
3017	X	South African Airways—**SAFRICAN**
3018	X	Varig (Brazil)—**VARIG**
3020	X	Air India—**AIR-INDI**
3021	X	Air Algerie—**AIRALGER**
3022	X	Philippine Airlines—**PHILIPP**
3023	X	Mexicana—**MEXICANA**
3024	X	Pakistan International—**PAKISTAN**
3025	X	Air New Zealand Limited International—**AIR NZ**
3026	X	Emirates Airlines—**EMIRATES**
3027	X	Union de Transports Aeriens—**UTAAIR**
3028	X	Air Malta—**AIRMALTA**
3029	X	Sabena—**SABENA**
3030	X	Aerolineas Argentinas—**AERO ARG**
3031	X	Olympic Airways—**OLYMPICA**
3032	X	El Al—**EL AL**
3033	X	Ansett Airlines—**ANSETT**
3034	X	Australian Airways (TAA)—**TAA**
3035	X	Tap (Portugal)—**TAP**
3036	X	VASP (Brazil)—**VASP**
3037	X	EgyptAir—**EGYPTAIR**
3038	X	Kuwait Airways—**KUWAIT**
3039	X	Avianca—**AVIANCA**
3040	X	Gulf Air (Bahrain)—**GULF AIR**
3041	X	Balkan—Bulgarian Airlines—**BALKAN**

MCC	TCC	Description
3042	X	Finnair—**FINNAIR**
3043	X	Aer Lingus—**AERLING**
3044	X	Air Lanka—**AIR LANKA**
3045	X	Nigeria Airways—**NIGERIA**
3046	X	Cruzerio do Sul (Brazil)—**CRUZERIO**
3047	X	THY (Turkey)—**THY**
3048	X	Royal Air Maroc—**AIRMARO**
3049	X	Tunis Air—**TUNIS AI**
3050	X	Icelandair—**ICELANDA**
3051	X	Austrian Airlines—**AUSTRIAN**
3052	X	Lan-Chile—**LANCHILE**
3053	X	AVIACO (Spain)—**AVIACO**
3054	X	LADECO (Chile)—**LADECO**
3055	X	LAB (Bolivia)—**LAB**
3056	X	Quebecaire—**QUEBECAI**
3057	X	East-West Airlines (Australia)—**E/WAIRL**
3058	X	Delta—**DELTA**
3060	X	Northwest Airlines—**NWA AIR**
3061	X	Continental—**CONTINEN**
3063	X	U.S. Airways—**USAIRWYS**
3064	X	Adria Airways—**ADRIA AIR**
3065	X	Air Inter—**AIRINTER**
3066	X	Southwest Airlines—**SOUTHWES**
3067	X	Vanguard Airlines—**VANGUARD**
3071	X	Air British Columbia—**AIR B R C**
3075	X	Singapore Airlines—**SINGAPOR**
3076	X	Aeromexico—**AEROMEXI**
3077	X	Thai Airways—**THAIAIRW**
3078	X	China Airlines—**CHINAAIR**
3081	X	Nordair—**NORDAIR**
3082	X	Korean Airlines—**KOREAN**

MCC	TCC	Description
3083	X	Air Afrique—**AIRAFRIQUE**
3084	X	Eva Airways—**EVA AIR**
3085	X	Midwest Express Airlines—**MIDWEST EXP**
3086	X	Carnival Airlines—**CARNIVALAIR**
3087	X	Metro Airlines—**METROAI**
3088	X	Croatia Air—**CROATIA AIR**
3089	X	Transaero—**TRANSAERO**
3090	X	Uni Airways—**UNIAIR**
3092	X	Midway Airlines—**MIDWAY**
3094	X	Zambia Airways—**ZAMBIA A**
3096	X	Air Zimbabwe—**AIRZIMBA**
3097	X	Spanair—**SPANAIR**
3098	X	Asiana Airlines—**ASIANA AIR**
3099	X	Cathay Pacific—**CATHAYPA**
3100	X	Malaysian Airline System—**MALAY AI**
3102	X	Iberia—**IBERIA**
3103	X	Garuda (Indonesia)—**GARUDA**
3106	X	Braathens S.A.F.E. (Norway)—**BRAATHEN**
3110	X	Wings Airways—**WINGSAIR**
3111	X	British Midland—**BRITISH M**
3112	X	Windward Island—**WINDWARD**
3115	X	Tower Air—**TOWERAIR**
3117	X	Venezolana International de Aviacion—**VIASA**
3118	X	Valley Airlines—**VALLEYA**
3125	X	Tan Airlines—**TAN AIR**
3126	X	Talair—**TALAIR**
3127	X	Taca International—**TACA INT**
3129	X	Surinam Airways—**SURINAM**
3130	X	Sunworld International Airways—**SUNWORLD**
3132	X	Frontier Airlines—**FRONTIER AIR**
3133	X	Sunbelt Airlines—**SUNBELT**

MCC	TCC	Description
3135	X	Sudan Airways—**SUDANAIR**
3136	X	Qatar Airways Company W.L.L.—**QATAR AIR**
3137	X	Singleton Air—**SINGLETO**
3138	X	Simmons Airlines—**SIMMONS**
3143	X	Scenic Airlines—**SCENIC A**
3144	X	Virgin Atlantic—**VIR ATL**
3145	X	San Juan Airlines—**SAN JUAN**
3146	X	Luxair—**LUXAIR**
3148	X	Air Littoral, S.A.—**LITTORAL**
3151	X	Air Zaire—**AIRZAIRE**
3154	X	Princeville—**PRINCEVI**
3156	X	GO FLY Ltd.—**GOFLY**
3159	X	Provincetown-Boston Airways—**PBA**
3161	X	All Nippon Airways—**ANAAIR**
3164	X	Norontair—**NORONTAI**
3165	X	New York Helicopter—**NY HELI**
3170	X	Mount Cook—**MT COOK**
3171	X	Canadian Airlines—**CANADIAN**
3172	X	Nation Air—**NATIONAI**
3174	X	JetBlue Airways—**JETBLUE**
3175	X	Middle East Air—**MIDEASTA**
3176	X	Metroflight Airlines—**METROFLT**
3178	X	Mesa Air—**MESA AIR**
3181	X	Malev Hungarian Airlines—**MALEV**
3182	X	LOT - Polish Airlines—**LOT**
3184	X	LIAT—**LIAT**
3185	X	LAV Linea Aeropostal Venezolana—**LAV**
3186	X	LAP Lineas Aereas Paraguayas—**LAP**
3187	X	LACSA (Costa Rica)—**LACSA**
3190	X	Jugoslav Air—**JUGOSLAV**
3191	X	Island Airlines—**ISLANDAI**

MCC	TCC	Description
3192	X	Iran Air—**IRANAIR**
3193	X	Indian Airlines—**INDIAN A**
3196	X	Hawaiian Air—**HAWAIIAN**
3197	X	Havasu Airlines—**HAVASUAI**
3200	X	Guyana Airways—**GUYANA A**
3203	X	Golden Pacific Air—**GOLDPACA**
3204	X	Freedom Airlines—**FREEDOM**
3206	X	China Eastern Airlines—**CHINEASTAIR**
3212	X	Dominicana de Aviacion—**DOMINICA**
3215	X	Dan Air Services—**DANAIRSE**
3216	X	Cumberland Airlines—**CUMBERLN**
3217	X	CSA Ceskoslovenske Aerolinie—**CSA**
3218	X	Crown Air—**CROWN AI**
3219	X	Compania Panamena de Aviacion (Copa)—**COPA**
3220	X	Compania Faucett—**COMPANIA**
3221	X	Transportes Aeros Militares Ecuatorianos—**TAME AIR**
3222	X	Command Airways—**COMMAND**
3223	X	Comair—**COMAIR**
3228	X	Cayman Airways—**CAYMANAI**
3229	X	SAETA (Sociedad Ecuatorianas De Transportes Aereo)—**SAETAAIR**
3231	X	SAHSA (Servicio Aero de Honduras)—**SAHSA**
3233	X	Capitol Air—**CAPITOL**
3234	X	BWIA International—**BWIA**
3235	X	Brockway Air—**BROCKWAY**
3238	X	Bemidji Airlines—**BEMIDJI**
3239	X	Bar Harbor Airlines—**BARHARBO**
3240	X	Bahamasair—**BAHAMASA**
3241	X	Aviateca (Guatemala)—**AVIATECA**
3242	X	Avensa—**AVENSA**
3243	X	Austrian Air Service—**AUSTRAIR**
3251	X	Aloha Airlines—**ALOH AI**

MCC	TCC	Description
3252	X	ALM Antilean Airlines—**ALM**
3253	X	America West—**AMERWEST**
3254	X	US Air Shuttle—**USAIRSHUTL**
3256	X	Alaska Airlines Inc.—**ALASKA A**
3259	X	American Trans Air—**ATA AIR**
3261	X	Air China—**AIR CHINA**
3262	X	Reno Air, Inc.—**RENO AIR**
3263	X	Aero Servicio Carabobo—**ASC AIRLINE**
3266	X	Air Seychelles—**AIR SEYC**
3267	X	Air Panama International—**AIR PANA**
3280	X	Air Jamaica—**AIR JAMA**
3282	X	Air Djibouti—**AIR DJIB**
3284	X	Aero Virgin Islands—**AERVIRGI**
3285	X	Aero Peru—**AEROPERU**
3286	X	Aero. Nicaraguenses—**AERO NIC**
3287	X	Aero Coach Aviation—**AEROCOAC**
3292	X	Cyprus Airways—**CYPRUSA**
3293	X	Ecuatoriana—**ECUATORI**
3294	X	Ethiopian Airlines—**ETHIOPIA**
3295	X	Kenya Airways—**KENYAAIR**
3297	X	Tarom Romanian Air Transport—**TAROM AIR**
3298	X	Air Mauritius—**AIRMAURI**
3299	X	Wideroes Flyveselskap—**WIDEROES**
4011	X	Railroads—Freight
4111	X	Transportation—Suburban and Local Commuter Passenger, including Ferries
4112	X	Passenger Railways
4131	X	Bus Lines
4511	X	Air Carriers, Airlines—not elsewhere classified
4722	X	Travel Agencies and Tour Operators
4784	X	Bridge and Road Fees, Tolls

MCC	TCC	Description
4789	X	Transportation Services—not elsewhere classified
-Z-		
6011	Z	Member Financial Institution—Automated Cash Disbursements

Glossary

This chapter defines various terms, concepts, acronyms, and abbreviations used in this document. These definitions appear for convenience only and are not to be used or otherwise relied on for any legal or technical purpose. MasterCard specifically reserves the right to amend any definition appearing herein and to interpret and apply all such definitions in its sole discretion as MasterCard deems fit.

ACH

See Automated Clearing House system.

acquirer

A member that maintains the merchant relationship and acquires the data relating to a transaction from the merchant or card acceptor.

acquirer processor system (APS)

The computer and communications system providing acquirer processing functions for a member. The APS is a single system that processes on behalf of multiple institutions or individual members. The APS may directly or indirectly support terminal devices at merchant locations. In most cases, the terms “acquirer” and “acquirer processor system” are interchangeable.

adjustment

A financial message sent from an acquirer processor to an issuer processor, indicating that a transaction did not complete as originally settled.

advice

A message that notifies a party of an action that has been taken, requiring no response.

advisement

See settlement advisement.

affiliate member

A type of MasterCard member that participates indirectly through an association member or a principal member in the activities of this corporation (for example, by issuing MasterCard® cards or by accepting transaction records from merchants).

APS

See acquirer processor system.

ATM

See automated teller machine.

ATM access fee

A fee charged by an acquirer in connection with a cash withdrawal or cash advance. Transaction initiated at that acquirer's ATM, where the MDS adds a fee to the amount of the transaction transmitted to the issuer.

ATM authorization

The approval or guarantee given by the card issuer (or its agent) to the acquirer (and/or card acceptor).

Automated Clearing House System (ACH)

One of the group of processing institutions that have networked together to exchange (clear and settle) electronic transactions.

automated teller machine (ATM)

An unattended, self-service terminal that provides cash withdrawals and other basic banking functions such as accepting deposits, ordering transfers among accounts, loan payments, and account balance inquiries. The ATM is activated by a card and requires a cardholder validation method (CVM) such as a PIN.

bank identification number (BIN)

A unique six-digit number, of which the first position is a 5, assigned by MasterCard International to identify member transactions and accounts. The BIN is the first six digits of a cardholder account number, and it is the second through seventh position of the 23-digit acquirer's reference number. Also known as the prefix.

Banknet® telecommunications network

The MasterCard worldwide telecommunications network. The primary "data transport" communications facility that links all MasterCard customers and MasterCard data processing centers into a single online financial network.

BIN

See bank identification number.

bulk data

A batch data transfer file that is transmitted through the Banknet Bulk Data Facility.

cardholder

The authorized user of a card issued by a licensed member.

cardholder number

The Primary Account Number embossed or printed on the card, or encoded in Track 2 on the card’s magnetic stripe.

chargeback

A procedure where an issuer charges all or part of the amount of an interchange transaction back to the acquirer in accordance with MasterCard regulations.

Cirrus®

Cirrus is the brand under which MasterCard International Incorporated operates the international ATM sharing network accepting MasterCard, Maestro®, and Cirrus® brands.

CIS

See Customer Interface Specification.

clearing

The process of exchanging financial transaction details between an acquirer and an issuer to facilitate posting of a cardholder's account and reconciliation of a customer's settlement position.

communication key

See key encryption key or key exchange key (KEK).

Coordinated Universal Time (UTC)

The time scale kept by time labs around the world using atomic clocks that is the basis for the worldwide system of civil time.

corresponding member (down-stream member)

A member that does not perform online message processing, clearing and/or settlement directly with the MasterCard® Debit Switch. A third party, called a processor, performs these services through the MDS on behalf of the corresponding member.

CPS

See customer processor system.

credit card

A plastic card bearing an account number assigned to a cardholder with a credit limit that can be used to purchase goods and services and to obtain cash disbursements on credit, for which a cardholder is subsequently billed by an issuer for repayment of the credit extended at once or on an installment basis.

credit line

An available credit limit applied to a specific account maintained for a bank at another bank or for a cardholder at the issuer.

credit slip (voucher)

A form establishing a refund or price adjustment to be credited to a cardholder's account for the sale of goods, services, or other items of value.

credit transaction

A transaction initiated by a merchant to effect a credit to a cardholder's account. It provides details of funds acknowledged as payable by the acquirer, the card acceptor, or both to the issuer.

cryptogram

The output from the process of transforming cleartext into ciphertext for security or privacy.

cryptographic key

A parameter used in conjunction with an algorithm for the purposes of validation, authentication, encipherment, or decipherment.

currency conversion

The process by which the transaction currency is converted into the currency of settlement or the currency of the issuer for facilitating transaction authorization, clearing, and settlement reporting.

customer

Any entity that participates in one or more MasterCard programs. Customers may be financial institutions, intermediate financial networks, payment processing associations, third-party processors, or any other entity that has a business relationship with MasterCard and a requirement to interface to MasterCard systems or communications facilities.

Customer Interface Specification (CIS)

MasterCard message formats adopted from the International Organization for Standardization (ISO) 8583 message standards that are used to communicate financial and network information between financial institutions.

customer processor system (CPS)

A computer and communications system that links a customer directly to the MasterCard® Debit Switch (MDS) network. A CPS may perform any combination of acquirer, issuer, or intermediate network facility processing functions.

data element

A component of an IPM message that is one of the 128 components defined in the ISO 8583--1993 specification. (Equivalent to a "field" in the INET format.)

Data Encryption Standard (DES)

A cryptographic algorithm adopted by the National Bureau of Standards for data security. Encryption scrambles PINs (personal identification numbers) and transaction data for safe transmission.

DDA

See demand deposit account.

DE

See data element.

debit card

A plastic card used to initiate a debit transaction. In general, these transactions are used primarily to purchase goods and services and to obtain cash, for which the cardholder's asset account is debited by the issuer.

debit MasterCard® card

An electronic financial service offered by MasterCard International that allows MasterCard cards to be used to purchase services and goods with a subsequent cardholder debit to a related asset account maintained by a financial institution; frequently referred to as an offline debit product.

debit transaction

A transaction initiated when a card is presented as payment for goods and services or at an ATM for cash withdrawals. The transaction amount is automatically deducted from the cardholder's demand deposit or other bank account.

decryption

The removal of the effects of a secret encryption code or scrambling from data, restoring the code to an intelligible form.

demand deposit account (DDA)

Any checking or shared draft account through which funds are payable on demand.

DES

See Data Encryption Standard.

down-stream member

See corresponding member.

dual control

A process of using two or more people, operating in concert, to protect cryptographic keys whereby no single person is able to access or use the keys.

dynamic key

A working key which is changed through online Network Management Messages and does not require manual intervention to change. The processor’s system ideally changes the key once every 24 hours or once every 2,500 transactions, whichever occurs first.

EBCDIC

See extended binary-coded decimal interchange code.

emergency communication key

A temporary communication key to be used by a processor and the MDS when it is determined that the current communication key used between a processor and the MDS system is either corrupt or out of sync.

encryption

The technique of modifying a known bit stream on a transmission line so that it appears to be a random sequence of bits to an unauthorized observer. It often is done automatically in the terminal or computer before data is transmitted.

end-to-end

The logical transmission path from the point of origin of a message to its final destination.

end-to-end key management

Encryption of data between its origination point and final destination, using a key unique to the end-to-end pair. Data is encrypted at the source and not decrypted until reaching the final destination. Each point of origin must maintain a unique key for each final destination it transmits to, and each final destination must maintain a unique key for each point of entry it can receive from.

extended binary coded decimal interchange code (EBCDIC)

An eight-bit code that makes possible the representation of 256 characters.

financial institution

Any commercial bank, federal or state savings and loan association, federal or state savings bank, or credit union.

financial network

Electronic funds transfer (EFT) network for financial products or services.

floor limit

A maximum monetary amount, established in accordance with particular criteria, at or above which the merchant must obtain authorization before completing the transaction.

GCMS

See Global Clearing Management System.

Global Clearing Management System (GCMS)

A centralized clearing facility owned and operated by MasterCard for the daily processing and routing of financial transactions between MasterCard and its members.

hardware encryption

A physically secure device (PSD) that encrypts sensitive data in and never appears in clear text, except at the point of entry and at the final destination. At each INF, a PSD decrypts and reencrypts cypher text.

ICA number

See MasterCard ICA number.

IDF

See Institution Definition File.

INF

See intermediate network facility.

Institution Definition File (IDF)

The computer-resident table used to identify members in the MDS.

A file, which allows the user to verify valid product participation codes as an acquirer. The file is driven by the routing and transit number (institution number.)

Institution Routing Table (IRT)

The computer-resident table of members' card prefixes or BINs used for transaction_ routing, sometimes referred to as the FIT.

integrated circuit card (ICC)

An International Organization for Standardization (ISO) standard card with an embedded integrated circuit chip. An ICC can contain one or more integrated circuits that perform processing or memory functions. Also called a smart card or chip card.

integrated circuit card interchange

The exchange of payment and/or fund transfer related to transaction data or the exchange of transaction data between acquirers and issuers in accordance with MasterCard bylaws and rules.

Integrated Product Messages (IPM)

A format for processing of clearing activity and interchange-related activity. The message format uses a variable-length and variable-format structure based on the ISO 8583-1993 standard.

Integrated Service for Intracurrency Settlement (ISIS)

See intracurrency settlement.

interchange fee

A fee applied to an interchange transaction, applicable to the two members participating in the transaction as issuer and acquirer.

intermediate network facility (INF)

Any message processing entity positioned between an acquirer or issuer. INF processors may act as acquirer processing systems (APS), issuer processing systems (IPS), or both.

intracurrency settlement

The process by which MasterCard advises credits or debits to a member's account at the MasterCard-designated local intracurrency settlement bank. Funds exchanged are in the member's currency. All participating MasterCard members in the specified country settle their MasterCard activity within the country through the MasterCard local intracountry settlement bank.

IPM

See Integrated Product Messages.

IPS

See issuer processing system.

IRT

See Institution Routing Table.

ISIS

See intracurrency settlement.

issuer

A member that issues MasterCard® cards.

issuer processing system (IPS)

The computer and communications system providing issuer processing functions for a member. The IPS may be a single system that processes for an individual customer or on behalf of multiple institutions, or may be a single system processing for an individual member.

key encryption key, or key exchange key (KEK)

Encryption key used for the sole purpose of encrypting other keys. Typically, a KEK is used to encrypt a PIN.

key management

The secure generation, allocation, distribution, and replacement of keys for a cryptosystem.

magnetic stripe

The magnetically encoded stripe on the bank card plastic that contains information pertinent to the cardholder account. The physical and magnetic characteristics of the magnetic stripe are specified in ISO Standards 7810, 7811, and 7813.

master key

A key used by an INF to encrypt working keys and communication keys for safe storage on a database.

MasterCard® ATM Network

Identifies the single electronic infrastructure and set of operating rules that support automatic teller machine (ATM) access for cards displaying either the MasterCard, Cirrus, or Maestro logos. Participation in the MasterCard®/Maestro®/Cirrus® ATM Network is the worldwide ATM/POS service that MasterCard offers to its members.

MasterCard Debit Hotcard Facility

An account management system used for debit MasterCard issuers to block cards that are lost or stolen or to indicate special account status.

MasterCard® Debit Switch (MDS)

The MasterCard Tandem-based electronic funds transfer (EFT) network processing system. This system provides debit card processing support for several MasterCard programs including MasterCard®/Cirrus® ATM Network, debit MasterCard®, and the Maestro™ point-of-sale debit program.

MasterCard ICA number

See MasterCard Interbank Card Association Number.

MasterCard Interbank Card Association Number (MasterCard ICA Number)

Unique four-digit identification number assigned by MasterCard to identify a member or processing endpoint. (Originally, the Interbank Card Association [ICA] number.)

MasterCard interface processor (MIP)

Any front-end communications processor that is placed on-site at a MasterCard customer's facility by MasterCard for the purpose of providing access to the Banknet® telecommunications network.

MasterCard settlement bank

Bank that maintains the MasterCard net settlement accounts and effects funds transfers with the Member Clearing Bank. Transfer agent accounts can reside at the settlement bank or elsewhere within a national or international payment network acceptable to MasterCard and its settlement bank.

MasterCard® card

A payment card issued by a member of MasterCard International which bears the MasterCard symbol and includes a 16-digit account number, a cardholder name, the identity of the issuer, a magnetic stripe, and other defined design features. Reference the *MasterCard® Card Design Standards* manual for specifications. Types of MasterCard cards include the Gold MasterCard® Card, MasterCard Corporate Card, MasterCard® Corporate Purchasing Card, MasterCard BusinessCard® card, MasterCard® Standard Card, and MasterCard Corporate Fleet Card®.

MDS

See MasterCard® Debit Switch.

member

A corporation or other organization that has been approved as and has entered into an agreement with MasterCard to be a MasterCard member.

merchant

A retailer, or any other person, firm, or corporation that, pursuant to a merchant agreement, agrees to MasterCard cards when properly presented.

message

A set of data elements used to exchange information between institutions (or their agents). No communications (header, trailer, protocol, or character code) or security implications are assumed or identified.

MIP

See MasterCard interface processor.

Mondex

An electronic cash system that operates with a single global technology that allows for cross-border payments. It allows up to five different currencies to be carried on the card at any one time in separate electronic pockets.

NICS™

A utility that members and processors use to access MasterCard® Debit Switch (MDS) database information to perform a variety of back office functions that support the management of their debit processing.

non-interactive message

An electronic message that does not have a requirement for direct endpoint-to-endpoint communication in real time. For example, all Advice and Advice Response messages are non-interactive messages.

non-posting transaction

An electronic funds transfer (EFT) message that does not contain sufficient information to permit posting against a cardholder's account.

PAN

See Primary Account Number or Cardholder Number.

personal identification number (PIN)

A four to 12-character secret alphanumeric code that enables an issuer to authenticate the cardholder for the purpose of approving an ATM or POS transaction occurring at a point-of-interaction device.

physically secure device (PSD)

A hardware device that cannot be successfully penetrated to disclose all or part of any PIN or key resident within such device. Penetration of a physically secure device shall cause the automatic and immediate erasure of all PINs, keys, and all useful residue of PINs and keys contained within the device.

PIN

See personal identification number.

PIN offset

The difference between the customer's chosen or assigned PIN and the PIN-derived based on a specific algorithm that is applied against the primary account number (PAN). If used, PIN offset information is encoded on track 2 of the card's magnetic stripe.

point-of-interaction (POI)

The location where a MasterCard card transaction occurs (formerly point of sale).

point-of-service (POS)

The location where a transaction is contracted.

POS terminal

A device placed at the point-of-service to process authorization requests and communicate transaction information either directly to a larger remote computer, or through a network.

posting transaction

An electronic funds transfer (EFT) message that contains sufficient information to permit posting against a cardholder's account.

preauthorization

A preauthorization is a non-posting authorization transaction that will be followed by a separate financial transaction within a specific time interval. The subsequent financial transaction will be a posting transaction against the cardholder's account. Preauthorizations are used primarily in POI situations where the cardholder wants to obtain "advance approval" or “verification” that sufficient funds are available to make a subsequent purchase using a debit card.

prefix (see also BIN)

The initial sequence of numbers from a bankcard's primary account number (PAN) used to identify the issuer uniquely. All PANs (and therefore all prefixes) must conform to ISO specifications for identifying financial institutions.

Primary Account Number (PAN)

The number that is embossed, encoded, or both, on a MasterCard card that identifies the issuer and the particular cardholder account. The PAN consists of a major industry identifier, issuer identifier, individual account identifier, and check digit.

processor

An organization that is affiliated with MasterCard International Incorporated. The processor provides authorization and/or clearing and settlement services on behalf of a member.

purchase

A financial transaction, if approved, in which funds are debited from an issuer and credited to an acquirer in payment for goods or services.

representment

A financial transaction originated by an acquirer to partially or wholly recover funds charged back to the acquirer by an issuer in a prior chargeback action.

A financial message sent from an acquirer processor to an issuer processor reversing a chargeback. This indicates that a customer's EFT claim is invalid, and requires supporting documentation.

request

A message originating an interactive series of messages.

retrieval request

The request for either an original or legible copy of the transaction information document or substitute draft as identified in the electronic record.

reversal

A message informing the receiver that the sender has canceled a previous record.

reversal credit

A credit arising from the reversal of a previous debit.

reversal debit

A debit arising from the reversal of a previous credit.

reversal transfer

A debit and a credit arising from the reversal of a previous transfer.

routing

The directional flow of messages by which the acquirer and issuer communicate with each other directly or through intermediate network facilities, which may act as an agent for the original parties involved in the message flow.

SAF

See store-and-forward.

Sanity Check

A check done by a PSD or by an INF security software module to ensure against working key corruption. This check is done by verifying that the clear text is in the expected PIN Block format.

scrip

A preprinted sales voucher for a transaction for which preauthorization has been obtained. This may be a credit transaction also, in which case only an authorization is required, not a preauthorization.

second presentment

A financial transaction originated by an acquirer to partially or wholly recover funds charged back to the acquirer by an issuer in a prior chargeback action.

settlement

The process by which MasterCard members exchange financial data and value resulting from sales transactions, cash disbursements, or merchandise credits.

settlement advisement

A message that identifies the net settlement position of one or more ICAs (or members) and is transmitted to the destination specified by the member Net Settlement Questionnaire (NSQ) form.

settlement bank

See MasterCard settlement bank.

settlement institution

A customer-designated financial institution responsible for maintaining settlement accounts (on behalf of the customer) for reimbursing other parties to a transaction, such as the acquirer or issuer.

SIP

See Stand-In processing.

SNA

See Systems Network Architecture.

software encryption

An INF using software code encrypts sensitive data. The data is in clear text within the processor itself.

Split Knowledge

A condition under which two or more parties separately and confidentially have custody of components of a single key that individually convey no knowledge of the resultant cryptographic key.

Sponsored Member

A type of MasterCard member that participates indirectly through an association member or a principal member in the activities of this corporation (for example, by issuing MasterCard® cards or by accepting transaction records from merchants.)

Stand-In processing (SIP)

An authorization subsystem within the MasterCard® Debit Switch that provides authorization services when the issuer is unavailable or does not respond within the allotted time. Stand-In accesses numerous files, including the Negative Card Facility, VIP Card Facility, and the Velocity file. Information in those files affects the determination of an authorization response given by Stand-In.

Stand-In VIP Card Facility

An account management system used by Stand-In processing (SIP) card issuers to indicate special account status and to set individual card level parameters.

static key

A working key that is typically manually loaded using dual control at both sites. A static working key has an indefinite life cycle.

store-and-forward (SAF)

A process by which transaction messages that could not be immediately delivered are automatically saved and then transmitted later to the intended receiver.

subelement

A component of a CIS message that is defined by and proprietary to MasterCard. In the CIS format, subelements have tags and length indicators. Subelements are included in data elements (DE); for example, the MasterCard Promotion Code is subelement tag 95 of Additional Data--Private Use (DE 48).

subfield

A smaller component of a data element (DE) or subelement. For example, Banknet Data (DE 63) has two subfields: Financial Network Code and Banknet Reference Number.

Switch Serial Number

The MDS-assigned trace number.

Systems Network Architecture

In IBM networks, the layered logical structure, formats, protocols, and procedures that govern information transmission.

Tandem® Terminal Emulation Software

Software that allows a PC terminal to emulate a Tandem® terminal.

transaction

Action between a cardholder and the acquirer or a cardholder that results in activity on the cardholder account.

Action between a cardholder and a merchant or a cardholder and a member that results in activity on the cardholder account.

transaction fee

A fee charged by the acquirer for transaction activity at one of the acquirer's terminals.

UTC

See Coordinated Universal Time.

working key

A cryptogram used by two processing points to encrypt data within a message or to encrypt other keys. Working Key in this manual is a generic term that would encompass other key types such as communications key. Each processing point is responsible for maintaining the secrecy of the key.

zone

The logical transmission path between two INFs.

zoned or zone key management

Encryption of data between zones using a key that is unique to the INF pair. Data that must be transmitted through several zones is decrypted and reencrypted at each INF. Each INF must maintain a unique Communication Key for each INF that it transmits to or receives from.

Forms

Institution Definition File

MasterCard International

To be Completed by Member

Indicate Action to be Taken: ☐ Add ☐ Modify ☐ Delete *(Attach approved notice to leave network)*
☐ Transfer *(New principal member must forward completed form to current principal member for approval)*

New Principal Member Contact: ____________________ Phone: ________
Current Principal Member Institution: ____________________ Phone: ________
Current Principal Member Approval (Print): ____________________ Date: ________
Current Principal Member Approval (Signature): ____________________ Date: ________

Effective Date (mm/dd/yy): ☐☐/☐☐/☐☐

For MasterCard Use:

Institution (R&T) Number: ☐☐☐☐☐☐☐☐☐ ☐☐☐☐☐☐☐☐☐

Institution Name: ____________________ Member ID/ICA Number: ☐☐☐☐

City: ____________________ State: ________

Numeric Currency Code: ____________ Numeric Country Code: ____________

Processor No.: ____________ Processor Name: ____________________

Principal Member Name: ____________________ Intracountry Settlement (Y/N): ☐

Acquirer-Only Participation (Y/N): ☐

Acquirer Product Participation: ☐ Cirrus® ☐ MasterCard ☐ Maestro®
☐ Plus ☐ Visa

Additional Comments:

To be Completed by Principal Member

Prepared by: ____________________ Date: ________ Phone: ________
Approved by: ____________________ Date: ________ Phone: ________

To be Completed by MasterCard

Approved by: ________ Date: ________ Principal ID: ________ CMAA Date: ________
Entered by: ________ Date: ________ Verified by: ________ Date: ________

Institution Routing Table

MasterCard International

To be Completed by Member

Indicate Action to be Taken: ☐ Add ☐ Modify ☐ Delete *(Attach approved notice to leave network)*
☐ Transfer *(New principal member must forward completed form to current principal member for approval)*

New Principal Member Contact: ______________________ Phone: __________
Current Principal Member Institution: ______________________ Phone: __________
Current Principal Member Approval (Print): ______________________ Date: __________
Current Principal Member Approval (Signature): ______________________ Date: __________

Effective Date (mm/dd/yy): ☐☐/☐☐/☐☐

For MasterCard Use:

Prefix Number (BIN): ☐☐☐☐☐☐☐☐☐☐☐ ☐☐☐☐☐☐☐☐☐

Institution (R&T) Number: ☐☐☐☐☐☐☐☐☐ ICA No.: ☐☐☐☐

Institution Name: ______________________ Card Type (DR or CR): __________

State: __________ Numeric Currency Code: __________ Numeric Country Code: __________

Processor No.: __________ Processor Name: ______________________

Principal Member Name: ______________________ PIN for Purchase (Y/N): ☐

Encrypted PIN Key: ______________________ Intracurrency Settlement (Y/N): ☐

Product Participation: ☐ Cirrus® ☐ Plus ☐ MasterCard ☐ Visa

Credit Processor Number: ______________________ *(if different from above)*

POS Product Participation: ☐ Maestro ☐ debit MasterCard ☐ Cirrus Purchase

POS Debit Processor Number: __________ debit MasterCard Processor Number: __________

To be Completed by Principal Member

Prepared by: ______________________ Date: __________ Phone: __________

Approved by: ______________________ Date: __________ Phone: __________

To be Completed by Regional Office (or DPS)

☐☐☐ Region ID ☐☐ ☐☐☐☐ % MCCR ☐☐ ☐☐☐☐ % ICCR ☐☐☐ Secondary Currency Code

Region Ids: 001 U.S. 002 CAN. 003 A/P 004 EUR 005 ME/A 006 LA/C

To be Completed by MasterCard

Approved by: __________ Date: __________ Principal ID: __________ CMAA Date: __________
Entered by: __________ Date: __________ Verified by: __________ Date: __________

MDS Implementation Task List

MasterCard International

Financial Institution: ____________________

Processor Number: ____________________ **Member ID/ICA Number:** __________

No.	Task Description	Individual Responsible	Date Completed
1	Submit the following to the MasterCard Regional Office:		
	• Signed product contracts		
	• Net Settlement Questionnaire (NSQ) form, if needed		
	• Principal Member Questionnaire (PMQ) form, if needed		
2	Review local Ministry of Finance regulatory regulations as they apply to processing cross border debit card transactions, if necessary.		
3	Apply for necessary approvals from the Minister of Finance, if necessary.		
4	Review local PTT regulatory regulations as they apply to processing cross border debit card transactions, if necessary.		
5	Apply for necessary approvals from PTT, if necessary.		
6	Obtain relevant MasterCard manuals from the MasterCard Regional Office:		
	• MDS Settlement and Reports Manual		
	• MDS Programs and Services Manual		
	• Banknet Systems Manual		
	• Chargeback Guide		
	• Cirrus Worldwide Operating Rules		
	• Global Deposit Access Implementation Guide for Maestro and Cirrus		
	• Maestro Operating Rules		
	• MasterCard Test Simulator - Debit, User Guide		
	• MasterCard Member Testing Procedures - Debit		
	• Quick Reference Booklet		
	• Settlement Manual		
	• MasterCard Consolidated Billing System Manual		
	• Authorization System Manual		
	• Account Management User Manual		
	• NICS™ User's Guide		
7	Review Cirrus operating rules, if applicable.		
8	Review Maestro operating rules.		
9	Review settlement alternatives.		
10	Review bulk file retrieval options.		
11	Review message format interface specifications as defined in the MDS Online Specifications manual.		
12	Review marketing materials as they relate to card issuance, card design, and signage requirements. Submit card design to the MasterCard Regional Office for approval.		
13	Review processing and telecommunications environments for both test and production requirements.		
14	Complete the following forms (for test) and forward forms as indicated on form:		
	• Institution Definition File (IDF)		

No.	Task Description	Individual Responsible	Date Completed
	• Acquirer Functionality Form		
	• Institution Routing Table (IRT)		
	• Issuer Functionality Form		
	• Processor Information Questionnaire (PIQ)		
	• Communications Key Part Exchange Form		
	• NICS™ User ID Request Form		
15	Set target implementation dates for all tasks and submit proposed schedule to Debit Payment Systems—Implementation Support for approval.		
16	Schedule test time with Debit Payment Systems—Implementation Support.	MasterCard Regional Office	
17	Send Test Plan and Test Scripts to member and/or MasterCard Regional Office.	Debit Payment Systems—Implementation Support	
18	Meet with MasterCard Regional Office staff and/or Debit Payment Systems—Implementation Support to develop the following:		
	• customized test plan, if needed		
	• customized test scripts, if needed		
	• implementation plan		
	• contingency plans		
	• migration plan, if applicable		
19	Install and test telecommunications environment.		
20	Install and test processing environment.		
21	Set up test database including the following to be used in both integration and certification testing:		
	• test accounts		
	• test plastics		
22	Receive MDS test plastics.		
23	Receive final Test Plan and Test Scripts from MasterCard Regional Office.		
24	Meet with MasterCard Regional Office staff to review following:		
	• final test plan		
	• final test scripts		
	• steps for troubleshooting during test		
	• define vendor involvement		
	• scheduled test time		
	• test keys		
25	Load debit software on test port on MIP, if applicable.	MasterCard (St. Louis)	
26	Conduct end-to-end MIP communications handshake test with MDS, if applicable.		
27	Conduct end-to-end acquirer integration testing with MDS.		
28	Review results of end-to-end acquirer integration testing with MDS and make changes, as needed. Retrieve settlement files.		
29	Conduct end-to-end acquirer certification testing with MDS.		
30	Retrieve and balance settlement files form end-to-end acquirer certification testing with MDS.		
31	Conduct end-to-end issuer integration testing with MDS.		
32	Review results of end-to-end issuer integration testing with MDS and make changes, as needed.		
	• Retrieve settlement files.		
33	Conduct end-to-end issuer certification testing with MDS.		
34	Retrieve and balance settlement files from end-to-end issuer certification testing with MDS and submit to MasterCard staff the Implementation Date Request Form.		

No.	Task Description	Individual Responsible	Date Completed
35	Confirm to MasterCard Regional Office ability to retrieve and balance settlement files.		
36	Request implementation date and time via the Implementation Date Request Form.		
37	Send via courier production Communication Key component to MDS.		
38	Submit revised production forms, if needed:		
	• Institution Definition File (IDF)	MasterCard Regional Office	
	• Institution Routing Table (IRT)	MasterCard Regional Office	
	• Processor Information Questionnaire (PIQ)	MasterCard Regional Office	
	• Production Scheduling Form	MasterCard Regional Office	
39	Send via courier the production Communication Key component and check value to member.		
40	Confirm implementation date and time to MasterCard Regional Office.	Debit Payment Systems—Implementation Support	
41	Load debit software on production port on MIP, if needed.	MasterCard (St. Louis)	
42	Coordinates post-production testing.		
	• Debit Payment Systems—Implementation Support provides acquirer post-production test deck.	MasterCard Regional Office	
	• Member provides issuer post-production test deck.	MasterCard Regional Office	
43	Confirm implementation date and time to member.	MasterCard Regional Office	
44	Conduct final pre-production conference call to review implementation details.		
45	Move certified software into production, as scheduled.		
46	Conduct post-production testing.		
47	Review post-production test results with all organizations.		
48	Conduct settlement training including:		
	• Install NICS™, if needed.		
	• NICS™ usage and exception item processing.		
	• Reconcilement.		
49	Conduct general operations training including:		
	• Trouble reporting and escalation procedures.		
	• Provide input to MOLLI for the ATM locator database.		

Implementation Task List Definitions (By Task)

Task 1—The member must submit the following documents, as indicated below, to the MasterCard Regional Office.

- Contract(s)—All Cirrus, Maestro or MasterCard contracts signed and fully executed, as appropriate.
- Net Settlement Questionnaire (NSQ)—Fully executed and signed if the financial institution is not a MasterCard principal member or it has not previously obtained a Member ID/ICA number. **If the financial institution has an existing Member ID/ICA number but they desire a different Member ID/ICA number for debit settlement it must submit a fully executed and signed NSQ.**
- Principal Member Questionnaire (PMQ)—Fully executed and signed if the financial institution is not a MasterCard principal member or it has not previously obtained a Member ID/ICA number. **If the financial institution has an existing Member ID/ICA number but they desire a different Member ID/ICA number for debit settlement, the PMQ process must be followed.**

Task 2 and Task 3—Members are encouraged to consult with legal counsel to establish applicable local Minister of Finance regulations for domestic and international access for debit cards, if applicable.

Task 4 and Task 5—Members are encouraged to consult with legal counsel to establish applicable local telecommunications regulatory regulations for domestic and international access for debit cards, if applicable.

Task 6—Members should obtain copies of the documents listed from the MasterCard Manuals and Publications Department in St. Louis. Manuals may also be obtained from the MasterCard Regional Office.

Task 7 and Task 8—Members should review all Cirrus and Maestro Operating Rules with their MasterCard Regional Office with the Debit Payments Systems—Implementation Support Manager or both.

Task 9 and Task 10—Members should review the settlement process with their MasterCard Regional Office with the Debit Payment Systems—Implementation Support Manager or both. This review should include:

- Settlement file retrieval (reports and data provided)
- Automatic reconcilement (file format and retrieval)
- NICS™ requirements
- Debit ISIS (Integrated System for Intracounty Settlement), if applicable
- Funds movement (ACH or GCMS)
- Billing Processes (MCBS)

Task 11—Members should review all aspects of the functionality to be offered with their MasterCard Regional Office with the Debit Payment Systems—Implementation Support Manager or both. This review should included the requirements for:

- Cirrus acquiring
- Cirrus issuing
- Maestro acquiring
- Maestro issuing
- Debit MasterCard issuing, if applicable

Task 12—Members should review all aspects of their marketing plan with their MasterCard Regional Office with the Debit Payment Systems—Implementation Support Manager or both. This review should include the following:

- Card issuance or re-issuance plan
- Encoding requirements
- Card design requirements
- Cirrus, Maestro logo placements, or both

Task 13—Members should review all aspects of their processing and telecommunications requirements for their MDS interface with their MasterCard Regional Office with the Debit Payment Systems—Implementation Support Manager or both. This review should include the following:

- Define MDS access (leased line or through the Banknet® telecommunications network)
- Establish capacity requirements (both hardware and software)
- Establish encryption requirements
- Define vendor participation, if applicable

Members are encouraged to maintain a permanent test environment after production implementation. A permanent test environment will facilitate the certification of additional functionality, such as the certification of upgraded hardware and software at member's site and future MIP changes or upgrades.

Task 14—Members should consult with their MasterCard Regional Office with the Debit Payment Systems—Implementation Support Manager or both to receive the latest version of these forms. At the same time, it will be established who completes each form.

Task 15 and Task 16—To be completed jointly by members and their MasterCard Regional Office and/or Debit Payment Systems—Implementation Support Manager.

Task 17—Your MasterCard Regional Office, the Debit Payment Systems—Implementation Support Manager or both will provide the Test Plan and Test Scripts for the specific implementation.

Task 18—Members will develop with their MasterCard Regional Office with the Debit Payment Systems—Implementation Support Manager, or with both, the following:

- Customized test plan and test scripts, if needed
- Implementation plan
- Contingency plan
- Migration plan, if applicable

Task 19 and Task 20—Members will establish their test environment (both processing and telecommunications). This may include installing, testing and certifying new hardware or software components prior to the start of testing with the MDS.

Task 25 and Task 26—Members using the Banknet network and a MIP for their telecommunications connectivity to the MDS, will complete these tasks prior to the start of testing with the MDS.

Task 27, Task 28, Task 29 and Task 30—Members will complete all aspects of the acquirer testing and certification process with the MDS.

Task 31, Task 32, Task 33 and Task 34—Members will complete all aspects of the issuer testing and certification process with the MDS.

Task 35 and Task 36—Members are expected to reconcile the MDS settlement reports generated during acquirer and issuer certification testing with their internal settlement reports. Members complete the MDS Certification and Implementation Date Request Form to confirm the following:

- Online interface is certified
- Product functionality is certified
- Ability to retrieve and reconcile MDS Settlement Reports
- Desired implementation date and time

Task 37, Task 38, Task 39 and Task 40—To be completed jointly by members and their MasterCard Regional Office, with the Debit Payment Systems—Implementation Support Manager, or with both.

Task 41—These tasks must be completed prior to production implementation on the MDS by CPS using the Banknet network and a MIP for their telecommunications connectivity to the MDS.

Task 42 and Task 44—Members will review Implementation Day Activities and Post-Production Test requirements with their MasterCard Regional Office, with the Debit Payment Systems—Implementation Support Manager, or with both.

Task 43—These tasks must be completed prior to production implementation on the MDS by CPS using the Banknet network and a MIP for their telecommunications connectivity to the MDS.

Task 44—Members will review Implementation Day Activities and Post-Production Test requirements with their MasterCard Regional Office, with the Debit Payment Systems—Implementation Support Manager, or with both.

Task 45—Members and MasterCard (MDS and St. Louis, as required) move the certified software and load the required database entries into the production environment at the established implementation date and time. Members must contact the MDS to initiate the production handshake and sign-on. **Members' implementation must occur within sixty (60) calendar days of certification. If the implementation is to take place after the sixty (60) calendar days period has expired, the member will be required to re-certify.**

Task 46—Conduct Post-Production Testing as agreed.

Task 47—MasterCard Regional Office or Debit Payment Systems—Implementation Support Manager will coordinate the final Post-Implementation Conference Call. The primary purpose of this Conference Call is to review any problems encountered during the Post-Production Testing and to establish a Plan of Action to correct any remaining problems.

Task 48—MasterCard Regional Office or Debit Payment Systems—Implementation Support Manager conduct settlement training.

Task 49—MasterCard Regional Office or Debit Payment Systems—Implementation Support Manager conduct general operations training.